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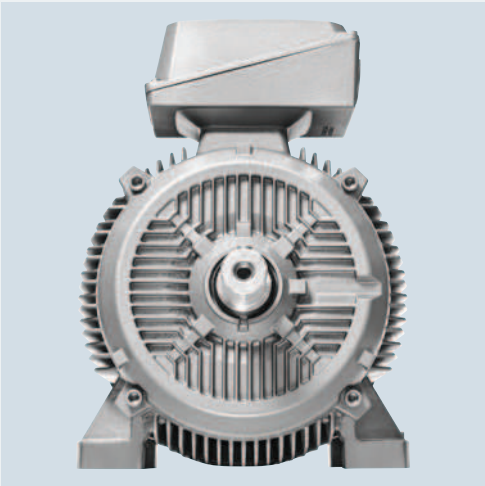
Motors
SIMOTICS
Low-Voltage Motors
Type series 1LE1, 1PC1, 1MB1

Catalog D 81.1 · 2014



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SIMOTICS GP/SD 1LE1/1PC1 Standard Motors

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Overview

Increasing energy costs have resulted in greater emphasis on the power consumption of drive systems. It is extremely important to utilize the full potential for minimizing energy consumption here to secure competitiveness today and in the future. The environment will also profit from reduced energy consumption.

This is the reason that already today we are developing a new generation of low-voltage motors. Innovative rotors create the best requisites for motors with a high degree of efficiency. IE1 and IE2 motors with the same output have the same dimensions. The new motors for IE2, IE3 and IE4 offer considerable energy savings and protect our environment. We also consider environmental sustainability during production to preserve resources. Potting compounds and coatings are, for example, solvent-free.

The modular mounting concept provides total flexibility. Each motor is based on a uniform concept for all markets worldwide. Our motors are manufactured in accordance with the latest ecological concepts and are launched on the market step by step.

The new 1LE1 motor family is therefore one of the most compact in the world, because it is manufactured using innovative technology. For an optimized design, a compound of highly conductive materials is used in the rotor (up to frame size 200). This results in minimum rotor losses and an excellent starting and switching response.

The design of the 1LE1 motors ensures maximum flexibility and minimum installation costs. Users benefit from integral eyebolts, screw-on feet, reinforced bearing plates with optimum mechanical properties and easily accessible terminal boxes. Encoders, brakes and separately driven fans can also be added without any problems. Smaller inventories make stockkeeping easier, so motor suppliers can respond to customer requirements more quickly.

The 1LE1/1PC1 motor family comprises two main series:

- SIMOTICS GP for general purpose applications:
  Motors with an aluminum housing

  SIMOTICS GP 1LE1/1PC1 motors with an aluminum housing are suitable for a wide range of standard drive tasks in the industrial environment. Thanks to their particular low weight, they are predestined for applications in pumps, fans and compressors. But they also reliably fulfill their tasks in conveyor systems and lifting gear.

Brief overview	
Output and voltage range:	0.37 ... 22 kW for all commonly used voltages
Frame sizes and types of construction:	80 ... 160 in all common types of construction
Rated speed:	750 ... 3600 rpm
Number of poles:	2, 4, 6, 8
Efficiency classes:	• IE1 (Standard Efficiency) • IE2 (High Efficiency) • IE3 (Premium Efficiency) • NEE (NEMA Energy Efficient, according to NEMA MG, Table 12-11) • NPE (NEMA Premium Efficient, according to NEMA MG, Table 12-12)

- SIMOTICS SD for severe duty applications:
  Motors with cast-iron housing

  SIMOTICS SD 1LE1 motors with a cast-iron housing are extremely rugged and are therefore the first choice for applications under harsh environmental conditions. They master dust or vibration in mills and mixers as well as the corrosive atmosphere in the petrochemical industry. Their design supports optimized heat dissipation and offers the same handling as the general purpose variants.

Brief overview	
Output and voltage range:	0.18 ... 200 kW for all commonly used voltages
Frame sizes and types of construction:	71 ... 315 in all common types of construction
Rated speed:	750 ... 3600 rpm
Number of poles:	2, 4, 6, 8
Efficiency classes:	• IE1 (Standard Efficiency) • IE2 (High Efficiency) • IE3 (Premium Efficiency) • NEE (NEMA Energy Efficient, according to NEMA MG, Table 12-11) • NPE (NEMA Premium Efficient, according to NEMA MG, Table 12-12)

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Overview (continued)

High efficiency energy-saving motors for a positive energy balance

Depending on requirements, energy-saving motors for a positive energy balance are available that are compliant with the legal requirements applicable in the European economic area in accordance with EU Directive 640/2009 as well as for the North American market in accordance with US federal law EISA (Energy Independence Security Act).

Motors with increased output and compact construction (1LE1)

Motors with increased output and compact construction can be used to advantage in confined spaces. For a slightly longer overall length, the output is at least as high as that of the next largest frame size. These compact motors are also optimized for efficiency. They are offered in IE2 and IE3 and therefore reduce operating costs.

Motors without fan cover and without external fan (1LE1 with order code F90)

Forced-air cooled motors with surface cooling without fan cover and without external fan are mainly used for driving fans.

Motors with reduced output without fan cover and without external fan (1PC1)

Naturally cooled motors with surface cooling without fan cover and without external fan are suitable for the following operating conditions:

- Types of duty with adequate cooling times (e.g. temporary duty for positioning drives)
- Environmental conditions that demand compact installation space (e.g. in motors with a stopping function)

Requirements which make an external fan disadvantageous, e.g. simple cleaning in the food industry, textile industry.

Preferred and Express motors

The most popular basic versions of motor series 1LE1 are available under special terms as so-called "Preferred motors". Most of the "Preferred motors" are also available with a shorter delivery time as so-called "Express motors".

The standard delivery time for "Express motors" is 1 to 2 days from the time of clarification of the order at the factory until dispatch from the factory. To determine the delivery date at the customer site, the appropriate shipping time must be added.

The complete range is covered by Price List D 81.1 P Part 1 "Preferred and Express Motors".

Benefits

There is considerable potential in the new 1LE1/1PC1 series of low-voltage motors. As a consistent further development of existing motors, the 1LE1/1PC1 motors offer numerous advantages.

Greater efficiency

Innovative rotor technology and manufacturing technology has been implemented for the IE1, IE2 and IE3 high efficiency motor variants. The energy-efficient motors are therefore considerably more compact.

The energy saving potential and life cycle costs of the new motors can be calculated with the SinaSave software. The SinaSave program can be downloaded from the Internet using the following link: www.siemens.com/sinasave

The 1LE1 motors also impress customers with their extremely long life and their weight-optimized design has a positive effect on the stability of the equipment unit.

A wider range of applications

The motors are certified for worldwide use and satisfy high standards of quality (confirmed, for example, by CSA [1], UL [2], CQC [3]).

Improved design

The optimized housing in modern EMC design has an attractive appearance and enhances functionality. The rotatable, accessible terminal boxes, integral eyebolts, screw-on feet and reinforced bearing plates ensure this.

Greater output

For the same frame size, the high-performance motors offer one complete rated output level more. We are also consistently implementing energy efficiency improvements here, too. The motors are offered (based on the categories of IEC 60034-30) in various efficiency classes.

More flexibility

The optimized design of the motors makes installation easier in general. Encoders, brakes and separately driven fans can be retrofitted easily. Terminal boxes and feet for flexible mounting can be selected. Smaller inventories make stockkeeping easier and motor suppliers can respond to customer requirements more quickly. Optimized manufacturing processes support fast availability. All motors up to 500 V can be operated either directly on the line or converter-fed.

For general purpose applications: SIMOTICS GP motors with an aluminum housing

Particularly user friendly

The previously introduced, well-proven, obliquely partitioned terminal box is being implemented consistently throughout the entire motor series.

Special export line

For exporting to NAFTA, the Eagle Line is available. The motors are supplied with the electrical values stamped on the rating plate in accordance with EISA requirements.

Greater output

If the motor has to be extremely compact because there is insufficient space for a standard motor, a motor with increased output could be the solution. In efficiency class IE2, these motors allow the outputs of a standard motor to be achieved in the next smallest frame size.

1) Canadian Standard Association
2) Underwriters Laboratories Inc.
3) China Quality Certification

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Benefits (continued)

For severe duty applications:
SIMOTICS SD motors with a cast-iron housing

The right motor for various challenges

The following lines are available for severe duty applications:

- Basic Line (1LE15): rugged, reliable motors for machine construction
- Performance Line (1LE16): motors for the process industry with reinforced bearings and a rugged coating – for requirements that extend beyond the Basic Line
- "Eagle Line": motors for exporting to NAFTA; they fulfill the requirements of UL and CSA and are supplied with the electrical values stamped on the rating plate in accordance with EISA requirements

Comparison: Basic Line versus Performance Line

Function	Basic Line	Performance Line
Bearing size	62 (63 from frame size 280 upwards)	63
Relubrication	Optional (standard from frame size 280 upwards)	Standard from frame size 160 upwards (optional for frame size 100 to 132)
Paint system	Standard coating, corrosion class C2 [1)]	Special coating, corrosion class C3 [1)]
Drainage	Drain plugs	T drains
Rating plate	Plastic	Steel
Motor protection	Optional	PTC
Fan cover	Plastic	Steel
Warranty	12 months	36 months

Compact design

The size of a motor is often an important aspect in the case of machines. For this reason, the 1LE1 motors in IE2 and IE3 are no longer than their predecessors in the 1LG series in IE2.

Another highlight: some of the IE3 motors fit in the same housing as the IE2 motors. The efficiency classes naturally do not differ with regard to shaft height, so that the mechanical interface to the equipment unit remains the same. This also supports a largely problem-free efficiency upgrade to IE3 – without the need to adapt the mechanical design of a machine.

Greater output

In severe duty applications, increased output motors can also be the right solution if sufficient space is not available for a standard motor. Because these motors offer the same output in the next smallest frame size.

Application

As soon as the range of motors and options is complete, it will be possible to use the 1LE1/1PC1 motors from Siemens in all areas and sectors of industry due to their numerous options. They are suitable both for special environmental conditions such as those that predominate in the chemical or petrochemical industry as well as for most climatic requirements such as those of offshore applications.

Their large range of line voltages enables them to be used all over the world.

The wide field of implementation includes the following applications:

- Pumps
- Fans
- Compressors
- Conveyor systems such as cranes, belts and lifting gear
- High-bay warehouses
- Packaging machines
- Automation and drives
- Manufacturing industry
- General machine construction

Motors with a cast-iron housing are particularly suitable for the following severe duty applications:

- Petrochemical industry
- Pharmaceuticals
- Chemical industry
- Printing industry
- Process industry

[1)] C2 and C3 are corrosion classes according to DIN EN ISO 12944. The corrosion protection must be selected in accordance with the expected corrosiveness of the environment at the installation location as well as the required service life. Five corrosion classes are defined in the above-mentioned standard, ranging from a non-corrosive indoor atmosphere (C1) to a highly corrosive industrial or marine environment (C5 I and C5 M).

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Technical specifications

Overview of technical specifications

This table lists the most important technical specifications. For more information and details, see Catalog Section 1 "Introduction".

Type of motor	SIMOTICS GP/SD 1LE1/1PC1 IEC Low-Voltage Motors
Connection types	Star connection/delta connection The connection type to be used can be established from the Article No. supplements for the required motor.
Number of poles	2, 4, 6, 8
Frame sizes	71 M ... 315 L
Rated output	0.18 ... 200 kW (1LE1 motor series)/0.3 ... 9 kW (1PC1 motor series)
Frequencies	50 Hz and 60 Hz
Versions	Self-ventilated 1LE1 energy-saving motors with: • IE1 (Standard Efficiency) • IE2 (High Efficiency) • IE3 (Premium Efficiency) • NEE (NEMA Energy Efficient, according to NEMA MG, Table 12-11) • NPE (NEMA Premium Efficient, according to NEMA MG, Table 12-12) Self-ventilated 1LE1 motors with increased output and: • IE1 (Standard Efficiency) • IE2 (High Efficiency) Forced-air cooled 1LE1 motors without external fan and fan cover with: • IE1 (Standard Efficiency) • IE2 (High Efficiency) Naturally cooled 1PC1 motors without external fan and fan cover with: • IE1 (Standard Efficiency) • IE2 (High Efficiency)
Marking	IEC 60034-30 IE1, IE2, IE3: 2, 4 and 6-pole US Energy Independence Security Act EISA: 2, 4, 6 and 8-pole
Rated speed (synchronous speed)	750 ... 3000 rpm
Rated torque	2.6 ... 1703 Nm (1LE1 motor series)
Insulation of the stator winding in accordance with EN 60034-1 (IEC 60034-1)	Temperature class 155 (F), utilized acc. to temperature class 130 (B) (also for motors with increased output) DURIGNIT IR 2000 insulation system
Degree of protection according to EN 60034-5 (IEC 60034-5)	IP55 as standard
Cooling according to EN 60034-6 (IEC 60034-6)	• Self-ventilated (1LE1 motor series) frame size 80 M to 315 L (IC 411), • Forced-air cooled (1LE1 motor series with order code **F90**) frame size 80 M to 160 L (IC 418) • Naturally cooled (1PC1 motor series) frame size 100 L to 160 L (IC 410)
Admissible coolant temperature and site altitude	–20 ... +40 °C as standard, site altitude up to 1000 m above sea level. See "Coolant temperature and site altitude" in Catalog Section 1 "Introduction".
Standard voltages according to EN 60038 (IEC 60038)	50 Hz: 230 V, 400 V, 500 V, 690 V The voltage to be used can be found in the "Selection and ordering data" for the required motor.
Type of construction according to EN 60034-7 (IEC 60034-7)	• Without flange: IM B3, IM B6, IM B7, IM B8, IM V5 without protective cover, IM V6, IM V5 with protective cover • With flange: IM B5, IM V1, IM V3, IM B35 • With standard flange and special flange (next larger flange): IM B14, IM V19, IM V18, IM B34
Paint finish Suitability of paint finish for climate group according to IEC 60721, Part 2-1	Standard: color RAL 7030 stone gray See "Paint finish" in Catalog Section 1 "Introduction".
Vibration severity level according to EN 60034-14 (IEC 60034-14)	Level A (normal – without special vibration requirements) Optionally: level B (with special vibration requirements) See "Balance and vibration quantity" in Catalog Section 1 "Introduction".
Shaft extension according to DIN 748 (IEC 60072)	Balance type: half-key balancing as standard See "Balance and vibration quantity" in Catalog Section 1 "Introduction".
Sound pressure level according to DIN EN ISO 1680 (tolerance +3 dB)	The sound pressure level is listed in the selection and ordering data for the required motor.
Weights	The weight is listed in the selection and ordering data for the required motor.
Modular mounting concept	Rotary pulse encoder, brake, separately driven fan or prepared for mountings
Consistent series concept	• Cast housing feet, screwed-on feet available as an option and retrofittable • Terminal box obliquely partitioned and rotatable through 4 x 90° • Bearings at DE and NDE are of identical design, reinforced bearings available as an option
Options	See "Supplements to article numbers and special versions"

More information

For further information, please get in touch with your local Siemens contact.

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- product or
- sector

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Converter-fed operation

Overview

Converter-fed operation up to 500 V +10 % line voltage

The standard insulation of 1LE1 motors is designed such that converter-fed operation is possible at line voltages up to $U_N \leq 500$ V. Compliance with the following limit values is essential (voltage values are peak values): $\hat{U}_{phase\text{-}to\text{-}phase} \leq 1500$ V, $\hat{U}_{phase\text{-}to\text{-}ground} \leq 1100$ V, voltage rise times of $t_s > 0.1$ µs. Operation of 1LE1 motors at higher voltage peaks (e.g. on converters with controlled input, e.g. AFE, ALM) requires motors with higher insulation resistance.

Please inquire in this case. For motors with protruding connection cables (order codes R20, R21, R22, R23 and R24) please inquire in the case of converter-fed operation.

During installation, the EMC guidelines must be complied with

Note:

When motors are operated on SINAMICS converters additional losses occur which, depending on the admissible winding temperature, can make it necessary to reduce the torque. The admissible torque values can be obtained from the SIZER configuring tool. The lowest frequency specified there is 5 Hz. For stationary converter-fed operation at lower frequencies, particularly in the case of frame sizes < 100, it is necessary to inquire at the Quotation Center.

Benefits

Motors operating with frequency converters offer the user numerous advantages.

The motors feature the future-oriented insulation system DURIGNIT IR 2000 (IR = Inverter Resistant).
The DURIGNIT IR 2000 insulating system consists of high-quality enamel wires and insulating sheet materials in conjunction with temperature-resistant resin impregnation.

The motors specially developed for operation on a frequency converter with special insulation are converter-compatible at 690 V (+10 %).

Application

The motors can be used in numerous drive applications with variable-speed drives when they are combined with converters from the MICROMASTER and SINAMICS spectrum.

The wide field of implementation includes the following applications:

- Conveyor systems such as cranes, belts and lifting gear
- High-bay warehouses
- Packaging machines
- Automation and drives

Their large range of line voltages enables them to be used all over the world.

Technical specifications

General note

All the data listed in the catalog is applicable for a 50 Hz line supply. With converter-fed operation, the torque reduction factors for constant torque and drives for fans, pumps and compressors must be configured using the "SIZER for Siemens Drives" engineering tool. Higher noise levels must be expected at frequencies other than 50 Hz for motors operating with converters due to the harmonic content of the supply.

Mechanical limit speeds

When the motor is operated at its rated frequency, it is important to note that the maximum speeds are limited by the limits for the roller bearings, critical rotor speed and rigidity of the rotating parts (see Table on page 1/44).

Motor protection

A motor protection function can be implemented using the I^2t sensing circuit implemented in the converter software.

If required, more precise motor protection can be afforded by direct temperature measurement using KTY-84 sensors or PTC thermistors in the motor winding. Some converters from Siemens determine the motor temperature using the resistance of the temperature sensor. They can be set to a required temperature for alarm and tripping.

Insulation

The insulation of 1LE motors is designed such that converter-fed operation is possible at voltages up to 500 V +10 %. $\hat{U}_{phase\text{-}to\text{-}phase} \leq 1500$ V, $\hat{U}_{phase\text{-}to\text{-}ground} \leq 1100$ V, voltage rise times of $t_s > 0.1$ µs.

All motors with voltage codes 22 and 34 must be operated on a converter under these conditions. For converter-fed operation with the outputs specified in the catalog, the motors are used according to temperature class 155 (F), i.e. in this case neither a service factor > 1 nor an increased coolant temperature is possible (order codes N01, N02 and N03 cannot be ordered).

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Selection and ordering data

The article number consists of a combination of digits and letters and is divided into three hyphenated blocks to provide a better overview, e.g.:
1LE1001-1DB22-2CB5-Z
H00

The first block (positions 1 to 7) identifies the motor type; the second block (positions 8 to 12) defines the motor frame size and length, the number of poles and in some cases the frequency/output; and in the third block (positions 13 to 16), the frequency/output, type of construction and other design features are encoded.

For deviations in the second and third block from the catalog codes, either **-Z** or **90** should be used as appropriate.

Ordering data:

- Complete Article No. and order code(s) or plain text
- If a quotation has been requested, please specify the quotation number in addition to the Article No.
- When ordering a complete motor as a spare part, please specify the works serial No. for the previously supplied motor as well as the Article No.

Structure of the Article No.:	Position:	1	2	3	4	5	6	7	-	8	9	10	11	12	-	13	14	15	16	
Positions 1 to 4: Digit, letter, letter, digit	• Self-ventilated by fan mounted on and driven by rotor • Forced-air cooled by air flow from the fan to be driven with option extension **F90**	**1**	**L**	**E**	**1**															
	• Naturally cooled without external fan and fan cover	**1**	**P**	**C**	**1**															
Position 5: Digit	Aluminum housing					**0**														
	Cast-iron housing Basic Line					**5**														
	Cast-iron housing Performance Line					**6**														
Positions 6 to 7: 2 digits	Motors with High Efficiency IE2						**0**	**1**												
	Motors with Standard Efficiency IE1						**0**	**2**												
	Motors with Premium Efficiency IE3						**0**	**3**												
	Pole-changing motors with one winding connected in Dahlander circuit						**1**	**1**												
	Pole-changing motors with two windings						**1**	**2**												
	NEMA Energy Efficient MG1 motors, Table 12-11 – Eagle Line						**2**	**1**												
	NEMA Premium Efficient MG1 motors, Table 12-12 – Eagle Line						**2**	**3**												
Positions 8, 9 and 11: Digit, letter, digit	**Motor frame size** (frame size as a combination of shaft height and overall length, encoded)									**0** ... **3**	**A** ... **E**		**0** ... **6**							
Position 10: Letter	**No. of poles** A: 2-pole, B: 4-pole, C: 6-pole, D: 8-pole, J: 4/2-pole const. load torque, L: 8/4-pole const. load torque, P: 4/2-pole square-law load torque, Q: 6/4-pole square-law load torque, R: 8/4-pole square-law Load torque											**A** ... **R**								
Positions 12 and 13: 2 digits	**Voltage, circuit and frequency** (encoded with two digits, 9-0 requires order code M.. (e. g. M1Y))													**0** ... **9**		**0** ... **8**				
Position 14: Letter	**Type of construction** (encoded with A ... V)																**A** ... **V**			
Position 15: Letter	**Motor protection** (encoded with A ... Z; Z requires order code Q.. (e. g. Q2A))																	**A** ... **Z**		
Position 16: Digit	**Terminal box position** 4: Terminal box top, 5: Terminal box right, 6: Terminal box left, 7: Terminal box below																		**4** ... **7**	
	Special order versions: encoded – additional order code required not encoded – additional plain text required																			**- Z**

Ordering example

Selection criteria	Requirement	Structure of the Article No.
Motor type 1LE1	Standard motor with High Efficiency IE2, IP55 degree of protection, aluminum housing	**1LE1001-■■■■■-■■■■**
Motor frame size/No. of poles/Speed	160 M/4-pole/1500 rpm	**1LE1001-1DB2■-■■■■**
Rated output	11 kW	
Voltage and frequency	230 VΔ/400 VY, 50 Hz	**1LE1001-1DB22-2■■■**
Type of construction with special version	IM V5 with protective cover [1]	**1LE1001-1DB22-2C■■-Z H00**
Motor protection	Motor protection with PTC thermistor with 3 embedded temperature sensors for tripping	**1LE1001-1DB22-2CB■-Z H00**
Terminal box position	Terminal box right (viewed from DE)	**1LE1001-1DB22-2CB5-Z H00**

[1] Standard without protective cover – the protective cover is defined with order code **H00** and must be ordered in addition to the Article No. with **-Z** and this order code.

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SIMOTICS GP 1LE1 Standard Motors

Motors with High Efficiency IE2

Self-ventilated or forced-air cooled motors
Aluminum series 1LE1001

IE2

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	cosφ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Aluminum series **1LE1001** – *IE2 version in accordance with IEC 60034-30* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)		kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) or with order code F90 forced-air cooled without external fan and fan cover (IC 416)																				
• Efficiency: High Efficiency IE2, service factor (SF) 1.15																				
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz 1)																				
0.75	**0.86**	**80 M**	2805	2.6	IE2	IE2	77.4	79.5	78.8	0.84	1.67	1.9	4.9	2.3	60	71	**1LE1001-0DA2■-■■■■**	9.0	0.00080	16
1.1	**1.27**	**80 M**	2835	3.7	IE2	IE2	79.6	81.3	80.8	0.83	2.40	2.7	6.0	3.1	60	71	**1LE1001-0DA3■-■■■■**	11	0.0011	16
1.5	**1.75**	**90 S**	2885	5.0	IE2	IE2	81.3	82.3	80.8	0.84	3.15	2.7	6.9	3.6	65	77	**1LE1001-0EA0■-■■■■**	13	0.0017	16
2.2	**2.55**	**90 L**	2890	7.3	IE2	IE2	83.2	83.9	82.3	0.85	4.5	2.5	7.1	3.7	65	77	**1LE1001-0EA4■-■■■■**	15	0.0021	16
3	**3.45**	**100 L**	2905	9.9	IE2	IE2	84.6	85.2	84.7	0.84	6.1	2.3	7.0	3.3	67	79	**1LE1001-1AA4■-■■■■**	21	0.0044	16
4	**4.55**	**112 M**	2950	13	IE2	IE2	85.8	86.7	86.1	0.86	7.8	2.4	7.4	3.3	69	81	**1LE1001-1BA2■-■■■■**	27	0.0092	16
5.5	**6.3**	**132 S**	2950	18	IE2	IE2	87.0	88.0	87.4	0.87	10.5	1.8	6.6	2.9	68	80	**1LE1001-1CA0■-■■■■**	39	0.020	16
7.5	**8.6**	**132 S**	2950	24	IE2	IE2	88.1	88.7	88.6	0.87	14.1	2.2	7.5	3.1	68	80	**1LE1001-1CA1■-■■■■**	43	0.024	16
11	**12.6**	**160 M**	2955	36	IE2	IE2	89.4	90.0	89.1	0.87	20.5	2.1	7.4	3.2	70	82	**1LE1001-1DA2■-■■■■**	67	0.045	16
15	**17.3**	**160 M**	2955	48	IE2	IE2	90.3	90.9	90.3	0.88	27	2.4	7.6	3.4	70	82	**1LE1001-1DA3■-■■■■**	75	0.053	16
18.5	**21.3**	**160 L**	2955	60	IE2	IE2	90.9	91.2	90.4	0.88	33.5	2.9	7.9	3.6	70	82	**1LE1001-1DA4■-■■■■**	84	0.061	16
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz 1)																				
0.55	**0.63**	**80 M**	1440	3.7	–	–	78.1	78.9	76.1	0.74	1.37	2.2	5.3	3.1	53	64	**1LE1001-0DB2■-■■■■**	10	0.0017	16
0.75	**0.86**	**80 M**	1440	5.0	IE2	IE2	79.6	80.2	78.0	0.76	1.79	2.2	5.6	3.1	53	64	**1LE1001-0DB3■-■■■■**	11	0.0021	16
1.1	**1.27**	**90 S**	1425	7.4	IE2	IE2	81.4	81.7	79.9	0.78	2.5	2.3	5.6	2.9	56	68	**1LE1001-0EB0■-■■■■**	13	0.0028	16
1.5	**1.75**	**90 L**	1435	10	IE2	IE2	82.8	83.5	82.0	0.79	3.3	2.6	6.4	3.4	56	68	**1LE1001-0EB4■-■■■■**	16	0.0036	16
2.2	**2.55**	**100 L**	1455	14	IE2	IE2	84.3	85.1	84.3	0.81	4.65	2.1	6.9	3.3	60	72	**1LE1001-1AB4■-■■■■**	21	0.0086	16
3	**3.45**	**100 L**	1455	20	IE2	IE2	85.5	86.7	86.0	0.82	6.2	2.0	6.9	3.1	60	72	**1LE1001-1AB5■-■■■■**	25	0.011	16
4	**4.55**	**112 M**	1460	26	IE2	IE2	86.6	87.3	86.5	0.81	8.2	2.5	7.1	3.2	58	70	**1LE1001-1BB2■-■■■■**	29	0.014	16
5.5	**6.3**	**132 S**	1465	36	IE2	IE2	87.7	89.0	87.7	0.80	11.3	2.3	6.9	2.9	64	76	**1LE1001-1CB0■-■■■■**	42	0.027	16
7.5	**8.6**	**132 M**	1465	49	IE2	IE2	88.7	90.3	88.8	0.83	14.7	2.3	6.9	2.9	64	76	**1LE1001-1CB2■-■■■■**	49	0.034	16
11	**12.6**	**160 M**	1470	71	IE2	IE2	89.8	90.9	90.8	0.85	21	2.1	6.7	2.8	65	77	**1LE1001-1DB2■-■■■■**	71	0.065	16
15	**17.3**	**160 L**	1475	97	IE2	IE2	90.6	91.3	91.0	0.85	28	2.3	7.3	3.0	65	77	**1LE1001-1DB4■-■■■■**	83	0.083	16

Voltages				**Motor protection**	No. of poles	Frame size	Motor type	Version					Order code(s)
Frame sizes 80 M to 90 L 2)													
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	PTC thermistor with 1 temp. sensor	2, 4	80 M ... 90 L	1LE1001-0D ... -0E	**Standard**	**2**	**2**			–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ		2, 4	80 M ... 90 L	1LE1001-0D ... -0E	**Standard**	**3**	**4**			–
50 Hz	400 VY	60 Hz 1)	460 VY	Without	2, 4	80 M ... 90 L	1LE1001-0D ... -0E	**Standard**	**0**	**2**	**A**		–
Frame sizes 100 L to 160 L: use of the 4 x 90° rotatable terminal box													
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	Any	2, 4	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**2**	**2**			–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ	Any	2, 4	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**3**	**4**			–
50 Hz	500 VY			Any	2, 4	100 L ... 160 L	1LE1001-1A ... -1D	Without add. charge	**2**	**7**			–
50 Hz	500 VΔ			Any	2, 4	100 L ... 160 L	1LE1001-1A ... -1D	Without add. charge	**4**	**0**			–
Further voltages 1)				For price information, code numbers, order codes and descriptions, see from Page 2/38					**9**	**0**			. . .

Types of construction		No. of poles	Frame size	Motor type	Version			Order code(s)
Without flange	IM B3 3)	2, 4	80 M ... 160 L	1LE1001-0D ... -1D	**Standard**	**A**		–
With flange	IM B5 3)	2, 4	80 M ... 160 L	1LE1001-0D ... -1D	With additional charge	**F**		–
With standard flange	IM B14 3)	2, 4	80 M ... 160 L	1LE1001-0D ... -1D	With additional charge	**K**		–
Further types of construction	For price information, code letters and descriptions, see from Page 2/41					■		. . .

Motor protection	No. of poles	Frame size	Motor type	Version			Order code(s)
Frame sizes 100 L to 160 L: use of the 4 x 90° rotatable terminal box							
Without	2, 4	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**A**		–
PTC thermistor with 3 temperature sensors	2, 4	100 L ... 160 L	1LE1001-1A ... -1D	With additional charge	**B**		–
Further motor protection	For price information, code letters and descriptions, see from Page 2/49				■		. . .

Terminal box position	No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top	2, 4	80 M ... 160 L	1LE1001-0D ... -1D	**Standard**	**4**	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/51					

Special versions	No. of poles	Frame size	Motor type		Order code(s)
Forced-air cooled motors without ext. fan/fan cover (IC 416)	2, 4	80 M ... 160 L	1LE1001-0D ... -1D	**1LE1001-. . . . ■-■■■■-Z**	**F90 +. . .+. . .+. . .**
Options	For price information, order codes and descriptions, see from Page 2/53			**1LE1001-. . . . ■-■■■■-Z**	**. . .+. . .+. . .+. . .**

1) Operating values at rated output for 60 Hz are available on request.

2) For converter-fed operation of shaft heights 80 and 90, ordering with PTC thermistors and their connection to the converter is recommended.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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Selection and ordering data (continued)

Operating values at rated output

P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	cosφ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	**Aluminum series 1LE1001** – *IE2 version in accordance with IEC 60034-30* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)		kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) or with order code F90 forced-air cooled without external fan and fan cover (IC 416) • Efficiency: High Efficiency IE2, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz 1)																				
0.37	**0.43**	**80 M**	925	3.85	–	–	71.4	71.5	66.5	0.69	1.08	2.1	4.0	2.4	42	53	**1LE1001-0DC2■-■■■■**	9	0.0017	16
0.55	**0.63**	**80 M**	935	5.6	–	–	74.0	74.0	70.5	0.66	1.63	2.5	4.4	2.9	42	53	**1LE1001-0DC3■-■■■■**	12	0.0025	16
0.75	**0.86**	**90 S**	925	7.7	IE2	IE2	75.9	76.0	73.0	0.70	2.05	2.0	4.1	2.5	43	55	**1LE1001-0EC0■-■■■■**	13	0.0030	16
1.1	**1.27**	**90 L**	935	11.2	IE2	–	78.1	78.5	75.0	0.70	2.90	2.2	4.4	2.6	43	55	**1LE1001-0EC4■-■■■■**	16	0.0040	16
1.5	**1.75**	**100 L**	970	15	IE2	IE2	79.8	80.2	79.0	0.73	3.7	2.0	6.2	2.9	59	71	**1LE1001-1AC4■-■■■■**	25	0.011	16
2.2	**2.55**	**112 M**	965	22	IE2	IE2	81.8	82.5	81.3	0.75	5.2	2.1	6.0	3.1	57	69	**1LE1001-1BC2■-■■■■**	29	0.014	16
3	**3.45**	**132 S**	970	30	IE2	IE2	83.3	84.0	82.8	0.74	7.0	1.6	5.6	2.6	63	75	**1LE1001-1CC0■-■■■■**	38	0.024	13
4	**4.55**	**132 M**	970	39	IE2	IE2	84.6	85.8	85.0	0.78	8.7	1.6	5.6	2.5	63	75	**1LE1001-1CC2■-■■■■**	43	0.029	13
5.5	**6.3**	**132 M**	970	54	IE2	IE2	86.0	87.4	87.0	0.77	12	1.9	6.1	2.8	63	75	**1LE1001-1CC3■-■■■■**	52	0.037	16
7.5	**8.6**	**160 M**	975	73	IE2	IE2	87.2	88.0	87.3	0.74	16.8	1.9	4.7	2.2	67	79	**1LE1001-1DC2■-■■■■**	77	0.075	16
11	**12.6**	**160 L**	975	108	IE2	IE2	88.7	89.6	89.2	0.76	23.5	1.9	4.8	2.2	67	79	**1LE1001-1DC4■-■■■■**	93	0.098	16
8-pole: 750 rpm at 50 Hz, 900 rpm at 60 Hz 1)																				
0.75	**0.86**	**100 L**	725	9.9	–	–	68.3	65.8	59.3	0.58	2.75	1.6	4.0	2.8	60	72	**1LE1001-1AD4■-■■■■**	21	0.0086	13
1.1	**1.27**	**100 L**	725	14	–	–	68.3	65.4	58.9	0.58	4.0	1.8	4.1	2.8	60	72	**1LE1001-1AD5■-■■■■**	25	0.011	13
1.5	**1.75**	**112 M**	720	20	–	–	75.8	76.0	73.0	0.67	4.25	1.4	4.2	2.4	63	75	**1LE1001-1BD2■-■■■■**	29	0.014	13
2.2	**2.55**	**132 S**	725	29	–	–	78.8	79.3	77.2	0.65	6.2	1.4	4.3	2.1	63	75	**1LE1001-1CD0■-■■■■**	41	0.027	10
3	**3.45**	**132 M**	730	39	–	–	82.7	83.0	80.9	0.65	8.1	1.4	5.0	2.4	63	75	**1LE1001-1CD2■-■■■■**	49	0.035	10
4	**4.55**	**160 M**	730	52	–	–	81.9	82.6	81.7	0.67	10.5	1.6	3.7	1.9	63	75	**1LE1001-1DD2■-■■■■**	69	0.065	13
5.5	**6.3**	**160 M**	730	72	–	–	83.8	84.3	83.1	0.67	14.1	1.7	3.9	2	63	75	**1LE1001-1DD3■-■■■■**	82	0.083	13
7.5	**8.6**	**160 L**	730	98	–	–	85.3	86.5	86.1	0.7	18.1	1.6	3.8	1.9	63	75	**1LE1001-1DD4■-■■■■**	94	0.098	13

Voltages				No. of poles	Frame size	Motor type	Version					Order code(s)
Frame sizes 80 M to 90 L 2)												
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	PTC thermistor with 1 temp. sensor	6	80 M ... 90 L	1LE1001-0D ... -0E	**Standard**	**2**	**2**		–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ		6	80 M ... 90 L	1LE1001-0D ... -0E	**Standard**	**3**	**4**		–
50 Hz	400 VY	60 Hz 1)	460 VY	Without	6	80 M ... 90 L	1LE1001-0D ... -0E	**Standard**	**0**	**2**	**A**	–
Frame sizes 100 L to 160 L: use of the 4 x 90° rotatable terminal box												
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY		6, 8	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**2**	**2**		–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ		6, 8	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**3**	**4**		–
50 Hz	500 VY				6, 8	100 L ... 160 L	1LE1001-1A ... -1D	Without add. charge	**2**	**7**		–
50 Hz	500 VΔ				6, 8	100 L ... 160 L	1LE1001-1A ... -1D	Without add. charge	**4**	**0**		–
Further voltages 1)				For price information, code numbers, order codes and descriptions, see from Page 2/38					**9**	**0**		. . .

Types of construction			No. of poles	Frame size	Motor type	Version		Order code(s)
Without flange	IM B3 3)		6, 8	80 M ... 160 L	1LE1001-0D ... -1D	**Standard**	**A**	–
With flange	IM B5 3)		6, 8	80 M ... 160 L	1LE1001-0D ... -1D	With additional charge	**F**	–
With standard flange	IM B14 3)		6, 8	80 M ... 160 L	1LE1001-0D ... -1D	With additional charge	**K**	–
Further types of construction		For price information, code letters and descriptions, see from Page 2/41					■	. . .

Motor protection		No. of poles	Frame size	Motor type	Version		Order code(s)
Frame sizes 100 L to 160 L: use of the 4 x 90° rotatable terminal box							
Without		6, 8	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**A**	–
PTC thermistor with 3 temperature sensors		6, 8	100 L ... 160 L	1LE1001-1A ... -1D	With additional charge	**B**	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/49					■	. . .

Terminal box position		No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top		6, 8	80 M ... 160 L	1LE1001-0D ... -1D	**Standard**	**4**	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/51						

Special versions		No. of poles	Frame size	Motor type		Order code(s)
Forced-air cooled motors without ext. fan/fan cover (IC 416)		6, 8	80 M ... 160 L	1LE1001-0D ... -1D	**1LE1001-. . . . ■-■■■■-Z**	**F90 +. . .+. . .+. . .**
Options	For price information, order codes and descriptions, see from Page 2/53				**1LE1001-. . . . ■-■■■■-Z**	**. . .+. . .+. . .+. . .**

1) Operating values at rated output for 60 Hz are available on request.

2) For converter-fed operation of shaft heights 80 and 90, ordering with PTC thermistors and their connection to the converter is recommended.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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Self-ventilated motors
Aluminum series 1LE1001 with increased output

IE2

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz [1]	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	cosφ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Aluminum series 1LE1001 – *IE2 version in accordance with IEC 60034-30 with increased output* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)		kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) • Efficiency: High Efficiency IE2, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz [1]																				
4	**4.55**	**100 L**	2905	13	IE2	IE2	85.8	87.2	87.0	0.86	7.8	2.5	7.6	3.5	67	79	**1LE1001-1AA6■-■■■■**	26	0.0054	16
5.5	**6.3**	**112 M**	2950	18	IE2	IE2	87.0	87.5	87.2	0.89	10.3	2.2	7.7	3.3	69	81	**1LE1001-1BA6■-■■■■**	34	0.012	16
11	**12.6**	**132 M**	2950	36	IE2	IE2	89.4	90.2	90.3	0.89	20	2.3	7.9	3.2	68	80	**1LE1001-1CA6■-■■■■**	57	0.031	16
22	**25.3**	**160 L**	2955	71	IE2	IE2	91.3	91.7	91.3	0.89	39	3.1	8.4	3.7	70	82	**1LE1001-1DA6■-■■■■**	94	0.068	16
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz [1]																				
4	**4.55**	**100 L**	1460	26	IE2	IE2	86.6	87.4	86.7	0.80	8.3	2.2	7.5	3.5	60	72	**1LE1001-1AB6■-■■■■**	30	0.014	16
5.5	**6.3**	**112 M**	1460	36	IE2	IE2	87.7	88.1	87.4	0.81	11.2	2.5	7.1	3.2	58	70	**1LE1001-1BB6■-■■■■**	34	0.017	16
11	**12.6**	**132 M**	1465	72	IE2	IE2	89.8	90.6	90.4	0.84	21	2.6	7.7	3.1	64	76	**1LE1001-1CB6■-■■■■**	64	0.046	16
18.5	**21.3**	**160 L**	1475	120	IE2	IE2	91.2	91.7	91.6	0.85	34.5	2.5	7.7	3.3	65	77	**1LE1001-1DB6■-■■■■**	100	0.099	16
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz [1]																				
2.2	**2.55**	**100 L**	965	22	IE2	IE2	81.8	82.5	81.5	0.76	5.1	1.9	5.7	2.9	59	71	**1LE1001-1AC6■-■■■■**	30	0.014	16
3	**3.45**	**112 M**	960	30	IE2	IE2	83.3	84.1	83.6	0.79	6.6	2.1	6.0	3.1	57	69	**1LE1001-1BC6■-■■■■**	34	0.017	16
7.5	**8.6**	**132 M**	970	74	IE2	IE2	87.2	87.8	87.3	0.77	16.1	2.1	6.5	3.0	63	75	**1LE1001-1CC6■-■■■■**	64	0.046	16
15	**17.3**	**160 L**	975	147	IE2	IE1	89.7	90.3	89.7	0.75	32.0	2	5.2	2.4	67	79	**1LE1001-1DC6■-■■■■**	115	0.12	16

Voltages				No. of poles	Frame size	Motor type	Version			Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz [1]	460 VY	2, 4, 6	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**2**	**2**	–
50 Hz	400 VΔ/690 VY	60 Hz [1]	460 VΔ	2, 4, 6	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**3**	**4**	–
50 Hz	500 VY			2, 4, 6	100 L ... 160 L	1LE1001-1A ... -1D	Without add. charge	**2**	**7**	–
50 Hz	500 VΔ			2, 4, 6	100 L ... 160 L	1LE1001-1A ... -1D	Without add. charge	**4**	**0**	–
Further voltages [1]				For price information, code numbers, order codes and descriptions, see from Page 2/38				**9**	**0**	. . .

Types of construction		No. of poles	Frame size	Motor type	Version		Order code(s)
Without flange	IM B3 [2]	2, 4, 6	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**A**	–
With flange	IM B5 [2]	2, 4, 6	100 L ... 160 L	1LE1001-1A ... -1D	With additional charge	**F**	–
With standard flange	IM B14 [2]	2, 4, 6	100 L ... 160 L	1LE1001-1A ... -1D	With additional charge	**K**	–
Further types of construction		For price information, code letters and descriptions, see from Page 2/41				**■**	. . .

Motor protection	No. of poles	Frame size	Motor type	Version		Order code(s)
Without	2, 4, 6	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**A**	–
PTC thermistor with 3 temperature sensors	2, 4, 6	100 L ... 160 L	1LE1001-1A ... -1D	With additional charge	**B**	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/49				**■**	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top	2, 4, 6	100 L ... 160 L	1LE1001-1A ... -1D	**Standard**	**4**	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/51					

Special versions	No. of poles	Frame size	Motor type		Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/53			**1LE1001-. . . . ■-■■■■-Z**	. . .+. . .+. . .+. . .

[1] Operating values at rated output for 60 Hz are available on request.

[2] Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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IE2

Naturally cooled motors without external fan
Aluminum series 1PC1001

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class	η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	cosφ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	**Aluminum series 1PC1001** – *IE2 version in accordance with IEC 60034-30* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)		kg	kgm²	CL
• Cooling: naturally cooled without external fan (IC 410) • Efficiency: High Efficiency IE2, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz 1)																			
1.4	–	**100 L**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1AA4■-■■■■**	21	0.0044	13
1.6	–	**112 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1BA2■-■■■■**	27	0.0092	16
3.1	–	**132 S**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1CA0■-■■■■**	39	0.020	13
4.3	–	**132 S**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1CA1■-■■■■**	43	0.024	13
6.3	–	**160 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1DA2■-■■■■**	67	0.045	10
6.5	–	**160 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1DA3■-■■■■**	75	0.053	13
9	–	**160 L**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1DA4■-■■■■**	84	0.061	16
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz 1)																			
1.1	–	**100 L**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1AB4■-■■■■**	21	0.0086	13
1.5	–	**100 L**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1AB5■-■■■■**	25	0.011	13
2	–	**112 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1BB2■-■■■■**	29	0.014	13
2.6	–	**132 S**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1CB0■-■■■■**	42	0.027	13
4	–	**132 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1CB2■-■■■■**	49	0.034	13
6	–	**160 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1DB2■-■■■■**	71	0.065	10
6.2	–	**160 L**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1DB4■-■■■■**	83	0.083	16
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz 1)																			
0.85	–	**100 L**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1AC4■-■■■■**	25	0.011	10
1.2	–	**112 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1BC2■-■■■■**	29	0.014	10
1.5	–	**132 S**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1CC0■-■■■■**	38	0.024	7
2.5	–	**132 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1CC2■-■■■■**	43	0.029	7
2.7	–	**132 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1CC3■-■■■■**	52	0.037	13
5	–	**160 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1DC2■-■■■■**	77	0.075	10
6.5	–	**160 L**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1DC4■-■■■■**	93	0.098	10
8-pole: 750 rpm at 50 Hz, 900 rpm at 60 Hz 1)																			
0.37	–	**100 L**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1AD4■-■■■■**	21	0.0086	10
0.55	–	**100 L**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1AD5■-■■■■**	25	0.011	10
0.75	–	**112 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1BD2■-■■■■**	29	0.014	7
1.1	–	**132 S**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1CD0■-■■■■**	41	0.027	7
1.5	–	**132 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1CD2■-■■■■**	49	0.035	7
2.4	–	**160 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1DD2■-■■■■**	69	0.065	10
3.3	–	**160 M**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1DD3■-■■■■**	82	0.083	10
4.6	–	**160 L**	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	**1PC1001-1DD4■-■■■■**	94	0.098	10

Voltages				No. of poles	Frame size	Motor type	Version				Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	2, 4, 6, 8	100 L ... 160 L	1PC1001-1A ... -1D	**Standard**	**2**	**2**		–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ	2, 4, 6, 8	100 L ... 160 L	1PC1001-1A ... -1D	**Standard**	**3**	**4**		–
50 Hz	500 VY			2, 4, 6, 8	100 L ... 160 L	1PC1001-1A ... -1D	Without add. charge	**2**	**7**		–
50 Hz	500 VΔ			2, 4, 6, 8	100 L ... 160 L	1PC1001-1A ... -1D	Without add. charge	**4**	**0**		–
Further voltages 1)				For price information, code numbers, order codes and descriptions, see from Page 2/38				**9**	**0**		. . .
Types of construction				No. of poles	Frame size	Motor type	Version				Order code(s)
Without flange		IM B3 2)		2, 4, 6, 8	100 L ... 160 L	1PC1001-1A ... -1D	**Standard**			**A**	–
With flange		IM B5 2)		2, 4, 6, 8	100 L ... 160 L	1PC1001-1A ... -1D	With additional charge			**F**	–
With standard flange		IM B14 2)		2, 4, 6, 8	100 L ... 160 L	1PC1001-1A ... -1D	With additional charge			**K**	–
Further types of construction				For price information, code letters and descriptions, see from Page 2/41						■	. . .
Motor protection				No. of poles	Frame size	Motor type	Version				Order code(s)
Without				2, 4, 6, 8	100 L ... 160 L	1PC1001-1A ... -1D	**Standard**			**A**	–
PTC thermistor with 3 temperature sensors				2, 4, 6, 8	100 L ... 160 L	1PC1001-1A ... -1D	With additional charge			**B**	–
Further motor protection				For price information, code letters and descriptions, see from Page 2/49						■	. . .
Terminal box position				No. of poles	Frame size	Motor type	Version				Order code(s)
Terminal box at top				2, 4, 6, 8	100 L ... 160 L	1PC1001-1A ... -1D	**Standard**			**4**	–
Further terminal box positions				For price information, code numbers and descriptions, see from Page 2/51							
Special versions				No. of poles	Frame size	Motor type					Order code(s)
Options				For price information, order codes and descriptions, see from Page 2/53			**1LE1001-. . . . ■-■■■■-Z**				. . .+. . .+. . .+. . .

Note: The rated outputs and weights may change slightly after they have been checked. Further electrical data can be calculated and supplied on receipt of order.

1) Operating values at rated output for 60 Hz are available on request.

2) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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SIMOTICS SD 1LE1 Standard Motors

Motors with High Efficiency IE2

Self-ventilated motors
Cast-iron series 1LE1501/1LE1601 Basic/Performance Line

IE2

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz [1]	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	$\cos\varphi$ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Cast-iron series: 1LE1501 – *Basic Line*; 1LE1601 – *Performance Line* IE2 version in accordance with IEC 60034-30. Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) • Efficiency: High Efficiency IE2, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz [1]																				
0.37	0.5	71 M	2770	1.3	IE2		69.5	70.5	68.5	0.81	0.95	2.5	4.1	2.5	52	63	▲ 1LE1 5 01-0CA2 ■-■■■■	11.5	0.00035	16
0.55	0.75	71 M	2780	1.9	IE2		74.1	75.0	73.1	0.80	1.34	2.6	4.6	2.6	52	63	▲ 1LE1 5 01-0CA3 ■-■■■■	13	0.00045	16
0.75	1	80 M	2805	2.6	IE2		77.4	79.5	78.8	0.84	1.67	1.9	4.9	2.3	60	71	▲ 1LE1 5 01-0DA2 ■-■■■■	16	0.00080	16
1.1	1.5	80 M	2835	3.7	IE2		79.6	81.3	80.8	0.83	2.40	2.7	6.0	3.1	60	71	▲ 1LE1 5 01-0DA3 ■-■■■■	18	0.0011	16
1.5	2	90 S	2885	5.0	IE2		81.3	82.3	80.8	0.84	3.15	2.7	6.9	3.6	65	77	▲ 1LE1 5 01-0EA0 ■-■■■■	23	0.0017	16
2.2	3	90 L	2890	7.3	IE2		83.2	83.9	82.3	0.85	4.5	2.5	7.1	3.7	65	77	▲ 1LE1 5 01-0EA4 ■-■■■■	25.5	0.0021	16
3	3.45	100 L	2905	9.9	IE2	IE2	84.6	85.2	84.7	0.84	6.1	2.3	7.0	3.3	67	79	1LE1 ■ 01-1AA4 ■-■■■■	32	0.0044	16
4	4.55	112 M	2950	13	IE2	IE2	85.8	86.7	86.1	0.86	7.8	2.4	7.4	3.3	69	81	1LE1 ■ 01-1BA2 ■-■■■■	39	0.0092	16
5.5	6.3	132 S	2950	18	IE2	IE2	87.0	88.0	87.4	0.87	10.5	1.8	6.6	2.9	68	80	1LE1 ■ 01-1CA0 ■-■■■■	57	0.020	16
7.5	8.6	132 S	2950	24	IE2	IE2	88.1	88.7	88.6	0.87	14.1	2.2	7.5	3.1	68	80	1LE1 ■ 01-1CA1 ■-■■■■	61	0.024	16
11	12.6	160 M	2955	36	IE2	IE2	89.4	90.0	89.1	0.87	20.5	2.1	7.4	3.2	70	82	1LE1 ■ 01-1DA2 ■-■■■■	96	0.045	16
15	17.3	160 M	2955	48	IE2	IE2	90.3	90.9	90.3	0.88	27	2.4	7.6	3.4	70	82	1LE1 ■ 01-1DA3 ■-■■■■	104	0.053	16
18.5	21.3	160 L	2955	60	IE2	IE2	90.9	91.2	90.4	0.88	33.5	2.9	7.9	3.6	70	82	1LE1 ■ 01-1DA4 ■-■■■■	113	0.061	16
22	24.5	180 M	2940	71	IE2	IE2	91.3	91.8	91.4	0.87	40.5	2.7	7.4	3.6	69	83	1LE1 ■ 01-1EA2 ■-■■■■	145	0.069	16
30	33.5	200 L	2955	97	IE2	IE2	92.0	92.3	91.7	0.87	54	2.5	6.9	3.3	72	85	1LE1 ■ 01-2AA4 ■-■■■■	200	0.13	16
37	41.5	200 L	2960	119	IE2	IE2	92.5	92.8	92.3	0.88	66	2.7	7.4	3.5	69	82	1LE1 ■ 01-2AA5 ■-■■■■	225	0.15	16
45	51	225 M	2965	145	IE2	IE2	92.9	93.1	92.5	0.88	79	2.7	7.8	3.7	76	89	1LE1 ■ 01-2BA2 ■-■■■■	295	0.23	16
55	62	250 M	2970	177	IE2	IE2	93.2	93.3	92.4	0.88	97	2.3	6.8	3.1	76	89	1LE1 ■ 01-2CA2 ■-■■■■	360	0.40	13
75	84	280 S	2978	240	IE2	IE2	93.8	93.6	92.4	0.86	137	2.5	7.2	3.2	76	89	1LE1 ■ 01-2DA0 ■-■■■■	490	0.71	13
90	101	280 M	2975	289	IE2	IE2	94.1	94.2	93.5	0.88	157	2.5	7.1	3.1	76	89	1LE1 ■ 01-2DA2 ■-■■■■	530	0.83	13
110	123	315 S	2982	352	IE2	IE2	94.3	94.2	93.3	0.90	187	2.4	7.3	3.0	77	91	1LE1 ■ 01-3AA0 ■-■■■■	720	1.3	13
132	148	315 M	2982	423	IE2	IE2	94.6	94.7	94.1	0.91	220	2.4	7.2	3.1	77	91	1LE1 ■ 01-3AA2 ■-■■■■	880	1.6	13
160	180	315 L	2982	512	IE2	IE2	94.8	94.9	94.3	0.92	265	2.3	7.0	3.1	80	95	1LE1 ■ 01-3AA4 ■-■■■■	930	1.8	13
200	224	315 L	2982	640	IE2	IE2	95.0	95.2	94.8	0.92	330	2.5	7.3	3.0	80	95	1LE1 ■ 01-3AA5 ■-■■■■	1130	2.2	13

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects						
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	up to 500 V	12 months	5					
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6					

Voltages [2]			No. of poles	Frame size	Motor type	Version						Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz [1]	460 VY	2	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**	2	2			–
50 Hz	400 VΔ/690 VY	60 Hz [1]	460 VΔ	2	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**	3	4			–
50 Hz	500 VY			2	71 M ... 315 L	1LE1■01-0C … -3A	Without add. charge	2	7			–
50 Hz	500 VΔ			2	71 M ... 315 L	1LE1■01-0C … -3A	Without add. charge	4	0			–
Further voltages [1]	For price information, code numbers, order codes and descriptions, see from Page 2/40							9	0			. . .

Types of construction		No. of poles	Frame size	Motor type	Version						Order code(s)
Without flange	IM B3 [3]	2	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**			A			–
With flange	IM B5 [3]	2	71 M ... 315 M	1LE1■01-0C … -3A	With additional charge			F			–
With standard flange	IM B14 [3]	2	71 M ... 160 L	1LE1■01-0C … -1D	With additional charge			K			–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45							■			. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version						Order code(s)
Without	Only possible for **Basic Line**	2	71 M ... 315 L	1LE1501-0C … -3A	**Standard**				A		–
PTC thermistor with 3 temperature sensors	**Basic Line**	2	71 M ... 315 L	1LE1501-0C … -3A	With additional charge				B		–
	Performance Line	2	71 M ... 315 L	1LE1601-0C … -3A	**Standard**				B		–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50								■		. . .

Terminal box position	No. of poles	Frame size	Motor type	Version						Order code(s)
Terminal box at top	2	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**					4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52									

Special versions	No. of poles	Frame size	Motor type	Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58		1LE1■01-. . . . ■-■■■■ -Z	. . .+. . .+. . .+. . .

1) Operating values at rated output for 60 Hz are available on request.

2) Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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IE2

Self-ventilated motors
Cast-iron series 1LE1501/1LE1601 Basic/Performance Line

Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz [1]	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	$\cos\varphi$ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Cast-iron series: 1LE1501 – *Basic Line*; 1LE1601 – *Performance Line IE2 version in accordance with IEC 60034-30*. Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411)																				
• Efficiency: High Efficiency IE2, service factor (SF) 1.15																				
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz [1]																				
0.25	**0.33**	**71 M**	1395	1.7	IE2		68.5	68.2	63.8	0.69	0.76	2.4	3.7	2.5	44	55	▲ **1LE1 5 01-0CB2 ■-■■■■**	12	0.00076	16
0.37	**0.5**	**71 M**	1380	2.6	IE2		72.7	73.2	70.2	0.72	1.02	2.3	3.8	2.4	44	55	▲ **1LE1 5 01-0CB3 ■-■■■■**	13	0.00095	16
0.55	**0.75**	**80 M**	1440	3.7	IE2		78.1	78.9	76.1	0.74	1.37	2.2	5.3	3.1	53	64	▲ **1LE1 5 01-0DB2 ■-■■■■**	17	0.0017	16
0.75	**1**	**80 M**	1440	5.0	IE2		79.6	80.2	78.0	0.76	1.79	2.2	5.6	3.1	53	64	▲ **1LE1 5 01-0DB3 ■-■■■■**	18.5	0.0021	16
1.1	**1.5**	**90 S**	1425	7.4	IE2		81.4	81.7	79.9	0.78	2.5	2.3	5.6	2.9	56	68	▲ **1LE1 5 01-0EB0 ■-■■■■**	23	0.0028	16
1.5	**2**	**90 L**	1435	10	IE2		82.8	83.5	82.0	0.79	3.3	2.6	6.4	3.4	56	68	▲ **1LE1 5 01-0EA4 ■-■■■■**	25	0.0036	16
2.2	**2.55**	**100 L**	1455	14	IE2	IE2	84.3	85.1	84.3	0.81	4.65	2.1	6.9	3.3	60	72	**1LE1 ■ 01-1AB4 ■-■■■■**	32	0.0086	16
3	**3.45**	**100 L**	1455	20	IE2	IE2	85.5	86.7	86.0	0.82	6.2	2.0	6.9	3.1	60	72	**1LE1 ■ 01-1AB5 ■-■■■■**	37	0.011	16
4	**4.55**	**112 M**	1460	26	IE2	IE2	86.6	87.3	86.5	0.81	8.2	2.5	7.1	3.2	58	70	**1LE1 ■ 01-1BB2 ■-■■■■**	46	0.014	16
5.5	**6.3**	**132 S**	1465	36	IE2	IE2	87.7	89.0	87.7	0.80	11.3	2.3	6.9	2.9	64	76	**1LE1 ■ 01-1CB0 ■-■■■■**	61	0.027	16
7.5	**8.6**	**132 M**	1465	49	IE2	IE2	88.7	90.3	88.8	0.83	14.7	2.3	6.9	2.9	64	76	**1LE1 ■ 01-1CB2 ■-■■■■**	75	0.034	16
11	**12.6**	**160 M**	1470	71	IE2	IE2	89.8	90.9	90.8	0.85	21	2.1	6.7	2.8	65	77	**1LE1 ■ 01-1DB2 ■-■■■■**	96	0.065	16
15	**17.3**	**160 L**	1475	97	IE2	IE2	90.6	91.3	91.0	0.85	28	2.3	7.3	3.0	65	77	**1LE1 ■ 01-1DB4 ■-■■■■**	104	0.083	16
18.5	**21.3**	**180 M**	1465	121	IE2	IE2	91.2	92.0	91.9	0.84	35	2.5	7.2	3.4	61	74	**1LE1 ■ 01-1EB2 ■-■■■■**	160	0.12	16
22	**25.3**	**180 L**	1465	143	IE2	IE2	91.6	92.2	91.9	0.84	41.5	2.6	7.3	3.5	62	75	**1LE1 ■ 01-1EB4 ■-■■■■**	170	0.13	16
30	**34.5**	**200 L**	1470	195	IE2	IE2	92.3	92.8	92.6	0.84	56	2.5	6.7	3.3	64	77	**1LE1 ■ 01-2AB5 ■-■■■■**	230	0.20	16
37	**42.5**	**225 S**	1470	240	IE2	IE2	92.7	93.5	93.5	0.88	65	2.3	6.6	2.9	66	79	**1LE1 ■ 01-2BB0 ■-■■■■**	280	0.42	16
45	**52**	**225 M**	1475	291	IE2	IE2	93.1	93.8	93.7	0.87	80	2.5	6.9	3.1	66	79	**1LE1 ■ 01-2BB2 ■-■■■■**	305	0.46	16
55	**63**	**250 M**	1480	355	IE2	IE2	93.5	93.9	93.5	0.85	100	2.7	6.8	3.0	66	79	**1LE1 ■ 01-2CB2 ■-■■■■**	385	0.75	16
75	**86**	**280 S**	1485	482	IE2	IE2	94.0	94.2	93.8	0.87	132	2.5	6.8	3.0	71	85	**1LE1 ■ 01-2DB0 ■-■■■■**	550	1.3	16
90	**104**	**280 M**	1486	578	IE2	IE2	94.2	94.3	93.6	0.87	159	2.6	7.3	3.1	71	85	**1LE1 ■ 01-2DB2 ■-■■■■**	570	1.4	16
110	**127**	**315 S**	1490	705	IE2	IE2	94.5	94.6	94.0	0.86	195	2.7	7.4	3.0	72	86	**1LE1 ■ 01-3AB0 ■-■■■■**	740	2.0	16
132	**152**	**315 M**	1490	846	IE2	IE2	94.7	94.9	94.6	0.87	230	2.7	7.1	2.9	75	89	**1LE1 ■ 01-3AB2 ■-■■■■**	870	2.3	16
160	**184**	**315 L**	1490	1025	IE2	IE2	94.9	95.0	94.5	0.87	280	2.8	7.2	3.1	76	91	**1LE1 ■ 01-3AB4 ■-■■■■**	940	2.8	16
200	**230**	**315 L**	1490	1282	IE2	IE2	95.1	95.3	94.7	0.87	350	3.1	7.5	3.2	77	92	**1LE1 ■ 01-3AB5 ■-■■■■**	1140	3.5	16

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects					
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5				
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6				

Voltages [2]			No. of poles	Frame size	Motor type	Version					Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz [1]	460 VY	4	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**	2	2		–
50 Hz	400 VΔ/690 VY	60 Hz [1]	460 VΔ	4	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**	3	4		–
50 Hz	500 VY			4	71 M ... 315 L	1LE1■01-0C … -3A	Without add. charge	2	7		–
50 Hz	500 VΔ			4	71 M ... 315 L	1LE1■01-0C … -3A	Without add. charge	4	0		–
Further voltages [1]	For price information, code numbers, order codes and descriptions, see from Page 2/40							9	0		. . .

Types of construction		No. of poles	Frame size	Motor type	Version			Order code(s)
Without flange	IM B3 [3]	4	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**	A		–
With flange	IM B5 [3]	4	71 M ... 315 M	1LE1■01-0C … -3A	With additional charge	F		–
With standard flange	IM B14 [3]	4	71 M ... 160 L	1LE1■01-0C … -1D	With additional charge	K		–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45					■		. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version			Order code(s)
Without	Only possible for **Basic Line**	4	71 M ... 315 L	1LE1501-0C … -3A	**Standard**	A		–
PTC thermistor with 3 temperature sensors	**Basic Line**	4	71 M ... 315 L	1LE1501-0C … -3A	With additional charge	B		–
	Performance Line	4	71 M ... 315 L	1LE1601-0C … -3A	**Standard**	B		–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50					■		. . .

Terminal box position	No. of poles	Frame size	Motor type	Version			Order code(s)
Terminal box at top	4	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**	4		–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52						

Special versions	No. of poles	Frame size	Motor type				Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58			**1LE1■01-. . . . ■-■■■■-Z**			. . .+. . .+. . .+. . .

[1] Operating values at rated output for 60 Hz are available on request.

[2] Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

[3] Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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Self-ventilated motors
Cast-iron series 1LE1501/1LE1601 Basic/Performance Line

IE2

Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz [1]	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	$\cos\varphi$ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Cast-iron series 1LE1501 – *Basic Line* 1LE1601 – *Performance Line IE2 version in accordance with IEC 60034-30* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411)																				
• Efficiency: High Efficiency IE2, service factor (SF) 1.15																				
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz [1]																				
0.18	0.25	71 M	875	2	IE2		56.6	57.0	53.5	0.68	0.68	2.2	2.5	2.3	39	50	▲ 1LE1 5 01-0CC2 ■-■■■■	11.5	0.0008	16
0.25	0.33	71 M	870	2.7	IE2		61.6	62.7	60.0	0.70	0.84	2.3	2.6	2.3	39	50	▲ 1LE1 5 01-0CC3 ■-■■■■	12.5	0.0010	16
0.37	0.5	80 M	925	3.8	IE2		71.4	71.5	66.5	0.69	1.08	2.1	4.0	2.4	42	53	▲ 1LE1 5 01-0DC2 ■-■■■■	16.5	0.0017	16
0.55	0.75	80 M	935	5.6	IE2		74.0	74.0	70.5	0.66	1.63	2.5	4.4	2.9	42	53	▲ 1LE1 5 01-0DC3 ■-■■■■	18.5	0.0025	16
0.75	1	90 S	925	7.7	IE2		75.9	76.0	73.0	0.70	2.05	2.0	4.1	2.5	43	55	▲ 1LE1 5 01-0EC0 ■-■■■■	23	0.0030	16
1.1	1.5	90 L	935	11	IE2		78.1	78.5	75.0	0.70	2.90	2.2	4.4	2.6	43	55	▲ 1LE1 5 01-0EC4 ■-■■■■	26.5	0.0040	16
1.5	1.75	100 L	970	15	IE2	IE2	79.8	80.2	79.0	0.73	3.7	2.0	6.2	2.9	59	71	1LE1 ■ 01-1AC4 ■-■■■■	36	0.011	16
2.2	2.55	112 M	965	22	IE2	IE2	81.8	82.5	81.3	0.75	5.2	2.1	6.0	3.1	57	69	1LE1 ■ 01-1BC2 ■-■■■■	41	0.014	16
3	3.45	132 S	970	30	IE2	IE2	83.3	84.0	82.8	0.74	7.0	1.6	5.6	2.6	63	75	1LE1 ■ 01-1CC0 ■-■■■■	56	0.024	13
4	4.55	132 M	970	39	IE2	IE2	84.6	85.8	85.0	0.78	8.7	1.6	5.6	2.5	63	75	1LE1 ■ 01-1CC2 ■-■■■■	61	0.029	13
5.5	6.3	132 M	970	54	IE2	IE2	86.0	87.4	87.0	0.77	12	1.9	6.1	2.8	63	75	1LE1 ■ 01-1CC3 ■-■■■■	70	0.037	16
7.5	8.6	160 M	975	73	IE2	IE2	87.2	88.0	87.3	0.74	16.8	1.9	4.7	2.2	67	79	1LE1 ■ 01-1DC2 ■-■■■■	106	0.075	16
11	12.6	160 L	975	108	IE2	IE2	88.7	89.6	89.2	0.76	23.5	1.9	4.8	2.2	67	79	1LE1 ■ 01-1DC4 ■-■■■■	122	0.098	16
15	18	180 L	975	147	IE2	IE2	89.7	90.1	90.2	0.78	31	2.5	6.0	3.1	57	70	1LE1 ■ 01-1EC4 ■-■■■■	155	0.17	16
18.5	22	200 L	978	181	IE2	IE1	90.4	91.3	91.2	0.82	36	2.4	5.8	2.6	57	71	1LE1 ■ 01-2AC4 ■-■■■■	200	0.25	16
22	26.5	200 L	978	215	IE2	IE1	90.9	91.6	91.2	0.82	42.5	2.5	6.2	2.6	61	74	1LE1 ■ 01-2AC5 ■-■■■■	220	0.30	16
30	36	225 M	980	292	IE2	IE1	91.7	92.5	92.3	0.83	57	2.5	5.6	2.7	65	78	1LE1 ■ 01-2BC2 ■-■■■■	300	0.58	16
37	44.5	250 M	982	360	IE2	IE1	92.2	93.1	93.1	0.83	70	2.8	6.0	2.5	62	77	1LE1 ■ 01-2CC2 ■-■■■■	370	0.86	16
45	54	280 S	985	436	IE2	IE1	92.7	93.4	93.2	0.84	83	2.7	6.3	2.6	65	79	1LE1 ■ 01-2DC0 ■-■■■■	460	1.1	16
55	66	280 M	985	533	IE2	IE1	93.1	93.9	94.0	0.86	99	2.5	6.4	2.6	65	79	1LE1 ■ 01-2DC2 ■-■■■■	510	1.4	16
75	90	315 S	988	725	IE2	IE1	93.7	94.0	93.6	0.84	138	2.5	6.7	2.8	65	79	1LE1 ■ 01-3AC0 ■-■■■■	660	2.1	16
90	108	315 M	988	870	IE2	IE1	94.0	94.3	93.6	0.84	165	2.6	6.9	2.8	65	79	1LE1 ■ 01-3AC2 ■-■■■■	730	2.5	16
110	132	315 L	988	1063	IE2	IE1	94.3	94.6	94.5	0.86	196	2.7	7.0	2.8	68	82	1LE1 ■ 01-3AC4 ■-■■■■	940	3.6	16
132	158	315 L	988	1276	IE2	IE2	94.6	94.9	94.7	0.86	235	3.0	7.5	2.9	69	84	1LE1 ■ 01-3AC5 ■-■■■■	990	4.0	16
160	192	315 L	988	1546	IE2	IE2	94.8	94.7	94.4	0.86	285	3.1	7.7	3.3	69	84	1LE1 ■ 01-3AC6 ■-■■■■	1160	4.7	16

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects				
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5			
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6			

Voltages [2]			No. of poles	Frame size	Motor type	Version				Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz [1]	460 VY	6	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**	2	2	–
50 Hz	400 VΔ/690 VY	60 Hz [1]	460 VΔ	6	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**	3	4	–
50 Hz	500 VY			6	71 M ... 315 L	1LE1■01-0C … -3A	Without add. charge	2	7	–
50 Hz	500 VΔ			6	71 M ... 315 L	1LE1■01-0C … -3A	Without add. charge	4	0	–
Further voltages [1]		For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0	. . .

Types of construction		No. of poles	Frame size	Motor type	Version			Order code(s)
Without flange	IM B3 [3]	6	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**		A	–
With flange	IM B5 [3]	6	71 M ... 315 M	1LE1■01-0C … -3A	With additional charge		F	–
With standard flange	IM B14 [3]	6	71 M ... 160 L	1LE1■01-0C … -1D	With additional charge		K	–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45						■	. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version			Order code(s)
Without	Only possible for **Basic Line**	6	71 M ... 315 L	1LE1501-0C … -3A	**Standard**		A	–
PTC thermistor with 3 temperature sensors	**Basic Line**	6	71 M ... 315 L	1LE1501-0C … -3A	With additional charge		B	–
	Performance Line	6	71 M ... 315 L	1LE1601-0C … -3A	**Standard**		B	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50						■	. . .

Terminal box position		No. of poles	Frame size	Motor type	Version			Order code(s)
Terminal box at top		6	71 M ... 315 L	1LE1■01-0C … -3A	**Standard**		4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52							

Special versions		No. of poles	Frame size	Motor type				Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58				1LE1■01-. . . .	■-■■■■	-Z	. . .+. . .+. . .+. . .

1) Operating values at rated output for 60 Hz are available on request.

2) Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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IE2

Self-ventilated motors
Cast-iron series 1LE1501/1LE1601 Basic/Performance Line

Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class	η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	cosφ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Cast-iron series 1LE1501 – *Basic Line* 1LE1601 – *Performance Line* *IE2 version in accordance with IEC 60034-30* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz 60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm²	CL
• Cooling: self-ventilated (IC 411) • Efficiency: High Efficiency IE2, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
8-pole: 750 rpm at 50 Hz, 900 rpm at 60 Hz 1)																			
0.75	0.86	100 L	725	9.9	– –	68.3	65.8	59.3	0.58	2.8	1.6	4.0	2.8	60	72	1LE1■01-1AD4■-■■■■	32	0.0086	13
1.1	1.3	100 L	725	14	– –	68.3	65.4	58.9	0.58	4.0	1.8	4.1	2.8	60	72	1LE1■01-1AD5■-■■■■	36	0.011	13
1.5	1.75	112 M	720	20	– –	75.8	76.0	73.0	0.67	4.25	1.4	4.2	2.4	63	75	1LE1■01-1BD2■-■■■■	51	0.014	13
2.2	2.55	132 S	725	29	– –	78.8	79.3	77.2	0.65	6.2	1.4	4.3	2.1	63	75	1LE1■01-1CD0■-■■■■	59	0.027	10
3	3.45	132 M	730	39	– –	82.7	83.0	80.9	0.65	8.1	1.4	5.0	2.4	63	75	1LE1■01-1CD2■-■■■■	67	0.035	10
4	4.55	160 M	730	52	– –	81.9	82.6	81.7	0.67	10.5	1.6	3.7	1.9	63	75	1LE1■01-1DD2■-■■■■	98	0.065	13
5.5	6.3	160 M	730	72	– –	83.8	84.3	83.1	0.67	14.1	1.7	3.9	2	63	75	1LE1■01-1DD3■-■■■■	111	0.083	13
7.5	8.6	160 L	730	98	– –	85.3	86.5	86.1	0.7	18.1	1.6	3.8	1.9	63	75	1LE1■01-1DD4■-■■■■	123	0.098	13
11	13.2	180 L	720	146	– –	86.6	87.6	87.1	0.70	26	2.3	4.9	2.6	65	78	1LE1■01-1ED4■-■■■■	155	0.20	13
15	18	200 L	718	200	– –	88.9	90.8	91.2	0.76	32	2.4	5.4	2.8	55	69	1LE1■01-2AD5■-■■■■	220	0.34	13
18.5	22	225 S	730	242	– –	89.0	89.9	89.5	0.78	38.5	2.2	5.4	2.7	59	72	1LE1■01-2BD0■-■■■■	250	0.43	13
22	26.5	225 M	730	288	– –	90.3	91.3	91.1	0.80	44	2.3	5.5	2.7	58	71	1LE1■01-2BD2■-■■■■	270	0.50	13
30	36	250 M	732	391	– –	91.3	92.2	92.0	0.80	59	2.4	5.6	2.7	60	73	1LE1■01-2CD2■-■■■■	370	0.86	13
37	44.5	280 S	736	480	– –	91.9	92.5	92.1	0.78	75	2.3	5.4	2.4	63	77	1LE1■01-2DD0■-■■■■	460	1.10	13
45	54	280 M	738	582	– –	92.4	92.8	92.4	0.79	89	2.5	5.7	2.5	66	80	1LE1■01-2DD2■-■■■■	510	1.40	13
55	66	315 S	740	710	– –	92.9	93.3	92.9	0.80	107	2.2	5.8	2.6	69	83	1LE1■01-3AD0■-■■■■	640	2.00	13
75	90	315 M	738	970	– –	93.5	94.4	94.5	0.81	143	2.3	5.9	2.7	69	84	1LE1■01-3AD2■-■■■■	710	2.50	13
90	108	315 L	740	1161	– –	93.5	94.3	94.4	0.83	167	2.2	5.8	2.5	69	84	1LE1■01-3AD4■-■■■■	860	3.10	13
110	132	315 L	740	1419	– –	94.2	95.0	95.1	0.82	205	2.7	6.7	2.9	74	88	1LE1■01-3AD5■-■■■■	980	3.90	13
132	158	315 L	740	1703	– –	94.4	94.8	94.4	0.81	250	2.9	7.2	3.3	76	90	1LE1■01-3AD6■-■■■■	1060	4.50	16

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects						
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5					
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6					

Voltages 2)			No. of poles	Frame size	Motor type	Version					Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	8	100 L ... 315 L	1LE1■01-1A … -3A	**Standard**	2	2		–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ	8	100 L ... 315 L	1LE1■01-1A … -3A	**Standard**	3	4		–
50 Hz	500 VY			8	100 L ... 315 L	1LE1■01-1A … -3A	Without add. charge	2	7		–
50 Hz	500 VΔ			8	100 L ... 315 L	1LE1■01-1A … -3A	Without add. charge	4	0		–
Further voltages 1)	For price information, code numbers, order codes and descriptions, see from Page 2/40							9	0		. . .

Types of construction		No. of poles	Frame size	Motor type	Version			Order code(s)
Without flange	IM B3 3)	8	100 L ... 315 L	1LE1■01-1A … -3A	**Standard**		A	–
With flange	IM B5 3)	8	100 L ... 315 M	1LE1■01-1A … -3A	With additional charge		F	–
With standard flange	IM B14 3)	8	100 L ... 160 L	1LE1■01-1A … -1D	With additional charge		K	–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45						■	. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version			Order code(s)
Without	Only possible for **Basic Line**	8	100 L ... 315 L	1LE1501-1A … -3A	**Standard**		A	–
PTC thermistor with 3 temperature sensors	**Basic Line**	8	100 L ... 315 L	1LE1501-1A … -3A	With additional charge		B	–
	Performance Line	8	100 L ... 315 L	1LE1601-1A … -3A	**Standard**		B	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50						■	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version			Order code(s)
Terminal box at top	8	100 L ... 315 L	1LE1■01-1A … -3A	**Standard**		4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52						

Special versions	No. of poles	Frame size	Motor type		Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58			1LE1■01-. . . . ■-■■■■	-Z . . .+. . .+. . .+. . .

1) Operating values at rated output for 60 Hz are available on request.

2) Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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SIMOTICS SD 1LE1 Standard Motors

Motors with High Efficiency IE2

Self-ventilated motors
Cast-iron series 1LE1501/1LE1601 with increased output

IE2

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz [1]	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	$\cos\varphi$ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Cast-iron series: 1LE1501 – *Basic Line*; 1LE1601 – *Performance Line*; *IE2 version in accordance with IEC 60034-30 with increased output*; Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) • Efficiency: High Efficiency IE2, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz [1]																				
4	4.55	100 L	2905	13	IE2	IE2	85.8	87.2	87.0	0.86	7.8	2.5	7.6	3.5	67	79	1LE1 ■ 01-1AA6 ■-■■■■	45	0.0054	16
5.5	6.3	112 M	2950	18	IE2	IE2	87.0	87.5	87.2	0.89	10.3	2.2	7.7	3.3	69	81	1LE1 ■ 01-1BA6 ■-■■■■	53	0.012	16
11	12.6	132 M	2950	36	IE2	IE2	89.4	90.2	90.3	0.89	20	2.3	7.9	3.2	68	80	1LE1 ■ 01-1CA6 ■-■■■■	80	0.031	16
22	25.3	160 L	2955	71	IE2	IE2	91.3	91.7	91.3	0.89	39	3.1	8.4	3.7	70	82	1LE1 ■ 01-1DA6 ■-■■■■	126	0.068	16
30	33.5	180 L	2940	97	IE2	IE2	92.0	92.5	92.2	0.89	53	2.3	7.8	3.4	70	83	1LE1 ■ 01-1EA6 ■-■■■■	180	0.09	16
45	51	200 L	2950	146	IE2	IE2	92.9	93.4	93.1	0.87	81	2.5	7.1	3.2	72	85	1LE1 ■ 01-2AA6 ■-■■■■	245	0.18	16
55	62	225 M	2960	177	IE2	IE2	93.2	93.6	93.2	0.86	99	2.5	7.0	3.3	76	89	1LE1 ■ 01-2BA6 ■-■■■■	320	0.26	16
75	84	250 M	2970	241	IE2	IE2	93.8	93.6	92.6	0.84	137	2.2	7.0	3.3	75	89	1LE1 ■ 01-2CA6 ■-■■■■	390	0.46	13
110	123	280 M	2978	353	IE2	IE2	94.3	94.5	94.1	0.90	187	2.9	8.5	3.6	80	91	1LE1 ■ 01-2DA6 ■-■■■■	650	1.20	16
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz [1]																				
4	4.55	100 L	1460	26	IE2	IE2	86.6	87.4	86.7	0.80	8.3	2.2	7.5	3.5	60	72	1LE1 ■ 01-1AB6 ■-■■■■	46	0.014	16
5.5	6.3	112 M	1460	36	IE2	IE2	87.7	88.1	87.4	0.81	11.2	2.5	7.1	3.2	58	70	1LE1 ■ 01-1BB6 ■-■■■■	58	0.017	16
11	12.6	132 M	1465	72	IE2	IE2	89.8	90.6	90.4	0.84	21	2.6	7.7	3.1	64	76	1LE1 ■ 01-1CB6 ■-■■■■	80	0.046	16
18.5	21.3	160 L	1475	120	IE2	IE2	91.2	91.7	91.6	0.85	34.5	2.5	7.7	3.3	65	77	1LE1 ■ 01-1DB6 ■-■■■■	116	0.099	16
30	34.5	180 L	1465	196	IE2	IE2	92.3	93.0	92.9	0.81	58	2.5	7.3	3.3	64	77	1LE1 ■ 01-1EB6 ■-■■■■	185	0.16	16
37	42.5	200 L	1470	240	IE2	IE2	92.7	93.6	93.8	0.84	69	2.4	7.0	3.0	62	75	1LE1 ■ 01-2AB6 ■-■■■■	240	0.25	16
55	63	225 M	1475	356	IE2	IE2	93.5	94.2	94.1	0.84	101	2.5	5.8	2.7	69	82	1LE1 ■ 01-2BB6 ■-■■■■	320	0.47	16
75	86	250 M	1480	484	IE2	IE2	94.0	94.5	94.3	0.86	134	2.3	6.2	2.8	74	87	1LE1 ■ 01-2CB6 ■-■■■■	440	0.85	13
110	127	280 M	1485	707	IE2	IE2	94.5	94.9	94.8	0.87	193	2.5	6.9	3.0	73	87	1LE1 ■ 01-2DB6 ■-■■■■	680	1.70	13

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects				
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5			
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6			

Voltages [2]			No. of poles	Frame size	Motor type	Version				Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz [1] 460 VY	2, 4	100 L ... 280 M	1LE1■01-1A … -2D	**Standard**	2	2		–
50 Hz	400 VΔ/690 VY	60 Hz [1] 460 VΔ	2, 4	100 L ... 280 M	1LE1■01-1A … -2D	**Standard**	3	4		–
50 Hz	500 VY		2, 4	100 L ... 280 M	1LE1■01-1A … -2D	Without add. charge	2	7		–
50 Hz	500 VΔ		2, 4	100 L ... 280 M	1LE1■01-1A … -2D	Without add. charge	4	0		–
Further voltages [1]	For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0		. . .

Types of construction		No. of poles	Frame size	Motor type	Version			Order code(s)
Without flange	IM B3 [3]	2, 4	100 L ... 280 M	1LE1■01-1A … -2D	**Standard**	A		–
With flange	IM B5 [3]	2, 4	100 L ... 280 M	1LE1■01-1A … -2D	With additional charge	F		–
With standard flange	IM B14 [3]	2, 4	100 L ... 280 M	1LE1■01-1A … -2D	With additional charge	K		–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45					■		. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version			Order code(s)
Without	Only possible for **Basic Line**	2, 4	100 L ... 280 M	1LE1501-1A … -2D	**Standard**	A		–
PTC thermistor with 3 temperature sensors	**Basic Line**	2, 4	100 L ... 280 M	1LE1501-1A … -2D	With additional charge	B		–
	Performance Line	2, 4	100 L ... 280 M	1LE1601-1A … -2D	**Standard**	B		–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50					■		. . .

Terminal box position	No. of poles	Frame size	Motor type	Version			Order code(s)
Terminal box at top	2, 4	100 L ... 280 M	1LE1■01-1A … -2D	**Standard**	4		–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52						

Special versions	No. of poles	Frame size	Motor type		Order code(s)
Forced-air cooled motors without ext. fan/fan cover (IC 416)	2, 4	100 L ... 280 M	1LE1■01-1A … -2D	1LE1■01-. . . . ■-■■■■-Z	F90 +. . .+. . .+. . .
Options	For price information, order codes and descriptions, see from Page 2/58			1LE1■01-. . . . ■-■■■■-Z	. . .+. . .+. . .+. . .

[1] Operating values at rated output for 60 Hz are available on request.

[2] Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

[3] Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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IE2

Self-ventilated motors
Cast-iron series 1LE1501/1LE1601 with increased output

Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class	η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	cosφ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Cast-iron series 1LE1501 – *Basic Line* 1LE1601 – *Performance Line* *IE2 version in accordance with IEC 60034-30 with increased output* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz 60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) **• Efficiency: High Efficiency IE2, service factor (SF) 1.15** **• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)**																			
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz 1)																			
2.2	**2.55**	**100 L**	965	22	IE2 IE2	81.8	82.5	81.5	0.76	5.1	1.9	5.7	2.9	59	71	**1LE1■01-1AC6■-■■■■**	49	0.014	16
3	**3.45**	**112 M**	960	30	IE2 IE2	83.3	84.1	83.6	0.79	6.6	2.1	6.0	3.1	57	69	**1LE1■01-1BC6■-■■■■**	53	0.017	16
7.5	**8.6**	**132 M**	970	74	IE2 IE2	87.2	87.8	87.3	0.77	16.1	2.1	6.5	3.0	63	75	**1LE1■01-1CC6■-■■■■**	87	0.046	16
15	**17.3**	**160 L**	975	147	IE2 IE1	89.7	90.3	89.7	0.75	32.0	2	5.2	2.4	67	79	**1LE1■01-1DC6■-■■■■**	147	0.12	16
18.5	**22**	**180 L**	975	181	IE2 IE2	90.4	91.1	90.8	0.77	38.5	2.3	6.0	2.9	67	80	**1LE1■01-1EC6■-■■■■**	165	0.21	16
30	**34.5**	**200 L**	975	294	IE2 IE2	91.7	92.5	92.5	0.77	61	2.6	6.3	2.7	62	75	**1LE1■01-2AC6■-■■■■**	245	0.38	16
37	**44.5**	**225 M**	978	361	IE2 IE1	92.2	93.0	92.9	0.83	70	2.5	6.3	2.9	64	77	**1LE1■01-2BC6■-■■■■**	325	0.67	16
45	**54**	**250 M**	985	436	IE2 IE1	92.7	93.7	94.0	0.84	83	2.4	6.6	2.7	67	81	**1LE1■01-2CC6■-■■■■**	410	1.00	16
75	**90**	**280 M**	986	726	IE2 IE2	93.7	94.3	94.4	0.85	136	3.2	7.0	2.9	66	80	**1LE1■01-2DC6■-■■■■**	570	1.80	16
8-pole: 750 rpm at 50 Hz, 900 rpm at 60 Hz 1)																			
15	**18**	**180 L**	720	199	IE2 IE2	87.9	88.9	88.2	0.73	33.5	2.2	4.9	2.5	67	75	**1LE1■01-1ED6■-■■■■**	190	0.26	13
18.5	**22**	**200 L**	720	245	IE2 IE2	88.6	89.9	90.1	0.78	38.5	2.6	5.8	3.0	65	72	**1LE1■01-2AD6■-■■■■**	250	0.42	13
30	**36**	**225 M**	732	391	IE2 IE2	90.8	92.0	92.1	0.76	63	2.8	6.1	3.2	62	76	**1LE1■01-2BD6■-■■■■**	325	0.67	16
37	**44.5**	**250 M**	730	484	IE2 IE2	91.6	92.6	92.7	0.83	70	2.3	5.5	2.6	63	77	**1LE1■01-2CD6■-■■■■**	405	1.00	13
55	**66**	**280 M**	736	714	IE2 IE2	92.9	93.4	93.0	0.8	107	2.5	5.9	2.5	70	81	**1LE1■01-2DD6■-■■■■**	550	1.60	13

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects					
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5				
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6				

Voltages 2)			No. of poles	Frame size	Motor type	Version				Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	6, 8	100 L ... 280 M	1LE1■01-1A ... -2D	**Standard**	2	2	–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ	6, 8	100 L ... 280 M	1LE1■01-1A ... -2D	**Standard**	3	4	–
50 Hz	500 VY			6, 8	100 L ... 280 M	1LE1■01-1A ... -2D	Without add. charge	2	7	–
50 Hz	500 VΔ			6, 8	100 L ... 280 M	1LE1■01-1A ... -2D	Without add. charge	4	0	–
Further voltages 1)		For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0	. . .

Types of construction		No. of poles	Frame size	Motor type	Version		Order code(s)
Without flange	IM B3 3)	6, 8	100 L ... 280 M	1LE1■01-1A ... -2D	**Standard**	A	–
With flange	IM B5 3)	6, 8	100 L ... 280 M	1LE1■01-1A ... -2D	With additional charge	F	–
With standard flange	IM B14 3)	6, 8	100 L ... 280 M	1LE1■01-1A ... -2D	With additional charge	K	–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45					■	. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version		Order code(s)
Without	Only possible for **Basic Line**	6, 8	100 L ... 280 M	1LE1501-1A ... -2D	**Standard**	A	–
PTC thermistor with 3 temperature sensors	**Basic Line**	6, 8	100 L ... 280 M	1LE1501-1A ... -2D	With additional charge	B	–
	Performance Line	6, 8	100 L ... 280 M	1LE1601-1A ... -2D	**Standard**	B	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50					■	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top	6, 8	100 L ... 280 M	1LE1■01-1A ... -2D	**Standard**	4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52					

Special versions	No. of poles	Frame size	Motor type		Order code(s)
Forced-air cooled motors without ext. fan/fan cover (IC 416)	6, 8	100 L ... 280 M	1LE1■01-1A ... -2D	**1LE1■01-. . . . ■-■■■■-Z**	**F90 +. . .+. . .+. . .**
Options	For price information, order codes and descriptions, see from Page 2/58			**1LE1■01-. . . . ■-■■■■-Z**	. . .+. . .+. . .+. . .

1) Operating values at rated output for 60 Hz are available on request.

2) Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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SIMOTICS GP 1LE1 Standard Motors

Motors with Premium Efficiency IE3

Self-ventilated motors
Aluminum series 1LE1003

IE3

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	cos φ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Aluminum series 1LE1003 – *IE3 version in accordance with IEC 60034-30* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) • Efficiency: Premium Efficiency IE3, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz 1)																				
0.75	0.86	80 M	2850	2.5	IE3	IE3	80.7	82.0	81.5	0.86	1.56	2.6	6.2	3.0	60	71	1LE1003-0DA2■-■■■■	11	0.0011	16
1.1	1.3	80 M	2885	3.6	IE3	IE3	82.7	82.7	81.7	0.85	2.25	2.8	7.4	3.8	60	71	1LE1003-0DA3■-■■■■	12	0.0013	16
1.5	1.75	90 S	2910	4.9	IE3	IE3	84.2	84.5	83.5	0.86	3.00	2.7	8.1	4.2	65	77	1LE1003-0EA0■-■■■■	15	0.0021	16
2.2	2.55	90 L	2920	7.2	IE3	IE3	85.9	86.8	86.1	0.88	4.2	2.6	8.3	4.0	65	77	1LE1003-0EA4■-■■■■	19	0.0031	16
3	3.45	100 L	2920	9.8	IE3	IE2	87.1	87.1	86.1	0.88	5.6	2.8	8.0	4.3	67	79	1LE1003-1AA4■-■■■■	26	0.0054	16
4	4.55	112 M	2950	12.9	IE3	IE2	88.1	88.1	87.1	0.89	7.4	1.9	7.5	3.9	69	81	1LE1003-1BA2■-■■■■	34	0.012	16
5.5	6.3	132 S	2950	17.8	IE3	IE3	89.2	89.2	88.2	0.90	9.9	1.8	7.4	3.6	68	80	1LE1003-1CA0■-■■■■	43	0.024	16
7.5	8.6	132 S	2950	24.3	IE3	IE3	90.1	90.1	89.1	0.92	13.1	1.9	8.3	3.9	68	80	1LE1003-1CA1■-■■■■	57	0.031	16
11	12.6	160 M	2955	35.5	IE3	IE3	91.2	91.2	90.2	0.89	19.6	2.4	7.9	3.8	70	82	1LE1003-1DA2■-■■■■	75	0.053	16
15	18	160 M	2960	48.4	IE3	IE3	91.9	91.9	90.9	0.87	27.0	2.7	8.7	4.3	70	82	1LE1003-1DA3■-■■■■	84	0.061	16
18.5	22	160 L	2955	60.0	IE3	IE3	92.4	92.4	91.4	0.90	32.0	2.8	9.0	4.2	70	82	1LE1003-1DA4■-■■■■	94	0.068	16
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz 1)																				
0.55	0.63	80 M	1440	3.6	–	–	81.3	82.0	80.2	0.78	1.25	2.1	5.9	3.1	53	64	1LE1003-0DB2■-■■■■	11	0.0021	16
0.75	0.86	80 M	1450	4.9	IE3	IE3	82.5	82.3	80.0	0.75	1.75	2.7	7.1	3.9	53	64	1LE1003-0DB3■-■■■■	14	0.0029	16
1.1	1.3	90 S	1440	7.3	IE3	IE3	84.1	84.6	83.5	0.78	2.4	2.9	6.9	3.6	56	68	1LE1003-0EB0■-■■■■	16	0.0036	16
1.5	1.75	90 L	1445	9.9	IE3	IE3	85.3	85.9	84.9	0.80	3.15	2.6	7.2	2.7	56	68	1LE1003-0EB4■-■■■■	19	0.0049	16
2.2	2.55	100 L	1465	14.3	IE3	IE2	86.7	86.7	85.7	0.83	4.4	2.1	7.6	3.6	60	72	1LE1003-1AB4■-■■■■	30	0.014	16
3	3.45	100 L	1460	19.6	IE3	IE3	87.7	87.7	86.7	0.83	5.9	2.3	7.3	3.7	60	72	1LE1003-1AB5■-■■■■	30	0.014	16
4	4.55	112 M	1460	26.0	IE3	IE3	88.6	88.6	87.6	0.82	7.9	2.4	7.1	3.7	58	70	1LE1003-1BB2■-■■■■	34	0.017	16
5.5	6.3	132 S	1470	35.7	IE3	IE2	89.6	89.6	88.6	0.84	10.5	2.1	7.2	3.4	64	76	1LE1003-1CB0■-■■■■	64	0.046	16
7.5	8.6	132 M	1470	48.7	IE3	IE2	90.4	90.4	89.4	0.84	14.3	2.4	7.4	3.5	64	76	1LE1003-1CB2■-■■■■	64	0.046	16
11	12.6	160 M	1475	71.0	IE3	IE3	91.4	91.4	90.4	0.84	20.5	2.2	6.9	3.2	65	77	1LE1003-1DB2■-■■■■	83	0.083	16
15	17.3	160 L	1475	97.0	IE3	IE3	92.1	92.1	91.1	0.82	28.5	2.5	8.5	3.8	65	77	1LE1003-1DB4■-■■■■	100	0.99	16

Voltages				Motor protection	No. of poles	Frame size	Motor type	Version					Order code(s)
Frame sizes 80 M to 90 L: use of the 360° freely rotatable terminal box for 2 and 4-pole motors 2)													
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	PTC thermistor with 1 temp. sensor	2, 4	80 M ... 90 L	1LE1003-0D ... -0E	**Standard**	**2**	**2**			–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ		2, 4	80 M ... 90 L	1LE1003-0D ... -0E	**Standard**	**3**	**4**			–
50 Hz	400 VY	60 Hz 1)	460 VY	Without	2, 4	80 M ... 90 L	1LE1003-0D ... -0E	**Standard**	**0**	**2**	**A**		–
Frame sizes 100 L to 160 L: use of the 4 x 90° rotatable terminal box													
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	Any	2, 4	100 L ... 160 L	1LE1003-1A ... -1D	**Standard**	**2**	**2**			–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ	Any	2, 4	100 L ... 160 L	1LE1003-1A ... -1D	**Standard**	**3**	**4**			–
50 Hz	500 VY			Any	2, 4	100 L ... 160 L	1LE1003-1A ... -1D	Without add. charge	**2**	**7**			–
50 Hz	500 VΔ			Any	2, 4	100 L ... 160 L	1LE1003-1A ... -1D	Without add. charge	**4**	**0**			–
Further voltages 1)				For price information, code numbers, order codes and descriptions, see from Page 2/38					**9**	**0**			. . .

Types of construction			No. of poles	Frame size	Motor type	Version				Order code(s)
Without flange	IM B3 3)		2, 4	80 M ... 160 L	1LE1003-0D ... -1D	**Standard**		**A**		–
With flange	IM B5 3)		2, 4	80 M ... 160 L	1LE1003-0D ... -1D	With add. charge		**F**		–
With standard flange	IM B14 3)		2, 4	80 M ... 160 L	1LE1003-0D ... -1D	With add. charge		**K**		–
Further types of construction		For price information, code letters and descriptions, see from Page 2/41						■		. . .

Motor protection	No. of poles	Frame size	Motor type	Version				Order code(s)
Frame sizes 100 L to 160 L: use of the 4 x 90° rotatable terminal box								
Without	2, 4	80 L ... 160 L	1LE1003-0D ... -1D	**Standard**		**A**		–
PTC thermistor with 3 temperature sensors	2, 4	80 L ... 160 L	1LE1003-0D ... -1D	With add. charge		**B**		–
Further motor protection	For price information, code letters and descriptions, see from Page 2/49					■		. . .

Terminal box position	No. of poles	Frame size	Motor type	Version				Order code(s)
Terminal box at top	2, 4	80 M ... 160 L	1LE1003-0D ... -1D	**Standard**			**4**	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/51							

Special versions	No. of poles	Frame size	Motor type					Order code(s)
Forced-air cooled motors without ext. fan/fan cover (IC 416)	2, 4	80 M ... 160 L	1LE1003-0D ... -1D	1LE1003-. . . . ■-■■■■			-Z	F90 +. . .+. . .+. . .
Options	For price information, order codes and descriptions, see from Page 2/53			1LE1003-. . . . ■-■■■■			-Z	. . .+. . .+. . .+. . .

1) Operating values at rated output for 60 Hz are available on request.

2) For converter-fed operation of shaft heights 80 and 90, ordering with PTC thermistors and their connection to the converter is recommended.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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Selection and ordering data (continued)

Operating values at rated output																Aluminum series 1LE1003 – IE3 version in accordance with IEC 60034-30			
P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	$\cos\varphi$ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) • Efficiency: Premium Efficiency IE3, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz 1)																			
0.37	0.43	80 M	940	3.8	–	–	74.8	74.3	70.5	0.66	1.08	2.3	4.2	2.7	42	53	1LE1003-0DC2■-■■■■	12	0.0025	13
0.55	0.63	80 M	935	5.6	–	–	77.2	77.2	75.5	0.67	1.53	2.5	4.5	2.8	42	53	1LE1003-0DC3■-■■■■	14	0.0031	13
0.75	0.86	90 S	945	7.6	IE3	IE3	78.9	80.0	78.5	0.70	1.96	2.2	4.6	2.6	43	55	1LE1003-0EC0■-■■■■	16	0.0040	13
1.1	1.3	90 L	940	11.0	IE3	–	81.0	81.0	79.5	0.69	2.85	2.3	4.6	2.7	43	55	1LE1003-0EC4■-■■■■	19	0.0048	13
1.5	1.75	100 L	970	15.0	IE3	IE2	82.5	82.5	81.5	0.76	3.45	1.9	6.9	3.0	59	71	1LE1003-1AC4■-■■■■	30	0.014	13
2.2	2.55	112 M	970	22.0	IE3	IE2	84.3	84.3	83.3	0.8	4.7	2.3	6.8	3.4	59	71	1LE1003-1BC2■-■■■■	29	0.014	13
3	3.45	132 S	970	29.4	IE3	IE2	85.6	85.6	84.6	0.77	6.6	1.7	5.2	2.6	63	75	1LE1003-1CC0■-■■■■	43	0.029	13
4	4.55	132 M	970	39.3	IE3	IE2	86.8	86.8	85.8	0.77	8.6	1.9	5.7	2.9	63	75	1LE1003-1CC2■-■■■■	52	0.037	13
5.5	6.3	132 M	970	54.0	IE3	IE2	88.0	88.0	87.0	0.78	11.6	1.9	5.9	2.9	63	75	1LE1003-1CC3■-■■■■	52	0.037	13
7.5	8.6	160 M	980	73.0	IE3	IE2	89.1	89.9	89.3	0.76	16.0	1.9	4.9	2.3	67	79	1LE1003-1DC2■-■■■■	93	0.098	13
11	12.6	160 L	975	108.0	IE3	IE2	90.3	91.1	90.7	0.77	23.0	1.9	5	2.3	67	79	1LE1003-1DC4■-■■■■	115	0.12	13

Voltages				Motor protection	No. of poles	Frame size	Motor type	Version					Order code(s)
Frame sizes 80 M to 90 L: use of the 360° freely rotatable terminal box for 2 and 4-pole motors 2)													
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	PTC thermistor with 1 temp. sensor	6	80 M ... 90 L	1LE1003-0D ... -0E	**Standard**	2	2			–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ		6	80 M ... 90 L	1LE1003-0D ... -0E	**Standard**	3	4			–
50 Hz	400 VY	60 Hz 1)	460 VY	Without	6	80 M ... 90 L	1LE1003-0D ... -0E	**Standard**	0	2	A		–
Frame sizes 100 L to 160 L: use of the 4 x 90° rotatable terminal box													
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	Any	6	100 L ... 160 L	1LE1003-1A ... -1D	**Standard**	2	2			–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ	Any	6	100 L ... 160 L	1LE1003-1A ... -1D	**Standard**	3	4			–
50 Hz	500 VY			Any	6	100 L ... 160 L	1LE1003-1A ... -1D	Without add. charge	2	7			–
50 Hz	500 VΔ			Any	6	100 L ... 160 L	1LE1003-1A ... -1D	Without add. charge	4	0			–
Further voltages 1)				For price information, code numbers, order codes and descriptions, see from Page 2/38					9	0			. . .

Types of construction			No. of poles	Frame size	Motor type	Version					Order code(s)
Without flange	IM B3 3)		6	80 M ... 160 L	1LE1003-0D ... -1D	**Standard**		A			–
With flange	IM B5 3)		6	80 M ... 160 L	1LE1003-0D ... -1D	With additional charge		F			–
With standard flange	IM B14 3)		6	80 M ... 160 L	1LE1003-0D ... -1D	With additional charge		K			–
Further types of construction		For price information, code letters and descriptions, see from Page 2/41						■			. . .

Motor protection	No. of poles	Frame size	Motor type	Version					Order code(s)
Frame sizes 100 L to 160 L: use of the 4 x 90° rotatable terminal box									
Without	6	80 L ... 160 L	1LE1003-0D ... -1D	**Standard**		A			–
PTC thermistor with 3 temperature sensors	6	80 L ... 160 L	1LE1003-0D ... -1D	With additional charge		B			–
Further motor protection	For price information, code letters and descriptions, see from Page 2/49					■			. . .

Terminal box position	No. of poles	Frame size	Motor type	Version					Order code(s)
Terminal box at top	6	80 M ... 160 L	1LE1003-0D ... -1D	**Standard**			4		–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/51								

Special versions	No. of poles	Frame size	Motor type						Order code(s)
Forced-air cooled motors without ext. fan/fan cover (IC 416)	6	80 M ... 160 L	1LE1003-0D ... -1D	1LE1003-. . . . ■-■■■■ -Z					F90 +. . .+. . .+. . .
Options	For price information, order codes and descriptions, see from Page 2/53			1LE1003-. . . . ■-■■■■ -Z					. . .+. . .+. . .+. . .

1) Operating values at rated output for 60 Hz are available on request.

2) For converter-fed operation of shaft heights 80 and 90, ordering with PTC thermistors and their connection to the converter is recommended.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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SIMOTICS SD 1LE1 Standard Motors

Motors with Premium Efficiency IE3

Self-ventilated motors
Cast-iron series 1LE1503/1LE1603 Basic/Performance Line

IE3

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	cosφ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Cast-iron series 1LE1503 – *Basic Line* 1LE1603 – *Performance Line* *IE3 version in accordance with IEC 60034-30* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) • Efficiency: Premium Efficiency IE3, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz 1)																				
0.37	0.5	71 M	2850	1.2	IE3		73.8	73.2	69.6	0.76	0.95	3.5	5.8	3.5	52	63	▲ 1LE1 5 03-0CA2 ■-■■■■	13	0.00045	16
0.55	0.75	71 M	2860	1.8	IE3		77.8	77.4	74.3	0.76	1.34	3.5	6.1	3.5	52	63	▲ 1LE1 5 03-0CA3 ■-■■■■	14.5	0.00056	16
0.75	1	80 M	2850	2.5	IE3		80.7	82.0	81.5	0.86	1.56	2.6	6.2	3.0	60	71	▲ 1LE1 5 03-0DA2 ■-■■■■	18	0.0011	16
1.1	1.5	80 M	2885	3.6	IE3		82.7	82.7	81.7	0.85	2.25	2.8	7.4	3.8	60	71	▲ 1LE1 5 03-0DA3 ■-■■■■	21	0.0013	16
1.5	2	90 S	2910	4.9	IE3		84.2	84.5	83.5	0.86	3	2.7	8.1	4.2	65	77	▲ 1LE1 5 03-0EA0 ■-■■■■	25.5	0.0021	16
2.2	3	90 L	2920	7.2	IE3		85.9	86.8	86.1	0.88	4.2	2.6	8.3	4.0	65	77	▲ 1LE1 5 03-0EA4 ■-■■■■	32	0.0031	16
3	3.45	100 L	2920	9.8	IE3	IE3	87.1	87.1	86.1	0.88	5.6	2.8	8.0	4.3	67	79	1LE1 ■ 03-1AA4 ■-■■■■	36	0.0054	16
4	4.55	112 M	2950	12.9	IE3	IE3	88.1	88.1	87.1	0.89	7.4	1.9	7.5	3.9	69	81	1LE1 ■ 03-1BA2 ■-■■■■	45	0.012	16
5.5	6.3	132 S	2950	17.8	IE3	IE3	89.2	89.2	88.2	0.90	9.9	1.8	7.4	3.6	68	80	1LE1 ■ 03-1CA0 ■-■■■■	58	0.024	16
7.5	8.6	132 S	2950	24.3	IE3	IE3	90.1	90.1	89.1	0.92	13.1	1.9	8.3	3.9	68	80	1LE1 ■ 03-1CA1 ■-■■■■	73	0.031	16
11	12.6	160 M	2955	35.5	IE3	IE3	91.2	91.2	90.2	0.89	19.6	2.4	7.9	3.8	70	82	1LE1 ■ 03-1DA2 ■-■■■■	100	0.053	16
15	17.3	160 M	2960	48.4	IE3	IE3	91.9	91.9	90.9	0.87	27.0	2.7	8.7	4.3	70	82	1LE1 ■ 03-1DA3 ■-■■■■	110	0.061	16
18.5	21.3	160 L	2955	60.0	IE3	IE3	92.4	92.4	91.4	0.90	32.0	2.8	9.0	4.2	70	82	1LE1 ■ 03-1DA4 ■-■■■■	127	0.068	16
22	24.5	180 M	2950	71	IE3	IE3	92.7	93.0	92.4	0.89	38.5	2.5	7.5	3.5	73	80	1LE1 ■ 03-1EA2 ■-■■■■	160	0.080	16
30	33.5	200 L	2955	97	IE3	IE3	93.3	93.7	93.3	0.87	53	2.5	6.6	3.3	73	80	1LE1 ■ 03-2AA4 ■-■■■■	225	0.13	16
37	41.5	200 L	2955	120	IE3	IE3	93.7	94.1	93.8	0.88	65	2.5	6.6	3.2	74	81	1LE1 ■ 03-2AA5 ■-■■■■	250	0.16	16
45	51	225 M	2960	145	IE3	IE3	94.0	94.5	94.4	0.89	78	2.4	6.9	3.3	73	87	1LE1 ■ 03-2BA2 ■-■■■■	315	0.26	16
55	62	250 M	2975	177	IE3	IE3	94.3	94.5	93.9	0.89	95	2.3	6.7	3.1	73	87	1LE1 ■ 03-2CA2 ■-■■■■	385	0.46	13
75	84	280 S	2975	241	IE3	IE3	94.7	94.8	94.1	0.89	128	2.4	6.8	3.0	74	88	1LE1 ■ 03-2DA0 ■-■■■■	510	0.77	13
90	101	280 M	2975	289	IE3	IE3	95.0	95.1	94.6	0.90	152	2.4	7.2	3.1	74	88	1LE1 ■ 03-2DA2 ■-■■■■	590	0.94	13
110	123	315 S	2982	352	IE3	IE3	95.2	95.4	94.9	0.91	183	2.4	7.1	3.1	75	89	1LE1 ■ 03-3AA0 ■-■■■■	750	1.4	13
132	148	315 M	2982	423	IE3	IE3	95.4	95.5	95.2	0.91	220	2.5	7.2	3.1	75	89	1LE1 ■ 03-3AA2 ■-■■■■	880	1.6	13
160	180	315 L	2982	512	IE3	IE3	95.6	95.7	95.2	0.92	265	2.8	7.8	3.3	77	91	1LE1 ■ 03-3AA4 ■-■■■■	980	1.9	13
200	224	315 L	2982	640	IE3	IE3	95.8	95.9	95.5	0.92	330	2.5	7.2	3.0	77	91	1LE1 ■ 03-3AA5 ■-■■■■	1150	2.3	13

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects				
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	up to 500 V	12 months	5			
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6			

Voltages 2)			No. of poles	Frame size	Motor type	Version				Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	2	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	2	2	–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ	2	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	3	4	–
50 Hz	500 VY			2	71 M ... 315 L	1LE1■03-0C … -3A	Without add. charge	2	7	–
50 Hz	500 VΔ			2	71 M ... 315 L	1LE1■03-0C … -3A	Without add. charge	4	0	–
Further voltages 1)		For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0	. . .

Types of construction		No. of poles	Frame size	Motor type	Version			Order code(s)
Without flange	IM B3 3)	2	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	A		–
With flange	IM B5 3)	2	71 M ... 315 M	1LE1■03-0C … -3A	With additional charge	F		–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45					■		. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version		Order code(s)
Without	Only possible for **Basic Line**	2	71 M ... 315 L	1LE1503-0C … -3A	**Standard**	A	–
PTC thermistor with 3 temperature sensors	**Basic Line**	2	71 M ... 315 L	1LE1503-0C … -3A	With additional charge	B	–
	Performance Line	2	71 M ... 315 L	1LE1603-0C … -3A	**Standard**	B	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50					■	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top	2	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52					

Special versions	No. of poles	Frame size	Motor type		Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58			1LE1■03-. . . . ■-■■■■ -Z	. . .+. . .+. . .+. . .

1) Operating values at rated output for 60 Hz are available on request.

2) Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	cosφ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Cast-iron series 1LE1503 – *Basic Line* 1LE1603 – *Performance Line* *IE3 version in accordance with IEC 60034-30* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) • Efficiency: Premium Efficiency IE3, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz 1)																				
0.25	0.33	71 M	1395	1.7	IE3		73.5	73.6	70.3	0.72	0.68	2.5	4.2	2.6	44	55	▲ 1LE1 5 03-0CB2 ■-■■■■	13	0.00095	16
0.37	0.5	71 M	1410	2.5	IE3		77.3	76.5	72.7	0.69	1	2.6	4.5	2.8	44	55	▲ 1LE1 5 03-0CB3 ■-■■■■	16	0.0014	16
0.55	0.75	80 M	1440	3.6	IE3		81.3	82.0	80.2	0.78	1.25	2.1	5.9	3.1	53	64	▲ 1LE1 5 03-0DB2 ■-■■■■	18.5	0.0021	16
0.75	1	80 M	1450	4.9	IE3		82.5	82.3	80.0	0.75	1.75	2.7	7.1	3.9	53	64	▲ 1LE1 5 03-0DB3 ■-■■■■	22.5	0.0029	16
1.1	1.5	90 S	1440	7.3	IE3		84.1	84.6	83.5	0.78	2.4	2.9	6.9	3.6	56	68	▲ 1LE1 5 03-0EB0 ■-■■■■	25	0.0036	16
1.5	2	90 L	1445	9.9	IE3		85.3	85.9	84.9	0.80	3.15	2.6	7.2	2.7	56	68	▲ 1LE1 5 03-0EB4 ■-■■■■	31	0.0049	16
2.2	2.55	100 L	1465	14.3	IE3	IE3	86.7	86.7	85.7	0.83	4.4	2.1	7.6	3.6	60	72	1LE1 ■ 03-1AB4 ■-■■■■	40	0.014	16
3	3.45	100 L	1460	19.6	IE3	IE3	87.7	87.7	86.7	0.83	5.9	2.3	7.3	3.7	60	72	1LE1 ■ 03-1AB5 ■-■■■■	40	0.014	16
4	4.55	112 M	1460	26	IE3	IE3	88.6	88.6	87.6	0.82	7.9	2.4	7.1	3.7	58	70	1LE1 ■ 03-1BB2 ■-■■■■	46	0.017	16
5.5	6.3	132 S	1470	35.7	IE3	IE3	89.6	89.6	88.6	0.84	10.5	2.1	7.2	3.4	64	76	1LE1 ■ 03-1CB0 ■-■■■■	74	0.046	16
7.5	8.6	132 M	1470	48.7	IE3	IE3	90.4	90.4	89.4	0.84	14.3	2.4	7.4	3.5	64	76	1LE1 ■ 03-1CB2 ■-■■■■	80	0.046	16
11	12.6	160 M	1475	71.0	IE3	IE3	91.4	91.4	90.4	0.82	21.0	2.2	6.9	3.2	65	77	1LE1 ■ 03-1DB2 ■-■■■■	109	0.083	16
15	17.3	160 L	1475	97	IE3	IE3	92.1	92.1	91.1	0.82	28.5	2.5	8.5	3.8	65	77	1LE1 ■ 03-1DB4 ■-■■■■	127	0.099	16
18.5	21.3	180 M	1470	120	IE3	IE3	92.6	93.2	93.2	0.82	35	2.5	6.9	3.3	66	73	1LE1 ■ 03-1EB2 ■-■■■■	165	0.13	16
22	25.3	180 L	1470	143	IE3	IE3	93.0	93.7	93.7	0.83	41	2.5	6.8	3.3	68	75	1LE1 ■ 03-1EB4 ■-■■■■	170	0.14	16
30	34.5	200 L	1470	195	IE3	IE3	93.6	94.3	94.4	0.84	55	2.6	6.9	3.1	65	72	1LE1 ■ 03-2AB5 ■-■■■■	240	0.22	16
37	42.5	225 S	1478	239	IE3	IE3	93.9	94.5	94.4	0.86	66	2.5	6.4	2.7	65	78	1LE1 ■ 03-2BB0 ■-■■■■	285	0.42	16
45	52	225 M	1478	291	IE3	IE3	94.2	94.9	95.1	0.86	80	2.6	6.4	2.7	65	78	1LE1 ■ 03-2BB2 ■-■■■■	320	0.47	16
55	63	250 M	1482	354	IE3	IE3	94.6	95.1	95.0	0.87	96	2.5	6.8	2.9	66	79	1LE1 ■ 03-2CB2 ■-■■■■	420	0.85	16
75	86	280 S	1485	482	IE3	IE3	95.0	95.3	95.0	0.86	133	2.5	6.9	3.0	69	83	1LE1 ■ 03-2DB0 ■-■■■■	570	1.4	16
90	104	280 M	1485	579	IE3	IE3	95.2	95.5	95.3	0.87	157	2.6	7.2	3.0	70	84	1LE1 ■ 03-2DB2 ■-■■■■	670	1.7	16
110	127	315 S	1488	706	IE3	IE3	95.4	95.8	95.5	0.87	191	2.6	6.8	2.9	70	84	1LE1 ■ 03-3AB0 ■-■■■■	760	2.2	16
132	152	315 M	1490	846	IE3	IE3	95.6	95.9	95.9	0.87	230	2.8	7.3	3.0	73	87	1LE1 ■ 03-3AB2 ■-■■■■	960	2.9	16
160	184	315 L	1490	1025	IE3	IE3	95.8	96.1	96.1	0.87	275	2.9	7.3	3.1	73	87	1LE1 ■ 03-3AB4 ■-■■■■	990	3.1	16
200	230	315 L	1488	1284	IE3	IE3	96.0	96.3	96.1	0.88	340	3.2	7.4	3.0	73	87	1LE1 ■ 03-3AB5 ■-■■■■	1190	3.7	16

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects					
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5				
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6				

Voltages 2)			No. of poles	Frame size	Motor type	Version					Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	4	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	2	2		–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ	4	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	3	4		–
50 Hz	500 VY			4	71 M ... 315 L	1LE1■03-0C … -3A	Without add. charge	2	7		–
50 Hz	500 VΔ			4	71 M ... 315 L	1LE1■03-0C … -3A	Without add. charge	4	0		–
Further voltages 1)		For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0		. . .

Types of construction		No. of poles	Frame size	Motor type	Version			Order code(s)
Without flange	IM B3 3)	4	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	A		–
With flange	IM B5 3)	4	71 M ... 315 M	1LE1■03-0C … -3A	With additional charge	F		–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45					■		. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version			Order code(s)
Without	Only possible for **Basic Line**	4	71 M ... 315 L	1LE1503-0C … -3A	**Standard**	A		–
PTC thermistor with 3 temperature sensors	**Basic Line**	4	71 M ... 315 L	1LE1503-0C … -3A	With additional charge	B		–
	Performance Line	4	71 M ... 315 L	1LE1603-0C … -3A	**Standard**	B		–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50					■		. . .

Terminal box position		No. of poles	Frame size	Motor type	Version			Order code(s)
Terminal box at top		4	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	4		–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52							

Special versions		No. of poles	Frame size	Motor type				Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58				1LE1■03-	■-■■■■	-Z	. . .+. . .+. . .+. . .

1) Operating values at rated output for 60 Hz are available on request.

2) Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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Self-ventilated motors
Cast-iron series 1LE1503/1LE1603 Basic/Performance Line

IE3

Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz [1]	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class		η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	cosφ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Cast-iron series: 1LE1503 – *Basic Line*; 1LE1603 – *Performance Line IE3 version in accordance with IEC 60034-30*; Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm	50 Hz	60 Hz	%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) • Efficiency: Premium Efficiency IE3, service factor (SF) 1.15 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																				
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz [1]																				
0.18	**0.25**	**71 M**	885	1.9	IE3		63.9	64.6	60.8	0.69	0.59	2.3	2.8	2.3	39	50	▲ **1LE1 5 03-0CC2 ■-■■■■**	12.5	0.0010	16
0.25	**0.33**	**71 M**	900	2.7	IE3		68.6	69.3	65.7	0.67	0.79	2.7	3.1	2.8	39	50	▲ **1LE1 5 03-0CC3 ■-■■■■**	15.5	0.0015	16
0.37	**0.5**	**80 M**	940	3.8	IE3		74.8	74.3	70.5	0.66	1.08	2.3	4.2	2.7	42	53	▲ **1LE1 5 03-0DC2 ■-■■■■**	18.5	0.0025	13
0.55	**0.75**	**80 M**	935	5.6	IE3		77.2	77.2	75.5	0.67	1.53	2.5	4.5	2.8	42	53	▲ **1LE1 5 03-0DC3 ■-■■■■**	22.5	0.0031	13
0.75	**1**	**90 S**	945	7.6	IE3		78.9	80.0	78.5	0.70	1.96	2.2	4.6	2.6	43	55	▲ **1LE1 5 03-0EC0 ■-■■■■**	26.5	0.0040	13
1.1	**1.5**	**90 L**	940	11.0	IE3		81.0	81.0	79.5	0.69	2.85	2.3	4.6	2.7	43	55	▲ **1LE1 5 03-0EC4 ■-■■■■**	32	0.0048	13
1.5	**1.75**	**100 L**	970	15	IE3	IE2	82.5	82.5	81.5	0.76	3.45	1.9	6.9	3.0	59	71	**1LE1 ■ 03-1AC4 ■-■■■■**	34	0.014	13
2.2	**2.55**	**112 M**	970	22	IE3	IE2	84.3	84.3	83.3	0.80	4.7	2.3	6.8	3.4	59	71	**1LE1 ■ 03-1BC2 ■-■■■■**	47	0.014	13
3	**3.45**	**132 S**	970	29.4	IE3	IE2	85.6	85.6	84.6	0.77	6.6	1.7	5.2	2.6	63	75	**1LE1 ■ 03-1CC0 ■-■■■■**	68	0.029	13
4	**4.55**	**132 M**	970	39.3	IE3	IE2	86.8	86.8	85.8	0.77	8.6	1.9	5.7	2.9	63	75	**1LE1 ■ 03-1CC2 ■-■■■■**	68	0.037	13
5.5	**6.3**	**132 M**	970	54.0	IE3	IE2	88.0	88.0	87.0	0.78	11.6	1.9	5.9	2.9	63	75	**1LE1 ■ 03-1CC3 ■-■■■■**	81	0.037	13
7.5	**8.6**	**160 M**	980	73.0	IE3	IE2	89.1	89.9	89.3	0.76	16.0	1.9	4.9	2.3	67	79	**1LE1 ■ 03-1DC2 ■-■■■■**	120	0.098	13
11	**12.6**	**160 L**	975	108	IE3	IE2	90.3	91.1	90.7	0.77	23.0	1.9	5	2.3	67	79	**1LE1 ■ 03-1DC4 ■-■■■■**	149	0.122	13
15	**18**	**180 L**	975	147	IE3	IE2	91.2	92.4	92.6	0.80	29.5	2.3	5.9	2.8	61	68	**1LE1 ■ 03-1EC4 ■-■■■■**	180	0.19	16
18.5	**22**	**200 L**	978	181	IE3	IE2	91.7	92.5	92.5	0.79	37	2.5	5.6	2.6	64	71	**1LE1 ■ 03-2AC4 ■-■■■■**	215	0.28	16
22	**26.5**	**200 L**	978	215	IE3	IE2	92.2	93.1	93.3	0.79	43.5	2.5	5.6	2.6	61	68	**1LE1 ■ 03-2AC5 ■-■■■■**	230	0.32	16
30	**36**	**225 M**	982	292	IE3	IE2	92.9	93.6	93.5	0.83	56	2.6	6.6	3.0	64	77	**1LE1 ■ 03-2BC2 ■-■■■■**	325	0.67	16
37	**44.5**	**250 M**	985	359	IE3	IE2	93.3	94.0	94.0	0.85	67	2.7	7.0	2.9	62	75	**1LE1 ■ 03-2CC2 ■-■■■■**	405	1.0	16
45	**54**	**280 S**	988	435	IE3	IE2	93.7	94.3	94.2	0.85	82	3.0	6.8	2.8	60	74	**1LE1 ■ 03-2DC0 ■-■■■■**	510	1.4	16
55	**66**	**280 M**	988	532	IE3	IE2	94.1	94.5	94.2	0.85	99	3.2	7.2	3.0	60	74	**1LE1 ■ 03-2DC2 ■-■■■■**	560	1.6	16
75	**90**	**315 S**	990	723	IE3	IE3	94.6	94.7	94.1	0.84	136	2.6	7.5	3.1	63	78	**1LE1 ■ 03-3AC0 ■-■■■■**	750	2.6	16
90	**108**	**315 M**	991	867	IE3	IE2	94.9	95.1	94.7	0.85	161	2.5	6.7	2.8	63	78	**1LE1 ■ 03-3AC2 ■-■■■■**	890	3.1	16
110	**132**	**315 L**	991	1060	IE3	IE2	95.1	95.3	95.1	0.84	199	2.8	7.2	3.0	63	78	**1LE1 ■ 03-3AC4 ■-■■■■**	990	3.9	16
132	**158**	**315 L**	991	1272	IE3	IE2	95.4	95.3	94.5	0.84	240	2.7	7.2	3.0	67	82	**1LE1 ■ 03-3AC5 ■-■■■■**	1110	4.4	16
160	**192**	**315 L**	991	1542	IE3	IE2	95.6	95.8	95.4	0.83	290	3.3	7.7	3.5	67	82	**1LE1 ■ 03-3AC6 ■-■■■■**	1160	4.6	16

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects				
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	**5**			
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	**6**			

Voltages [2]			No. of poles	Frame size	Motor type	Version			Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz [1]	6	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	**2**	**2**	–
50 Hz	400 VΔ/690 VY	60 Hz [1]	6	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	**3**	**4**	–
50 Hz	500 VY		6	71 M ... 315 L	1LE1■03-0C … -3A	Without add. charge	**2**	**7**	–
50 Hz	500 VΔ		6	71 M ... 315 L	1LE1■03-0C … -3A	Without add. charge	**4**	**0**	–
Further voltages [1]	For price information, code numbers, order codes and descriptions, see from Page 2/40						**9**	**0**	. . .

Types of construction		No. of poles	Frame size	Motor type	Version		Order code(s)
Without flange	IM B3 [3]	6	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	**A**	–
With flange	IM B5 [3]	6	71 M ... 315 M	1LE1■03-0C … -3A	With additional charge	**F**	–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45					■	. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version		Order code(s)
Without	Only possible for **Basic Line**	6	71 M ... 315 L	1LE1503-0C … -3A	**Standard**	**A**	–
PTC thermistor with 3 temperature sensors	**Basic Line**	6	71 M ... 315 L	1LE1503-0C … -3A	With additional charge	**B**	–
	Performance Line	6	71 M ... 315 L	1LE1603-0C … -3A	**Standard**	**B**	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50					■	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top	6	71 M ... 315 L	1LE1■03-0C … -3A	**Standard**	**4**	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52					

Special versions	No. of poles	Frame size	Motor type			Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58			**1LE1■03-. . . .**	**■-■■■■-Z**	. . .+. . .+. . .+. . .

1) Operating values at rated output for 60 Hz are available on request.

2) Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

3) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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Please observe legal minimum efficiencies in the EEA! IE1

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz [1]	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class	η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	$\cos\varphi$ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Aluminum series 1LE1002 – *IE1 version in accordance with IEC 60034-30* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)		kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) or with order code F90 forced-air cooled without external fan and fan cover (IC 416) • Efficiency: Standard Efficiency IE1, service factor (SF) 1.1 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz [1]																			
3	**3.45**	**100 L**	2835	10	IE1	81.5	82.8	82.1	0.87	6.1	3.2	6.2	2.9	67	79	**1LE1002-1AA4■-■■■■**	20	0.0034	16
4	**4.55**	**112 M**	2930	13	IE1	83.1	83.8	82.2	0.86	8.1	2.7	7.3	3.7	69	81	**1LE1002-1BA2■-■■■■**	25	0.0067	16
5.5	**6.3**	**132 S**	2905	18	IE1	84.7	85.7	85.0	0.89	10.5	1.9	5.6	2.5	68	80	**1LE1002-1CA0■-■■■■**	35	0.013	16
7.5	**8.6**	**132 S**	2925	24	IE1	86.0	86.9	85.8	0.87	14.5	2.1	6.3	3.2	68	80	**1LE1002-1CA1■-■■■■**	40	0.016	16
11	**12.6**	**160 M**	2925	36	IE1	87.6	87.6	86.1	0.85	21.5	2.0	5.8	2.6	70	82	**1LE1002-1DA2■-■■■■**	60	0.030	16
15	**17.3**	**160 M**	2930	49	IE1	88.7	89.0	88.0	0.84	29	2.5	6.1	3.1	70	82	**1LE1002-1DA3■-■■■■**	68	0.036	16
18.5	**21.3**	**160 L**	2935	60	IE1	89.3	90.0	89.7	0.86	35	2.5	7.0	3.2	70	82	**1LE1002-1DA4■-■■■■**	78	0.044	16
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz [1]																			
2.2	**2.55**	**100 L**	1425	15	IE1	79.7	80.5	78.5	0.81	4.9	2.2	5.1	2.3	60	72	**1LE1002-1AB4■-■■■■**	18	0.0059	16
3	**3.45**	**100 L**	1425	20	IE1	81.5	83.0	82.3	0.85	6.3	2.4	5.4	2.6	60	72	**1LE1002-1AB5■-■■■■**	22	0.0078	16
4	**4.55**	**112 M**	1435	27	IE1	83.1	84.5	84.0	0.85	8.2	2.2	5.3	2.6	58	70	**1LE1002-1BB2■-■■■■**	27	0.010	16
5.5	**6.3**	**132 S**	1450	36	IE1	84.7	85.7	84.9	0.82	11.2	2.3	5.7	2.7	64	76	**1LE1002-1CB0■-■■■■**	38	0.019	16
7.5	**8.6**	**132 M**	1450	49	IE1	86.0	86.9	86.3	0.82	15.2	2.6	6.6	3.1	64	76	**1LE1002-1CB2■-■■■■**	44	0.024	16
11	**12.6**	**160 M**	1460	72	IE1	87.6	88.0	86.6	0.82	22	2.3	6.4	3.1	65	77	**1LE1002-1DB2■-■■■■**	62	0.044	16
15	**17.3**	**160 L**	1460	98	IE1	88.7	89.3	88.3	0.82	30	2.5	7.0	3.4	65	77	**1LE1002-1DB4■-■■■■**	73	0.056	16
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz [1]																			
1.5	**1.75**	**100 L**	940	15	IE1	75.2	76.0	72.4	0.74	3.9	2.0	4.0	2.2	59	71	**1LE1002-1AC4■-■■■■**	19	0.0065	16
2.2	**2.55**	**112 M**	930	23	IE1	77.7	78.8	76.9	0.75	5.4	2.3	4.1	2.5	57	69	**1LE1002-1BC2■-■■■■**	25	0.0092	16
3	**3.45**	**132 S**	955	30	IE1	79.7	80.2	77.7	0.74	7.3	2.0	4.6	2.6	63	75	**1LE1002-1CC0■-■■■■**	34	0.017	16
4	**4.55**	**132 M**	950	40	IE1	81.4	82.9	82.1	0.76	9.3	2.1	4.7	2.5	63	75	**1LE1002-1CC2■-■■■■**	39	0.021	16
5.5	**6.3**	**132 M**	950	55	IE1	83.1	84.6	84.0	0.75	12.7	2.5	5.2	2.8	63	75	**1LE1002-1CC3■-■■■■**	48	0.027	16
7.5	**8.6**	**160 M**	970	74	IE1	84.7	85.4	85.0	0.73	17.5	2.1	5.5	2.9	67	79	**1LE1002-1DC2■-■■■■**	72	0.056	16
11	**12.6**	**160 L**	965	109	IE1	86.4	86.4	85.4	0.77	24	1.9	5.9	2.7	67	79	**1LE1002-1DC4■-■■■■**	92	0.078	16
8-pole: 750 rpm at 50 Hz, 900 rpm at 60 Hz [1]																			
0.75	**0.86**	**100 L**	705	10	–	62.6	60.8	53.9	0.62	3.0	1.9	3.0	2.2	60	72	**1LE1002-1AD4■-■■■■**	17	0.0056	16
1.1	**1.27**	**100 L**	705	15	–	65.5	64.2	60.0	0.63	3.9	2.0	3.2	2.3	60	72	**1LE1002-1AD5■-■■■■**	22	0.0078	16
1.5	**1.75**	**112 M**	700	20	–	71.6	72.2	68.5	0.65	4.7	1.6	3.3	1.9	63	75	**1LE1002-1BD2■-■■■■**	29	0.0094	16
2.2	**2.55**	**132 S**	715	29	–	76.8	77.4	75.2	0.66	6.3	1.7	3.9	2.4	63	75	**1LE1002-1CD0■-■■■■**	37	0.019	16
3	**3.45**	**132 M**	715	40	–	76.6	77.8	75.8	0.66	8.6	1.8	3.9	2.2	63	75	**1LE1002-1CD2■-■■■■**	44	0.024	16
4	**4.55**	**160 M**	720	53	–	78.3	78.5	75.6	0.69	10.7	1.7	3.8	2.3	63	75	**1LE1002-1DD2■-■■■■**	60	0.044	16
5.5	**6.3**	**160 M**	720	73	–	81.7	82.5	81.4	0.70	13.9	1.6	4.0	2.2	63	75	**1LE1002-1DD3■-■■■■**	72	0.056	16
7.5	**8.6**	**160 L**	715	100	–	83.5	84.5	83.6	0.70	18.5	1.7	3.8	2.2	63	75	**1LE1002-1DD4■-■■■■**	91	0.077	16

Voltages				No. of poles	Frame size	Motor type	Version				Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz [1]	460 VY	2, 4, 6, 8	100 L ... 160 L	1LE1002-1A ... -1D	**Standard**	**2**	**2**		–
50 Hz	400 VΔ/690 VY	60 Hz [1]	460 VΔ	2, 4, 6, 8	100 L ... 160 L	1LE1002-1A ... -1D	**Standard**	**3**	**4**		–
50 Hz	500 VY			2, 4, 6, 8	100 L ... 160 L	1LE1002-1A ... -1D	Without add. charge	**2**	**7**		–
50 Hz	500 VΔ			2, 4, 6, 8	100 L ... 160 L	1LE1002-1A ... -1D	Without add. charge	**4**	**0**		–
Further voltages [1]				For price information, code numbers, order codes and descriptions, see from Page 2/38				**9**	**0**		. . .
Types of construction				No. of poles	Frame size	Motor type	Version				Order code(s)
Without flange		IM B3 [2]		2, 4, 6, 8	100 L ... 160 L	1LE1002-1A ... -1D	**Standard**			**A**	–
With flange		IM B5 [2]		2, 4, 6, 8	100 L ... 160 L	1LE1002-1A ... -1D	With additional charge			**F**	–
With standard flange		IM B14 [2]		2, 4, 6, 8	100 L ... 160 L	1LE1002-1A ... -1D	With additional charge			**K**	–
Further types of construction				For price information, code letters and descriptions, see from Page 2/41						■	. . .
Motor protection				No. of poles	Frame size	Motor type	Version				Order code(s)
Without				2, 4, 6, 8	100 L ... 160 L	1LE1002-1A ... -1D	**Standard**			**A**	–
PTC thermistor with 3 temperature sensors				2, 4, 6, 8	100 L ... 160 L	1LE1002-1A ... -1D	With additional charge			**B**	–
Further motor protection				For price information, code letters and descriptions, see from Page 2/49						■	. . .
Terminal box position				No. of poles	Frame size	Motor type	Version				Order code(s)
Terminal box at top				2, 4, 6, 8	100 L ... 160 L	1LE1002-1A ... -1D	**Standard**			**4**	–
Further terminal box positions				For price information, code numbers and descriptions, see from Page 2/51							
Special versions				No. of poles	Frame size	Motor type					Order code(s)
Forced-air cooled motors without ext. fan/fan cover (IC 416)					100 L ... 160 L	1LE1002-1A ... -1D	**1LE1002- . . . . ■-■■■■-Z**				**F90 +. . .+. . .+. . .**
Options				For price information, order codes and descriptions, see from Page 2/53			**1LE1002- . . . . ■-■■■■-Z**				**. . .+. . .+. . .+. . .**

[1] Operating values at rated output for 60 Hz are available on request.

[2] Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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SIMOTICS GP 1LE1 Standard Motors

Motors with Standard Efficiency IE1

Self-ventilated motors
Aluminum series 1LE1002 with increased output

IE1 Please observe legal minimum efficiencies in the EEA!

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class	η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	$\cos\varphi$ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Aluminum series 1LE1002 – *IE1 version in accordance with IEC 60034-30 with increased output* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)		kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) • Efficiency: Standard Efficiency IE1, (SF) 1.1 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz 1)																			
4	**4.55**	**100 L**	2850	13	IE1	83.1	84.8	84.5	0.85	8.2	4.5	7.0	4.1	67	79	**1LE1002-1AA6■-■■■■**	25	0.0044	16
5.5	**6.3**	**112 M**	2935	18	IE1	84.7	85.5	84.7	0.86	10.9	2.9	7.5	3.8	69	81	**1LE1002-1BA6■-■■■■**	31	0.0085	16
11	**12.6**	**132 M**	2920	36	IE1	87.6	89.0	88.8	0.90	20	2.8	7.5	3.7	68	80	**1LE1002-1CA6■-■■■■**	53	0.022	16
22	**24.5**	**160 L**	2935	72	IE1	89.9	90.6	90.3	0.90	39	2.8	7.5	3.2	70	82	**1LE1002-1DA6■-■■■■**	85	0.049	16
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz 1)																			
4	**4.55**	**100 L**	1435	27	IE1	83.1	83.8	82.8	0.81	8.6	3.2	6.5	3.1	60	72	**1LE1002-1AB6■-■■■■**	27	0.010	16
5.5	**6.3**	**112 M**	1420	37	IE1	84.7	86.5	86.4	0.81	11.6	3.0	5.8	3.1	58	70	**1LE1002-1BB6■-■■■■**	33	0.012	16
11	**12.6**	**132 M**	1450	72	IE1	87.6	88.8	88.7	0.84	21.5	2.5	7.2	3.0	64	76	**1LE1002-1CB6■-■■■■**	58	0.033	16
18.5	**21.3**	**160 L**	1460	121	IE1	89.3	90.4	89.9	0.85	35	2.7	7.2	3.2	65	77	**1LE1002-1DB6■-■■■■**	85	0.068	16
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz 1)																			
2.2	**2.55**	**100 L**	930	22	IE1	77.7	78.5	77.5	0.78	5.2	2.0	4.0	2.2	59	71	**1LE1002-1AC6■-■■■■**	24	0.0084	16
3	**3.45**	**112 M**	945	30	IE1	79.7	79.7	76.6	0.72	7.5	2.5	4.6	2.6	57	69	**1LE1002-1BC6■-■■■■**	32	0.013	16
7.5	**8.6**	**132 M**	950	75	IE1	84.7	84.2	82.6	0.74	17.3	2.8	5.3	3.0	63	75	**1LE1002-1CC6■-■■■■**	54	0.032	16
15	**17.3**	**160 L**	965	148	IE1	87.7	88.2	86.8	0.75	33	2.9	6.0	3.4	67	79	**1LE1002-1DC6■-■■■■**	109	0.094	16

Voltages				No. of poles	Frame size	Motor type	Version				Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	2, 4, 6	100 L ... 160 L	1LE1002-1A ... -1D	**Standard**	**2**	**2**		–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ	2, 4, 6	100 L ... 160 L	1LE1002-1A ... -1D	**Standard**	**3**	**4**		–
50 Hz	500 VY			2, 4, 6	100 L ... 160 L	1LE1002-1A ... -1D	Without add. charge	**2**	**7**		–
50 Hz	500 VΔ			2, 4, 6	100 L ... 160 L	1LE1002-1A ... -1D	Without add. charge	**4**	**0**		–
Further voltages 1)				For price information, code numbers, order codes and descriptions, see from Page 2/38				**9**	**0**		. . .
Types of construction				No. of poles	Frame size	Motor type	Version				Order code(s)
Without flange		IM B3 2)		2, 4, 6	100 L ... 160 L	1LE1002-1A ... -1D	**Standard**			**A**	–
With flange		IM B5 2)		2, 4, 6	100 L ... 160 L	1LE1002-1A ... -1D	With additional charge			**F**	–
With standard flange		IM B14 2)		2, 4, 6	100 L ... 160 L	1LE1002-1A ... -1D	With additional charge			**K**	–
Further types of construction				For price information, code letters and descriptions, see from Page 2/41						■	. . .
Motor protection				No. of poles	Frame size	Motor type	Version				Order code(s)
Without				2, 4, 6	100 L ... 160 L	1LE1002-1A ... -1D	**Standard**			**A**	–
PTC thermistor with 3 temperature sensors				2, 4, 6	100 L ... 160 L	1LE1002-1A ... -1D	With additional charge			**B**	–
Further motor protection				For price information, code letters and descriptions, see from Page 2/49						■	. . .
Terminal box position				No. of poles	Frame size	Motor type	Version				Order code(s)
Terminal box at top				2, 4, 6	100 L ... 160 L	1LE1002-1A ... -1D	**Standard**			**4**	–
Further terminal box positions				For price information, code numbers and descriptions, see from Page 2/51							
Special versions				No. of poles	Frame size	Motor type					Order code(s)
Options				For price information, order codes and descriptions, see from Page 2/53			**1LE1002- . . . . ■-■■■■**			**-Z**	. . .+. . .+. . .+. . .

1) Operating values at rated output for 60 Hz are available on request.

2) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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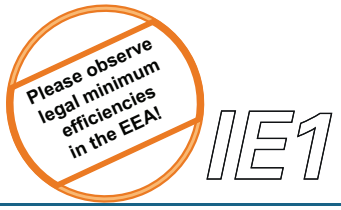

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz 1)	Frame size	Operating values at rated output: n_{rated}, 50 Hz	T_{rated}, 50 Hz	IE class	η_{rated}, 50 Hz, 4/4	η_{rated}, 50 Hz, 3/4	η_{rated}, 50 Hz, 2/4	$\cos\varphi$ rated, 50 Hz, 4/4	I_{rated}, 50 Hz, 400 V	T_{LR}/T_{rated}, 50 Hz	I_{LR}/I_{rated}, 50 Hz	T_B/T_{rated}, 50 Hz	L_{pfA}, 50 Hz	L_{WA}, 50 Hz	Aluminum series 1PC1002 – *IE1 version in accordance with IEC 60034-30* Article No.	$m_{IM\ B3}$	J	Torque class
kW	kW	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)		kg	kgm^2	CL
• Cooling: naturally cooled without external fan (IC 410) • Efficiency: Standard Efficiency IE1, service factor (SF) 1.1 • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz 1)																			
1.2	–	100 L	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1AA4 ■-■■■■	20	0.0034	16
1.6	–	112 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1BA2 ■-■■■■	25	0.0067	13
2.2	–	132 S	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1CA0 ■-■■■■	35	0.013	10
3	–	132 S	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1CA1 ■-■■■■	40	0.016	13
4.4	–	160 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1DA2 ■-■■■■	60	0.030	13
6	–	160 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1DA3 ■-■■■■	68	0.036	16
7.4	–	160 L	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1DA4 ■-■■■■	78	0.044	16
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz 1)																			
0.88	–	100 L	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1AB4 ■-■■■■	18	0.0059	13
1.2	–	100 L	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1AB5 ■-■■■■	22	0.0078	13
1.6	–	112 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1BB2 ■-■■■■	27	0.010	13
2.2	–	132 S	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1CB0 ■-■■■■	38	0.019	13
3	–	132 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1CB2 ■-■■■■	44	0.024	16
4.4	–	160 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1DB2 ■-■■■■	62	0.044	13
6	–	160 L	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1DB4 ■-■■■■	73	0.056	16
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz 1)																			
0.6	–	100 L	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1AC4 ■-■■■■	19	0.0065	10
0.88	–	112 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1BC2 ■-■■■■	25	0.0092	13
1.2	–	132 S	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1CC0 ■-■■■■	34	0.017	10
1.6	–	132 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1CC2 ■-■■■■	39	0.021	13
2.2	–	132 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1CC3 ■-■■■■	48	0.027	13
3	–	160 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1DC2 ■-■■■■	72	0.056	13
4.4	–	160 L	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1DC4 ■-■■■■	92	0.078	13
8-pole: 750 rpm at 50 Hz, 900 rpm at 60 Hz 1)																			
0.3	–	100 L	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1AD4 ■-■■■■	17	0.0056	10
0.44	–	100 L	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1AD5 ■-■■■■	22	0.0078	10
0.6	–	112 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1BD2 ■-■■■■	25	0.0094	10
0.88	–	132 S	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1CD0 ■-■■■■	37	0.019	10
1.2	–	132 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1CD2 ■-■■■■	44	0.024	10
1.6	–	160 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1DD2 ■-■■■■	60	0.044	10
2.2	–	160 M	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1DD3 ■-■■■■	72	0.056	10
3	–	160 L	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	–	–	1PC1002-1DD4 ■-■■■■	91	0.077	10

Voltages				No. of poles	Frame size	Motor type	Version			Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz 1)	460 VY	2, 4, 6, 8	100 L ... 160 L	1PC1002-1A ... -1D	**Standard**	**2**	**2**	–
50 Hz	400 VΔ/690 VY	60 Hz 1)	460 VΔ	2, 4, 6, 8	100 L ... 160 L	1PC1002-1A ... -1D	**Standard**	**3**	**4**	–
50 Hz	500 VY			2, 4, 6, 8	100 L ... 160 L	1PC1002-1A ... -1D	Without add. charge	**2**	**7**	–
50 Hz	500 VΔ			2, 4, 6, 8	100 L ... 160 L	1PC1002-1A ... -1D	Without add. charge	**4**	**0**	–
Further voltages 1)				For price information, code numbers, order codes and descriptions, see from Page 2/38				**9**	**0**	. . .

Types of construction		No. of poles	Frame size	Motor type	Version		Order code(s)
Without flange	IM B3 2)	2, 4, 6, 8	100 L ... 160 L	1PC1002-1A ... -1D	**Standard**	**A**	–
With flange	IM B5 2)	2, 4, 6, 8	100 L ... 160 L	1PC1002-1A ... -1D	With additional charge	**F**	–
With standard flange	IM B14 2)	2, 4, 6, 8	100 L ... 160 L	1PC1002-1A ... -1D	With additional charge	**K**	–
Further types of construction		For price information, code letters and descriptions, see from Page 2/41				■	. . .

Motor protection	No. of poles	Frame size	Motor type	Version		Order code(s)
Without	2, 4, 6, 8	100 L ... 160 L	1PC1002-1A ... -1D	**Standard**	**A**	–
PTC thermistor with 3 temperature sensors	2, 4, 6, 8	100 L ... 160 L	1PC1002-1A ... -1D	With additional charge	**B**	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/49				■	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top	2, 4, 6, 8	100 L ... 160 L	1PC1002-1A ... -1D	**Standard**	**4**	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/51					

Special versions	No. of poles	Frame size	Motor type		Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/53			**1PC1002-. . . . ■-■■■■-Z**	. . .+. . .+. . .+. . .

Note: The rated outputs and weights may change slightly after they have been checked. Further electrical data can be calculated and supplied on receipt of order.

1) Operating values at rated output for 60 Hz are available on request.

2) Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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SIMOTICS GP 1LE1 Standard Motors – Eagle Line

NEMA Energy Efficient MG1 motors, Table 12-11

Self-ventilated or forced-air cooled motors
Aluminum series 1LE1021

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz	Frame size	Operating values at rated output: n_{rated}, 60 Hz	T_{rated}, 60 Hz	EISA CC No. CC032A	η_{rated}, 60 Hz, 4/4	η_{rated}, 60 Hz, 3/4	η_{rated}, 60 Hz, 2/4	$\cos\varphi$ rated, 60 Hz, 4/4	I_{rated}, 60 Hz, 460 V	T_{LR}/T_{rated}, 60 Hz	I_{LR}/I_{rated}, 60 Hz	T_B/T_{rated}, 60 Hz	L_{pfA}, 60 Hz	L_{WA}, 60 Hz	Aluminum series 1LE1021 – *NEMA Energy Efficient version* Article No.	$m_{IM\ B3}$	J	Torque class
kW	hp	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)		kg	kgm²	CL
• Cooling: self-ventilated (IC 411) or with order code F90 forced-air cooled without external fan and fan cover (IC 416)																			
• Efficiency: NEMA Energy Efficient, UL, CSA and service factor (SF) 1.15 – for operation in the USA and Canada, not admissible for exporting to Mexico																			
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz																			
0.75	1	80 M	3445	2.1	–	75.5	75.5	72.5	0.83	1.5	2.1	6.0	3.0	64	75	1LE1021-0DA2■-■■■■	9	0.0008	16
1.1	1.5	80 M	3465	3.0	–	82.5	82.5	81.5	0.82	2.05	3.1	7.2	3.8	64	75	1LE1021-0DA3■-■■■■	11	0.0011	16
1.5	2	90 S	3505	4.1	–	84.0	84.0	83.0	0.82	2.75	3.1	8.5	4.5	69	81	1LE1021-0EA0■-■■■■	13	0.0017	16
2.2	3	90 L	3510	6.0	–	85.5	85.5	84.5	0.83	3.9	3.0	8.7	4.6	69	81	1LE1021-0EA4■-■■■■	15	0.0021	16
3	4	100 L	3520	8.1	–	87.5	87.3	86.2	0.83	5.2	2.6	8.1	3.8	71	83	1LE1021-1AA4■-■■■■	21	0.0044	16
4	5	112 M	3565	9.9	✓	87.5	87.4	85.8	0.84	6.3	2.9	9.3	4.0	73	85	1LE1021-1BA2■-■■■■	27	0.0092	16
5.5	7.5	132 S	3555	15	✓	88.5	88.3	88.1	0.86	9.1	2.0	7.6	3.3	72	84	1LE1021-1CA0■-■■■■	39	0.020	16
7.5	10	132 S	3560	20	✓	89.5	89.6	89.6	0.87	12.1	2.3	8.2	3.6	72	84	1LE1021-1CA1■-■■■■	43	0.024	16
11	15	160 M	3560	30	✓	90.2	89.6	87.8	0.86	17.8	2.4	8.2	3.6	77	89	1LE1021-1DA2■-■■■■	67	0.045	16
15	20	160 M	3565	40	✓	90.2	89.9	88.0	0.87	24	2.8	8.4	3.9	77	89	1LE1021-1DA3■-■■■■	75	0.053	16
18.5	25	160 L	3565	50	✓	91.0	90.5	89.4	0.87	29.5	3.3	8.9	4.1	77	89	1LE1021-1DA4■-■■■■	84	0.061	16
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz																			
0.55	0.75	80 M	1750	3.0	–	80.0	80.0	79.0	0.74	1.17	2.4	5.7	3.3	55	66	1LE1021-0DB2■-■■■■	10	0.0017	16
0.75	1	80 M	1750	4.1	–	82.5	82.5	81.5	0.72	1.58	2.5	6.8	3.8	55	66	1LE1021-0DB3■-■■■■	11	0.0021	16
1.1	1.5	90 S	1740	6.0	–	84.0	84.0	83.0	0.74	2.2	2.7	7.0	3.6	58	70	1LE1021-0EB0■-■■■■	13	0.0028	16
1.5	2	90 L	1745	8.2	–	84.0	84.0	83.0	0.75	3.0	2.9	7.5	4.0	58	70	1LE1021-0EB4■-■■■■	16	0.0036	16
2.2	3	100 L	1760	12	–	87.5	87.5	86.5	0.78	4.05	2.5	8.1	3.9	62	74	1LE1021-1AB4■-■■■■	21	0.0086	16
3	4	100 L	1765	16	–	87.5	88.3	87.1	0.79	5.4	2.4	8.3	3.7	62	74	1LE1021-1AB5■-■■■■	25	0.011	16
4	5	112 M	1770	20	✓	87.5	87.0	86.0	0.77	6.9	3.0	8.7	4.0	62	74	1LE1021-1BB2■-■■■■	29	0.014	16
5.5	7.5	132 S	1770	30	✓	89.5	89.6	88.3	0.78	9.9	2.6	8.0	3.3	68	80	1LE1021-1CB0■-■■■■	42	0.027	16
7.5	10	132 M	1770	40	✓	89.5	90.3	89.5	0.82	12.8	2.7	8.0	3.4	68	80	1LE1021-1CB2■-■■■■	49	0.034	16
11	15	160 M	1775	59	✓	91.0	91.3	90.5	0.84	18.1	2.5	7.7	3.2	69	81	1LE1021-1DB2■-■■■■	71	0.065	16
15	20	160 L	1780	80	✓	91.0	90.7	89.9	0.84	24.5	2.6	8.5	3.4	69	81	1LE1021-1DB4■-■■■■	83	0.083	16

Voltages (≤ 600 V) [1]				No. of poles	Frame size	Motor type	Version				Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz	460 VY	2, 4	80 M ... 160 L	1LE1021-0D ... -1D	**Standard**	**2**	**2**		–
50 Hz	400 VΔ	60 Hz	460 VΔ	2, 4	80 M ... 160 L	1LE1021-0D ... -1D	**Standard**	**3**	**4**		–
50 Hz	500 VY			2, 4	80 M ... 160 L	1LE1021-0D ... -1D	Without add. charge	**2**	**7**		–
50 Hz	500 VΔ			2, 4	80 M ... 160 L	1LE1021-0D ... -1D	Without add. charge	**4**	**0**		–
Further voltages				For price information, code numbers, order codes and descriptions, see from Page 2/38				**9**	**0**		. . .

Types of construction [2]		No. of poles	Frame size	Motor type	Version			Order code(s)
With flange	IM B5 [3]	2, 4	80 M ... 160 L	1LE1021-0D ... -1D	With additional charge		**F**	–
With standard flange	IM B14 [3]	2, 4	80 M ... 160 L	1LE1021-0D ... -1D	With additional charge		**K**	–
Further types of construction		For price information, code letters and descriptions, see from Page 2/41					■	. . .

Motor protection	No. of poles	Frame size	Motor type	Version			Order code(s)
Without	2, 4	80 M ... 160 L	1LE1021-0D ... -1D	**Standard**		**A**	–
PTC thermistor with 3 temperature sensors	2, 4	80 M ... 160 L	1LE1021-0D ... -1D	With additional charge		**B**	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/49					■	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version			Order code(s)
Terminal box at top	2, 4	80 M ... 160 L	1LE1021-0D ... -1D	**Standard**		**4**	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/51						

Special versions	No. of poles	Frame size	Motor type		Order code(s)
Forced-air cooled motors without ext. fan/fan cover (IC 416)	2, 4	80 M ... 160 L	1LE1021-0D ... -1D	**1LE1021-. . . . ■-■■■■-Z**	**F90** +. . .+. . .+. . .
Options	For price information, order codes and descriptions, see from Page 2/53			**1LE1021-. . . . ■-■■■■-Z**	. . .+. . .+. . .+. . .

– Not required
✓ available

[1] Operating voltages only ≤ 600 V admissible in accordance with MG1 Table 12-11.

[2] Types of construction with feet are not possible for 2-pole, 4-pole and 6-pole motors ≤ 200 hp in accordance with MG1 Table 12-11.

[3] Types derived from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz	Frame size	Operating values at rated output: n_{rated}, 60 Hz	T_{rated}, 60 Hz	EISA CC No. CC032A	η_{rated}, 60 Hz, 4/4	η_{rated}, 60 Hz, 3/4	η_{rated}, 60 Hz, 2/4	$\cos\varphi$ rated, 60 Hz, 4/4	I_{rated}, 60 Hz, 460 V	T_{LR}/T_{rated}, 60 Hz	I_{LR}/I_{rated}, 60 Hz	T_B/T_{rated}, 60 Hz	L_{pfA}, 60 Hz	L_{WA}, 60 Hz	Aluminum series 1LE1021 – *NEMA Energy Efficient version* Article No.	$m_{IM\ B3}$	J	Torque class
kW	hp	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)		kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) or with order code F90 forced-air cooled without external fan and fan cover (IC 416)																			
• Efficiency: NEMA Energy Efficient, UL, CSA and service factor (SF) 1.15 – for operation in the USA and Canada, not admissible for exporting to Mexico																			
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz																			
0.37	**0.5**	**80 M**	1140	3.1	–	75.3	74.1	69.5	0.63	0.98	2.3	4.6	2.9	45	56	**1LE1021-0DC2■-■■■■**	9	0.0017	16
0.55	**0.75**	**80 M**	1135	4.6	–	77.0	77.5	74.0	0.61	1.47	2.9	5.2	3.6	45	56	**1LE1021-0DC3■-■■■■**	12	0.0025	16
0.75	**1**	**90 S**	1155	6.2	–	80.0	80.0	79.0	0.69	1.98	2.2	5.3	3.0	46	58	**1LE1021-0EC0■-■■■■**	16	0.0040	16
1.1	**1.5**	**100 L**	1175	12.0	–	85.5	85.5	84.5	0.73	2.8	2.3	6.8	3.3	62	74	**1LE1021-1AC3■-■■■■**	25	0.011	16
1.5	**2**	**100 L**	1175	12	–	86.5	86.0	84.4	0.69	3.15	2.3	7.0	3.4	62	74	**1LE1021-1AC4■-■■■■**	25	0.011	16
2.2	**3**	**112 M**	1170	18	✓	87.5	87.4	85.9	0.73	4.3	2.3	7.3	3.4	60	72	**1LE1021-1BC2■-■■■■**	29	0.014	16
3	**4**	**132 S**	1175	24	–	87.5	87.6	85.9	0.70	6.1	1.8	6.5	3.0	67	79	**1LE1021-1CC0■-■■■■**	38	0.024	13
4	**5**	**132 M**	1180	30	✓	87.5	88.3	87.0	0.73	7.3	2.1	6.6	3.2	67	79	**1LE1021-1CC2■-■■■■**	43	0.029	13
5.5	**7.5**	**132 M**	1175	45	✓	89.5	89.7	88.7	0.74	10.4	2.0	7.1	3.2	67	79	**1LE1021-1CC3■-■■■■**	52	0.037	16
7.5	**10**	**160 M**	1180	61	✓	89.5	89.6	88.4	0.73	14.4	2.1	5.4	2.5	70	82	**1LE1021-1DC2■-■■■■**	77	0.075	16
11	**15**	**160 L**	1180	89	✓	90.2	90.5	89.5	0.74	20.5	2.2	5.5	2.5	70	82	**1LE1021-1DC4■-■■■■**	93	0.098	16

Voltages (≤ 600 V) [1]				No. of poles	Frame size	Motor type	Version					Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz	460 VY	6	80 M ... 160 L	1LE1021-0D ... -1D	**Standard**	**2**	**2**			–
50 Hz	400 VΔ	60 Hz	460 VΔ	6	80 M ... 160 L	1LE1021-0D ... -1D	**Standard**	**3**	**4**			–
50 Hz	500 VY			6	80 M ... 160 L	1LE1021-0D ... -1D	Without add. charge	**2**	**7**			–
50 Hz	500 VΔ			6	80 M ... 160 L	1LE1021-0D ... -1D	Without add. charge	**4**	**0**			–
Further voltages				For price information, code numbers, order codes and descriptions, see from Page 2/38				**9**	**0**			. . .
Types of construction [2]				No. of poles	Frame size	Motor type	Version					Order code(s)
With flange		IM B5 [3]		6	80 M ... 160 L	1LE1021-0D ... -1D	With additional charge			**F**		–
With standard flange		IM B14 [3]		6	80 M ... 160 L	1LE1021-0D ... -1D	With additional charge			**K**		–
Further types of construction				For price information, code letters and descriptions, see from Page 2/41						■		. . .
Motor protection				No. of poles	Frame size	Motor type	Version					Order code(s)
Without				6	80 M ... 160 L	1LE1021-0D ... -1D	**Standard**				**A**	–
PTC thermistor with 3 temperature sensors				6	80 M ... 160 L	1LE1021-0D ... -1D	With additional charge				**B**	–
Further motor protection				For price information, code letters and descriptions, see from Page 2/49							■	. . .
Terminal box position				No. of poles	Frame size	Motor type	Version					Order code(s)
Terminal box at top				6	80 M ... 160 L	1LE1021-0D ... -1D	**Standard**				**4**	–
Further terminal box positions				For price information, code numbers and descriptions, see from Page 2/51								
Special versions				No. of poles	Frame size	Motor type						Order code(s)
Forced-air cooled motors without ext. fan/fan cover (IC 416)				6	100 L ... 160 L	1LE1021-0D ... -1D	**1LE1021- . . . . ■-■■■■ -Z**					**F90** +. . .+. . .+. . .
Options				For price information, order codes and descriptions, see from Page 2/53			**1LE1021- . . . . ■-■■■■ -Z**					. . .+. . .+. . .+. . .

– Not required
✓ available

1) Operating voltages only ≤ 600 V admissible in accordance with MG1 Table 12-11.

2) Types of construction with feet are not possible for 2-pole, 4-pole and 6-pole motors ≤ 200 hp in accordance with MG1 Table 12-11.

3) Types derived from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (**H03**) and stamping of the type on the rating plate. The basic type IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (**H03**), the type must be specified.

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SIMOTICS SD 1LE1 Standard Motors – Eagle Line

NEMA Energy Efficient MG1 motors, Table 12-11

Self-ventilated motors
Cast-iron series 1LE1521/1LE1621 Basic/Performance Line

IE2

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz	Frame size	Operating values at rated output: n_{rated}, 60 Hz	T_{rated}, 60 Hz	EISA CC No. CC032A	η_{rated}, 60 Hz, 4/4	η_{rated}, 60 Hz, 3/4	η_{rated}, 60 Hz, 2/4	$\cos\varphi$ rated, 60 Hz, 4/4	I_{rated}, 60 Hz, 460 V	T_{LR}/T_{rated}, 60 Hz	I_{LR}/I_{rated}, 60 Hz	T_B/T_{rated}, 60 Hz	L_{pfA}, 60 Hz	L_{WA}, 60 Hz	Cast-iron series 1LE1521 – *Basic Line* 1LE1621 – *Performance Line* *NEMA Energy Efficient version* Article No.	$m_{IM\ B3}$	J	Torque class
kW	hp	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm²	CL
• Cooling: self-ventilated (IC 411)																			
• Efficiency: NEMA Energy Efficient, UL, CSA and service factor (SF) 1.15 – for operation in the USA and Canada, not admissible for exporting to Mexico																			
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz																			
0.37	0.5	71 M	3410	1.0	–	72.0	71.2	67.6	0.77	0.84	2.9	5.1	3.0	57	68	▲ 1LE1 5 21-0CA2 ■-■■■■	11.5	0.00035	16
0.55	0.75	71 M	3420	1.5	–	74.0	73.3	69.2	0.76	1.23	3.4	5.4	3.4	57	68	▲ 1LE1 5 21-0CA3 ■-■■■■	13	0.00045	16
0.75	1	80 M	3445	2.1	✓	75.5	75.5	72.5	0.83	1.5	2.1	6.0	3.0	64	75	▲ 1LE1 5 21-0DA2 ■-■■■■	16	0.0008	16
1.1	1.5	80 M	3465	3.0	✓	82.5	82.5	81.5	0.82	2.05	3.1	7.2	3.8	64	75	▲ 1LE1 5 21-0DA3 ■-■■■■	18	0.0011	16
1.5	2	90 S	3505	4.1	✓	84.0	84.0	83.0	0.82	2.75	3.1	8.5	4.5	69	81	▲ 1LE1 5 21-0EA0 ■-■■■■	23	0.0017	16
2.2	3	90 L	3510	6.0	✓	85.5	85.5	84.5	0.83	3.9	3.0	8.7	4.6	69	81	▲ 1LE1 5 21-0EA4 ■-■■■■	25.5	0.0021	16
3	4	100 L	3520	8.1	–	87.5	87.3	86.2	0.83	5.2	2.6	8.1	3.8	71	83	1LE1 ■ 21-1AA4 ■-■■■■	32	0.0044	16
4	5	112 M	3565	9.9	✓	87.5	87.4	85.8	0.84	6.3	2.9	9.3	4.0	73	85	1LE1 ■ 21-1BA2 ■-■■■■	39	0.0092	16
5.5	7.5	132 S	3555	15	✓	88.5	88.3	88.1	0.86	9.1	2.0	7.6	3.3	72	84	1LE1 ■ 21-1CA0 ■-■■■■	57	0.020	16
7.5	10	132 S	3560	20	✓	89.5	89.6	89.6	0.87	12.1	2.3	8.2	3.6	72	84	1LE1 ■ 21-1CA1 ■-■■■■	61	0.024	16
11	15	160 M	3560	30	✓	90.2	89.6	87.8	0.86	17.8	2.4	8.2	3.6	77	89	1LE1 ■ 21-1DA2 ■-■■■■	96	0.045	16
15	20	160 M	3565	40	✓	90.2	89.9	88.0	0.87	24	2.8	8.4	3.9	77	89	1LE1 ■ 21-1DA3 ■-■■■■	104	0.053	16
18.5	25	160 L	3555	50	✓	91.0	90.5	89.4	0.87	29.5	3.3	8.9	4.1	77	89	1LE1 ■ 21-1DA4 ■-■■■■	113	0.061	16
22	30	180 M	3550	60	✓	91.0	91.0	89.6	0.86	36	3.0	8.4	4.1	74	88	1LE1 ■ 21-1EA2 ■-■■■■	145	0.069	16
30	40	200 L	3565	80	✓	91.7	91.2	89.6	0.86	44.5	2.9	7.7	3.8	76	89	1LE1 ■ 21-2AA4 ■-■■■■	200	0.13	16
37	50	200 L	3565	100	✓	92.4	92.1	91.0	0.87	58	3.3	8.1	3.8	78	91	1LE1 ■ 21-2AA5 ■-■■■■	225	0.15	16
45	60	225 M	3570	120	✓	93.0	92.7	91.3	0.88	69	3.1	8.7	3.8	77	90	1LE1 ■ 21-2BA2 ■-■■■■	295	0.23	16
55	75	250 M	3575	149	–	93.0	92.5	91.0	0.89	85	2.4	7.4	3.5	80	94	1LE1 ■ 21-2CA2 ■-■■■■	360	0.40	13
75	100	280 S	3580	199	–	93.6	92.9	91.1	0.87	115	2.8	7.7	3.5	81	95	1LE1 ■ 21-2DA0 ■-■■■■	490	0.71	13
90	125	280 M	3578	249	✓	94.5	94.2	93.1	0.88	141	2.7	7.9	3.4	81	95	1LE1 ■ 21-2DA2 ■-■■■■	530	0.83	13
110	150	315 S	3585	298	✓	94.5	94.0	92.5	0.90	165	2.6	7.9	3.3	82	96	1LE1 ■ 21-3AA0 ■-■■■■	720	1.3	13
132	175	315 M	3585	348	–	95.0	94.7	93.6	0.91	190	2.7	8.1	3.4	82	96	1LE1 ■ 21-3AA2 ■-■■■■	880	1.6	13
150	200	315 L	3585	397	✓	95.0	94.6	93.3	0.92	215	2.9	8.3	3.5	84	99	1LE1 ■ 21-3AA4 ■-■■■■	930	1.8	13
185	250	315 L	3585	497	✓	95.4	95.2	94.2	0.92	265	3.5	8.9	3.5	84	99	1LE1 ■ 21-3AA5 ■-■■■■	1130	2.2	13

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects							
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5						
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6						

Voltages (≤ 600 V) [1]			No. of poles	Frame size	Motor type	Version						Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz 460 VY	2	71 M ... 315 L	1LE1■21-0C ... -3A	**Standard**	2	2				–
50 Hz	400 VΔ	60 Hz 460 VΔ	2	71 M ... 315 L	1LE1■21-0C ... -3A	**Standard**	3	4				–
50 Hz	500 VY		2	71 M ... 315 L	1LE1■21-0C ... -3A	Without add. charge	2	7				–
50 Hz	500 VΔ		2	71 M ... 315 L	1LE1■21-0C ... -3A	Without add. charge	4	0				–
Further voltages	For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0				. . .

Types of construction [2]		No. of poles	Frame size	Motor type	Version					Order code(s)
Without flange	IM B3 [3]	2	315 L > 200 hp	1LE1■21-3AA5	**Standard**			A		–
With flange	IM B5 [3]	2	71 M ... 315 M	1LE1■21-0C ... -3A	With additional charge			F		–
With standard flange	IM B14 [3]	2	71 M ... 160 L	1LE1■21-0C ... -1D	With additional charge			K		–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45							■		. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version					Order code(s)
Without	Only possible for **Basic Line**	2	71 M ... 315 L	1LE1521-0C ... -3A	**Standard**			A		–
PTC thermistor with 3 temperature sensors	**Basic Line**	2	71 M ... 315 L	1LE1521-0C ... -3A	With additional charge			B		–
	Performance Line	2	71 M ... 315 L	1LE1621-0C ... -3A	**Standard**			B		–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50							■		. . .

Terminal box position		No. of poles	Frame size	Motor type	Version					Order code(s)
Terminal box at top		2	71 M ... 315 L	1LE1■21-0C ... -3A	**Standard**			4		–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52									

Special versions		No. of poles	Frame size	Motor type						Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58				1LE1■21-. . . .	■	■	■	■	-Z . . .+. . .+. . .+. . .

– Not required
✓ available

[1] Operating voltages only ≤ 600 V admissible in accordance with MG1 Table 12-11. Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

[2] Types of construction with feet are not possible for 2-pole, 4-pole and 6-pole motors ≤ 200 hp in accordance with MG1 Table 12-11.

[3] Types derived from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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SIMOTICS SD 1LE1 Standard Motors – Eagle Line

NEMA Energy Efficient MG1 motors, Table 12-11

Self-ventilated motors
Cast-iron series 1LE1521/1LE1621 Basic/Performance Line

Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz	Frame size	Operating values at rated output: n_{rated}, 60 Hz	T_{rated}, 60 Hz	EISA CC No. CC032A	η_{rated}, 60 Hz, 4/4	η_{rated}, 60 Hz, 3/4	η_{rated}, 60 Hz, 2/4	$\cos\varphi$ rated, 60 Hz, 4/4	I_{rated}, 60 Hz, 460 V	T_{LR}/T_{rated}, 60 Hz	I_{LR}/I_{rated}, 60 Hz	T_B/T_{rated}, 60 Hz	L_{pfA}, 60 Hz	L_{WA}, 60 Hz	Cast-iron series 1LE1521 – *Basic Line* 1LE1621 – *Performance Line* *NEMA Energy Efficient version* Article No.	$m_{IM\ B3}$	J	Torque class
kW	hp	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411)																			
• Efficiency: NEMA Energy Efficient, UL, CSA and service factor (SF) 1.15 – for operation in the USA and Canada, not admissible for exporting to Mexico																			
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz																			
0.25	0.33	71 M	1715	1.4	–	70.0	68.5	63.5	0.64	0.7	2.8	4.4	3.1	47	58	▲ 1LE1 5 21-0CB2 ■-■■■■	12	0.00076	16
0.37	0.5	71 M	1705	2.1	–	72.0	71.5	67.5	0.67	0.96	2.8	4.4	2.8	47	58	▲ 1LE1 5 21-0CB3 ■-■■■■	13	0.00095	16
0.55	0.75	80 M	1750	3.0	–	80.0	80.0	79.0	0.74	1.17	2.4	5.7	3.3	55	66	▲ 1LE1 5 21-0DB2 ■-■■■■	17	0.0017	16
0.75	1	80 M	1750	4.1	–	82.5	82.5	81.5	0.72	1.58	2.5	6.8	3.8	55	66	▲ 1LE1 5 21-0DB3 ■-■■■■	18.5	0.0021	16
1.1	1.5	90 S	1740	6.0	–	84.0	84.0	83.0	0.74	2.2	2.7	7.0	3.6	58	70	▲ 1LE1 5 21-0EB0 ■-■■■■	23	0.0028	16
1.5	2	90 L	1745	8.2	–	84.0	84.0	83.0	0.75	3.0	2.9	7.5	4.0	58	70	▲ 1LE1 5 21-0EB4 ■-■■■■	25	0.0036	16
2.2	3	100 L	1760	12	–	87.5	87.5	86.5	0.78	4.05	2.5	8.1	3.9	62	74	1LE1 ■ 21-1AB4 ■-■■■■	32	0.0086	16
3	4	100 L	1765	16	–	87.5	88.3	87.1	0.79	5.4	2.4	8.3	3.7	62	74	1LE1 ■ 21-1AB5 ■-■■■■	37	0.011	16
4	5	112 M	1770	20	✓	87.5	87.0	86.0	0.77	6.9	3.0	8.7	4.0	62	74	1LE1 ■ 21-1BB2 ■-■■■■	46	0.014	16
5.5	7.5	132 S	1770	30	✓	89.5	89.6	88.3	0.78	9.9	2.6	8.0	3.3	68	80	1LE1 ■ 21-1CB0 ■-■■■■	61	0.027	16
7.5	10	132 M	1770	40	✓	89.5	90.3	89.5	0.82	12.8	2.7	8.0	3.4	68	80	1LE1 ■ 21-1CB2 ■-■■■■	75	0.034	16
11	15	160 M	1775	59	✓	91.0	91.3	90.5	0.84	18.1	2.5	7.7	3.2	69	81	1LE1 ■ 21-1DB2 ■-■■■■	96	0.065	16
15	20	160 L	1780	80	✓	91.0	90.7	89.9	0.84	24.5	2.6	8.5	3.4	69	81	1LE1 ■ 21-1DB4 ■-■■■■	104	0.083	16
18.5	25	180 M	1770	101	✓	92.4	92.6	91.8	0.83	31	2.8	7.7	3.9	64	77	1LE1 ■ 21-1EB2 ■-■■■■	160	0.12	16
22	30	180 L	1770	121	✓	92.4	92.5	91.8	0.83	36.5	3.0	8.4	3.9	63	76	1LE1 ■ 21-1EB4 ■-■■■■	170	0.13	16
30	40	200 L	1778	160	✓	93.0	92.9	92.2	0.84	48	3.2	8.2	3.7	66	79	1LE1 ■ 21-2AB5 ■-■■■■	230	0.20	16
37	50	225 S	1778	200	–	93.0	93.2	92.5	0.87	58	2.7	7.2	3.3	69	82	1LE1 ■ 21-2BB0 ■-■■■■	280	0.42	16
45	60	225 M	1778	240	✓	93.6	93.8	93.1	0.86	70	3.0	7.6	3.5	69	83	1LE1 ■ 21-2BB2 ■-■■■■	305	0.46	16
55	75	250 M	1785	299	–	94.1	94.1	93.3	0.84	89	3.1	7.3	3.3	69	83	1LE1 ■ 21-2CB2 ■-■■■■	385	0.75	16
75	100	280 S	1788	398	–	94.5	94.3	93.2	0.87	114	2.7	7.6	3.2	79	92	1LE1 ■ 21-2DB0 ■-■■■■	550	1.3	16
90	125	280 M	1788	498	✓	94.5	94.3	93.3	0.87	142	2.9	8.1	3.4	78	92	1LE1 ■ 21-2DB2 ■-■■■■	570	1.4	16
110	150	315 S	1790	597	✓	95.0	94.8	93.8	0.86	172	3.1	8.0	3.3	79	93	1LE1 ■ 21-3AB0 ■-■■■■	740	2.0	16
132	175	315 M	1790	696	–	95.0	94.8	94.0	0.86	200	3.1	7.8	3.2	79	93	1LE1 ■ 21-3AB2 ■-■■■■	870	2.3	16
150	200	315 L	1790	796	✓	95.0	94.7	93.5	0.87	225	3.5	8.6	3.6	80	95	1LE1 ■ 21-3AB4 ■-■■■■	940	2.8	16
185	250	315 L	1792	994	✓	95.0	94.7	93.6	0.86	285	4.3	9.3	3.9	84	98	1LE1 ■ 21-3AB5 ■-■■■■	1140	3.5	16

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects						
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5					
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6					

Voltages (≤ 600 V) [1]			No. of poles	Frame size	Motor type	Version					Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz	460 VY	4	71 M ... 315 L	1LE1■21-0C … -3A	**Standard**	2	2		–
50 Hz	400 VΔ	60 Hz	460 VΔ	4	71 M ... 315 L	1LE1■21-0C … -3A	**Standard**	3	4		–
50 Hz	500 VY			4	71 M ... 315 L	1LE1■21-0C … -3A	Without add. charge	2	7		–
50 Hz	500 VΔ			4	71 M ... 315 L	1LE1■21-0C … -3A	Without add. charge	4	0		–
Further voltages		For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0		. . .

Types of construction [2]		No. of poles	Frame size	Motor type	Version			Order code(s)
Without flange	IM B3 [3]	4	315 L > 200 hp	1LE1■21-3AB5	**Standard**		A	–
With flange	IM B5 [3]	4	71 M ... 315 M	1LE1■21-0C … -3A	With additional charge		F	–
With standard flange	IM B14 [3]	4	71 M ... 160 L	1LE1■21-0C … -1D	With additional charge		K	–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45						Z	. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version			Order code(s)
Without	Only possible for **Basic Line**	4	71 M ... 315 L	1LE1521-0C … -3A	**Standard**		A	–
PTC thermistor with 3 temperature sensors	**Basic Line**	4	71 M ... 315 L	1LE1521-0C … -3A	With additional charge		B	–
	Performance Line	4	71 M ... 315 L	1LE1621-0C … -3A	**Standard**		B	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50						Z	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version			Order code(s)
Terminal box at top	4	71 M ... 315 L	1LE1■21-0C … -3A	**Standard**		4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52						

Special versions	No. of poles	Frame size	Motor type				Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58			1LE1■21-. . . .	■-■■■■	-Z	. . .+. . .+. . .+. . .

– Not required
✓ available

[1] Operating voltages only ≤ 600 V admissible in accordance with MG1 Table 12-11. Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

[2] Types of construction with feet are not possible for 2-pole, 4-pole and 6-pole motors ≤ 200 hp in accordance with MG1 Table 12-11.

[3] Types derived from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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SIMOTICS SD 1LE1 Standard Motors – Eagle Line

NEMA Energy Efficient MG1 motors, Table 12-11

Self-ventilated motors
Cast-iron series 1LE1521/1LE1621 Basic/Performance Line

IE2

Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz	Frame size	Operating values at rated output: n_{rated}, 60 Hz	T_{rated}, 60 Hz	EISA CC No. CC032A	η_{rated}, 60 Hz, 4/4	η_{rated}, 60 Hz, 3/4	η_{rated}, 60 Hz, 2/4	cosφ rated, 60 Hz, 4/4	I_{rated}, 60 Hz, 460 V	T_{LR}/T_{rated}, 60 Hz	I_{LR}/I_{rated}, 60 Hz	T_B/T_{rated}, 60 Hz	L_{pfA}, 60 Hz	L_{WA}, 60 Hz	Cast-iron series 1LE1521 – *Basic Line* 1LE1621 – *Performance Line* *NEMA Energy Efficient version* Article No.	$m_{IM\ B3}$	J	Torque class
kW	hp	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm²	CL
• Cooling: self-ventilated (IC 411)																			
• Efficiency: NEMA Energy Efficient, UL, CSA and service factor (SF) 1.15 – for operation in the USA and Canada, not admissible for exporting to Mexico																			
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz																			
0.18	**0.25**	**71 M**	1105	1.6	–	55.0	54.0	49.0	0.61	0.67	2.9	2.7	2.9	42	53	▲ **1LE1 5 21-0CC2 ■-■■■■**	11.5	0.00080	16
0.25	**0.33**	**71 M**	1100	2.2	–	59.5	59.0	55.1	0.64	0.82	2.7	3.0	2.7	42	53	▲ **1LE1 5 21-0CC3 ■-■■■■**	12.5	0.00100	16
0.37	**0.5**	**80 M**	1140	3.1	–	75.3	74.1	69.5	0.63	0.98	2.3	4.6	2.9	45	56	▲ **1LE1 5 21-0DC2 ■-■■■■**	16.5	0.0017	16
0.55	**0.75**	**80 M**	1135	4.6	–	77.0	77.5	74.0	0.61	1.47	2.9	5.2	3.6	45	56	▲ **1LE1 5 21-0DC3 ■-■■■■**	18.5	0.0025	16
0.75	**1**	**90 S**	1155	6.2	–	80.0	80.5	78.0	0.68	1.98	2.2	5.3	3.0	46	58	▲ **1LE1 5 21-0EC0 ■-■■■■**	26	0.0040	16
1.1	**1.5**	**90 L**	1175	12	–	85.5	85.5	84.5	0.73	2.8	2.3	6.8	3.3	62	74	▲ **1LE1 5 21-0EC4 ■-■■■■**	32	0.011	16
1.5	**2**	**100 L**	1175	12	–	86.5	86.0	84.4	0.69	3.15	2.3	7.0	3.4	62	74	**1LE1 ■ 21-1AC4 ■-■■■■**	36	0.011	16
2.2	**3**	**112 M**	1170	18	✓	87.5	87.4	85.9	0.73	4.3	2.3	7.3	3.4	60	72	**1LE1 ■ 21-1BC2 ■-■■■■**	41	0.014	16
3	**4**	**132 S**	1175	24	–	87.5	87.6	85.9	0.70	6.1	1.8	6.5	3.0	67	79	**1LE1 ■ 21-1CC0 ■-■■■■**	56	0.024	13
4	**5**	**132 M**	1180	30	✓	87.5	88.3	87.0	0.73	7.3	2.1	6.6	3.2	67	79	**1LE1 ■ 21-1CC2 ■-■■■■**	61	0.029	13
5.5	**7.5**	**132 M**	1175	45	✓	89.5	89.7	88.7	0.74	10.4	2.0	7.1	3.2	67	79	**1LE1 ■ 21-1CC3 ■-■■■■**	70	0.037	16
7.5	**10**	**160 M**	1180	61	✓	89.5	89.6	88.4	0.73	14.4	2.1	5.4	2.5	70	82	**1LE1 ■ 21-1DC2 ■-■■■■**	106	0.075	16
11	**15**	**160 L**	1180	89	✓	90.2	90.5	89.5	0.74	20.5	2.2	5.5	2.5	70	82	**1LE1 ■ 21-1DC4 ■-■■■■**	122	0.098	16
15	**20**	**180 L**	1178	121	✓	90.2	90.2	89.0	0.77	27	2.8	6.9	3.4	59	72	**1LE1 ■ 21-1EC4 ■-■■■■**	155	0.17	16
18.5	**25**	**200 L**	1182	151	✓	91.7	92.1	91.5	0.81	31.5	2.6	6.7	3.0	58	71	**1LE1 ■ 21-2AC4 ■-■■■■**	200	0.25	16
22	**30**	**200 L**	1182	181	✓	91.7	92.1	91.5	0.81	38	3.0	7.4	3.0	62	76	**1LE1 ■ 21-2AC5 ■-■■■■**	220	0.30	16
30	**40**	**225 M**	1182	241	✓	93.0	93.3	92.6	0.83	48.5	2.9	7.0	3.1	66	79	**1LE1 ■ 21-2BC2 ■-■■■■**	300	0.58	16
37	**50**	**250 M**	1185	301	–	93.0	93.3	92.6	0.83	61	3.3	7.3	2.8	66	79	**1LE1 ■ 21-2CC2 ■-■■■■**	370	0.86	16
45	**60**	**280 S**	1188	360	–	93.6	93.8	93.1	0.84	71	3.1	7.4	3.0	67	81	**1LE1 ■ 21-2DC0 ■-■■■■**	460	1.1	16
55	**75**	**280 M**	1188	450	–	93.6	93.9	93.4	0.85	88	3.1	7.2	2.9	67	81	**1LE1 ■ 21-2DC2 ■-■■■■**	510	1.4	16
75	**100**	**315 S**	1190	599	✓	94.1	94.1	93.2	0.83	120	2.7	7.5	3.0	67	82	**1LE1 ■ 21-3AC0 ■-■■■■**	670	2.1	16
90	**125**	**315 M**	1190	748	✓	94.1	94.4	93.5	0.84	148	2.9	7.6	3.1	68	83	**1LE1 ■ 21-3AC2 ■-■■■■**	730	2.5	16
110	**150**	**315 L**	1190	898	✓	95.0	95.0	94.6	0.85	174	3.3	8.1	3.2	69	84	**1LE1 ■ 21-3AC4 ■-■■■■**	940	3.6	16
132	**175**	**315 L**	1190	1048	–	95.0	95.0	94.4	0.85	205	3.7	9.2	3.6	69	84	**1LE1 ■ 21-3AC5 ■-■■■■**	990	4.0	16
150	**200**	**315 L**	1192	1195	✓	95.0	94.9	94.2	0.85	230	4.3	9.6	3.8	72	87	**1LE1 ■ 21-3AC6 ■-■■■■**	1160	4.7	16

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects						
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5					
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6					

Voltages (≤ 600 V) [1]			No. of poles	Frame size	Motor type	Version					Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz	460 VY	6	71 M ... 315 L	1LE1■21-0C … -3A	**Standard**	2	2		–
50 Hz	400 VΔ	60 Hz	460 VΔ	6	71 M ... 315 L	1LE1■21-0C … -3A	**Standard**	3	4		–
50 Hz	500 VY			6	71 M ... 315 L	1LE1■21-0C … -3A	Without add. charge	2	7		–
50 Hz	500 VΔ			6	71 M ... 315 L	1LE1■21-0C … -3A	Without add. charge	4	0		–
Further voltages		For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0		. . .

Types of construction [2]		No. of poles	Frame size	Motor type	Version			Order code(s)
With flange	IM B5 [3]	6	71 M ... 315 M	1LE1■21-0C … -3A	With additional charge		F	–
With standard flange	IM B14 [3]	6	71 M ... 160 L	1LE1■21-0C … -1D	With additional charge		K	–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45						Z	. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version			Order code(s)
Without	Only possible for **Basic Line**	6	71 M ... 315 L	1LE1521-0C … -3A	**Standard**		A	–
PTC thermistor with 3 temperature sensors	**Basic Line**	6	71 M ... 315 L	1LE1521-0C … -3A	With additional charge		B	–
	Performance Line	6	71 M ... 315 L	1LE1621-0C … -3A	**Standard**		B	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50						Z	. . .

Terminal box position		No. of poles	Frame size	Motor type	Version			Order code(s)
Terminal box at top		6	71 M ... 315 L	1LE1■21-0C … -3A	**Standard**		4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52							

Special versions		No. of poles	Frame size	Motor type				Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58				**1LE1■21-. . . .**	■	**-■■■■-Z**	. . .+. . .+. . .+. . .

– Not required
✓ available

[1] Operating voltages only ≤ 600 V admissible in accordance with MG1 Table 12-11. Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

[2] Types of construction with feet are not possible for 2-pole, 4-pole and 6-pole motors ≤ 200 hp in accordance with MG1 Table 12-11.

[3] Types derived from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz	Frame size	Operating values at rated output: n_{rated}, 60 Hz	T_{rated}, 60 Hz	EISA CC No. CC032A	η_{rated}, 60 Hz, 4/4	η_{rated}, 60 Hz, 3/4	η_{rated}, 60 Hz, 2/4	$\cos\varphi$ rated, 60 Hz, 4/4	I_{rated}, 60 Hz, 460 V	T_{LR}/T_{rated}, 60 Hz	I_{LR}/I_{rated}, 60 Hz	T_B/T_{rated}, 60 Hz	L_{pfA}, 60 Hz	L_{WA}, 60 Hz	Cast-iron series 1LE1521 – *Basic Line* 1LE1621 – *Performance Line* *NEMA Energy Efficient version* Article No.	$m_{IM\ B3}$	J	Torque class
kW	hp	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm²	CL
• Cooling: self-ventilated (IC 411)																			
• Efficiency: NEMA Energy Efficient, UL, CSA and service factor (SF) 1.15 – for operation in the USA and Canada, not admissible for exporting to Mexico																			
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
8-pole: 750 rpm at 50 Hz, 900 rpm at 60 Hz																			
11	15	180 L	875	122	✓	88.5	88.8	87.7	0.69	23	2.6	5.6	2.9	66	79	1LE1 ■ 21-1ED4 ■-■■■■	155	0.20	13
15	20	200 L	875	163	✓	89.5	90.7	90.9	0.74	28	2.8	6.3	3.3	59	72	1LE1 ■ 21-2AD5 ■-■■■■	220	0.34	13
18.5	25	225 S	885	201	✓	89.5	89.7	88.6	0.75	35	2.5	6.3	3.1	60	73	1LE1 ■ 21-2BD0 ■-■■■■	250	0.43	13
22	30	225 M	885	241	✓	91.0	91.3	90.4	0.78	39.5	2.5	6.4	3.0	61	74	1LE1 ■ 21-2BD2 ■-■■■■	270	0.50	13
30	40	250 M	885	322	✓	91.0	91.3	90.5	0.79	52	2.7	6.4	3.0	61	75	1LE1 ■ 21-2CD2 ■-■■■■	370	0.86	13
37	50	280 S	890	400	–	91.7	91.8	90.9	0.77	66	2.5	6.1	2.6	64	78	1LE1 ■ 21-2DD0 ■-■■■■	460	1.10	16
45	60	280 M	890	480	–	91.7	91.7	90.8	0.78	79	2.7	6.5	2.7	64	78	1LE1 ■ 21-2DD2 ■-■■■■	510	1.40	16
55	75	315 S	890	600	–	93.0	93.0	92.1	0.79	96	2.4	6.6	2.9	70	84	1LE1 ■ 21-3AD0 ■-■■■■	640	2.00	13
75	100	315 M	890	800	✓	93.0	93.3	92.9	0.80	126	2.5	6.7	3.0	73	87	1LE1 ■ 21-3AD2 ■-■■■■	710	2.50	16
90	125	315 L	890	1001	✓	93.6	93.9	93.6	0.81	154	2.4	6.5	2.8	74	88	1LE1 ■ 21-3AD4 ■-■■■■	860	3.10	13
110	150	315 L	891	1199	✓	93.6	93.9	93.6	0.81	185	2.8	7.2	3.2	72	86	1LE1 ■ 21-3AD5 ■-■■■■	980	3.90	16
132	175	315 L	892	1398	–	94.1	94.2	93.7	0.80	220	3.2	7.9	3.7	78	93	1LE1 ■ 21-3AD6 ■-■■■■	1060	4.50	16

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects					
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5				
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6				

Voltages (≤ 600 V) [1]			No. of poles	Frame size	Motor type	Version					Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz	460 VY	6	180 L ... 315 L	1LE1■21-1E … -3A	**Standard**	2	2		–
50 Hz	400 VΔ	60 Hz	460 VΔ	6	180 L ... 315 L	1LE1■21-1E … -3A	**Standard**	3	4		–
50 Hz	500 VY			6	180 L ... 315 L	1LE1■21-1E … -3A	Without add. charge	2	7		–
50 Hz	500 VΔ			6	180 L ... 315 L	1LE1■21-1E … -3A	Without add. charge	4	0		–
Further voltages		For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0		. . .

Types of construction [2]		No. of poles	Frame size	Motor type	Version				Order code(s)
With flange	IM B5 [3]	6	180 L ... 315 M	1LE1■21-1E … -3A	With additional charge			F	–
With standard flange	IM B14 [3]	6	180 L ... 160 L	1LE1■21-1E … -1D	With additional charge			K	–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45							Z	. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version				Order code(s)
Without	Only possible for **Basic Line**	6	180 L ... 315 L	1LE1521-1E … -3A	**Standard**			A	–
PTC thermistor with 3 temperature sensors	**Basic Line**	6	180 L ... 315 L	1LE1521-1E … -3A	With additional charge			B	–
	Performance Line	6	180 L ... 315 L	1LE1621-1E … -3A	**Standard**			B	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50							Z	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version					Order code(s)
Terminal box at top	6	180 L ... 315 L	1LE1■21-1E … -3A	**Standard**				4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52								

Special versions	No. of poles	Frame size	Motor type						Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58			1LE1■21-. . . .	■-■■■■	-Z	. . .+. . .+. . .+. . .		

– Not required
✓ available

[1] Operating voltages only ≤ 600 V admissible in accordance with MG1 Table 12-11. Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

[2] Types of construction with feet are not possible for 2-pole, 4-pole and 6-pole motors ≤ 200 hp in accordance with MG1 Table 12-11.

[3] Types derived from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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SIMOTICS GP 1LE1 Standard Motors – Eagle Line

NEMA Premium Efficient MG1 motors, Table 12-12

Self-ventilated or forced-air cooled motors
Aluminum series 1LE1023

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz	Frame size	Operating values at rated output: n_{rated}, 60 Hz	T_{rated}, 60 Hz	EISA CC No. CC032A	η_{rated}, 60 Hz, 4/4	η_{rated}, 60 Hz, 3/4	η_{rated}, 60 Hz, 2/4	cos φ rated, 60 Hz, 4/4	I_{rated}, 60 Hz, 460 V	T_{LR}/T_{rated}, 60 Hz	I_{LR}/I_{rated}, 60 Hz	T_B/T_{rated}, 60 Hz	L_{pfA}, 60 Hz	L_{WA}, 60 Hz	Aluminum series 1LE1023 – *NEMA Premium Efficient version* Article No.	$m_{IM\ B3}$	J	Torque class
kW	hp	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm²	CL
• Cooling: self-ventilated (IC 411) or with order code F90 forced-air cooled without external fan and fan cover (IC 416)																			
• Efficiency: NEMA Premium Efficient, UL, CSA and service factor (SF) 1.15 – for operation in the USA, Canada and Mexico																			
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz																			
0.75	**1**	**80 M**	3480	2.1	✓	77.0	78.0	76.0	0.84	1.46	3.0	7.1	3.6	64	75	**1LE1023-0DA2■-■■■■**	11	0,0011	16
1.1	**1.5**	**80 M**	3500	3	✓	84.0	84.0	83.0	0.83	1.98	3.3	8.4	4.5	64	75	**1LE1023-0DA3■-■■■■**	12	0,0013	16
1.5	**2**	**90 S**	3525	4.1	✓	85.5	85.0	82.5	0.84	2.60	3.1	9.8	4.9	69	81	**1LE1023-0EA0■-■■■■**	15	0,0021	16
2.2	**3**	**90 L**	3530	6	✓	86.5	86.3	84.5	0.87	3.65	3.0	9.6	4.9	69	81	**1LE1023-0EA4■-■■■■**	19	0,0031	16
3	**4**	**100 L**	3525	8.1	–	88.5	88.5	87.5	0.87	4.90	3.8	9.7	5.5	71	83	**1LE1023-1AA4■-■■■■**	26	0,0054	16
4	**5**	**112 M**	3565	9.9	✓	88.5	88.5	87.5	0.87	6.0	3.8	10.0	5.6	73	85	**1LE1023-1BA2■-■■■■**	34	0,012	16
5.5	**7.5**	**132 S**	3555	15	✓	89.5	89.5	88.5	0.90	8.6	2.1	8.6	4.4	72	84	**1LE1023-1CA0■-■■■■**	43	0,024	16
7.5	**10**	**132 S**	3555	20	✓	90.2	90.2	89.2	0.91	11.5	2.4	9.5	4.7	72	84	**1LE1023-1CA1■-■■■■**	57	0,031	16
11	**15**	**160 M**	3560	30	✓	91.0	91.0	90.0	0.88	17.2	2.8	8.5	4.3	77	89	**1LE1023-1DA2■-■■■■**	75	0,053	16
15	**20**	**160 M**	3565	40	✓	91.0	91.0	90.0	0.86	24	3.1	9.7	4.8	77	89	**1LE1023-1DA3■-■■■■**	84	0,061	16
18.5	**25**	**160 L**	3560	50	✓	91.7	91.7	90.7	0.90	28	3.1	9.4	4.4	77	89	**1LE1023-1DA4■-■■■■**	94	0,068	16
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz																			
0.55	**0.75**	**80 M**	1750	3	–	82.5	82.2	79.4	0.74	1.15	2.7	6.9	3.8	55	66	**1LE1023-0DB2■-■■■■**	11	0,0021	16
0.75	**1**	**80 M**	1760	4.1	✓	85.5	84.5	81.0	0.71	1.53	3.1	8.3	4.7	55	66	**1LE1023-0DB3■-■■■■**	14	0,0029	16
1.1	**1.5**	**90 S**	1750	6	✓	86.5	86.3	84.1	0.75	2.10	3.4	8.2	4.4	58	70	**1LE1023-0EB0■-■■■■**	16	0,0036	16
1.5	**2**	**90 L**	1755	8.2	✓	86.5	87.0	85.0	0.77	2.85	3.0	8.4	4.3	58	70	**1LE1023-0EB4■-■■■■**	19	0,0049	16
2.2	**3**	**100 L**	1770	12	–	89.5	89.5	88.5	0.81	3.80	3.5	9.6	5.1	62	74	**1LE1023-1AB4■-■■■■**	30	0,014	16
3	**4**	**100 L**	1760	16	–	89.5	89.5	88.5	0.82	5.1	3.1	9.5	4.6	62	74	**1LE1023-1AB5■-■■■■**	30	0,014	16
4	**5**	**112 M**	1770	20	✓	89.5	89.5	88.5	0.80	6.5	2.9	8.2	4.3	62	74	**1LE1023-1BB2■-■■■■**	34	0,017	16
5.5	**7.5**	**132 S**	1780	30	✓	91.7	91.7	90.7	0.83	9.1	2.9	9.5	4.4	68	80	**1LE1023-1CB0■-■■■■**	64	0,046	16
7.5	**10**	**132 M**	1770	40	✓	91.7	91.7	90.7	0.83	12.4	2.7	9.6	4.2	68	80	**1LE1023-1CB2■-■■■■**	64	0,046	16
11	**15**	**160 M**	1775	59	✓	92.4	92.4	91.4	0.83	18	3.0	8.9	3.8	69	81	**1LE1023-1DB2■-■■■■**	83	0,083	16
15	**20**	**160 L**	1780	80	✓	93.0	93.0	91.5	0.81	25	2.9	9.5	4.3	69	81	**1LE1023-1DB4■-■■■■**	100	0,099	16
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz																			
0.37	**0.5**	**80 M**	1150	3.1	–	78.5	77.5	73.0	0.61	0.97	2.7	5.0	3.3	45	56	**1LE1023-0DC2■-■■■■**	12	0,0025	13
0.55	**0.75**	**80 M**	1145	4.6	–	81.7	81.3	78.0	0.63	1.34	2.8	5.3	3.4	45	56	**1LE1023-0DC3■-■■■■**	14	0,0031	13
0.75	**1**	**90 S**	1155	6.2	✓	82.5	82.3	79.5	0.65	1.76	2.4	5.3	3.1	46	58	**1LE1023-0EC0■-■■■■**	16	0,0040	13
1.1	**1.5**	**100 L**	1175	8.9	–	87.5	87.5	86.5	0.71	2.2	2.4	7.0	3.8	62	74	**1LE1023-1AC3■-■■■■**	25	0,014	13
3	**4**	**132 S**	1175	24	✓	89.5	89.5	88.5	0.76	5.5	1.9	7.6	3.4	67	79	**1LE1023-1CC0■-■■■■**	52	0,037	13
4	**5**	**132 M**	1175	30	✓	89.5	89.5	88.5	0.76	6.8	2.2	7.9	3.7	67	79	**1LE1023-1CC2■-■■■■**	52	0,037	13
5.5	**7.5**	**132 M**	1175	45	✓	91.0	91.0	90.0	0.76	10	2.2	7.5	3.5	67	79	**1LE1023-1CC3■-■■■■**	52	0,037	13
7.5	**10**	**160 M**	1180	61	✓	91.0	91.0	89.8	0.75	13.8	2.4	5.9	2.6	70	82	**1LE1023-1DC2■-■■■■**	93	0,098	13
11	**15**	**160 L**	1180	89	✓	91.7	91.9	91.0	0.75	20.0	2.3	5.8	2.6	70	82	**1LE1023-1DC4■-■■■■**	115	0,12	13

Voltages (≤ 600 V) [1]				No. of poles	Frame size	Motor type	Version			Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz	460 VY	2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	**Standard**	**2**	**2**	–
50 Hz	400 VΔ	60 Hz	460 VΔ	2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	**Standard**	**3**	**4**	–
50 Hz	500 VY			2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	Without add. charge	**2**	**7**	–
50 Hz	500 VΔ			2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	Without add. charge	**4**	**0**	–
Further voltages				For price information, code numbers, order codes and descriptions, see from Page 2/38				**9**	**0**	. . .

Types of construction		No. of poles	Frame size	Motor type	Version		Order code(s)
Without flange	IM B3 [2]	2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	**Standard**	**A**	–
With flange	IM B5 [2]	2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	With additional charge	**F**	–
With standard flange	IM B14 [2]	2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	With additional charge	**K**	–
Further types of construction		For price information, code letters and descriptions, see from Page 2/41				■	. . .

Motor protection	No. of poles	Frame size	Motor type	Version		Order code(s)
Without	2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	**Standard**	**A**	–
PTC thermistor with 3 temperature sensors	2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	With additional charge	**B**	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/49				■	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top	2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	**Standard**	**4**	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/51					

Special versions	No. of poles	Frame size	Motor type		Order code(s)
Forced-air cooled motors without ext. fan/fan cover (IC 416)	2, 4, 6	80 M ... 160 L	1LE1023-0D ... -1D	**1LE1023- . . . . ■–■■■■ -Z**	**F90 +. . .+. . .+. . .**
Options	For price information, order codes and descriptions, see from Page 2/53			**1LE1023- . . . . ■–■■■■ -Z**	**. . .+. . .+. . .+. . .**

– Not required
✓ available

[1] Operating voltages only ≤ 600 V admissible in accordance with MG1 Table 12-12.

[2] Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (H03) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (H03), the type must be specified.

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SIMOTICS SD 1LE1 Standard Motors – Eagle Line

NEMA Premium Efficient MG1 motors, Table 12-12

Self-ventilated motors
Cast-iron series 1LE1523/1LE1623 Basic/Performance Line

Selection and ordering data

P_{rated}, 50 Hz	P_{rated}, 60 Hz	Frame size	Operating values at rated output: n_{rated}, 60 Hz	T_{rated}, 60 Hz	EISA CC No. CC032A	η_{rated}, 60 Hz, 4/4	η_{rated}, 60 Hz, 3/4	η_{rated}, 60 Hz, 2/4	cosφ rated, 60 Hz, 4/4	I_{rated}, 60 Hz, 460 V	T_{LR}/T_{rated}, 60 Hz	I_{LR}/I_{rated}, 60 Hz	T_B/T_{rated}, 60 Hz	L_{pfA}, 60 Hz	L_{WA}, 60 Hz	Cast-iron series 1LE1523 – *Basic Line* 1LE1623 – *Performance Line* *NEMA Premium Efficient version* Article No.	$m_{IM\ B3}$	J	Torque class
kW	hp	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm²	CL
• Cooling: self-ventilated (IC 411)																			
• Efficiency: NEMA Premium Efficient, UL, CSA and service factor (SF) 1.15 – for operation in the USA, Canada and Mexico																			
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
2-pole: 3000 rpm at 50 Hz, 3600 rpm at 60 Hz																			
0.37	0.5	71 M	3470	1.0	–	73.4	71.7	67.1	0.73	0.83	4.2	6.8	4.2	57	68	▲ 1LE1 5 23-0CA2 ■-■■■■	13	0.00045	16
0.55	0.75	71 M	3480	1.5	–	81.1	79.6	75.2	0.73	1.17	4.2	7.1	4.2	57	68	▲ 1LE1 5 23-0CA3 ■-■■■■	14.5	0.00056	16
0.75	1	80 M	3480	2.1	✓	77.0	78.0	76.0	0.84	1.46	3.0	7.1	3.6	64	75	▲ 1LE1 5 23-0DA2 ■-■■■■	18	0.0011	16
1.1	1.5	80 M	3500	3	✓	84.0	84.0	83.0	0.83	1.98	3.3	8.4	4.5	64	75	▲ 1LE1 5 23-0DA3 ■-■■■■	21	0.0013	16
1.5	2	90 S	3525	4.1	✓	85.5	85.0	82.5	0.84	2.60	3.1	9.8	4.9	69	81	▲ 1LE1 5 23-0EA0 ■-■■■■	25.5	0.0021	16
2.2	3	90 L	3530	6	✓	86.5	86.3	84.5	0.87	3.65	3.0	9.6	4.9	69	81	▲ 1LE1 5 23-0EA4 ■-■■■■	32	0.0031	16
3	4	100 L	3525	8.1	–	88.5	88.5	87.5	0.87	4.90	3.8	9.7	5.5	71	83	1LE1 ■ 23-1AA4 ■-■■■■	36	0.0054	16
4	5	112 M	3565	9.9	✓	88.5	88.5	87.5	0.87	6.0	3.8	10.0	5.6	73	85	1LE1 ■ 23-1BA2 ■-■■■■	45	0.012	16
5.5	7.5	132 S	3555	15	✓	89.5	89.5	88.5	0.90	8.6	2.1	8.6	4.4	72	84	1LE1 ■ 23-1CA0 ■-■■■■	58	0.024	16
7.5	10	132 S	3555	20	✓	90.2	90.2	89.2	0.91	11.5	2.4	9.5	4.7	72	84	1LE1 ■ 23-1CA1 ■-■■■■	73	0.031	16
11	15	160 M	3560	30	✓	91.0	91.0	90.0	0.88	17.2	2.8	8.5	4.3	77	89	1LE1 ■ 23-1DA2 ■-■■■■	100	0.053	16
15	20	160 M	3565	40	✓	91.0	91.0	90.0	0.86	24	3.1	9.7	4.8	77	89	1LE1 ■ 23-1DA3 ■-■■■■	110	0.061	16
18.5	25	160 L	3560	50	✓	91.7	91.7	90.7	0.90	28	3.1	9.4	4.4	77	89	1LE1 ■ 23-1DA4 ■-■■■■	127	0.068	16
22	30	180 M	3560	60	✓	91.7	91.4	90.0	0.89	34.5	2.8	8.3	3.9	78	85	1LE1 ■ 23-1EA2 ■-■■■■	160	0.080	16
30	40	200 L	3560	80	✓	92.4	92.2	91.4	0.87	46.5	2.9	7.6	3.6	78	86	1LE1 ■ 23-2AA4 ■-■■■■	225	0.13	16
37	50	200 L	3560	100	✓	93.0	92.8	91.6	0.88	57	2.8	7.5	3.6	79	86	1LE1 ■ 23-2AA5 ■-■■■■	250	0.16	16
45	60	225 M	3570	120	✓	93.6	93.7	93.1	0.88	68	2.7	7.6	3.5	75	89	1LE1 ■ 23-2BA2 ■-■■■■	315	0.26	16
55	75	250 M	3578	149	–	93.6	93.4	92.3	0.89	84	2.5	7.3	3.3	76	90	1LE1 ■ 23-2CA2 ■-■■■■	385	0.46	13
75	100	280 S	3578	199	–	94.1	93.9	92.7	0.89	112	2.7	7.6	3.2	78	92	1LE1 ■ 23-2DA0 ■-■■■■	510	0.77	13
90	125	280 M	3578	249	✓	95.0	94.8	93.8	0.90	137	2.7	8.1	3.3	78	92	1LE1 ■ 23-2DA2 ■-■■■■	590	0.94	13
110	150	315 S	3585	298	✓	95.0	94.8	93.8	0.91	162	2.6	8.0	3.3	79	93	1LE1 ■ 23-3AA0 ■-■■■■	750	1.4	13
132	175	315 M	3585	348	–	95.4	95.1	94.0	0.91	189	2.8	8.0	3.4	79	93	1LE1 ■ 23-3AA2 ■-■■■■	880	1.6	13
150	200	315 L	3588	397	✓	95.4	95.1	93.9	0.91	215	3.3	9.1	3.7	82	96	1LE1 ■ 23-3AA4 ■-■■■■	980	1.9	13
185	250	315 L	3586	497	–	95.8	95.7	94.8	0.92	265	3.5	8.5	3.5	82	96	1LE1 ■ 23-3AA5 ■-■■■■	1150	2.3	13

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects						
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	up to 500 V	12 months	5					
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6					

Voltages (≤ 600 V) [1]			No. of poles	Frame size	Motor type	Version					Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz	460 VY	2	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**	2	2		–
50 Hz	400 VΔ	60 Hz	460 VΔ	2	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**	3	4		–
50 Hz	500 VY			2	71 M ... 315 L	1LE1■23-0C … -3A	Without add. charge	2	7		–
50 Hz	500 VΔ			2	71 M ... 315 L	1LE1■23-0C … -3A	Without add. charge	4	0		–
Further voltages		For price information, code numbers, order codes and descriptions, see from Page 2/40					9	0			. . .

Types of construction		No. of poles	Frame size	Motor type	Version				Order code(s)
Without flange	IM B3 [2]	2	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**			A	–
With flange	IM B5 [2]	2	71 M ... 315 M	1LE1■23-0C … -3A	With additional charge			F	–
With standard flange	IM B14 [2]	2	71 M ... 160 L	1LE1■23-0C … -1D	With additional charge			K	–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45							■	. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version				Order code(s)
Without	Only possible for **Basic Line**	2	71 M ... 315 L	1LE1523-0C … -3A	**Standard**			A	–
PTC thermistor with 3 temperature sensors	**Basic Line**	2	71 M ... 315 L	1LE1523-0C … -3A	With additional charge			B	–
	Performance Line	2	71 M ... 315 L	1LE1623-0C … -3A	**Standard**			B	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50							■	. . .

Terminal box position		No. of poles	Frame size	Motor type	Version				Order code(s)
Terminal box at top		2	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**			4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52								

Special versions		No. of poles	Frame size	Motor type					Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58				1LE1■23-. . . .	■	■■■■	-Z	. . .+. . .+. . .+. . .

– Not required
✓ available

[1] Operating voltages only ≤ 600 V admissible in accordance with MG1 Table 12-12. Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

[2] Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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SIMOTICS SD 1LE1 Standard Motors – Eagle Line

NEMA Premium Efficient MG1 motors, Table 12-12

Self-ventilated motors
Cast-iron series 1LE1523/1LE1623 Basic/Performance Line

Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz	Frame size	Operating values at rated output: n_{rated}, 60 Hz	T_{rated}, 60 Hz	EISA CC No. CC032A	η_{rated}, 60 Hz, 4/4	η_{rated}, 60 Hz, 3/4	η_{rated}, 60 Hz, 2/4	$\cos\varphi$ rated, 60 Hz, 4/4	I_{rated}, 60 Hz, 460 V	T_{LR}/T_{rated}, 60 Hz	I_{LR}/I_{rated}, 60 Hz	T_B/T_{rated}, 60 Hz	L_{pfA}, 60 Hz	L_{WA}, 60 Hz	Cast-iron series 1LE1523 – *Basic Line* 1LE1623 – *Performance Line* *NEMA Premium Efficient version* Article No.	$m_{IM\ B3}$	J	Torque class
kW	hp	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm²	CL
• Cooling: self-ventilated (IC 411)																			
• Efficiency: NEMA Premium Efficient, UL, CSA and service factor (SF) 1.15 – for operation in the USA, Canada and Mexico																			
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
4-pole: 1500 rpm at 50 Hz, 1800 rpm at 60 Hz																			
0.25	**0.33**	**71 M**	1715	1.4	–	73.4	72.3	68.2	0.68	0.65	2.9	4.9	3.1	47	58	▲ **1LE1 5 23-0CB2 ■-■■■■**	13	0.00095	16
0.37	**0.5**	**71 M**	1725	2	–	78.2	76.6	71.9	0.65	0.90	2.9	5.4	3.3	47	58	▲ **1LE1 5 23-0CB3 ■-■■■■**	16	0.0014	16
0.55	**0.75**	**80 M**	1750	3	–	82.5	82.2	79.4	0.74	1.13	2.7	6.9	3.8	55	66	▲ **1LE1 5 23-0DB2 ■-■■■■**	18.5	0.0021	16
0.75	**1**	**80 M**	1760	4.1	✓	85.5	84.5	81.0	0.71	1.55	3.1	8.3	4.7	55	66	▲ **1LE1 5 23-0DB3 ■-■■■■**	22.5	0.0029	16
1.1	**1.5**	**90 S**	1750	6	✓	86.5	86.3	84.1	0.75	2.15	3.4	8.2	4.4	58	70	▲ **1LE1 5 23-0EB0 ■-■■■■**	25	0.0036	16
1.5	**2**	**90 L**	1755	8.2	✓	86.5	87.0	85.0	0.77	2.85	3.0	8.4	4.3	58	70	▲ **1LE1 5 23-0EB4 ■-■■■■**	31	0.0049	16
2.2	**3**	**100 L**	1770	12	–	89.5	89.5	88.5	0.81	3.80	3.5	9.6	5.1	62	74	**1LE1 ■ 23-1AB4 ■-■■■■**	40	0.014	16
3	**4**	**100 L**	1760	16	–	89.5	89.5	88.5	0.82	5.1	3.1	9.5	4.6	62	74	**1LE1 ■ 23-1AB5 ■-■■■■**	40	0.014	16
4	**5**	**112 M**	1770	20	✓	89.5	89.5	88.5	0.80	6.5	2.9	8.2	4.3	62	74	**1LE1 ■ 23-1BB2 ■-■■■■**	46	0.017	16
5.5	**7.5**	**132 S**	1780	30	✓	91.7	91.7	90.7	0.83	9.1	2.9	9.5	4.4	68	80	**1LE1 ■ 23-1CB0 ■-■■■■**	80	0.046	16
7.5	**10**	**132 M**	1770	40	✓	91.7	91.7	90.7	0.83	12.4	2.7	9.6	4.2	68	80	**1LE1 ■ 23-1CB2 ■-■■■■**	80	0.046	16
11	**15**	**160 M**	1775	59	✓	92.4	92.4	91.4	0.83	18	3.0	8.9	3.8	69	81	**1LE1 ■ 23-1DB2 ■-■■■■**	109	0.083	16
15	**20**	**160 L**	1780	80	✓	93.0	93.0	91.5	0.81	25	2.9	9.5	4.3	69	81	**1LE1 ■ 23-1DB4 ■-■■■■**	127	0.099	16
18.5	**25**	**180 M**	1775	100	✓	93.6	93.7	93.1	0.81	31	2.7	7.8	3.6	68	75	**1LE1 ■ 23-1EB2 ■-■■■■**	165	0.13	16
22	**30**	**180 L**	1775	120	✓	93.6	93.8	93.3	0.81	37	2.8	7.7	3.7	70	77	**1LE1 ■ 23-1EB4 ■-■■■■**	170	0.14	16
30	**40**	**200 L**	1778	160	✓	94.1	94.3	93.8	0.83	48	3.0	8.1	3.5	67	74	**1LE1 ■ 23-2AB5 ■-■■■■**	240	0.22	16
37	**50**	**225 S**	1782	200	–	94.5	94.7	94.2	0.85	58	2.8	7.5	3.0	66	80	**1LE1 ■ 23-2BB0 ■-■■■■**	285	0.42	16
45	**60**	**225 M**	1782	240	✓	95.0	95.3	94.9	0.84	70	2.9	7.2	3.0	67	81	**1LE1 ■ 23-2BB2 ■-■■■■**	320	0.47	16
55	**75**	**250 M**	1786	299	–	95.4	95.6	95.1	0.86	86	2.8	7.6	3.2	67	81	**1LE1 ■ 23-2CB2 ■-■■■■**	420	0.85	16
75	**100**	**280 S**	1788	398	–	95.4	95.3	94.5	0.85	115	2.8	7.7	3.3	77	91	**1LE1 ■ 23-2DB0 ■-■■■■**	570	1.4	16
90	**125**	**280 M**	1788	498	✓	95.4	95.5	94.9	0.87	141	2.9	8.0	3.3	79	93	**1LE1 ■ 23-2DB2 ■-■■■■**	670	1.7	16
110	**150**	**315 S**	1790	597	✓	95.8	95.9	95.4	0.86	170	3.0	7.5	3.1	73	87	**1LE1 ■ 23-3AB0 ■-■■■■**	760	2.2	16
132	**175**	**315 M**	1790	696	–	96.2	96.3	95.8	0.87	196	3.1	8.2	3.2	76	90	**1LE1 ■ 23-3AB2 ■-■■■■**	960	2.9	16
150	**200**	**315 L**	1791	796	✓	96.2	96.2	95.7	0.87	225	3.5	8.8	3.6	76	90	**1LE1 ■ 23-3AB4 ■-■■■■**	990	3.1	16
185	**250**	**315 L**	1791	994	–	96.2	96.2	95.5	0.87	280	3.9	9.0	3.6	78	93	**1LE1 ■ 23-3AB5 ■-■■■■**	1190	3.7	16

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects						
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5					
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6					

Voltages (≤ 600 V) [1]			No. of poles	Frame size	Motor type	Version						Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz	460 VY	4	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**	2	2			–
50 Hz	400 VΔ	60 Hz	460 VΔ	4	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**	3	4			–
50 Hz	500 VY			4	71 M ... 315 L	1LE1■23-0C … -3A	Without add. charge	2	7			–
50 Hz	500 VΔ			4	71 M ... 315 L	1LE1■23-0C … -3A	Without add. charge	4	0			–
Further voltages		For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0			. . .

Types of construction		No. of poles	Frame size	Motor type	Version					Order code(s)
Without flange	IM B3 [2]	4	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**			A		–
With flange	IM B5 [2]	4	71 M ... 315 M	1LE1■23-0C … -3A	With additional charge			F		–
With standard flange	IM B14 [2]	4	71 M ... 160 L	1LE1■23-0C … -1D	With additional charge			K		–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45							■		. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version					Order code(s)
Without	Only possible for **Basic Line**	4	71 M ... 315 L	1LE1523-0C … -3A	**Standard**			A		–
PTC thermistor with 3 temperature sensors	**Basic Line**	4	71 M ... 315 L	1LE1523-0C … -3A	With additional charge			B		–
	Performance Line	4	71 M ... 315 L	1LE1623-0C … -3A	**Standard**			B		–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50							■		. . .

Terminal box position		No. of poles	Frame size	Motor type	Version					Order code(s)
Terminal box at top		4	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**				4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52									

Special versions		No. of poles	Frame size	Motor type						Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58				**1LE1■23-. . . .**	■	■■■■		-Z	. . .+. . .+. . .+. . .

– Not required
✓ available

[1] Operating voltages only ≤ 600 V admissible in accordance with MG1 Table 12-12. Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

[2] Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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Selection and ordering data (continued)

P_{rated}, 50 Hz	P_{rated}, 60 Hz	Frame size	Operating values at rated output: n_{rated}, 60 Hz	T_{rated}, 60 Hz	EISA CC No. CC032A	η_{rated}, 60 Hz, 4/4	η_{rated}, 60 Hz, 3/4	η_{rated}, 60 Hz, 2/4	$\cos\varphi$ rated, 60 Hz, 4/4	I_{rated}, 60 Hz, 460 V	T_{LR}/T_{rated}, 60 Hz	I_{LR}/I_{rated}, 60 Hz	T_B/T_{rated}, 60 Hz	L_{pfA}, 60 Hz	L_{WA}, 60 Hz	Cast-iron series 1LE1523 – *Basic Line* 1LE1623 – *Performance Line* *NEMA Premium Efficient version* Article No.	$m_{IM\ B3}$	J	Torque class
kW	hp	FS	rpm	Nm		%	%	%		A				dB(A)	dB(A)	▲ New	kg	kgm^2	CL
• Cooling: self-ventilated (IC 411)																			
• Efficiency: NEMA Premium Efficient, UL, CSA and service factor (SF) 1.15 – for operation in the USA, Canada and Mexico																			
• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																			
6-pole: 1000 rpm at 50 Hz, 1200 rpm at 60 Hz																			
0.18	0.25	71 M	1110	1.5	–	67.5	66.3	61.0	0.63	0.53	2.8	3.5	2.9	42	53	▲ 1LE1 5 23-0CC2 ■-■■■■	12.5	0.001	16
0.25	0.33	71 M	1120	2.1	–	71.4	70.2	65.6	0.62	0.71	3.1	3.8	3.3	42	53	▲ 1LE1 5 23-0CC3 ■-■■■■	15.5	0.015	16
0.37	0.5	80 M	1150	3.1	–	78.5	77.5	73.0	0.61	0.97	2.7	5.0	3.3	45	56	▲ 1LE1 5 23-0DC2 ■-■■■■	18.5	0.0025	13
0.55	0.75	80 M	1145	4.6	–	81.7	81.3	78.0	0.63	1.34	2.8	5.3	3.4	45	56	▲ 1LE1 5 23-0DC3 ■-■■■■	22.5	0.0031	13
0.75	1	90 S	1155	6.2	✓	82.5	82.3	79.5	0.65	1.76	2.4	5.3	3.1	46	58	▲ 1LE1 5 23-0EC0 ■-■■■■	26.5	0.0040	13
1.1	1.5	90 L	1175	8.9	✓	87.5	87.5	86.5	0.71	2.2	2.4	7.0	3.8	62	74	▲ 1LE1 5 23-0EC4 ■-■■■■	30	0.014	13
3	4	132 S	1175	24	✓	89.5	89.5	88.5	0.76	5.5	1.9	7.6	3.4	67	79	1LE1 ■ 23-1CC0 ■-■■■■	31	0.037	13
4	5	132 M	1175	30	✓	89.5	89.5	88.5	0.76	6.8	2.2	7.9	3.7	67	79	1LE1 ■ 23-1CC2 ■-■■■■	68	0.037	13
5.5	7.5	132 M	1175	45	✓	91.0	91.0	90.0	0.76	10	2.2	7.5	3.5	67	79	1LE1 ■ 23-1CC3 ■-■■■■	81	0.037	13
7.5	10	160 M	1180	61	✓	91.0	91.0	89.8	0.75	13.8	2.4	5.9	2.6	70	82	1LE1 ■ 23-1DC2 ■-■■■■	128	0.098	13
11	15	160 L	1180	89	✓	91.7	91.9	91.0	0.75	20.0	2.3	5.8	2.6	70	82	1LE1 ■ 23-1DC4 ■-■■■■	149	0.12	13
15	20	180 L	1178	121	✓	91.7	92.0	91.5	0.79	26	2.5	6.8	3.0	61	68	1LE1 ■ 23-1EC4 ■-■■■■	180	0.19	16
18.5	25	200 L	1180	151	✓	93.0	93.2	92.6	0.78	32.5	2.8	6.5	3.0	64	71	1LE1 ■ 23-2AC4 ■-■■■■	215	0.28	16
22	30	200 L	1180	181	✓	93.0	93.6	93.5	0.79	38	2.6	6.3	2.8	63	70	1LE1 ■ 23-2AC5 ■-■■■■	230	0.32	16
30	40	225 M	1185	240	✓	94.1	94.4	94.1	0.82	48.5	2.9	7.6	3.3	66	79	1LE1 ■ 23-2BC2 ■-■■■■	325	0.67	16
37	50	250 M	1188	300	–	94.1	94.4	93.9	0.83	60	3.1	8.0	3.1	63	76	1LE1 ■ 23-2CC2 ■-■■■■	405	1.0	16
45	60	280 S	1190	359	–	94.5	94.6	94.1	0.83	72	3.3	7.7	3.1	66	80	1LE1 ■ 23-2DC0 ■-■■■■	510	1.4	16
55	75	280 M	1190	449	–	94.5	94.6	94.0	0.83	90	3.6	7.9	3.3	66	80	1LE1 ■ 23-2DC2 ■-■■■■	560	1.6	16
75	100	315 S	1192	598	✓	95.0	94.9	94.1	0.82	120	3.1	8.4	3.3	64	79	1LE1 ■ 23-3AC0 ■-■■■■	750	2.6	16
90	125	315 M	1192	747	✓	95.0	95.1	94.4	0.84	147	2.7	7.7	3.0	64	79	1LE1 ■ 23-3AC2 ■-■■■■	890	3.1	16
110	150	315 L	1192	896	✓	95.8	96.0	95.5	0.83	177	3.2	8.2	3.4	64	79	1LE1 ■ 23-3AC4 ■-■■■■	990	3.9	16
132	175	315 L	1192	1046	–	95.8	96.0	95.6	0.84	205	3.1	8.4	3.3	65	80	1LE1 ■ 23-3AC5 ■-■■■■	1110	4.4	16
150	200	315 L	1192	1195	✓	95.8	95.7	95.0	0.81	240	3.6	9.6	4.1	69	83	1LE1 ■ 23-3AC6 ■-■■■■	1160	4.6	16

	Relubrication	Motor protection	Fan cover	Bearing size	Converter-fed operation, motor mode	Liability for defects						
Basic Line	Optional (standard from FS 280 upwards)	Optional	Plastic	62 (63 from FS 280 upwards)	Up to 500 V	12 months	5					
Performance Line	Standard from FS 160 (optional for FS 100 to 132)	Standard PTC	Steel	63	Up to 500 V	36 months	6					

Voltages (≤ 600 V) [1]			No. of poles	Frame size	Motor type	Version					Order code(s)
50 Hz	230 VΔ/400 VY	60 Hz	460 VY	6	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**	2	2		–
50 Hz	400 VΔ	60 Hz	460 VΔ	6	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**	3	4		–
50 Hz	500 VY			6	71 M ... 315 L	1LE1■23-0C … -3A	Without add. charge	2	7		–
50 Hz	500 VΔ			6	71 M ... 315 L	1LE1■23-0C … -3A	Without add. charge	4	0		–
Further voltages		For price information, code numbers, order codes and descriptions, see from Page 2/40						9	0		. . .

Types of construction		No. of poles	Frame size	Motor type	Version			Order code(s)
Without flange	IM B3 [2]	6	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**	A		–
With flange	IM B5 [2]	6	71 M ... 315 M	1LE1■23-0C … -3A	With additional charge	F		–
With standard flange	IM B14 [2]	6	71 M ... 160 L	1LE1■23-0C … -1D	With additional charge	K		–
Further types of construction	For price information, code letters and descriptions, see from Page 2/45					■		. . .

Motor protection	Line	No. of poles	Frame size	Motor type	Version			Order code(s)
Without	Only possible for **Basic Line**	6	71 M ... 315 L	1LE1523-0C … -3A	**Standard**	A		–
PTC thermistor with 3 temperature sensors	**Basic Line**	6	71 M ... 315 L	1LE1523-0C … -3A	With additional charge	B		–
	Performance Line	6	71 M ... 315 L	1LE1623-0C … -3A	**Standard**	B		–
Further motor protection	For price information, code letters and descriptions, see from Page 2/50					■		. . .

Terminal box position		No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top		6	71 M ... 315 L	1LE1■23-0C … -3A	**Standard**	4	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/52						

Special versions		No. of poles	Frame size	Motor type		Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/58				1LE1■23-. . . . ■-■■■■-Z	. . .+. . .+. . .+. . .

– Not required
✓ available

[1] Operating voltages only ≤ 600 V admissible in accordance with MG1 Table 12-12. Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

[2] Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirement exists for stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

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Pole-changing motors

Self-ventilated motors
Aluminum series 1LE1011 for constant load torque

Selection and ordering data

			Operating values at rated output for N1								Operating values at rated output for N2								Aluminum series 1LE1011 – *One winding pole-changing for constant load torque* Article No.	*m* IM B3	*J*	Torque class
P **rated1, 50 Hz**	***P*** **rated2, 50 Hz**	**Frame size**	n_{rated1}, 50 Hz	T_{rated1}, 50 Hz	η_{rated1}, 50 Hz, 4/4	cos φ_{rated1}, 50 Hz, 4/4	I_{rated1}, 50 Hz, 400 V	T_{LR}/T_{rated1}, 50 Hz	I_{LR}/I_{rated1}, 50 Hz	T_{LR}/T_{rated1}, 50 Hz	n_{rated2}, 50 Hz	T_{rated2}, 50 Hz	η_{rated2}, 50 Hz, 4/4	cos φ_{rated2}, 50 Hz, 4/4	I_{rated2}, 50 Hz, 400 V	T_{LR}/T_{rated2}, 50 Hz	I_{LR}/I_{rated2}, 50 Hz	T_{LR}/T_{rated2}, 50 Hz			
kW	**kW**	**FS**	rpm	Nm	%		A				rpm	Nm	%		A					kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) • Line operation: Double pole-changing for constant load torque • Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)																						
4/2-pole: 1500/3000 rpm at 50 Hz with one winding connected in Dahlander circuit																						
1500 rpm	*3000 rpm*		*1500 rpm*								*3000 rpm*											
1.9	**2.4**	**100 L**	1390	13	72.0	0.87	4.40	1.7	4.1	1.8	2800	8.2	70.0	0.88	5.6	1.8	4.2	1.8	**1LE1011-1AJ4 ■-■■■■**	18	0.0059	13
2.5	**3.1**	**100 L**	1400	17	76.3	0.87	5.4	1.9	5.2	2.8	2840	10.0	77.3	0.90	6.4	2.1	5.2	2.9	**1LE1011-1AJ5 ■-■■■■**	22	0.0078	13
3.7	**4.4**	**112 M**	1420	25	79.4	0.86	7.8	1.8	4.9	2.3	2885	15.0	80.8	0.92	8.5	2.1	6.4	2.6	**1LE1011-1BJ2 ■-■■■■**	27	0.010	13
4.7	**5.9**	**132 S**	1440	31	82.0	0.84	9.8	1.6	5.6	2.7	2875	20.0	80.0	0.89	12.0	1.8	5.6	2.8	**1LE1011-1CJ0 ■-■■■■**	38	0.019	13
6.5	**8.0**	**132 M**	1435	43	82.0	0.86	13.3	1.7	5.4	2.6	2880	27.0	82.0	0.92	15.3	1.8	6.3	2.8	**1LE1011-1CJ2 ■-■■■■**	44	0.024	13
9.3	**11.5**	**160 M**	1440	62	84.5	0.87	18.3	1.7	5.7	2.8	2870	38.0	82.0	0.92	22.0	1.8	6.0	2.9	**1LE1011-1DJ2 ■-■■■■**	62	0.044	13
13.0	**16**	**160 L**	1450	86	87.0	0.85	25.5	1.6	6.0	2.3	2920	52.0	86.0	0.94	28.5	1.9	7.1	2.8	**1LE1011-1DJ6 ■-■■■■**	85	0.068	13
8/4-pole: 750/1500 rpm at 50 Hz with one winding connected in Dahlander circuit																						
750 rpm	*1500 rpm*		*750 rpm*								*1500 rpm*											
0.55	**1.1**	**100 L**	715	7.3	57.0	0.53	2.65	2.0	3.0	2.7	1425	7.4	77.7	0.87	2.35	1.7	4.6	2.1	**1LE1011-1AL4 ■-■■■■**	18	0.0059	10
0.9	**1.5**	**100 L**	700	12	64.2	0.64	3.15	1.5	2.9	2.0	1415	10.0	77.7	0.89	3.15	1.5	4.5	1.9	**1LE1011-1AL5 ■-■■■■**	22	0.0078	10
1.1	**1.9**	**112 M**	715	15	66.5	0.60	4.00	1.6	3.2	2.3	1440	13.0	80.9	0.87	3.90	1.6	5.4	2.3	**1LE1011-1BL2 ■-■■■■**	27	0.010	10
1.6	**3.2**	**132 S**	730	21	61.5	0.53	7.1	1.6	3.3	2.6	1450	21.0	82.3	0.87	6.5	1.4	5.0	2.1	**1LE1011-1CL0 ■-■■■■**	38	0.019	10
2.2	**4.4**	**132 M**	730	29	68.0	0.52	9.0	2.0	3.8	3.0	1450	29.0	84.5	0.88	8.5	1.5	5.5	2.3	**1LE1011-1CL2 ■-■■■■**	44	0.024	10
3.5	**7**	**160 M**	730	46	77.5	0.57	11.4	2.0	4.2	2.8	1450	46.0	84.0	0.90	13.4	1.6	5.2	2.2	**1LE1011-1DL2 ■-■■■■**	62	0.044	10
5.6	**11**	**160 L**	725	74	80.2	0.60	16.8	1.9	4.0	2.7	1445	73.0	84.4	0.90	21.0	1.5	5.1	2.2	**1LE1011-1DL4 ■-■■■■**	73	0.056	10

Voltages		No. of poles	Frame size	Motor type	Version			Order code(s)
50 Hz	230 V	4/2, 8/4	100 L ... 160 L	1LE1011-1A ... -1D	**Standard**	**2**	**2**	–
50 Hz	400 V	4/2, 8/4	100 L ... 160 L	1LE1011-1A ... -1D	**Standard**	**3**	**4**	–
50 Hz	500 V	4/2, 8/4	100 L ... 160 L	1LE1011-1A ... -1D	Without add. charge	**4**	**0**	–
50 Hz	690 V	4/2, 8/4	100 L ... 160 L	1LE1011-1A ... -1D	Without add. charge	**4**	**7**	–
Further voltages [1]		For price information, code numbers, order codes and descriptions, see from Page 2/39				**9**	**0**	. . .

Types of construction		No. of poles	Frame size	Motor type	Version		Order code(s)
Without flange	IM B3 [2]	4/2, 8/4	100 L ... 160 L	1LE1011-1A ... -1D	**Standard**	**A**	–
With flange	IM B5 [2]	4/2, 8/4	100 L ... 160 L	1LE1011-1A ... -1D	With additional charge	**F**	–
With standard flange	IM B14 [2]	4/2, 8/4	100 L ... 160 L	1LE1011-1A ... -1D	With additional charge	**K**	–
Further types of construction		For price information, code letters and descriptions, see from Page 2/41				■	. . .

Motor protection	No. of poles	Frame size	Motor type	Version		Order code(s)
Without	4/2, 8/4	100 L ... 160 L	1LE1011-1A ... -1D	**Standard**	**A**	–
PTC thermistor with 3 temperature sensors	4/2, 8/4	100 L ... 160 L	1LE1011-1A ... -1D	With additional charge	**B**	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/49				■	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top	4/2, 8/4	100 L ... 160 L	1LE1011-1A ... -1D	**Standard**	**4**	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/51					

Special versions			Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/53	**1LE1011-. . . . ■-■■■■-Z**	. . .+. . .+. . .+. . .

[1] Operating values at rated output for 60 Hz are available on request.

[2] Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (H03) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (H03), the type must be specified.

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Selection and ordering data

			Operating values at rated output for N1								Operating values at rated output for N2								Aluminum series **1LE1011** – *One winding* **1LE1012** – *Two windings pole-changing for square-law load torque* Article No.	*m*	*J*	Torque class
P **rated1, 50 Hz**	***P*** **rated2, 50 Hz**	**Frame size**	n_{rated1}, 50 Hz	T_{rated1}, 50 Hz	η_{rated1}, 50 Hz, 4/4	$\cos\varphi_{rated1}$, 50 Hz, 4/4	I_{rated1}, 50 Hz, 400 V	T_{LR}/T_{rated1}, 50 Hz	I_{LR}/I_{rated1}, 50 Hz	T_{LR}/T_{rated1}, 50 Hz	n_{rated2}, 50 Hz	T_{rated2}, 50 Hz	η_{rated2}, 50 Hz, 4/4	$\cos\varphi_{rated2}$, 50 Hz, 4/4	I_{rated2}, 50 Hz, 400 V	T_{LR}/T_{rated2}, 50 Hz	I_{LR}/I_{rated2}, 50 Hz	T_{LR}/T_{rated2}, 50 Hz		IM B3		
kW	**kW**	**FS**	rpm	Nm	%		A				rpm	Nm	%		A					kg	kgm^2	CL
• Cooling: self-ventilated (IC 411) **• Line operation: double pole-changing for square-law load torque, e.g. for driving fans** **• Insulation: thermal class 155 (temperature class F), IP55 degree of protection, utilization in accordance with thermal class 130 (temperature class B)**																						
4/2-pole: 1500/3000 rpm at 50 Hz with one winding connected in Dahlander circuit																						
1500 rpm	*3000 rpm*		*1500 rpm*								*3000 rpm*											
0.65	**2.4**	**100 L**	1415	4.4	75.0	0.86	1.45	1.6	4.1	1.8	2800	8.2	70.0	0.88	5.6	1.8	4.2	1.8	**1LE1011-1AP4■-■■■■**	18	0.0059	13
0.8	**3.1**	**100 L**	1435	5.3	79.0	0.85	1.72	1.9	5.2	2.8	2840	10.0	77.3	0.90	6.4	2.1	5.2	2.9	**1LE1011-1AP5■-■■■■**	22	0.0078	13
1.1	**4.4**	**112 M**	1455	7.2	83.4	0.85	2.25	2.2	6.1	2.5	2885	15.0	80.8	0.92	8.5	2.1	6.4	2.6	**1LE1011-1BP2■-■■■■**	27	0.010	13
1.45	**5.9**	**132 S**	1460	9.5	84.0	0.84	2.95	1.6	5.8	2.8	2875	20.0	80.0	0.89	12.0	1.8	5.6	2.8	**1LE1011-1CP0■-■■■■**	38	0.019	13
2.0	**8.0**	**132 M**	1455	13	85.0	0.85	4.00	1.8	5.6	2.8	2880	27.0	82.0	0.92	15.3	1.8	6.3	2.8	**1LE1011-1CP2■-■■■■**	44	0.024	13
2.9	**11.5**	**160 M**	1465	19	86.5	0.86	5.6	1.8	5.9	2.9	2870	38.0	82.0	0.92	22.0	1.8	6.0	2.9	**1LE1011-1DP2■-■■■■**	62	0.044	13
4.3	**16**	**160 L**	1455	28	87.0	0.85	8.4	1.6	6.0	2.3	2920	52.0	86.0	0.94	28.5	1.9	7.1	2.8	**1LE1011-1DP6■-■■■■**	85	0.068	13
6/4-pole: 1000/1500 rpm at 50 Hz with two windings																						
1000 rpm	*1500 rpm*		*1000 rpm*								*1500 rpm*											
0.6	**1.7**	**100 L**	970	5.9	55.5	0.62	2.50	1.7	3.4	2.7	1435	11.0	76.2	0.83	3.90	1.8	4.6	2.2	**1LE1012-1AQ4■-■■■■**	18	0.0059	10
0.8	**2.1**	**100 L**	955	8	64.2	0.77	2.35	1.2	3.4	2.0	1435	14.0	78.4	0.84	4.60	2.0	5.4	2.3	**1LE1012-1AQ5■-■■■■**	22	0.0078	10
0.9	**3.0**	**112 M**	975	8.8	64.7	0.66	3.05	1.6	3.9	2.5	1455	20.0	81.4	0.78	6.8	2.1	6.1	3.0	**1LE1012-1BQ2■-■■■■**	27	0.010	13
1.2	**3.9**	**132 S**	980	12	72.3	0.70	3.40	1.4	4.6	2.5	1455	26.0	83.1	0.83	8.2	1.5	5.7	2.4	**1LE1012-1CQ0■-■■■■**	38	0.019	10
1.7	**5.4**	**132 M**	980	17	74.1	0.71	4.65	1.7	5.0	2.5	1465	35.0	85.9	0.82	11.1	2.0	6.9	2.8	**1LE1012-1CQ2■-■■■■**	44	0.024	10
2.5	**7.2**	**160 M**	985	24	77.7	0.71	6.5	1.5	4.7	2.6	1470	47.0	86.9	0.85	14.1	1.8	6.3	2.7	**1LE1012-1DQ2■-■■■■**	62	0.044	10
3.7	**12.0**	**160 L**	985	36	82.4	0.69	9.4	2.3	6.2	3.5	1475	78.0	87.9	0.8	24.5	2.1	7.5	3.5	**1LE1012-1DQ4■-■■■■**	73	0.059	10
8/4-pole: 750/1500 rpm at 50 Hz with one winding connected in Dahlander circuit																						
750 rpm	*1500 rpm*		*750 rpm*								*1500 rpm*											
0.5	**2.0**	**100 L**	720	6.6	52.0	0.50	2.80	1.3	3.3	3.4	1440	13.0	82.0	0.79	4.45	3.0	7.5	4.0	**1LE1011-1AR4■-■■■■**	22	0.0078	7
0.65	**2.5**	**100 L**	715	8.7	56.0	0.58	2.90	1.0	3.2	2.6	1425	17.0	81.0	0.84	5.3	2.3	6.3	3.2	**1LE1011-1AR5■-■■■■**	22	0.0078	7
0.9	**3.6**	**112 M**	715	12	56.0	0.57	4.05	1.0	2.8	2.1	1430	24.0	82.0	0.84	7.5	1.9	5.6	2.4	**1LE1011-1BR2■-■■■■**	27	0.010	7
1.1	**4.7**	**132 S**	730	14	62.0	0.54	4.75	1.0	3.2	2.2	1430	31.0	82.0	0.86	9.6	1.7	5.2	2.2	**1LE1011-1CR0■-■■■■**	38	0.019	7
1.4	**6.4**	**132 M**	730	18	67.5	0.52	5.8	1.1	3.5	2.3	1440	42.0	84.5	0.87	12.6	1.9	5.7	2.3	**1LE1011-1CR2■-■■■■**	44	0.024	7
2.2	**9.5**	**160 M**	730	29	80.6	0.63	6.3	1.5	4.0	2.5	1465	62.0	86.1	0.84	19.0	2.0	6.3	2.7	**1LE1011-1DR2■-■■■■**	62	0.044	10
3.3	**14**	**160 L**	735	43	81.4	0.56	10.4	2.5	4.8	3.3	1475	91.0	85.8	0.73	32.5	2.5	7.2	3.8	**1LE1011-1DR4■-■■■■**	73	0.056	16

Voltages		No. of poles	Frame size	Motor type	Version			Order code(s)
50 Hz	230 V	4/2, 6/4, 8/4	100 L ... 160 L	1LE101■-1A ... -1D	**Standard**	**2**	**2**	–
50 Hz	400 V	4/2, 6/4, 8/4	100 L ... 160 L	1LE101■-1A ... -1D	**Standard**	**3**	**4**	–
50 Hz	500 V	4/2, 6/4, 8/4	100 L ... 160 L	1LE101■-1A ... -1D	Without add. charge	**4**	**0**	–
50 Hz	690 V	4/2, 6/4, 8/4	100 L ... 160 L	1LE101■-1A ... -1D	Without add. charge	**4**	**7**	–
Further voltages [1]		For price information, code numbers, order codes and descriptions, see from Page 2/39				**9**	**0**	. . .

Types of construction		No. of poles	Frame size	Motor type	Version		Order code(s)
Without flange	IM B3 [2]	4/2, 6/4, 8/4	100 L ... 160 L	1LE101■-1A ... -1D	**Standard**	**A**	–
With flange	IM B5 [2]	4/2, 6/4, 8/4	100 L ... 160 L	1LE101■-1A ... -1D	With additional charge	**F**	–
With standard flange	IM B14 [2]	4/2, 6/4, 8/4	100 L ... 160 L	1LE101■-1A ... -1D	With additional charge	**K**	–
Further types of construction		For price information, code letters and descriptions, see from Page 2/41				■	. . .

Motor protection	No. of poles	Frame size	Motor type	Version		Order code(s)
Without	4/2, 6/4, 8/4	100 L ... 160 L	1LE101■-1A ... -1D	**Standard**	**A**	–
PTC thermistor with 3 temperature sensors	4/2, 6/4, 8/4	100 L ... 160 L	1LE101■-1A ... -1D	With additional charge	**B**	–
Further motor protection	For price information, code letters and descriptions, see from Page 2/49				■	. . .

Terminal box position	No. of poles	Frame size	Motor type	Version		Order code(s)
Terminal box at top	4/2, 6/4, 8/4	100 L ... 160 L	1LE101■-1A ... -1D	**Standard**	**4**	–
Further terminal box positions	For price information, code numbers and descriptions, see from Page 2/51					

Special versions			Order code(s)
Options	For price information, order codes and descriptions, see from Page 2/53	**1LE101■-. . . . ■-■■■■**	**-Z** . . .+. . .+. . .

[1] Operating values at rated output for 60 Hz are available on request.

[2] Types derived from IM B3 (IM B6/7/8, IM V6 and IM V5), from IM B5 (IM V3 and IM V1) and from IM B14 (IM V19 and IM V18) are possible, provided that no requirements exist for condensation drainage holes (H03) and stamping of the type on the rating plate. The basic type IM B3, IM B5 or IM B14 is stamped as standard on the rating plate. When ordering with condensation drainage holes (H03), the type must be specified.

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Voltages
Aluminum series 1LE10, 1PC10

Selection and ordering data

Voltages	Voltage code 12th and 13th position of the Article No.		Additional identification code with order code and plain text if required	Motor category							
				Motor version	**Motor type (alum.)**	**Motor type – Frame size**					
						80	**90**	**100**	**112**	**132**	**160**
				High Efficiency IE2	1LE1001	**1LE1001** ①					
					1PC1001			**1PC1001** ②			
				IE3 Premium Efficiency	1LE1003	**1LE1003** ③					
				IE1 Standard Efficiency	1LE1002			**1LE1002** ④			
					1PC1002			**1PC1002** ⑤			
				NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line** ⑥					
				NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line** ⑦					
1LE10 . . - . . . . ■ - ■ . . .				Motor version	Motor type	Frame size					
1PC10 . . - . . . . ■ - ■ . . .			Order code			80	90	100	112	132	160
Voltage at 50 Hz or 60 Hz											
50 Hz 230 VΔ/400 VY, 60 Hz 460 VY	**2**	**2**	–	All	All	□	□	□	□	□	□
50 Hz 400 VΔ/690 VY, 60 Hz 460 VΔ [1]	**3**	**4**	–	All except ⑥ and ⑦		□	□	□	□	□	□
50 Hz 400 VΔ, 60 Hz 460 VΔ [1]				Only applicable for ⑥ and ⑦		□	□	□	□	□	□
50 Hz 400 VY, 60 Hz 460 VY [2] [3]	**0**	**2**		All	All	□	□	–	–	–	–
50 Hz 500 VY [2] 60 Hz 575 VΔ	**2**	**7**	–	All	All	○	○	○	○	○	○
50 Hz 500 VΔ 60 Hz 575 VΔ	**4**	**0**	–	All	All	–	–	○	○	○	○
50 Hz 220 VΔ/380 VY 60 Hz 440 VY	**2**	**1**	–	All	All	✓	✓	✓	✓	✓	✓
50 Hz 380 VΔ/660 VY [1], 60 Hz 440 VΔ	**3**	**3**	–	All except ⑥ and ⑦		✓	✓	✓	✓	✓	✓
50 Hz 380 VΔ [1]				Only applicable for ⑥ and ⑦		–	–	✓	✓	✓	✓
50 Hz 240 VΔ/415 VY, 60 Hz 480 VY	**2**	**3**	–	All	All	✓	✓	✓	✓	✓	✓
50 Hz 415 VΔ, 60 Hz 480 VΔ	**3**	**5**	–	All	All	✓	✓	✓	✓	✓	✓
Voltage at 60 Hz and required output at 60 Hz											
220 VΔ/380 VY; 50 Hz output	**9**	**0**	**M2A**	All	All	✓	✓	✓	✓	✓	✓
220 VΔ/380 VY; 60 Hz output [4]	**9**	**0**	**M1A**	All except ⑥ and ⑦		✓	✓	✓	✓	✓	✓
380 VΔ/660 VY; 50 Hz output [1]	**9**	**0**	**M2B**	All except ⑥ and ⑦		✓	✓	✓	✓	✓	✓
380 VΔ; 50 Hz output [1]				Only applicable for ⑥ and ⑦		–	–	✓	✓	✓	✓
380 VΔ/660 VY; 60 Hz output [1] [4]	**9**	**0**	**M1B**	All except ⑥ and ⑦		✓	✓	✓	✓	✓	✓
440 VY; 50 Hz output	**9**	**0**	**M2C**	All	All	✓	✓	✓	✓	✓	✓
440 VY; 60 Hz output [4]	**9**	**0**	**M1C**	All except ⑥ and ⑦		✓	✓	✓	✓	✓	✓
440 VΔ; 50 Hz output	**9**	**0**	**M2D**	All	All	✓	✓	✓	✓	✓	✓
440 VΔ; 60 Hz output [4]	**9**	**0**	**M1D**	All except ⑥ and ⑦		✓	✓	✓	✓	✓	✓
460 VY; 50 Hz output	**9**	**0**	**M2E**	All	All	✓	✓	✓	✓	✓	✓
460 VY; 60 Hz output [4]	**9**	**0**	**M1E**	All except ⑥ and ⑦		○	○	○	○	○	○
460 VΔ; 50 Hz output	**9**	**0**	**M2F**	All	All	✓	✓	✓	✓	✓	✓
460 VΔ; 60 Hz output [4]	**9**	**0**	**M1F**	All except ⑥ and ⑦		○	○	○	○	○	○
575 VY; 50 Hz output	**9**	**0**	**M2G**	All	All	✓	✓	✓	✓	✓	✓
575 VY; 60 Hz output [4]	**9**	**0**	**M1G**	All except ⑥ and ⑦		✓	✓	✓	✓	✓	✓
575 VΔ; 50 Hz output	**9**	**0**	**M2H**	All	All	✓	✓	✓	✓	✓	✓
575 VΔ; 60 Hz output [4]	**9**	**0**	**M1H**	All except ⑥ and ⑦		✓	✓	✓	✓	✓	✓
Voltage at 87 Hz and 87 Hz output											
400 VΔ [5]	**9**	**0**	**M3A**	All	All	✓	✓	✓	✓	✓	✓
Non-standard voltage and/or frequencies											
Non-standard winding [6]	**9**	**0**	**M1Y** • and identification code	All	All	✓	✓	✓	✓	✓	✓

□ Standard version
○ Without additional charge
• This order code only determines the price of the version – Additional plain text is required.
✓ With additional charge
– Not possible

1) For North America export versions Eagle Line 1LE1021 NEMA Energy Efficient and 1LE1023 NEMA Premium Efficient, voltages above 600 V will not be stamped.

2) Frame sizes 80 and 90 with voltage code 02 can only be supplied without motor protection (motor protection code A).

3) Delta connection is not possible.

4) Not admissible for North America export versions Eagle Line 1LE1021 NEMA Energy Efficient and 1LE1023 NEMA Premium Efficient.

5) Only possible for 4-pole, 6-pole and 8-pole motors. The operating data for converter-fed operation is also provided in a table on the rating plate.

6) Plain text must be specified in the order: Voltage between 200 and 690 V (voltages outside this range are available on request), frequency, circuit, for 60 Hz additionally required rated output in kW.

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Selection and ordering data

Voltages	Voltage code 12th and 13th position of the Article No.		Additional identification code with order code and plain text if required	Motor category					
				Motor version	**Motor type (alum.)**	**Motor type – Frame size**			
						100	**112**	**132**	**160**
				Pole-changing	1LE1011	**1LE1011**			
					1LE1012	**1LE1012**			
				Motor version	Motor type	Frame size			
1LE1. . . - . . . . ■ - ■ . . .			Order code			100	112	132	160
Voltage at 50 Hz and 50 Hz output									
230 V	**2**	**2**	**–**	All	All	□	□	□	□
400 V	**3**	**4**	**–**	All	All	□	□	□	□
500 V	**4**	**0**	**–**	All	All	○	○	○	○
690 V	**4**	**7**	**–**	All	All	○	○	○	○
Voltage at 60 Hz and required output									
220 V; 50 Hz output	**9**	**0**	**M5K**	All	All	✓	✓	✓	✓
220 V; 60 Hz output	**9**	**0**	**M5C**	All	All	✓	✓	✓	✓
380 V; 50 Hz output	**9**	**0**	**M5L**	All	All	✓	✓	✓	✓
380 V; 60 Hz output	**9**	**0**	**M5D**	All	All	✓	✓	✓	✓
440 V; 50 Hz output	**9**	**0**	**M5M**	All	All	✓	✓	✓	✓
440 V; 60 Hz output	**9**	**0**	**M5E**	All	All	✓	✓	✓	✓
460 V; 50 Hz output	**9**	**0**	**M5N**	All	All	✓	✓	✓	✓
460 V; 60 Hz output	**9**	**0**	**M5F**	All	All	✓	✓	✓	✓
575 V; 50 Hz output	**9**	**0**	**M5P**	All	All	✓	✓	✓	✓
575 V; 60 Hz output	**9**	**0**	**M5G**	All	All	✓	✓	✓	✓
Non-standard voltage and/or frequencies									
Non-standard winding [1]	**9**	**0**	**M1Y** • and identification code	All	All	✓	✓	✓	✓

□ Standard version
○ Without additional charge
• This order code only determines the price of the version – Additional plain text is required.
✓ With additional charge

[1] Plain text must be specified in the order: Voltage between 200 and 690 V (voltages outside this range are available on request), frequency, circuit, for 60 Hz additionally required rated output in kW.

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SIMOTICS SD 1LE1 Standard Motors

Supplements to article numbers and special versions

Voltages
Cast-iron series 1LE15 Basic Line, 1LE16 Performance Line

Selection and ordering data

Voltages	Voltage code 12th and 13th position of the Article No.		Additional identification code with order code and plain text if required	Motor category: Motor version	Motor type (cast-iron)	Motor type – Frame size: 71	80	90	100	112	132	160	180	200	225	250	280	315
				High Efficiency IE2	1LE1501	1LE1501 Basic Line ①											
					1LE1601				1LE1601 Performance Line ②								
				IE3 Premium Efficiency	1LE1503	1LE1503 Basic Line ③											
					1LE1603				1LE1603 Performance Line ④								
				NEMA Energy Efficient	1LE1521	1LE1521 Eagle Line Basic ⑤											
					1LE1621				1LE1621 Eagle Line Performance ⑥								
				NEMA Premium Efficient	1LE1523	1LE1523 Eagle Line Basic ⑦											
					1LE1623				1LE1623 Eagle Line Performance ⑧								
1LE1. . . - . . . . ■ - ■ . . .			Order code	Motor version	Motor type	Frame size: 71	80	90	100	112	132	160	180	200	225	250	280	315
Voltage at 50 Hz or 60 Hz																		
50 Hz 230 VΔ/400 VY, 60 Hz 460 VY [1]	**2**	**2**	–	All	All	□	□	□	□	□	□	□	□	□	□	□	□	□
50 Hz 400 VΔ/690 VY, 60 Hz 460 VΔ [2]	**3**	**4**	–	All except ⑤, ⑥, ⑦ and ⑧		□	□	□	□	□	□	□	□	□	□	□	□	□
50 Hz 400 VΔ, 60 Hz 460 VΔ [2]				Only applicable for ⑤, ⑥, ⑦ and ⑧		□	□	□	□	□	□	□	□	□	□	□	□	□
50 Hz 500 VY	**2**	**7**	–	All	All	□	□	□	○	○	○	○	○	○	○	○	○	○
50 Hz 500 VΔ	**4**	**0**	–	All	All	–	–	–	○	○	○	○	○	○	○	○	○	○
50 Hz 220 VΔ/380 VY, 60 Hz 440 VY [1]	**2**	**1**	–	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
50 Hz 380 VΔ/660 VY, 60 Hz 440 VΔ [2]	**3**	**3**	–	All except ⑤, ⑥, ⑦ and ⑧		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
50 Hz 380 VΔ [2]				Only applicable for ⑤, ⑥, ⑦ and ⑧		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
50 Hz 240 VΔ/415 VY, 60 Hz 480 VY [1]	**2**	**3**	–	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
50 Hz 415 VΔ, 60 Hz 480 VΔ	**3**	**5**	–	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Voltage at 60 Hz and required output																		
220 VΔ/380 VY; 50 Hz output [5]	**9**	**0**	**M2A**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
220 VΔ/380 VY; 60 Hz output [1] [3]	**9**	**0**	**M1A**	All except ⑤, ⑥, ⑦ and ⑧		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
380 VΔ/660 VY; 50 Hz output [2]	**9**	**0**	**M2B**	All except ⑤, ⑥, ⑦ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
380 VΔ; 50 Hz output [2]				Only applicable for ⑤, ⑥, ⑦ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
380 VΔ/660 VY; 60 Hz output [2] [3]	**9**	**0**	**M1B**	All except ⑤, ⑥, ⑦ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
440 VY; 50 Hz output	**9**	**0**	**M2C**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
440 VY; 60 Hz output [3]	**9**	**0**	**M1C**	All except ⑤, ⑥, ⑦ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
440 VΔ; 50 Hz output	**9**	**0**	**M2D**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
440 VΔ; 60 Hz output [3]	**9**	**0**	**M1D**	All except ⑤, ⑥, ⑦ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
460 VY; 50 Hz output	**9**	**0**	**M2E**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
460 VY; 60 Hz output [3]	**9**	**0**	**M1E**	All except ⑤, ⑥, ⑦ and ⑧		–	–	–	○	○	○	○	○	○	○	○	○	○
460 VΔ; 50 Hz output	**9**	**0**	**M2F**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
460 VΔ; 60 Hz output [3]	**9**	**0**	**M1F**	All except ⑤, ⑥, ⑦ and ⑧		–	–	–	○	○	○	○	○	○	○	○	○	○
575 VY; 50 Hz output	**9**	**0**	**M2G**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
575 VY; 60 Hz output [3]	**9**	**0**	**M1G**	All except ⑤, ⑥, ⑦ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
575 VΔ; 50 Hz output	**9**	**0**	**M2H**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
575 VΔ; 60 Hz output [3]	**9**	**0**	**M1H**	All except ⑤, ⑥, ⑦ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Voltage at 87 Hz and 87 Hz output																		
400 VΔ [4]	**9**	**0**	**M3A**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Non-standard voltage and/or frequencies																		
Non-standard winding [5]	**9**	**0**	**M1Y** • and identification code	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

□ Standard version
○ Without additional charge
• This order code only determines the price of the version – Additional plain text is required.
✓ With additional charge

1) Parallel supply lines are required in the case of connection to ≤ 240 V. For frame size 315 with connection to ≤ 240 V, due to the high current, a drilled, removable entry plate (Order code **R52**) or a larger terminal box (Order code **R50**) can be used. Order codes **R52** and **R50** alter the motor dimensions.

2) For North America export versions Eagle Line 1LE1521/1LE1621 NEMA Energy Efficient and 1LE1523/1LE1623 NEMA Premium Efficient, voltages above 600 V will not be stamped.

3) Not admissible for North America export versions Eagle Line 1LE1521/1LE1621 NEMA Energy Efficient and 1LE1523/1LE1623 NEMA Premium Efficient.

4) Only possible for 4-pole, 6-pole and 8-pole motors. The operating data for converter-fed operation is also provided in a table on the rating plate.

5) Plain text must be specified in the order: Voltage between 200 and 690 V (voltages outside this range are available on request), frequency, circuit, for 60 Hz additionally required rated output in kW.

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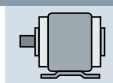
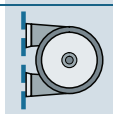
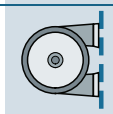
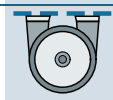
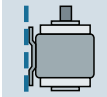
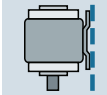
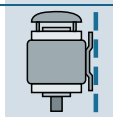
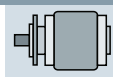
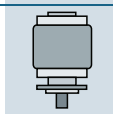
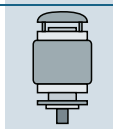
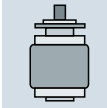
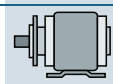
Selection and ordering data

Types of construction		Type of construction letter 14th position of the Article No.	For types of construction with order code(s) Article No. with additional identification code **-Z**	Motor category							
				Motor version	**Motor type (alum.)**	**Motor type – Frame size**					
						80	**90**	**100**	**112**	**132**	**160**
				High Efficiency IE2	1LE1001	**1LE1001** ①					
					1PC1001			**1PC1001** ②			
				IE3 Premium Efficiency	1LE1003	**1LE1003** ③					
				IE1 Standard Efficiency	1LE1002			**1LE1002** ④			
					1PC1002			**1PC1002** ⑤			
				NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line** ⑥					
				NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line** ⑦					
				Pole-changing	1LE1011			**1LE1011** ⑧			
					1LE1012			**1LE1012** ⑨			
1LE10 . . - . . . . . - ■ . . (-Z)				Motor version	Motor type	Frame size					
1PC10 . . - . . . . . - ■ . . (-Z)			Order code			80	90	100	112	132	160
Without flange											
IM B3 [1) 2) 3)]		**A**	–	All except ⑥		□	□	□	□	□	□
IM B6 [2) 3)]		**T**	–	All except ⑥		□	□	□	□	□	□
IM B7 [2) 3)]		**U**	–	All except ⑥		□	□	□	□	□	□
IM B8 [2) 3)]		**V**	–	All except ⑥		□	□	□	□	□	□
IM V6 [2) 3)]		**D**	–	All except ⑥		□	□	□	□	□	□
IM V5 without protective cover [2) 3)]		**C**	–	All except ⑥		□	□	□	□	□	□
IM V5 with protective cover [2) 3) 4) 5) 6)]		**C**	**H00**	All except ②, ⑤, ⑥ and in combination with order code F90		✓	✓	✓	✓	✓	✓
With flange		**acc. to DIN EN 50347 acc. to DIN 42948**				**FF165 A 200**	**FF165 A 200**	**FF215 A 250**	**FF215 A 250**	**FF265 A 300**	**FF300 A 350**
IM B5 [2) 7)]		**F**	–	All	All	✓	✓	✓	✓	✓	✓
IM V1 without protective cover [2)]		**G**	–	All	All	✓	✓	✓	✓	✓	✓
IM V1 with protective cover [2) 4) 5) 6)]		**G**	**H00**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
IM V3 [4)]		**H**	–	All	All	✓	✓	✓	✓	✓	✓
IM B35 [3)]		**J**	–	All except ⑥		✓	✓	✓	✓	✓	✓

For legends and footnotes, see Page 2/44.

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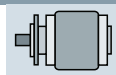
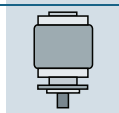
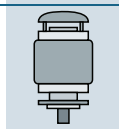
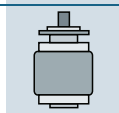
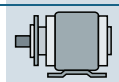
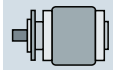
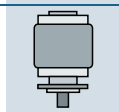
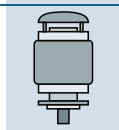
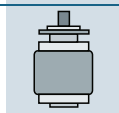
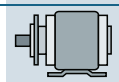
SIMOTICS GP 1LE1/1PC1 Standard Motors

Supplements to article numbers and special versions

Types of construction
Aluminum series 1LE10, 1PC10

Motor version	Motor type (alum.)	Motor type – Frame size 80	90	100	112	132	160
High Efficiency IE2	1LE1001	**1LE1001** ①					
	1PC1001			**1PC1001** ②			
IE3 Premium Efficiency	1LE1003	**1LE1003** ③					
IE1 Standard Efficiency	1LE1002			**1LE1002** ④			
	1PC1002			**1PC1002** ⑤			
NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line** ⑥					
NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line** ⑦					
Pole-changing	1LE1011			**1LE1011** ⑧			
	1LE1012			**1LE1012** ⑨			

Types of construction		Type of construction letter 14th position of the Article No.	For types of construction with order code(s) Article No. with additional identification code **-Z**	Motor version	Motor type	Frame size 80	90	100	112	132	160
1LE10 . . - . . . . . - . ■ . . (-Z)											
1PC10 . . - . . . . . - . ■ . . (-Z)			Order code								
With special flange next larger		**acc. to DIN EN 50347**				–	–	FF265	FF265	FF300	–
		acc. to DIN 42948				–	–	A 300	A 300	A 350	–
IM B5 2) 7)		**F**	**P01**	All	All	–	–	✓	✓	✓	–
IM V1 without protective cover 2)		**G**	**P01**	All	All	–	–	✓	✓	✓	–
IM V1 with protective cover 2) 4) 5) 6)		**G**	**P01+H00**	All except ②, ⑤ and in combination with order code F90		–	–	✓	✓	✓	–
IM V3 4)		**H**	**P01**	All	All	–	–	✓	✓	✓	–
IM B35 3)		**J**	**P01**	All except ⑥		–	–	✓	✓	✓	–
With special flange next smaller		**acc. to DIN EN 50347**				–	–	FF165	FF165	FF215	FF265
		acc. to DIN 42948				–	–	A 200	A 200	A 250	A 300
IM B5 2) 7)		**F**	**P02**	All	All	–	–	✓	✓	✓	✓
IM V1 without protective cover 2)		**G**	**P02**	All	All	–	–	✓	✓	✓	✓
IM V1 with protective cover 2) 4) 5) 6)		**G**	**P02+H00**	All except ②, ⑤ and in combination with order code F90		–	–	✓	✓	✓	✓
IM V3 4)		**H**	**P02**	All	All	–	–	✓	✓	✓	✓
IM B35 3)		**J**	**P02**	All except ⑥		–	–	✓	✓	✓	✓

For legends and footnotes, see Page 2/44.

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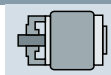
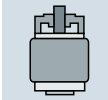
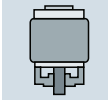
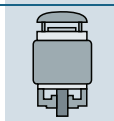
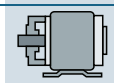
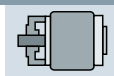
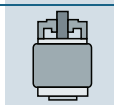
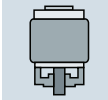
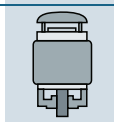
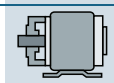
Types of construction		Type of construction letter 14th position of the Article No.	For types of construction with order code(s) Article No. with additional identification code **-Z**	Motor category							
				Motor version	**Motor type (alum.)**	**Motor type – Frame size**					
						80	**90**	**100**	**112**	**132**	**160**
				High Efficiency IE2	1LE1001	**1LE1001** ①					
					1PC1001			**1PC1001** ②			
				IE3 Premium Efficiency	1LE1003	**1LE1003** ③					
				IE1 Standard Efficiency	1LE1002			**1LE1002** ④			
					1PC1002			**1PC1002** ⑤			
				NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line** ⑥					
				NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line** ⑦					
				Pole-changing	1LE1011			**1LE1011** ⑧			
					1LE1012			**1LE1012** ⑨			
1LE10 . . - . . . . . - . ■ . . (-Z)				Motor version	Motor type	Frame size					
1PC10 . . - . . . . . - . ■ . . (-Z)			Order code			80	90	100	112	132	160
With standard flange		**acc. to DIN EN 50347** **acc. to DIN 42948**				**FT100** **C 120**	**FT115** **C 140**	**FT130** **C 160**	**FT130** **C 160**	**FT165** **C 200**	**FT215** **C 250**
IM B14 [2) 8)]		**K**	**–**	All	All	✓	✓	✓	✓	✓	✓
IM V19 [2)]		**L**	**–**	All	All	✓	✓	✓	✓	✓	✓
IM V18 without protective cover [2)]		**M**	**–**	All	All	✓	✓	✓	✓	✓	✓
IM V18 with protective cover 2) 4) 5) 6)		**M**	**H00**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
IM B34 [3)]		**N**	**–**	All except ⑥		✓	✓	✓	✓	✓	✓
With special flange next larger		**acc. to DIN EN 50347** **acc. to DIN 42948**				**FT130** **C 160**	**FT130** **C 160**	**FT165** **C 200**	**FT165** **C 200**	**FT215** **C 250**	**–** **–**
IM B14 [2) 8)]		**K**	**P01**	All	All	✓	✓	✓	✓	✓	–
IM V19 [2)]		**L**	**P01**	All	All	✓	✓	✓	✓	✓	–
IM V18 without protective cover [2)]		**M**	**P01**	All	All	✓	✓	✓	✓	✓	–
IM V18 with protective cover 2) 4) 5) 6)		**M**	**P01+H00**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	–
IM B34 [3)]		**N**	**P01**	All except ⑥		✓	✓	✓	✓	✓	–

For legends and footnotes, see Page 2/44.

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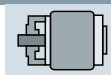
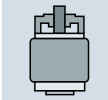
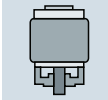
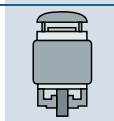
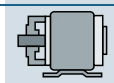
SIMOTICS GP 1LE1/1PC1 Standard Motors

Supplements to article numbers and special versions

Types of construction
Aluminum series 1LE10, 1PC10

Types of construction		Type of construction letter 14th position of the Article No.	For types of construction with order code(s) Article No. with additional identification code **-Z**	Motor category						
				Motor version	**Motor type (alum.)**	**Motor type – Frame size** 80	90	100	112	132	160
				High Efficiency IE2	1LE1001	**1LE1001** ①					
					1PC1001			**1PC1001** ②			
				IE3 Premium Efficiency	1LE1003	**1LE1003** ③					
				IE1 Standard Efficiency	1LE1002			**1LE1002** ④			
					1PC1002			**1PC1002** ⑤			
				NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line** ⑥					
				NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line** ⑦					
				Pole-changing	1LE1011			**1LE1011** ⑧			
					1LE1012			**1LE1012** ⑨			
1LE10 . . - . . . . . - . ■ . . (-Z)				Motor version	Motor type	Frame size 80	90	100	112	132	160
1PC10 . . - . . . . . - . ■ . . (-Z)			Order code								
With special flange next smaller		**acc. to DIN EN 50347**				**–**	**–**	**FT115**	**FT115**	**FT130**	**FT165**
		acc. to DIN 42948				**–**	**–**	**C 140**	**C 140**	**C 160**	**C 200**
IM B14 [2) 8)]		**K**	**P02**	All	All	–	–	O. R.	O. R.	O. R.	O. R.
IM V19 [2)]		**L**	**P02**	All	All	–	–	O. R.	O. R.	O. R.	O. R.
IM V18 without protective cover [2)]		**M**	**P02**	All	All	–	–	O. R.	O. R.	O. R.	O. R.
IM V18 with protective cover 2) 4) 5) 6)		**M**	**P02+H00**	All except ②, ⑤ and in combination with order code F90		–	–	O. R.	O. R.	O. R.	O. R.
IM B34 [3)]		**N**	**P02**	All except ⑥	–	–	–	O. R.	O. R.	O. R.	O. R.

□ Standard version
✓ With additional charge
– Not possible
O. R. Possible on request

1) The types of construction IM B6/7/8, IM V6 and IM V5 with/without protective cover are also possible as long as no condensation drainage holes (order code **H03**) and no stamping of these types of construction on the rating plate are required. As standard the type of construction IM B3 is then stamped on the rating plate. With type of construction IM V5 with protective cover, the protective cover has to be additionally ordered with order code **H00**. The protective cover is not stamped on the rating plate.

2) The type of construction is stamped on the rating plate. When ordering with condensation drainage holes (order code **H03**), if mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

3) For North America export version Eagle Line 1LE1021 NEMA Energy Efficient, types of construction with feet are not possible for 2-pole, 4-pole and 6-pole motors ≤ 200 hp in accordance with NEMA MG1 Table 12-11.

4) The "Second shaft extension" option (order code **L05**) is not possible.

5) In combination with an encoder it is not necessary to order the protective cover (order code **H00**), as this is delivered as a protection for the encoder as standard. In this case the protective cover is standard version (without additional charge).

6) Not possible for 1PC1 naturally cooled motors and 1LE1 forced-air cooled motors with order code **F90** without external fan and fan cover.

7) The types of construction IM V3 and IM V1 with/without protective cover are also possible as long as no condensation drainage holes (order code **H03**) and no stamping of these types of construction on the rating plate are required. As standard the type of construction IM B5 is then stamped on the rating plate. With type of construction IM V1 with protective cover, the protective cover has to be additionally ordered with order code **H00**. The protective cover is not stamped on the rating plate.

8) The types of construction IM V19 and IM V18 with/without protective cover are also possible as long as no condensation drainage holes (order code **H03**) and no stamping of these types of construction on the rating plate are required. As standard the type of construction IM B14 is then stamped on the rating plate. With type of construction IM V18 with protective cover, the protective cover has to be additionally ordered with order code **H00**. The protective cover is not stamped on the rating plate.

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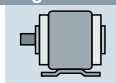
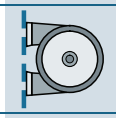
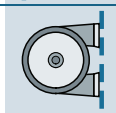
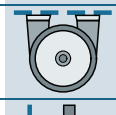
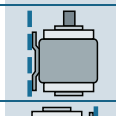
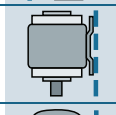
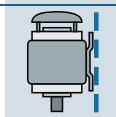
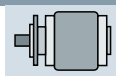
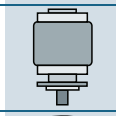
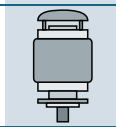
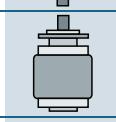
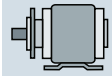
Selection and ordering data

Types of construction		Type of construction code 14th pos. of Article No.	For types of construction with order code(s) Article No. with additional identification code -Z	Motor category															
				Motor version	Motor type (cast-iron)	Motor type – Frame size 71	80	90	100	112	132	160	180	200	225	250	280	315 S/M	315 L 2-pole	315 L 4-, 6-, 8-pole
				High Efficiency IE2	1LE1501	1LE1501 Basic Line ①														
					1LE1601				1LE1601 Performance Line ②											
				IE3 Premium Efficiency	1LE1503	1LE1503 Basic Line ③														
					1LE1603				1LE1603 Performance Line ④											
				NEMA Energy Efficient	1LE1521	1LE1521 Eagle Line Basic ⑤														
					1LE1621				1LE1621 Eagle Line Performance ⑥											
				NEMA Premium Efficient	1LE1523	1LE1523 Eagle Line Basic ⑦														
					1LE1623				1LE1623 Eagle Line Performance ⑧											
				Motor version	Motor type	Frame size 71	80	90	100	112	132	160	180	200	225	250	280	315 S/M	315 L 2-pole	315 L 4-pole, 6-pole, 8-pole
1LE1. . . - . . . . . - . ■ . . (-Z)			Order code																	
Without flange																				
IM B3 1) 2) 3)		A	–	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		□	□	□	□	□	□	□	□	□	□	□	□	□	□	□
IM B6 2) 3)		T	–	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		□	□	□	□	□	□	□	□	□	□	□	□	□	□	□
IM B7 2) 3)		U	–	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		□	□	□	□	□	□	□	□	□	□	□	□	□	□	□
IM B8 2) 3)		V	–	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		□	□	□	□	□	□	□	□	□	□	□	□	□	□	□
IM V6 2) 3)		D	–	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		□	□	□	□	□	□	□	□	□	□	□	□	□	✓	□
IM V5 without protective cover 2) 3)		C	–	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		□	□	□	□	□	□	□	□	□	□	□	□	□	✓	□
IM V5 with protective cover 2) 3) 4) 5)		C	H00	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
With flange		acc. to DIN EN 50347 acc. to DIN 42948				FF130 A 160	FF165 A 200	FF165 A 200	FF215 A 250	FF215 A 250	FF265 A 300	FF300 A 350	FF300 A 350	FF350 A 400	FF400 A 450	FF500 A 550	FF500 A 550	FF600 A 660	FF600 A 660	FF600 A 660
IM B5 2) 6)		F	–	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	–	–
IM V1 without protective cover 2)		G	–	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
IM V1 with protective cover 2) 4) 5)		G	H00	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
IM V3 5)		H	–	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	–	–
IM B35 3)		J	–	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

For legends, see Page 2/47; for footnotes, see Page 2/48.

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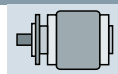
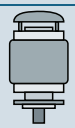
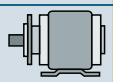
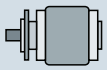
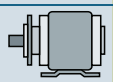
SIMOTICS SD 1LE1 Standard Motors

Supplements to article numbers and special versions

Types of construction
Cast-iron series 1LE15 Basic Line, 1LE16 Performance Line

Types of construction		Type of construc. code 14th pos. of Article No.	For types of construction with order code(s) Article No. with additional identification code -Z	Motor category																
				Motor version	**Motor type (cast-iron)**	**Motor type – Frame size** 71	80	90	100	112	132	160	180	200	225	250	280	315 S/M	315 L 2-pole	315 L 4-, 6-, 8-pole
				High Efficiency IE2	1LE1501	**1LE1501 Basic Line** ①														
					1LE1601				**1LE1601 Performance Line** ②											
				IE3 Premium Efficiency	1LE1503	**1LE1503 Basic Line** ③														
					1LE1603				**1LE1603 Performance Line** ④											
				NEMA Energy Efficient	1LE1521	**1LE1521 Eagle Line Basic** ⑤														
					1LE1621				**1LE1621 Eagle Line Performance** ⑥											
				NEMA Premium Efficient	1LE1523	**1LE1523 Eagle Line Basic** ⑦														
					1LE1623				**1LE1623 Eagle Line Performance** ⑧											
				Motor version	Motor type	Frame size 71	80	90	100	112	132	160	180	200	225	250	280	315 S/M	315 L 2-pole	315 L 4-pole, 6-pole, 8-pole
1LE1. . . - . . . . . - . ■ . . (-Z)			Order code																	
With special flange next larger		**acc. to DIN EN 50347**				–	–	–	FF265	FF265	FF300	–	–	–	–	–	–	–	–	–
		acc. to DIN 42948				–	–	–	A 300	A 300	A 350	–	–	–	–	–	–	–	–	–
IM B5 2) 6)		**F**	**P01**	All	All	–	–	–	✓	✓	✓	–	–	–	–	–	–	–	–	–
IM V1 without protective cover 2)		**G**	**P01**	All	All	–	–	–	✓	✓	✓	–	–	–	–	–	–	–	–	–
IM V1 with protective cover 2) 4) 5)		**G**	**P01+ H00**	All	All	–	–	–	✓	✓	✓	–	–	–	–	–	–	–	–	–
IM V3 5)		**H**	**P01**	All	All	–	–	–	✓	✓	✓	–	–	–	–	–	–	–	–	–
IM B35 3)		**J**	**P01**	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		–	–	–	✓	✓	✓	–	–	–	–	–	–	–	–	–
With special flange next smaller		**acc. to DIN EN 50347**				–	–	–	FF165	FF165	FF215	FF265	–	–	–	–	–	–	–	–
		acc. to DIN 42948				–	–	–	A 200	A 200	A 250	A 300	–	–	–	–	–	–	–	–
IM B5 2) 6)		**F**	**P02**	All	All	–	–	–	O.	R.	O.	R.	O.	R.	–	–	–	–	–	–
IM V1 without protective cover 2)		**G**	**-Z P02**	All	All	–	–	–	O.	R.	O.	R.	O.	R.	–	–	–	–	–	–
IM V1 with protective cover 2) 4) 5)		**G**	**P02+ H00**	All	All	–	–	–	O.	R.	O.	R.	O.	R.	–	–	–	–	–	–
IM V3 5)		**H**	**P02**	All	All	–	–	–	O.	R.	O.	R.	O.	R.	–	–	–	–	–	–
IM B35 3)		**J**	**P02**	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		–	–	–	O.	R.	O.	R.	O.	R.	–	–	–	–	–	–

For legends, see Page 2/47; for footnotes, see Page 2/48.

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Types of construction		Type of construc. code 14th pos. of Article No.	For types of construction with order code(s) Article No. with additional identification code **-Z**	Motor category																
				Motor version	**Motor type (cast-iron)**	**Motor type – Frame size** 71	80	90	100	112	132	160	180	200	225	250	280	315 S/M	315 L 2-pole	315 L 4-, 6-, 8-pole
				High Efficiency IE2	1LE1501	**1LE1501 Basic Line** ①														
					1LE1601				**1LE1601 Performance Line** ②											
				IE3 Premium Efficiency	1LE1503	**1LE1503 Basic Line** ③														
					1LE1603				**1LE1603 Performance Line** ④											
				NEMA Energy Efficient	1LE1521	**1LE1521 Eagle Line Basic** ⑤														
					1LE1621				**1LE1621 Eagle Line Performance** ⑥											
				NEMA Premium Efficient	1LE1523	**1LE1523 Eagle Line Basic** ⑦														
					1LE1623				**1LE1623 Eagle Line Performance** ⑧											
				Motor version	Motor type	Frame size 71	80	90	100	112	132	160	180	200	225	250	280	315 S/M	315 L 2-pole	315 L 4-pole, 6-pole, 8-pole
1LE1. . . - . . . . . - . ■ . . (-Z)			Order code																	
With flange		**acc. to DIN EN 50347**				–	–	–	FT130	FT130	FT165	FT215	–	–	–	–	–	–	–	–
		acc. to DIN 42948				–	–	–	C 160	C 160	C 200	C 250	–	–	–	–	–	–	–	–
IM B14 2) 7)		**K**	**–**	All	All	–	–	–	✓	✓	✓	✓	–	–	–	–	–	–	–	–
IM V19 2)		**L**	**–**	All	All	–	–	–	✓	✓	✓	✓	–	–	–	–	–	–	–	–
IM V18 without protective cover 2)		**M**	**–**	All	All	–	–	–	✓	✓	✓	✓	–	–	–	–	–	–	–	–
IM V18 with protective cover 2) 4) 5)		**M**	**H00**	All	All	–	–	–	✓	✓	✓	✓	–	–	–	–	–	–	–	–
IM B34 3)		**N**	**–**	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		–	–	–	✓	✓	✓	✓	–	–	–	–	–	–	–	–
With special flange next larger		**acc. to DIN EN 50347**				–	–	–	FT165	FT165	FT215	–	–	–	–	–	–	–	–	–
		acc. to DIN 42948				–	–	–	C 200	C 200	C 250	–	–	–	–	–	–	–	–	–
IM B14 2) 7)		**K**	**P01**	All	All	–	–	–	✓	✓	✓	–	–	–	–	–	–	–	–	–
IM V19 2)		**L**	**P01**	All	All	–	–	–	✓	✓	✓	–	–	–	–	–	–	–	–	–
IM V18 without protective cover 2)		**M**	**P01**	All	All	–	–	–	✓	✓	✓	–	–	–	–	–	–	–	–	–
IM V18 with protective cover 2) 4) 5)		**M**	**P01+H00**	All	All	–	–	–	✓	✓	✓	–	–	–	–	–	–	–	–	–
IM B34 3)		**N**	**P01**	All except ⑤, ⑥ 2, 4, 6-pole ≤ 200 hp and ⑦, ⑧ 8-pole ≤ 200 hp		–	–	–	✓	✓	✓	–	–	–	–	–	–	–	–	–

□ Standard version
✓ With additional charge
– Not possible
O. R. Possible on request

For footnotes, see Page 2/48.

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Types of construction
Cast-iron series 1LE15 Basic Line, 1LE16 Performance Line

1) The types of construction IM B6/7/8, IM V6 and IM V5 with/without protective cover are also possible as long as no stamping of these types of construction on the rating plate is required. As standard the type of construction IM B3 is then stamped on the rating plate. With type of construction IM V5 with protective cover, the protective cover has to be additionally ordered with order code **H00**. The protective cover is not stamped on the rating plate.

2) The type of construction is stamped on the rating plate. If mounted in a different position, the position must be specified to ensure that the condensation drainage holes are positioned correctly.

3) For North America export version Eagle Line 1LE1521/1LE1621 NEMA Energy Efficient, types of construction with feet are not possible for 2-pole, 4-pole and 6-pole motors ≤ 200 hp in accordance with NEMA MG1 Table 12-11.

4) In combination with an encoder it is not necessary to order the protective cover (order code **H00**), as this is delivered as a protection for the encoder as standard. In this case the protective cover is standard design (without extra price).

5) The "Second shaft extension" option (order code **L05**) is not possible.

6) The types of construction IM V3 and IM V1 with/without protective cover are also possible as long as no stamping of these types of construction on the rating plate is required. As standard the type of construction IM B5 is then stamped on the rating plate. With type of construction IM V1 with protective cover, the protective cover has to be additionally ordered with order code **H00**. The protective cover is not stamped on the rating plate.

7) The types of construction IM V19 and IM V18 with/without protective cover are also possible as long as no stamping of these types of construction on the rating plate is required. As standard the type of construction IM B14 is then stamped on the rating plate. With type of construction IM V18 with protective cover, the protective cover has to be additionally ordered with order code **H00**. The protective cover is not stamped on the rating plate.

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Selection and ordering data

Motor protection	Motor protection code 15th position in Article No.	Additional identification code with order code and plain text if required	Motor category							
			Motor version	**Motor type (alum.)**	**Motor type – Frame size**					
					80	**90**	**100**	**112**	**132**	**160**
			High Efficiency IE2	1LE1001	**1LE1001**					
				1PC1001			**1PC1001**			
			IE3 Premium Efficiency	1LE1003	**1LE1003**					
			IE1 Standard Efficiency	1LE1002			**1LE1002**			
				1PC1002			**1PC1002**			
			NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line**					
			NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line**					
			Pole-changing	1LE1011			**1LE1011**			
				1LE1012			**1LE1012**			
1LE10 . . - . . . . . - . . ■ .			Motor version	Motor type	Frame size					
1PC10 . . - . . . . . - . . ■ .		Order code			80	90	100	112	132	160
Motor protection (winding protection)										
Without motor protection [1]	**A**	**–**	All	All	□	□	□	□	□	□
Motor protection with PTC thermistors with 1 or 3 embedded temperature sensors for tripping [2]	**B**	**–**	All	All	✓	✓	✓	✓	✓	✓
Motor protection with PTC thermistors with 6 embedded temperature sensors for alarm and tripping [2]	**C**	**–**	All	All	✓	✓	✓	✓	✓	✓
Motor temperature detection with embedded temperature sensor KTY 84-130 [2]	**F**	**–**	All	All	✓	✓	✓	✓	✓	✓
Motor temperature detection with embedded temperature sensor 2 × KTY 84-130 [2]	**G**	**–**	All	All	✓	✓	✓	✓	✓	✓
Installation of 3 Pt100 resistance thermometers [2]	**H**	**–**	All	All	–	–	✓	✓	✓	✓
NTC thermistors for tripping	**Z**	**Q2A**	All	All	–	–	✓	✓	✓	✓
Temperature detectors for tripping [2]	**Z**	**Q3A**	All	All	✓	✓	✓	✓	✓	✓

□ Standard version
✓ With additional charge
– Not possible

[1] Frame sizes 80 and 90 with voltage code 02 can only be supplied without motor protection (motor protection code A).

[2] Evaluation with appropriate tripping unit (see Catalog IC 10) is recommended. For pole-changing motors, double the number of temperature sensors or temperature detectors is required. This also results in a double additional charge.

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Motor protection
Cast-iron series 1LE15 Basic Line, 1LE16 Performance Line

Selection and ordering data

Motor protection	Motor protection code 15th position in Article No.	Additional identification code with order code and plain text if required	Motor category: Motor version	Motor type (cast-iron)	Motor type – Frame size: 71	80	90	100	112	132	160	180	200	225	250	280	315
			High Efficiency IE2	1LE1501	1LE1501 Basic Line ①												
				1LE1601				1LE1601 Performance Line ②									
			IE3 Premium Efficiency	1LE1503	1LE1503 Basic Line ③												
				1LE1603				1LE1603 Performance Line ④									
			NEMA Energy Efficient	1LE1521	1LE1521 Eagle Line Basic ⑤												
				1LE1621				1LE1621 Eagle Line Performance ⑥									
			NEMA Premium Efficient	1LE1523	1LE1523 Eagle Line Basic ⑦												
				1LE1623				1LE1623 Eagle Line Performance ⑧									
1LE1. . . - . . . . . - . . ■ .		Order code	Motor version	Motor type	Frame size: 71	80	90	100	112	132	160	180	200	225	250	280	315
Motor protection (winding protection)																	
Without motor protection [1]	A	–	All except ②, ④, ⑥ and ⑧		□	□	□	□	□	□	□	□	□	□	□	□	□
Motor protection with PTC thermistor with 3 embedded temperature sensors for tripping [1] [2]	B	–	Standard version for Performance Line ②, ④, ⑥ and ⑧		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Motor protection with PTC thermistors with 6 embedded temperature sensors for alarm and tripping [2]	C	–	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Motor temperature detection with embedded temperature sensor KTY 84-130 [2]	F	–	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Motor temperature detection with embedded temperature sensor 2 × KTY 84-130 [2]	G	–	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Installation of 3 Pt100 resistance thermometers	H	–	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Installation of 6 Pt100 resistance thermometers	J	–	All	All	–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓
NTC thermistors for tripping	Z	Q2A	All	All	✓	✓	✓	✓	✓	✓	✓	–	–	–	–	–	–
Temperature detectors for tripping [2]	Z	Q3A	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

□ Standard version
✓ With additional charge
– Not possible

Note:

Options are available specifically for bearing protection –
For order codes and descriptions, see from Page 2/58.

[1] For the Performance Line, motor protection by means of PTC thermistors with 3 built-in temperature sensors for tripping (motor protection code B) is already included in the basic price. For the Performance Line, the option "without motor protection" (motor protection code A) is not possible.

[2] Evaluation with appropriate tripping unit (see Catalog IC 10) is recommended.

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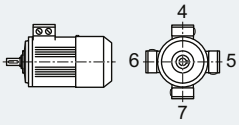
Selection and ordering data

Terminal box position	Terminal box position code 16th position of the Article No.	Additional identification code with order code and plain text if required	Motor category							
			**Motor version**	**Motor type (alum.)**	**Motor type – Frame size**					
					80	**90**	**100**	**112**	**132**	**160**
			High Efficiency IE2	1LE1001	**1LE1001**					
				1PC1001			**1PC1001**			
			IE3 Premium Efficiency	1LE1003	**1LE1003**					
			IE1 Standard Efficiency	1LE1002			**1LE1002**			
				1PC1002			**1PC1002**			
			NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line**					
			NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line**					
			Pole-changing	1LE1011			**1LE1011**			
				1LE1012			**1LE1012**			
1LE10 . . - . . . . . - . . . ■			Motor version	Motor type	Frame size					
1PC10 . . - . . . . . - . . . ■		Order code			80	90	100	112	132	160
Terminal box position										
Terminal box top [1]	**4**	–	All	All	□	□	□	□	□	□
Terminal box on RHS [2]	**5**	–	All	All	✓	✓	✓	✓	✓	✓
Terminal box on LHS [2]	**6**	–	All	All	✓	✓	✓	✓	✓	✓
Terminal box at bottom [2] [3]	**7**	–	All	All	–	–	✓	✓	✓	✓

□ Standard version
✓ With additional charge
– Not possible

[1] For types of construction with feet and flange-mounted with feet, cast feet are standard. Screwed-on feet are available with order code **H01**.

[2] For types of construction with feet and flange-mounted with feet, screwed-on feet are standard.
For types of construction with feet and flange-mounted with feet, cast feet are standard. Screwed-on feet are available with order code **H01**.

[3] Not generally possible for motors with feet.

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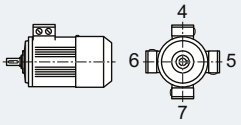
SIMOTICS SD 1LE1 Standard Motors

Supplements to article numbers and special versions

Terminal box position
Cast-iron series 1LE15 Basic Line, 1LE16 Performance Line

Selection and ordering data

Terminal box position	Terminal box position code 16th position of the Article No.	Additional identification code with order code and plain text if required	Motor category														
			**Motor version**	**Motor type (cast-iron)**	**Motor type – Frame size** 71	80	90	100	112	132	160	180	200	225	250	280	315
			High Efficiency IE2	1LE1501	**1LE1501 Basic Line** ①												
				1LE1601				**1LE1601 Performance Line** ②									
			IE3 Premium Efficiency	1LE1503	**1LE1503 Basic Line** ③												
				1LE1603				**1LE1603 Performance Line** ④									
			NEMA Energy Efficient	1LE1521	**1LE1521 Eagle Line Basic** ⑤												
				1LE1621				**1LE1621 Eagle Line Performance** ⑥									
			NEMA Premium Efficient	1LE1523	**1LE1523 Eagle Line Basic** ⑦												
				1LE1623				**1LE1623 Eagle Line Performance** ⑧									
1LE1. . . -. . . . . -. . . ■		Order code	Motor version	Motor type	Frame size 71	80	90	100	112	132	160	180	200	225	250	280	315
Terminal box position																	
Terminal box top [1]	**4**	–	All	All	□	□	□	□	□	□	□	□	□	□	□	□	□
Terminal box on RHS [2]	**5**	–	All	All	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Terminal box on LHS [2]	**6**	–	All	All	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Terminal box at bottom [3]	**7**	–	All	All	–	–	–	✓	✓	✓	✓	–	–	–	–	–	–

□ Standard version
✓ With additional charge
– Not possible

[1] For types of construction with feet and flange-mounted with feet, cast feet are standard. Screwed-on feet are available with order code **H01**.

[2] For types of construction with feet and flange-mounted with feet, screwed-on feet are standard. Except for frame sizes 225, 250, 280 and 315: in which case for types of construction with feet and flange-mounted with feet, cast feet are standard. Screwed-on feet are available with order code **H01**.

[3] Not generally possible for motors with feet.

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Selection and ordering data

Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category							
		Motor version	**Motor type (alum.)**	**Motor type – Frame size**					
				80	**90**	**100**	**112**	**132**	**160**
		High Efficiency IE2	1LE1001	**1LE1001** ①					
			1PC1001			**1PC1001** ②			
		IE3 Premium Efficiency	1LE1003	**1LE1003** ③					
		IE1 Standard Efficiency	1LE1002			**1LE1002** ④			
			1PC1002			**1PC1002** ⑤			
		NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line** ⑥					
		NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line** ⑦					
		Pole-changing	1LE1011			**1LE1011** ⑧			
			1LE1012			**1LE1012** ⑨			
1LE10 . . - . . . . . - . . . . -Z **1PC10 . . - . . . . . - . . . . -Z**	Order code	Motor version	Motor type	Frame size 80	90	100	112	132	160
Motor connection and terminal box									
External grounding	**H04**	All	All	✓	✓	✓	✓	✓	✓
Terminal box on NDE [3]	**H08**	All	All	✓	✓	✓	✓	✓	✓
Rotation of the terminal box through 90°, entry from DE [1]	**R10**	All	All	○	○	○	○	○	○
Rotation of the terminal box through 90°, entry from NDE	**R11**	All	All	○	○	○	○	○	○
Rotation of the terminal box through 180°	**R12**	All	All	○	○	○	○	○	○
One metal cable gland	**R15**	All	All	✓	✓	✓	✓	✓	✓
3 cables protruding, 0.5 m long [4][5]	**R20**	All except ⑧ and ⑨		✓	✓	✓	✓	✓	✓
3 cables protruding, 1,5 m long [4][5]	**R21**	All except ⑧ and ⑨		✓	✓	✓	✓	✓	✓
6 cables protruding, 0.5 m long [4]	**R22**	All	All	✓	✓	✓	✓	✓	✓
6 cables protruding, 1,5 m long [4]	**R23**	All	All	✓	✓	✓	✓	✓	✓
6 cables protruding, 3 m long [4]	**R24**	All	All	✓	✓	✓	✓	✓	✓
Reduction piece for M cable gland in accordance with British Standard, both cable entries mounted [2]	**R30**	All	All	–	–	✓	✓	✓	✓
Larger terminal box	**R50**	All, standard version for Eagle Line ⑥ and ⑦ < frame size 100		✓	✓	✓	✓	✓	✓
Motor connector Han-Drive 10e for 230 VΔ/400 VY [30]	**R70**	All	All	✓	✓	✓	✓	✓	–
Motor connector Han-Drive 10e EMC for 230 VΔ/400 VY [30]	**R71**	All	All	✓	✓	✓	✓	✓	–
Small motor connector CQ12 with EMC	**R72**	All	All	✓	✓	–	–	–	–
Small motor connector CQ12 without EMC	**R73**	All	All	✓	✓	–	–	–	–
Windings and insulation									
Temperature class 155 (F), utilized acc. to 155 (F), with service factor (SF)	**N01**	All	All	–	–	✓	✓	✓	✓
Temperature class 155 (F), utilized acc. to 155 (F), with increased output	**N02**	All	All	–	–	✓	✓	✓	✓
Temperature class 155 (F), utilized acc. to 155 (F), with increased coolant temperature	**N03**	All	All	✓	✓	✓	✓	✓	✓
Temperature class 155 (F), utilized acc. to 130 (B), coolant temperature 45 °C, derating approx. 4 %	**N05**	All	All	✓	✓	✓	✓	✓	✓
Temperature class 155 (F), utilized acc. to 130 (B), coolant temperature 50 °C, derating approx. 8 %	**N06**	All	All	✓	✓	✓	✓	✓	✓
Temperature class 155 (F), utilized acc. to 130 (B), coolant temperature 55 °C, derating approx. 13 %	**N07**	All	All	✓	✓	✓	✓	✓	✓
Temperature class 155 (F), utilized acc. to 130 (B), coolant temperature 60 °C, derating approx. 18 %	**N08**	All	All	✓	✓	✓	✓	✓	✓
Temperature class H	**N10** *New!*	All	All	✓	✓	–	–	–	–
Temperature class 180 (H) at rated output and max. CT 60 °C [6]	**N11**	All	All	✓	✓	✓	✓	✓	✓
Increased air humidity/temperature with 30 to 60 g water per m^3 of air	**N20**	All	All	✓	✓	✓	✓	✓	✓
Increased air humidity/temperature with 60 to 100 g water per m^3 of air	**N21**	All	All	✓	✓	✓	✓	✓	✓

For legends and footnotes, see Page 2/57.

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Supplements to article numbers and special versions

Options
Aluminum series 1LE10, 1PC10

Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category							
		Motor version	**Motor type (alum.)**	**Motor type – Frame size**					
				80	**90**	**100**	**112**	**132**	**160**
		High Efficiency IE2	1LE1001	**1LE1001** ①					
			1PC1001			**1PC1001** ②			
		IE3 Premium Efficiency	1LE1003	**1LE1003** ③					
		IE1 Standard Efficiency	1LE1002			**1LE1002** ④			
			1PC1002			**1PC1002** ⑤			
		NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line** ⑥					
		NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line** ⑦					
		Pole-changing	1LE1011			**1LE1011** ⑧			
			1LE1012			**1LE1012** ⑨			
1LE10 . . -. . . . . -. . . . -Z **1PC10 . . -. . . . . -. . . . -Z**	Order code	Motor version	Motor type	Frame size					
				80	90	100	112	132	160
Windings and insulation (continued)									
Temperature class 155 (F), utilized acc. to 130 (B), with higher coolant temperature and/or site altitude	**Y50** • and specified output, CT .. °C or SA …. m above sea level	All	All	✓	✓	✓	✓	✓	✓
Temperature class 155 (F), utilized according to 155 (F), other requirements	**Y52** • and identification code	All	All	✓	✓	✓	✓	✓	✓
Colors and paint finish									
Special finish in RAL 7030 stone gray		All	All	□	□	□	□	□	□
Unpainted (only cast-iron parts primed)	**S00**	All	All	○	○	○	○	○	○
Unpainted, only primed	**S01**	All	All	✓	✓	✓	✓	✓	✓
Special finish sea air resistant	**S03**	All	All	–	–	✓	✓	✓	✓
Special finish in special RAL colors: For RAL colors, see "Special finish in special RAL colors" (see Catalog Section 1 "Introduction")	**Y51** • and special finish RAL….	All	All	✓	✓	✓	✓	✓	✓
Special finish in other standard RAL colors: RAL 1002, 1013, 1015, 1019, 2003, 2004, 3000, 3007, 5007, 5009, 5010, 5012, 5015, 5017, 5018, 5019, 6011, 6019, 6021, 7000, 7001, 7004, 7011, 7016, 7022, 7031, 7032, 7033, 7035, 9001, 9002, 9005 (see Catalog Section 1 "Introduction")	**Y54** • and special finish RAL….	All	All	✓	✓	✓	✓	✓	✓
Modular technology – Basic versions [7]									
Mounting of holding brake (standard assignment) [8] [28]	**F01**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Mounting of brake for higher switching frequency (operating brake)	**F02**	All	All	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.
Mounting of separately driven fan [29]	**F70**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Mounting of 1XP8012-10 (HTL) rotary pulse encoder [9] [10]	**G01**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Mounting of 1XP8012-20 (TTL) rotary pulse encoder [9] [10]	**G02**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Modular technology – Additional versions									
Brake supply voltage 24 V DC	**F10**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Brake supply voltage 230 V AC, 50/60 Hz	**F11**	All except ②, ⑤ and in combination with order code F90		✓	✓	○	○	○	○
Brake supply voltage 400 V AC, 50/60 Hz	**F12**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Mechanical manual brake release with lever (no locking)	**F50**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Special technology [6]									
Mounting of LL 861 900 220 rotary pulse encoder [9]	**G04**	All except ②, ⑤ and in combination with order code F90		–	–	✓	✓	✓	✓
Mounting of HOG 9 D 1024 I rotary pulse encoder [9]	**G05**	All except ②, ⑤ and in combination with order code F90		–	–	✓	✓	✓	✓
Mounting of HOG 10 D 1024 I rotary pulse encoder [9]	**G06**	All except ②, ⑤ and in combination with order code F90		–	–	✓	✓	✓	✓
Mechanical design and degrees of protection									
Low-noise version for 2-pole motors with clockwise direction of rotation	**F77**	All except ②, ⑤ and in combination with order code F90		–	–	–	–	✓	✓
Low-noise version for 2-pole motors with counter-clockwise direction of rotation	**F78**	All except ②, ⑤ and in combination with order code F90		–	–	–	–	✓	✓
Prepared for mountings, center hole only [10]	**G40**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓

For legends and footnotes, see Page 2/57.

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Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category							
		Motor version	**Motor type (alum.)**	**Motor type – Frame size**					
				80	**90**	**100**	**112**	**132**	**160**
		High Efficiency IE2	1LE1001	**1LE1001** ①					
			1PC1001			**1PC1001** ②			
		IE3 Premium Efficiency	1LE1003	**1LE1003** ③					
		IE1 Standard Efficiency	1LE1002			**1LE1002** ④			
			1PC1002			**1PC1002** ⑤			
		NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line** ⑥					
		NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line** ⑦					
		Pole-changing	1LE1011			**1LE1011** ⑧			
			1LE1012			**1LE1012** ⑨			
1LE10 . . - . . . . . - . . . . -Z		Motor version	Motor type	Frame size					
1PC10 . . - . . . . . - . . . . -Z	Order code			80	90	100	112	132	160
Mechanical design and degrees of protection (continued)									
Prepared for mountings with D12 shaft [15]	**G41**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Prepared for mountings with D16 shaft [15]	**G42**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Protective cover for encoder (supplied loose – only for mountings with order codes G40, G41 and G4 [2]	**G43**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Protective cover [9] [11]	**H00**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Screwed-on (instead of cast) feet	**H01**	All	All	✓	✓	✓	✓	✓	✓
Vibration-proof version	**H02**	All	All	–	–	✓	✓	✓	✓
Condensation drainage holes [14]	**H03**	All	All	✓	✓	✓	✓	✓	✓
Rust-resistant screws (externally)	**H07**	All	All	–	–	✓	✓	✓	✓
Housing with screw mounting	**H10**	Only possible for ③, ⑥ and ⑦		✓	✓	–	–	–	–
IP65 degree of protection [13]	**H20**	All	All	✓	✓	✓	✓	✓	✓
IP56 degree of protection [12]	**H22**	All	All	✓	✓	✓	✓	✓	✓
Drive-end seal for flange-mounting motors, oil-tight to 0.1 bar [11]	**H23**	All	All	✓	✓	✓	✓	✓	✓
Next larger standard flange	**P01**	All	All	–	–	✓	✓	✓	–
Next smaller standard flange	**P02**	All	All	–	–	O. R.	O. R.	O. R.	O. R.
Coolant temperature and site altitude									
Coolant temperature –40 to +40 °C [16] [28]	**D03**	All	All	✓	✓	✓	✓	✓	✓
Coolant temperature –30 to +40 °C [16] [28]	**D04**	All	All	✓	✓	✓	✓	✓	✓
Designs in accordance with standards and specifications									
VIK version	**C02**	Only possible for ① and ③		–	–	✓	✓	✓	✓
IE1 motor without CE marking for export outside EEA (see EU Directive 640/200 [9]	**D22**	Only possible for ④ and ⑤		–	–	O	O	O	O
Electrical according to NEMA MG1-12 [18]	**D30**	All, standard version for ⑥ and ⑦		✓	✓	✓	✓	✓	✓
Design according to UL with "Recognition Mark" [19]	**D31**	All, standard version for ⑥ and ⑦		✓	✓	✓	✓	✓	✓
China Energy Efficiency Label	**D34**	Only possible for ① and ②		–	–	O	O	O	O
Canadian regulations (CSA) [17]	**D40**	All, standard version for ⑥ and ⑦		✓	✓	✓	✓	✓	✓
Train-compatible version	**L82**	All except ② and ⑤		✓	✓	✓	✓	✓	✓
Bearings and lubrication									
Located bearing DE	**L20**	All	All	✓	✓	✓	✓	✓	✓
Located bearing NDE	**L21**	All	All	✓	✓	✓	✓	✓	◘
Bearing design for increased cantilever forces	**L22**	All	All	–	–	✓	✓	✓	✓
Regreasing device [20]	**L23**	All	All	–	–	✓	✓	✓	✓
Special bearing for DE and NDE, bearing size 63	**L25**	All	All	–	–	✓	✓	✓	✓
Measuring nipple for SPM shock pulse measurement for bearing inspection [20]	**Q01**	All	All	–	–	✓	✓	✓	✓
Balance and vibration quantity									
Vibration severity grade A		All	All	◘	◘	◘	◘	◘	◘
Vibration quantity level B	**L00**	All	All	✓	✓	✓	✓	✓	✓
Half-key balancing (standard)		All	All	◘	◘	◘	◘	◘	◘
Balancing without key	**L01**	All	All	✓	✓	✓	✓	✓	✓
Full-key balancing	**L02**	All	All	✓	✓	✓	✓	✓	✓

For legends and footnotes, see Page 2/57.

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Supplements to article numbers and special versions

Options
Aluminum series 1LE10, 1PC10

Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category							
		Motor version	**Motor type (alum.)**	**Motor type – Frame size**					
				80	**90**	**100**	**112**	**132**	**160**
		High Efficiency IE2	1LE1001	**1LE1001** ①					
			1PC1001			**1PC1001** ②			
		IE3 Premium Efficiency	1LE1003	**1LE1003** ③					
		IE1 Standard Efficiency	1LE1002			**1LE1002** ④			
			1PC1002			**1PC1002** ⑤			
		NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line** ⑥					
		NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line** ⑦					
		Pole-changing	1LE1011			**1LE1011** ⑧			
			1LE1012			**1LE1012** ⑨			
1LE10 . . - . . . . . - . . . . -Z **1PC10 . . - . . . . . - . . . . -Z**	Order code	Motor version	Motor type	Frame size					
				80	90	100	112	132	160
Shaft and rotor									
Shaft extension with standard dimensions, without feather keyway	**L04**	All	All	–	–	✓	✓	✓	✓
Second standard shaft extension	**L05**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Standard shaft made of stainless steel	**L06**	All	All	✓	✓	✓	✓	✓	✓
Concentricity of shaft extension in accordance with DIN 42955 Tolerance R	**L07**	All	All	✓	✓	✓	✓	✓	✓
Concentricity of shaft extension, coaxiality and linear movement in accordance with DIN 42955 Tolerance R for flange-mounting motors	**L08**	All	All	✓	✓	✓	✓	✓	✓
Non-standard shaft extension, DE [21]	**Y58** • and identification code	All	All	✓	✓	✓	✓	✓	✓
Non-standard shaft extension, NDE [21]	**Y59** • and identification code	All	All	✓	✓	✓	✓	✓	✓
Heating and ventilation									
Sheet metal fan cover	**F74**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Fan cover for textile industry [22]	**F75**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Metal external fan [23] [29]	**F76**	All except ②, ⑤ and in combination with order code F90		✓	✓	✓	✓	✓	✓
Without external fan and without fan cover	**F90**	All except ②, ⑤, ⑧ and ⑨		✓	✓	✓	✓	✓	✓
Anti-condensation heating for 230 V	**Q02**	All	All	✓	✓	✓	✓	✓	✓
Anti-condensation heating for 115 V	**Q03**	All	All	–	–	✓	✓	✓	✓
Rating plate and extra rating plates									
Extra rating plate for voltage tolerance [24]	**B07**	All except ②, ⑤, ⑧, ⑨ and 8-pole motors		✓	✓	✓	✓	✓	✓
Second rating plate, loose [25]	**M10**	All	All	✓	✓	✓	✓	✓	✓
Rating plate, stainless steel	**M11**	All	All	✓	✓	✓	✓	✓	✓
Extra rating plate or rating plate with deviating rating plate data	**Y80** • and identification code	All	All	✓	✓	✓	✓	✓	✓
Extra rating plate with identification codes	**Y82** • and identification code	All	All	✓	✓	✓	✓	✓	✓
Additional information on rating plate and on package label (max. 20 characters)	**Y84** • and identification code	All	All	✓	✓	✓	✓	✓	✓
Adhesive label, supplied loose (Printed with: Article No., Serial No.: 2 lines of text)	**Y85** • and identification code	All	All	–	–	✓	✓	✓	✓
Packaging, safety notes, documentation and test certificates									
Printed German/English Operating Instructions (Compact) enclosed [27]		All	All	□	□	□	□	□	□
Printed German/English Operating Instructions (Compact) enclosed in each wire-lattice pallet [27]	**B01**	All	All	○	○	○	○	○	○
Acceptance test certificate 3.1 according to EN 10204 [26]	**B02**	All	All	✓	✓	✓	✓	✓	✓
Printed German/English operating instructions enclosed	**B04**	All	All	✓	✓	✓	✓	✓	✓
Document - Electrical data sheet	**B60**	All	All	✓	✓	✓	✓	✓	✓

For legends and footnotes, see Page 2/57.

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Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category							
		Motor version	**Motor type (alum.)**	**Motor type – Frame size** 80	90	100	112	132	160
		High Efficiency IE2	1LE1001	**1LE1001** ①					
			1PC1001			**1PC1001** ②			
		IE3 Premium Efficiency	1LE1003	**1LE1003** ③					
		IE1 Standard Efficiency	1LE1002			**1LE1002** ④			
			1PC1002			**1PC1002** ⑤			
		NEMA Energy Efficient	1LE1021	**1LE1021 Eagle Line** ⑥					
		NEMA Premium Efficient	1LE1023	**1LE1023 Eagle Line** ⑦					
		Pole-changing	1LE1011			**1LE1011** ⑧			
			1LE1012			**1LE1012** ⑨			
1LE10 . . -. . . . . -. . . . -Z **1PC10 . . -. . . . . -. . . . -Z**	Order code	Motor version	Motor type	Frame size 80	90	100	112	132	160
Packaging, safety notes, documentation and test certificates (continued)									
Document - Order dimensional drawing	**B61**	All	All	✓	✓	✓	✓	✓	✓
Type test with heat run for horizontal motors, with acceptance	**B83**	All	All	✓	✓	✓	✓	✓	✓
Wire-lattice pallet packaging	**B99**	All	All	○	○	○	○	○	○
Connected in star for dispatch	**M01**	All	All	–	–	✓	✓	✓	✓
Connected in delta for dispatch	**M02**	All	All	–	–	✓	✓	✓	✓

◘ Standard version
○ Without additional charge
• This order code only determines the price of the version – Additional plain text is required.
✓ With additional charge
O. R. Possible on request
– Not possible

1) With IM B5 flange, only possible in combination with **H08**.

2) Not possible in combination with order code **R15** "One metal cable gland".

3) With **H08**, feet dimensions C and CA differ from EN 50347! Further information is available in DT Configurator (see Appendix, "Tools and Configuring").

4) In combination with motor protection (15th position of the Article No.) or anti-condensation heating option, please inquire before ordering.

5) Not possible in combination with voltage code **22** or **34**.

6) Cannot be used for motors in UL version (order code **D31**). The grease lifetime specified in Catalog Section 1 "Introduction" refers to CT 40 °C. When the coolant temperature rises by 10 K, the grease service lifetime or relubrication interval is halved.

7) A second shaft extension is not possible. Please inquire for mounted brakes.

8) For order codes **F10**, **F11** and **F12**, the brake supply voltage must be specified or ordered.

9) All encoders are supplied with a protective cover as standard. The protective cover is omitted at the factory when a rotary pulse encoder is combined with a separately driven fan, because in this case the rotary pulse encoder is installed under the fan cover. In combination with a separately driven fan (order code **F70**) the 1XP8032-10 rotary pulse encoder is used instead of 1XP8012-10 or 1XP8032-20 is used instead of 1XP8012-20.

10) Motors that are prepared for additional mountings (order codes **G40**, **G41**, **G42**) are supplied without a protective cover as standard. If a protective cover is requested as a cover or mechanical protection for mountings provided by the customer, this can be ordered with order code **G43**.
Not possible in combination with order code **L00** vibration quantity level B.
In combination with a separately driven fan (order code **F70**) the 1XP803210 rotary pulse encoder is used instead of 1XP8012-10 or 1XP8032-20 is used instead of 1XP8012-20.

11) Order code **H00** provides mechanical protection for encoders.

12) Not possible in combination with brake 2LM8 – order code **F01**.

13) Not possible in combination with HOG 9 D 1024I rotary pulse encoder (order code **G05**) and/or brake 2LM8 (order code **F01**).

14) Supplied with the condensation drainage holes sealed at the drive end DE and non-drive end NDE (IP55, IP56, IP65). If the condensation drainage holes are required for motors of the IM B6, IM B7 or IM B8 type of construction (feet on side or top), the motors must be ordered in the respective type of construction and with order code **H03**, so that the condensation drainage holes will be placed in the correct position.

15) Motors that are prepared for additional mountings (order codes **G40**, **G41**, **G42**) are supplied without a protective cover as standard. If a protective cover is requested as a cover or mechanical protection for mountings provided by the customer, this can be ordered with order code **G43**. Not possible in combination with order code **L00** vibration quantity level B.

16) Not possible for type of construction IM V3.

17) CCC certification is required for
– 2-pole motors ≤2.2 kW
– 4-pole motors ≤1.1 kW
– 6-pole motors ≤0.75 kW
– 8-pole motors ≤0.55 kW

18) Possible up to 600 V max. The rated voltage is indicated on the rating plate without voltage range. Order codes **D30** and **D31** do not authorize importing into USA and Mexico. The North America export versions Eagle Line 1LE1021 NEMA Energy Efficient and 1LE1023 NEMA Premium Efficient are available for this purpose.

19) In connection with mountings, the respective technical specifications must be observed, please inquire before ordering.

20) Not possible when brake is mounted.

21) When motors are ordered that have a longer or shorter shaft extension than normal, the required position and length of the feather keyway must be specified in a sketch. It must be ensured that only feather keys in accordance with DIN 6885, Form A are permitted to be used. The feather keyway is positioned centrally on the shaft extension. The length is defined by the manufacturer normatively. Not valid for: Conical shafts, non-standard threaded journals, non-standard shaft tolerances, friction welded journals, extremely "thin" shafts, special geometry dimensions (e.g. square journals), hollow shafts. Valid for non-standard shaft extensions DE or NDE. The feather keys are supplied in every case. For order codes **Y58**, **Y59** and **L05**:
– Dimensions D and DA ≤ Inner diameter of roller bearing (see tables under "Dimensions")
– Dimensions E and EA ≤2 × length E (normal) of the shaft extension.

22) The special requirements of the textile industry regarding the sheet metal cover open up the possibility that a finger may be inserted between the cover and enclosure. The customer must implement appropriate measures to ensure that the installed system is "finger-safe".

23) Converter-fed operation is permitted for 1LE1 motors with metal external fans. The metal external fan is not possible in combination with the low-noise version – order code **F77** or **F78**.

24) Can be ordered for 230 VΔ/400 VY or 400 VΔ/690 VY (voltage code "**22**" or "**34**"). Not possible in combination with order code **D34**.

25) As adhesive label for frame sizes 80 and 90.

26) The delivery time for the factory test certificate may differ from the delivery time for the motor and it will be dispatched by email.

27) The Operating Instructions (compact) are available in PDF format for all official EU languages at http://support.automation.siemens.com/WW/view/en/40761976.

28) Not possible in combination with order code **N05**, **N06**, **N07**, **N08** and **N11**.

29) Order codes **F70** and **F76** cannot be combined.

30) When ordering with order code **R70** and **R71**, order code **R50** is included.

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SIMOTICS SD 1LE1 Standard Motors

Supplements to article numbers and special versions

Options
Cast-iron series 1LE15 Basic Line, 1LE16 Performance Line

Selection and ordering data

Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category														
		Motor version	**Motor type (cast-iron)**	**Motor type – Frame size**												
				71	**80**	**90**	**100**	**112**	**132**	**160**	**180**	**200**	**225**	**250**	**280**	**315**
		High Efficiency IE2	1LE1501	**1LE1501 Basic Line** ①												
			1LE1601				**1LE1601 Performance Line** ②									
		IE3 Premium Efficiency	1LE1503	**1LE1503 Basic Line** ③												
			1LE1603				**1LE1603 Performance Line** ④									
		NEMA Energy Efficient	1LE1521	**1LE1521 Eagle Line Basic** ⑤												
			1LE1621				**1LE1621 Eagle Line Performance** ⑥									
		NEMA Premium Efficient	1LE1523	**1LE1523 Eagle Line Basic** ⑦												
			1LE1623				**1LE1623 Eagle Line Performance** ⑧									
1LE1. . . -. . . . . -. . . . -Z	Order code	Motor version	Motor type	Frame size												
				71	80	90	100	112	132	160	180	200	225	250	280	315
Motor protection (bearing protection)																
Prepared for mounting a SIPLUS CMS 1000 vibration sensor	**Q05**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
2 × 3 temperature detectors for alarm and tripping	**Q32**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Installation of 1 Pt100 resistance thermometer in stator winding, two-wire circuit	**Q62**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Installation of 3 Pt100 resistance thermometers in stator winding, three-wire circuit [30]	**Q63**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Installation of 6 Pt100 resistance thermometers in stator winding, three-wire circuit [30]	**Q64**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Installation of 2 Pt100 screw-in resistance thermometers in basic circuit for rolling-contact bearings [2]	**Q72**	All	All	–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓
Installation of 2 Pt100 screw-in resistance thermometers in 3-wire circuit for rolling-contact bearings	**Q78**	All	All	–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓
Installation of 2 Pt100 double screw-in resistance thermometers in 3-wire circuit for rolling-contact bearings	**Q79**	All	All	–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓
Motor connection and terminal box																
External grounding	**H04**	All	All	✓	✓	✓	✓	✓	✓	✓	◘	◘	◘	◘	◘	◘
Terminal box on NDE [27]	**H08**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Second external grounding	**H70**	All	All	–	–	–	O. R.	O.R.	O.R.	O.R.	✓	✓	✓	✓	✓	✓
Rotation of the terminal box through 90°, entry from DE	**R10**	All	All	○	○	○	○	○	○	○	✓	✓	✓	✓	✓	✓
Rotation of the terminal box through 90°, entry from NDE	**R11**	All	All	○	○	○	○	○	○	○	✓	✓	✓	✓	✓	✓
Rotation of the terminal box through 180°	**R12**	All	All	○	○	○	○	○	○	○	✓	✓	✓	✓	✓	✓
One EMC cable gland	**R14**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
One metal cable gland	**R15**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
EMC cable gland, maximum configuration	**R16**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Stud terminal for cable connection, accessories pack (3 items)	**R17**	All	All	–	–	–	–	–	–	–	–	–	–	✓	✓	✓
Cable gland, maximum configuration	**R18**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Saddle terminal for connection without cable lug, accessories pack	**R19**	All	All	–	–	–	–	–	–	–	–	–	–	✓	✓	✓
3 cables protruding, 0.5 m long	**R20** *New!*	All	All	✓	✓	✓	✓	✓	✓	✓	–	–	–	–	–	–
3 cables protruding, 1.5 m long	**R21**	All	All	✓	✓	✓	✓	✓	✓	✓	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.
6 cables protruding, 0.5 m long	**R22** *New!*	All	All	✓	✓	✓	✓	✓	✓	✓	–	–	–	–	–	–
6 cables protruding, 1.5 m long	**R23**	All	All	✓	✓	✓	✓	✓	✓	✓	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.
6 cables protruding, 3 m long	**R24**	All	All	✓	✓	✓	✓	✓	✓	✓	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.
Reduction piece for M cable gland in accordance with British Standard, mounted on both cable entry openings	**R30** *New!*	All	All	–	–	–	✓	✓	✓	✓	–	–	–	–	–	–
Larger terminal box	**R50**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Terminal box without cable entry opening	**R51**	All	All	–	–	–	○	○	○	○	○	○	○	○	○	○
Drilled removable entry plate	**R52**	All	All	–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓

For legends, see Page 2/63; for footnotes, see Page 2/64.

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Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category															
		Motor version	**Motor type (cast-iron)**	**Motor type – Frame size**													
				71	**80**	**90**	**100**	**112**	**132**	**160**	**180**	**200**	**225**	**250**	**280**	**315**
		High Efficiency IE2	1LE1501	**1LE1501 Basic Line ①**													
			1LE1601				**1LE1601 Performance Line ②**										
		IE3 Premium Efficiency	1LE1503	**1LE1503 Basic Line ③**													
			1LE1603				**1LE1603 Performance Line ④**										
		NEMA Energy Efficient	1LE1521	**1LE1521 Eagle Line Basic ⑤**													
			1LE1621				**1LE1621 Eagle Line Performance ⑥**										
		NEMA Premium Efficient	1LE1523	**1LE1523 Eagle Line Basic ⑦**													
			1LE1623				**1LE1623 Eagle Line Performance ⑧**										
1LE1. . . -. . . . . -. . . . -Z	Order code	Motor version	Motor type	Frame size													
				71	80	90	100	112	132	160	180	200	225	250	280	315	
Motor connection and terminal boxes (continued)																	
Undrilled removable entry plate	**R53**	All	All	–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓	
Cast-iron auxiliary terminal box (small) 27)	**R62**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Silicon-free version	**R74**	All	All	–	–	–	□	□	□	□	□	□	□	✓	✓	✓	
Non-standard threaded through hole (NPT or G thread)	**Y61** • and identification code	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Windings and insulation																	
Temperature class 155 (F), utilized acc. to 155 (F), with service factor	**N01**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 155 (F), utilized acc. to 155 (F), with increased output	**N02**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 155 (F), utilized acc. to 155 (F), with increased coolant temperature	**N03**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 155 (F), utilized acc. to 130 (B), coolant temperature 45 °C, derating approx. 4 %	**N05**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 155 (F), utilized acc. to 130 (B), coolant temperature 50 °C, derating approx. 8 %	**N06**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 155 (F), utilized acc. to 130 (B), coolant temperature 55 °C, derating approx. 13 %	**N07**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 155 (F), utilized acc. to 130 (B), coolant temperature 60 °C, derating approx. 18 %	**N08**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 180 (H)	**N10** *New!*	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 180 (H) at rated output and max. CT 60 °C 4) 5)	**N11**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Increased air humidity/temperature with 30 to 60 g water per m^3 of air	**N20**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Increased air humidity/temperature with 60 to 100 g water per m^3 of air	**N21**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 155 (F), utilized acc. to 130 (B), with higher coolant temperature and/or site altitude	**Y50** • and specified output, CT .. °C or SA …. m above sea level	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 155 (F), utilized acc. to 155 (F), other requirements 5)	**Y52** • and identification code	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Temperature class 180 (H), utilized according to 155 (F)	**Y75** • and specified output, CT .. °C or SA …. m above sea level	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Colors and paint finish																	
Standard finish in RAL 7030 stone gray		Only possible for Basic Line ①, ③, ⑤, and ⑦		□	□	□	□	□	□	□	□	□	□	□	□	□	
Unpainted (only cast-iron parts primed)	**S00**	All	All	○	○	○	○	○	○	○	○	○	○	○	○	○	
Unpainted, only primed	**S01**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Special finish sea air resistant	**S03**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Special paint for use offshore	**S04**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Internal coating	**S05**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Special finish in RAL 7030 stone gray	**S10**	All, standard version for Performance Line ②, ④, ⑥ and ⑧		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Special finish in special RAL colors: For RAL colors, see "Special finish in special RAL colors" (see Catalog Section 1 "Introduction")	**Y51** • and special finish RAL….	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	

For legends, see Page 2/63; for footnotes, see Page 2/64.

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Supplements to article numbers and special versions

Options
Cast-iron series 1LE15 Basic Line, 1LE16 Performance Line

Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category **Motor version**	**Motor type (cast-iron)**	**Motor type – Frame size** 71 80 90 100 112 132 160 180 200 225 250 280 315
		High Efficiency IE2	1LE1501	**1LE1501 Basic Line** ①
			1LE1601	**1LE1601 Performance Line** ②
		IE3 Premium Efficiency	1LE1503	**1LE1503 Basic Line** ③
			1LE1603	**1LE1603 Performance Line** ④
		NEMA Energy Efficient	1LE1521	**1LE1521 Eagle Line Basic** ⑤
			1LE1621	**1LE1621 Eagle Line Performance** ⑥
		NEMA Premium Efficient	1LE1523	**1LE1523 Eagle Line Basic** ⑦
			1LE1623	**1LE1623 Eagle Line Performance** ⑧

1LE1. . . -. . . . . -. . . . -Z	Order code	Motor version	Motor type	Frame size 71 80 90 100 112 132 160 180 200 225 250 280 315
Colors and paint finish (continued)				
Standard finish in other standard RAL colors: RAL 1002, 1013, 1015, 1019, 2003, 2004, 3000, 3007, 5007, 5009, 5010, 5012, 5015, 5017, 5018, 5019, 6011, 6019, 6021, 7000, 7001, 7004, 7011, 7016, 7022, 7031, 7032, 7033, 7035, 9001, 9002, 9005 (see Catalog Section 1 "Introduction")	**Y53** • and standard finish RAL....	Only possible for Basic Line ①, ③, ⑤, and ⑦		✓ *) ✓ *) ✓ *) ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Special finish in other standard RAL colors: RAL 1002, 1013, 1015, 1019, 2003, 2004, 3000, 3007, 5007, 5009, 5010, 5012, 5015, 5017, 5018, 5019, 6011, 6019, 6021, 7000, 7001, 7004, 7011, 7016, 7022, 7031, 7032, 7033, 7035, 9001, 9002, 9005 (see Catalog Section 1 "Introduction")	**Y54** • and special finish RAL....	All	All	✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Modular technology – Basic versions [6]				
Mounting of holding brake (standard assignment) [7] [31] [32]	**F01**	All	All	– – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Backstop, counter-clockwise motion blocked, clockwise direction of rotation	**F40**	All	All	– – – – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Backstop, clockwise motion blocked, counter-clockwise direction of rotation	**F41**	All	All	– – – – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Mounting of separately driven fan [28]	**F70**	All	All	– – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Mounting of 1XP8012-10 (HTL) rotary pulse encoder [8] [9]	**G01**	All	All	✓ *) ✓ *) ✓ *) ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Mounting of 1XP8012-20 (TTL) rotary pulse encoder [8] [9]	**G02**	All	All	✓ *) ✓ *) ✓ *) ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Modular technology – Additional versions				
Brake supply voltage 24 V DC	**F10**	All	All	– – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Brake supply voltage 230 V AC, 50/60 Hz	**F11**	All	All	– – – ○ ○ ○ ○ ○ ○ ○ ○ ○ ○
Brake supply voltage 400 V AC, 50/60 Hz [32]	**F12**	All	All	– – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Mechanical manual brake release with lever (no locking)	**F50**	All	All	– – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Special technology [5]				
Mounting of LL 861 900 220 rotary pulse encoder [10]	**G04**	All	All	– – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Mounting of HOG 9 D 1024 I rotary pulse encoder [10]	**G05**	All	All	– – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Mounting of HOG 10 D 1024 I rotary pulse encoder [10]	**G06**	All	All	– – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Mounting of POG10D rotary pulse encoder (only in combination with separately driven fan or brake) 1)	**G07**	All	All	– – – – – – – ✓ ✓ ✓ ✓ ✓ ✓
Mounting of POG9 rotary pulse encoder (only in combination with separately driven fan or brake) [11]	**G08**	All	All	– – – – – – – ✓ ✓ ✓ ✓ ✓ ✓
Mounting of rotary pulse encoder HOG 10 DN 1024 I, terminal box moisture protection	**G15** *New!*	All except in combination with order code F90		– – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Mounting of rotary pulse encoder HOG 10 DN 1024 I, terminal box dust protection	**G16** *New!*	All except in combination with order code F90		– – – ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
Mounting of a special type of rotary pulse encoder	**Y70** • and identification code	All	All	– – – O. R. O. R. O. R. O. R. O. R. O. R. O. R. O. R. O. R. O. R.
Mounting of rotary pulse encoder HOG 10 DN 1024 I + FSL, (speed ... rpm), terminal box moisture protection	**Y74** • and requ. speed .... rpm	All except in combination with order code F90		– – – – – – – ✓ ✓ ✓ ✓ ✓ ✓

For legends, see Page 2/63; for footnotes, see Page 2/64.

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Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category														
		Motor version	**Motor type (cast-iron)**	**Motor type – Frame size**												
				71	**80**	**90**	**100**	**112**	**132**	**160**	**180**	**200**	**225**	**250**	**280**	**315**
		High Efficiency IE2	1LE1501	**1LE1501 Basic Line ①**												
			1LE1601				**1LE1601 Performance Line ②**									
		IE3 Premium Efficiency	1LE1503	**1LE1503 Basic Line ③**												
			1LE1603				**1LE1603 Performance Line ④**									
		NEMA Energy Efficient	1LE1521	**1LE1521 Eagle Line Basic ⑤**												
			1LE1621				**1LE1621 Eagle Line Performance ⑥**									
		NEMA Premium Efficient	1LE1523	**1LE1523 Eagle Line Basic ⑦**												
			1LE1623				**1LE1623 Eagle Line Performance ⑧**									
1LE1. . . - . . . . . - . . . . -Z	Order code	Motor version	Motor type	Frame size												
				71	80	90	100	112	132	160	180	200	225	250	280	315
Special technology [5] (continued)																
Mounting of rotary pulse encoder HOG 10 DN 1024 I + FSL, (speed ... rpm), terminal box dust protection	**Y76** • and requ. speed .... rpm	All except in combination with order code F90		–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓
Mounting of rotary pulse encoder HOG 10 DN 1024 I + ESL 93, (speed .... rpm), terminal box dust protection	**Y79** • and requ. speed (max 3) .... rpm	All except in combination with order code F90		–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓
Mechanical design and degrees of protection																
Low-noise version for 2-pole motors with clockwise direction of rotation	**F77**	All	All	–	–	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓
Low-noise version for 2-pole motors with counter-clockwise direction of rotation	**F78**	All	All	–	–	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓
Prepared for mountings, center hole only	**G40**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	□	□	□	□	□	□
Prepared for mountings with D12 shaft	**G41**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Prepared for mountings with D16 shaft	**G42**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Protective cover for encoder (supplied loose – only for mountings with order codes G40, G41 and G4[2])	**G43**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Protective cover [8] [10] [12]	**H00**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Screwed-on (instead of cast) feet	**H01**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Vibration resistant version (continuous vibration resistance Class 3M4 acc. to IEC721-3-3:199[4])	**H02**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	□	□
Condensation drainage holes	**H03**	All	All	✓ *)	✓ *)	✓ *)	□	□	□	□	□	□	□	□	□	□
Rust-resistant screws (externally)	**H07**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
IP65 degree of protection [14]	**H20**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
IP54 degree of protection	**H21**	All	All	–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓
IP56 degree of protection [15]	**H22**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Drive-end seal for flange-mounting motors, oil-tight to 0.1 bar [13] [29]	**H23**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Grounding brush for converter-fed operation	**L52**	All	All	–	–	–	–	–	–	–	–	–	–	–	✓	✓
Next larger standard flange	**P01**	All	All	–	–	–	✓	✓	✓	–	–	–	–	–	–	–
Coolant temperature and site altitude																
Coolant temperature –50 to +40 °C	**D02**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Coolant temperature –40 to +40 °C [16]	**D03**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Coolant temperature –30 to +40 °C [17]	**D04**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Designs in accordance with standards and specifications																
VIK version	**C02**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Electrical according to NEMA MG1-12 [18]	**D30**	All, standard version for Eagle Line ⑤, ⑥, ⑦ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Design according to UL with "Recognition Mark" [18]	**D31**	All, standard version for Eagle Line ⑤, ⑥, ⑦ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
China Energy Efficiency Label	**D34**	All	All	–	–	–	○	○	○	○	○	○	○	○	○	○
Canadian regulations (CSA) [7]	**D40**	All, standard version for Eagle Line ⑤, ⑥, ⑦ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Bearings and lubrication																
Regreasing device with M10X1 grease nipple according to DIN 71412-A	**L19** *New!*	All, for Performance Line ②, ④, ⑥ and ⑧ without add. charge		–	–	–	–	–	–	–	✓	✓	✓	✓	○	○
Located bearing DE	**L20**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Located bearing NDE	**L21**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	□	□	□	□	□	□	□
Bearing design for increased cantilever forces	**L22**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

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SIMOTICS SD 1LE1 Standard Motors

Supplements to article numbers and special versions

Options
Cast-iron series 1LE15 Basic Line, 1LE16 Performance Line

Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category															
		Motor version	**Motor type (cast-iron)**	**Motor type – Frame size**													
				71	**80**	**90**	**100**	**112**	**132**	**160**	**180**	**200**	**225**	**250**	**280**	**315**
		High Efficiency IE2	1LE1501	**1LE1501 Basic Line** ①													
			1LE1601				**1LE1601 Performance Line** ②										
		IE3 Premium Efficiency	1LE1503	**1LE1503 Basic Line** ③													
			1LE1603				**1LE1603 Performance Line** ④										
		NEMA Energy Efficient	1LE1521	**1LE1521 Eagle Line Basic** ⑤													
			1LE1621				**1LE1621 Eagle Line Performance** ⑥										
		NEMA Premium Efficient	1LE1523	**1LE1523 Eagle Line Basic** ⑦													
			1LE1623				**1LE1623 Eagle Line Performance** ⑧										
1LE1. . . -. . . . . -. . . . -Z	Order code	Motor version	Motor type	Frame size													
				71	80	90	100	112	132	160	180	200	225	250	280	315	
Bearings and lubrication (continued)																	
Regreasing device [1]	**L23**	All, standard version for Performance Line FS 160 and above ②, ④, ⑥ and ⑧		–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	□	□	
Hot bearing grease	**L24**	All	All	–	–	–	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	
Special bearing for DE and NDE, bearing size 63 [19]	**L25**	All, standard version for Performance Line ②, ④, ⑥ and ⑧		✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	□	□	
Reinforced bearings at both DE and NDE, DE bearing for increased cantilever forces	**L28**	All	All	–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓	
Increased max. speed	**L37** *New!*	All	All	–	–	–	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	
Bearing insulation DE	**L50**	All	All	–	–	–	–	–	–	–	–	–	✓	✓	✓	✓	
Bearing insulation NDE	**L51**	All	All	–	–	–	–	–	–	–	–	–	✓	✓	✓	✓	
Measuring nipple for SPM shock pulse measurement for bearing inspection [1]	**Q01**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Balance and vibration quantity																	
Vibration severity grade A		All	All	□	□	□	□	□	□	□	□	□	□	□	□	□	
Vibration quantity level B [20]	**L00**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Half-key balancing (standard)		All	All	□	□	□	□	□	□	□	□	□	□	□	□	□	
Balancing without feather key, feather key is supplied	**L01**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Full-key balancing	**L02**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Shaft and rotor																	
Shaft extension with standard dimensions, without feather keyway	**L04**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Second standard shaft extension	**L05**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Standard shaft made of stainless steel	**L06**	All	All	✓ *)	✓ *)	✓ *)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Concentricity of shaft extension in accordance with DIN 42955 Tolerance R	**L07**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Concentricity of shaft extension, coaxiality and linear movement in accordance with DIN 42955 Tolerance R for flange-mounting motors	**L08**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Non-standard shaft extension, DE [21]	**Y58** • and identification code	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Non-standard shaft extension, NDE [21]	**Y59** • and identification code	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Special shaft steel	**Y60** • and identification code	All	All	–	–	–	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	O. R.	
Heating and ventilation																	
Sheet metal fan cover	**F74**	All, standard version for Performance Line ②, ④, ⑥ and ⑧		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Metal external fan [22] [28]	**F76**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Without external fan and without fan cover	**F90** *New!*	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Anti-condensation heating for 230 V	**Q02**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Anti-condensation heating for 115 V	**Q03**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Separately driven fan with non-standard voltage and/or frequency	**Y81** • and identification code	All	All	–	–	–	–	–	–	–	–	–	✓	✓	✓	✓	
Rating plate and extra rating plates																	
Extra rating plate for voltage tolerance [23]	**B07**	All, with the exception of 8-pole motors		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Second rating plate, loose	**M10**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Rating plate, stainless steel	**M11**	All, for Performance Line ②, ④, ⑥ and ⑧ without add. charge		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Extra rating plate or rating plate with deviating rating plate data	**Y80** • and identification code	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	

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Special versions	Additional identification code **-Z** with order code and plain text if required	Motor category															
		Motor version	**Motor type (cast-iron)**	**Motor type – Frame size**													
				71	**80**	**90**	**100**	**112**	**132**	**160**	**180**	**200**	**225**	**250**	**280**	**315**	
		High Efficiency IE2	1LE1501	**1LE1501 Basic Line ①**													
			1LE1601				**1LE1601 Performance Line ②**										
		IE3 Premium Efficiency	1LE1503	**1LE1503 Basic Line ③**													
			1LE1603				**1LE1603 Performance Line ④**										
		NEMA Energy Efficient	1LE1521	**1LE1521 Eagle Line Basic ⑤**													
			1LE1621				**1LE1621 Eagle Line Performance ⑥**										
		NEMA Premium Efficient	1LE1523	**1LE1523 Eagle Line Basic ⑦**													
			1LE1623				**1LE1623 Eagle Line Performance ⑧**										
1LE1. . . -. . . . . -. . . . -Z	Order code	Motor version	Motor type	Frame size													
				71	80	90	100	112	132	160	180	200	225	250	280	315	
Rating plate and extra rating plates (continued)																	
Extra rating plate with identification codes	**Y82** • and identification code	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Additional information on rating plate and on package label (max. 20 characters)	**Y84** • and identification code	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Adhesive label, supplied loose (Printed with: Article No., Serial No.: 2 lines of text)	**Y85** • and identification code	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Extension of the liability for defects																	
Extension of the liability for defects by 12 months to a total of 24 months (2 years) from delivery [24]	**Q80**	All, for Performance Line ②, ④, ⑥ and ⑧ 36 months standard		–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓	
Extension of the liability for defects by 24 months to a total of 36 months (3 years) from delivery [24]	**Q82**	All, for Performance Line ②, ④, ⑥ and ⑧ 36 months standard		–	–	–	–	–	–	–	✓	✓	✓	✓	✓	✓	
Packaging, safety notes, documentation and test certificates																	
Acceptance test certificate 3.1 according to EN 10204 [25]	**B02**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Printed German/English Operating Instructions enclosed [26]	**B04**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Document - Electrical data sheet	**B60**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Document - Order dimensional drawing	**B61**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Standard test (routine test) with acceptance	**B65**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Type test with heat run for horizontal motors, without acceptance	**B82** *New!*	All	All	–	–	–	✓ *)	✓ *)	✓ *)	✓ *)	✓ *)	✓ *)	✓ *)	✓ *)	✓ *)	✓ *)	✓ *)
Type test with heat run for horizontal motors, with acceptance	**B83**	All	All	–	–	–	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Connected in star for dispatch	**M01**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Connected in delta for dispatch	**M02**	All	All	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	

- ◘ Standard version
- ○ Without additional charge
- • This order code only determines the price of the version – Additional plain text is required.
- ✓ With additional charge
- O. R. Possible on request
- – Not possible

For footnotes, see Page 2/64.

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*) Start of delivery ex stock planned for end of 2014.

1) Up to frame size 160 not possible when brake is mounted.

2) Evaluation with appropriate tripping unit (see Catalog IC 10) is recommended.

3) Parallel Whitworth threaded pipe DIN ISO 228 (DIN 259) BSPP (British Standard Pipe Parallel) Threaded pipe for connections not sealed in the thread (cylindrical), external = G.

4) Cannot be used for motors in UL version (order code **D31**). The grease lifetime specified in Catalog Section 1 "Introduction" refers to CT 40 °C. When the coolant temperature rises by 10 K, the grease service lifetime or relubrication interval is halved.

5) Not possible for 1LE15 and 1LE16 motors with increased output.

6) A second shaft extension is not possible. Please inquire for mounted brakes.

7) For order codes **F10**, **F11** and **F12**, the brake supply voltage must be specified or ordered.

8) The 1XP8 rotary pulse encoders are fitted with a protective cover as standard. The protective cover is omitted at the factory when a rotary pulse encoder is combined with a separately driven fan, because in this case the rotary pulse encoder is installed under the fan cover.

9) In combination with a separately driven fan (order code **F70**) the 1XP803210 rotary pulse encoder is used instead of 1XP8012-10 or 1XP8032-20 is used instead of 1XP8012-20.

10) LL and HOG rotary pulse encoders up to frame size 160 are fitted with a protective cover as standard. The protective cover is omitted at the factory when a rotary pulse encoder is combined with a separately driven fan, because in this case the rotary pulse encoder is installed under the fan cover.

11) Option (encoder mounting) is only possible for motors with a mounted separately driven fan or for naturally cooled motors (without a separately driven fan). This option can be used in combination with brakes of type KFB. This option cannot be used in combination with brakes of type 2LM8.

12) Order code **H00** provides mechanical protection for encoders.

13) Not possible for type of construction IM V3.

14) Not possible in combination with HOG 9 D 1024I rotary pulse encoder (order code **G05**) and/or brake 2LM8 (order code **F01**).

15) Not possible in combination with brake 2LM8 – order code **F01**.

16) In connection with mountings, the respective technical specifications must be observed, please inquire before ordering.

17) The rated voltage is indicated on the rating plate without voltage range. Order code **D40** does not authorize importing into Canada. The North America export versions Eagle Line 1LE1521/1LE1621 NEMA Energy Efficient and 1LE1523/1LE1623 NEMA Premium Efficient are available for this purpose.

18) Possible up to 600 V max. The rated voltage is indicated on the rating plate without voltage range. Order codes **D30** and **D31** do not authorize importing into USA and Mexico. The North America export versions Eagle Line 1LE1521/1LE1621 NEMA Energy Efficient and 1LE1523/1LE1623 NEMA Premium Efficient are available for this purpose.

19) For Performance Line motors (all frame sizes) and Basic Line motors (from frame size 280) in the standard version.

20) On request for 2-pole motors (concerns frame sizes 225 to 315).

21) When motors are ordered that have a longer or shorter shaft extension than normal, the required position and length of the feather keyway must be specified in a sketch. It must be ensured that only feather keys in accordance with DIN 6885, Form A are permitted to be used. The feather keyway is positioned centrally on the shaft extension. The length is defined by the manufacturer normatively. Not valid for: Conical shafts, non-standard threaded journals, non-standard shaft tolerances, friction welded journals, extremely "thin" shafts, special geometry dimensions (e.g. square journals), hollow shafts. Valid for non-standard shaft extensions DE or NDE. The feather keys are supplied in every case. For order codes **Y58**, **Y59** and **L05**:
– Dimensions D and DA ≤ Inner diameter of roller bearing (see tables under "Dimensions")
– Dimensions E and EA ≤2 × length E (normal) of the shaft extension.

22) Converter-fed operation is permitted for 1LE1 motors with metal external fans.

23) Can be ordered for 230 VΔ/400 VY or 400 VΔ/690 VY (voltage code "**22**" or "**34**"). Not possible for 8-pole motors and in combination with order code **D34**.

24) Wear parts (bearings) are excluded from the warranty extension.

25) The delivery time for the factory test certificate may differ from the delivery time for the motor.

26) The Operating Instructions (compact) are available in PDF format for all official EU languages at http://support.automation.siemens.com/WW/view/en/10803948/133300.

27) With **H08**, feet dimensions C and CA differ from EN 50347! Further information is available in DT Configurator (see Appendix, "Tools and Configuring").

28) Order codes **F70** and **F76** cannot be combined.

29) Not possible in combination with order codes **Q72** and **Q78**.

30) For frame sizes 100 to 132 only possible in combination with order code **R50**.

31) Not possible in combination with order codes **N05**, **N06**, **N07**, **N08** and **N11**.

32) For frame size 315, when combining order codes **F01** and **F12**, the rectifier for the brake will be supplied separately as a single part.

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Overview

Slide rails with fixing bolts and tensioning screws according to DIN 42923

Slide rails are used to tension the belt of a machine easily and conveniently when a belt tightener is not available. They are fixed to the base using stone bolts or foundation blocks.

The assignment of slide rails to motor size can be found in DIN 42923. For motors of frame sizes 355 to 450, there are no standardized slide rails (please inquire).

Available from:

Lütgert & Co. GmbH
Postfach 42 51
33276 Gütersloh, Germany
Tel. +49 (5241) 7407-0
Fax +49 (5241) 7407-90

www.luetgert-antriebe.de
E-mail: info@luetgert-antriebe.de

Foundation block acc. to DIN 799

The foundation blocks are inserted into the stone foundation and embedded in concrete. They are used for fixing machines of medium size, slide rails, pedestal bearings, baseframes, etc. After the fixing bolts have been unscrewed, the machine can be dragged without it having to be lifted.

When the machine is initially installed, the foundation block that is bolted to the machine (without washers) and fitted with taper pins is not embedded with concrete until the machine has been fully aligned. In this case, the machine is positioned 2 to 3 mm lower. The difference in shaft height is compensated by inserting shims on final installation. The taper pins safeguard the exact position of the machine when it is repeatedly removed and replaced without the need for realignment.

Available from:

Lütgert & Co. GmbH
Postfach 42 51
33276 Gütersloh, Germany
Tel. +49 (5241) 7407-0
Fax +49 (5241) 7407-90

www.luetgert-antriebe.de
E-mail: info@luetgert-antriebe.de

Taper pins to DIN 258 with threaded ends and constant taper lengths

Taper pins are used for components that are repeatedly removed. The drilled hole is ground conical using a conical reamer until the pin can be pushed in by hand until the cone shoulder lies approx. 3 to 4 mm above the rim of the hole.

It can then be driven in using a hammer until it is correctly seated. The pin is removed from the drilled hole by screwing on the nut and tightening it.

Standardized taper pins are available from general engineering suppliers.

Available from:

Otto Roth GmbH & Co. KG
Rutesheimer Straße 22
70499 Stuttgart, Germany
Tel. +49 (711) 1388-0
Fax. +49 (711) 1388-233

www.ottoroth.de
E-mail: info@ottoroth.de

Couplings

The motor from Siemens is connected to the machine or gear unit through a coupling. Siemens is an important coupling manufacturer with a wide range of products.

For standard applications, Siemens recommends that elastic couplings of types N-EUPEX and RUPEX or torsionally rigid couplings of types ARPEX and ZAPEX are used. For special applications, FLUDEX and ELPEX-S couplings are recommended.

Available from:

Siemens contact partner - ordering from catalog
Siemens MD 10.1 "FLENDER Standard Couplings"

or

SIEMENS AG
Kupplungswerk Mussum
Industriepark Bocholt
Schlavenhorst 100
46395 Bocholt, Germany
Tel. +49 (2871) 922185
Fax +49 (2871) 922579

www.siemens.com
E-mail: flendercouplings@siemens.com

More information

Spare motors and repair parts

- Supply commitment for spare motors and repair parts following delivery of the motor:
  - For up to 5 years, in the event of total motor failure, Siemens will supply a comparable motor with regard to the mounting dimensions and functions (the type series may vary).
  - Spare parts will be available for up to five years.
  - Within the time period of up to five years, Siemens will provide information about spare parts and will supply documents when required.
  - Replacement motors delivered after the active production of the machine series are also identified with "Spare motor" on the rating plate. Spare parts are offered only on request for these motors.
- When repair parts are ordered, the following details must be provided:
  - Designation and part number
  - Article No. and factory number of the motor.

  Example for ordering a fan cover 1LE1002, frame size 112 M, 4-pole:

  Fan cover No. 7.40, 1LE1002-1BB23-4AA4-Z, factory No. E1001/5236197_01_001
- For bearing types, see Catalog section 1 "Introduction".
- Repair parts for 1MJ6, 1MJ7, 1MJ8, 1MJ1, 1ME8, 1ML8 are available on request.
- For standard components, a supply commitment does not apply.
- Support – Hotline
  In Germany
  Tel.: +49 (180) 5050448

You will find telephone numbers for other countries on our Internet site:
www.siemens.com/automation/service&support

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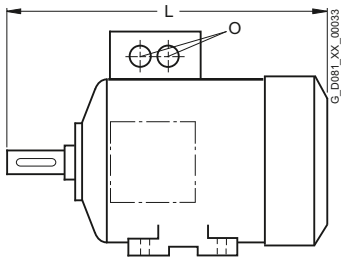
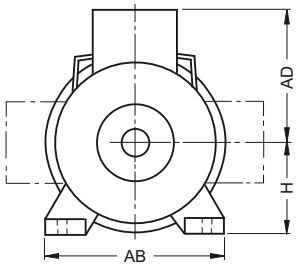
Dimensions

Overall dimensions

Overview





Frame size	Type	Dimension L	AD	H	AB	O
71 M	Cast-iron series, self-ventilated					
	1LE1501, 1LE1521, 1LE1503-, 1LE1523-0CA2, 0CB2, 0CC2	240	148	71	132	2 × M25 × 1.5
	1LE1503-, 1LE1523-0CA3, 0CB3, 0CC3	280	148	71	132	2 × M25 × 1.5
80 M	Aluminum series, self-ventilated					
	1LE1001	292	121	80	150	1 × M25 × 1.5
	Aluminum series, forced-air cooled or naturally cooled					
	1LE1001	253	121	80	150	1 × M25 × 1.5
	Cast-iron series, self-ventilated					
	1LE1501, 1LE1521, 1LE1503-, 1LE1523-0CA2, 0CB2, 0CC2	292	158	80	150	2 × M25 × 1.5
	1LE1503-, 1LE1523-0CA3, 0CB3, 0CC3	327	158	80	150	2 × M25 × 1.5
90 S/ 90 L	Aluminum series, self-ventilated					
	1LE1001	347	126	90	165	1 × M25 × 1.5
	Aluminum series, forced-air cooled or naturally cooled					
	1LE1001	295	126	90	165	1 × M25 × 1.5
	Cast-iron series, self-ventilated					
	1LE1501, 1LE1521, 1LE1503-, 1LE1523-0CA2; 0CB2, 0CC2	347	163	90	165	2 × M25 × 1.5
	1LE1503-, 1LE1523-0CA3, 0CB3, 0CC3	387	163	90	165	2 × M25 × 1.5
100 L	Aluminum series, self-ventilated					
	1LE1001, 1LE1002, 1LE1011, 1LE1012, 1LE1021	396 [1]	166	100	196	2 × M32 × 1.5
	Aluminum series, self-ventilated with increased output					
	1LE1001, 1LE1002	431 [1]	166	100	196	2 × M32 × 1.5
	Aluminum series, forced-air cooled or naturally cooled					
	1LE1001, 1PC1001, 1LE1002, 1PC1002, 1LE1021	322	166	100	196	2 × M32 × 1.5
	Aluminum series, self-ventilated					
	1LE1003, 1LE1023	431	166	100	196	2 × M32 × 1.5
	Aluminum series, forced-air cooled					
	1LE1023	357	166	100	196	2 × M32 × 1.5
	Cast-iron series, self-ventilated					
	1LE1501, 1LE1503, 1LE1521, 1LE1601, 1LE1603, 1LE1621	389	193	100	196	2 × M32 × 1.5
	1LE1523, 1LE1623	425	193	100	196	2 × M32 × 1.5

Frame size	Type	Dimension L	AD	H	AB	O
112 M	Aluminum series, self-ventilated					
	1LE1001, 1LE1002, 1LE1011, 1LE1012, 1LE1021	389 [1]	177	112	226	2 × M32 × 1.5
	Aluminum series, self-ventilated with increased output					
	1LE1001, 1LE1002	414 [1]	177	112	226	2 × M32 × 1.5
	Aluminum series, forced-air cooled or naturally cooled					
	1LE1001, 1PC1001, 1LE1002, 1PC1002, 1LE1021	311	177	112	226	2 × M32 × 1.5
	Aluminum series, self-ventilated					
	1LE1003, 1LE1023	414	177	112	226	2 × M32 × 1.5
	Aluminum series, forced-air cooled					
	1LE1023	336	177	112	226	2 × M32 × 1.5
	Cast-iron series, self-ventilated					
	1LE1501, 1LE1503, 1LE1521, 1LE1601, 1LE1603, 1LE1621	382	195	112	226	2 × M32 × 1.5
	1LE1523, 1LE1623	409	195	112	226	2 × M32 × 1.5
132 S/M	Aluminum series, self-ventilated					
	1LE1001, 1LE1002, 1LE1011, 1LE1012, 1LE1021	465 [1]	202	132	256	2 × M32 × 1.5
	Aluminum series, self-ventilated with increased output					
	1LE1001, 1LE1002	515 [1]	202	132	256	2 × M32 × 1.5
	Aluminum series, forced-air cooled or naturally cooled					
	1LE1001, 1PC1001, 1LE1002, 1PC1002, 1LE1021	381	202	132	256	2 × M32 × 1.5
	Aluminum series, self-ventilated					
	1LE1003-, 1LE1023-					
	1CA0, 1CC0, 1CC2	465	202	132	256	2 × M32 × 1.5
	1CA1, 1CB0, 1CB2, 1CC3	515	202	132	256	2 × M32 × 1.5
	Aluminum series, forced-air cooled					
	1LE1023-					
	1CA0, 1CC0, 1CC2	381	202	132	256	2 × M32 × 1.5
	1CA1, 1CB0, 1CB2, 1CC3	431	202	132	256	2 × M32 × 1.5
	Cast-iron series, self-ventilated					
	1LE1501, 1LE1503, 1LE1521, 1LE1601, 1LE1603, 1LE1621	457	215	132	256	2 × M32 × 1.5
	1LE1523-, LE1623-					
	1CA0, 1CC0, 1CC2	458	215	132	256	2 × M32 × 1.5
	1CA1, 1CB0, 1CB2, 1CC3	508	215	132	256	2 × M32 × 1.5

[1] The length is specified as far as the tip of the fan cover.

[2] Only for pole-changing types 1LE1011-1DP6 and 1LE1012-1DQ6 the dimension L is 664 mm.

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Overview (continued)

Frame size	Type	Dimension L	AD	H	AB	O
160 M/L	Aluminum series, self-ventilated					
	1LE1001, 1LE1002, 1LE1011, 1LE1012, 1LE1021	604 [1) 2)]	237	160	300	2 × M40 × 1.5
	Aluminum series, self-ventilated with increased output					
	1LE1001, 1LE1002	664 [1)]	237	160	300	2 × M40 × 1.5
	Aluminum series, forced-air cooled or naturally cooled					
	1LE1001, 1PC1001, 1LE1002, 1PC1002, 1LE1021	510	237	160	300	2 × M40 × 1.5
	Cast-iron series, self-ventilated					
	1LE1501, 1LE1503, 1LE1521, 1LE1601, 1LE1603, 1LE1621	594	265	160	300	2 × M40 × 1.5
160 M	Aluminum series, self-ventilated					
	1LE1003, 1LE1023	604	237	160	300	2 × M40 × 1.5
	Aluminum series, forced-air cooled					
	1LE1023	510	237	160	300	2 × M40 × 1.5
	Cast-iron series, self-ventilated					
	1LE1523, 1LE1623	596	261	160	300	2 × M40 × 1.5
160 L	Aluminum series, self-ventilated					
	1LE1003, 1LE1023	664	237	160	300	2 × M40 × 1.5
	Aluminum series, forced-air cooled					
	1LE1023	570	237	160	300	2 × M40 × 1.5
	Cast-iron series, self-ventilated					
	1LE1523, 1LE1623	656	261	160	300	2 × M40 × 1.5
180 M	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	1EA2, 1EB2	668	286	180	339	2 × M40 × 1.5
	1EA6	698				
	1LE15.3-, 1LE16.3-					
	1EB2	668	286	180	339	2 × M40 × 1.5
	1EA2	698				
180 L	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	1EC4, 1EC6	668	286	180	339	2 × M40 × 1.5
	1EB6	698				
	1LE15.3-, 1LE16.3-					
	1EC4	668	286	180	339	2 × M40 × 1.5
	1EB4	698				
200 L	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	2AA4, 2AA5, 2AB5, 2AC4, 2AC5	721	315	200	378	2 × M50 × 1.5
	2AA6	746				
	1LE15.3-, 1LE16.3-					
	2AA4, 2AC4	721	315	200	378	2 × M50 × 1.5
	2AA5, 2AB5, 2AC5	746				
225 S	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	2BB0, 2BD0	788	338	225	436	2 × M50 × 1.5
	1LE15.3-, 1LE16.3-					
	2BB0	788	338	225	436	2 × M50 × 1.5

Frame size	Type	Dimension L	AD	H	AB	O
225 M	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	2BA2, 2BA6	818	338	225	436	2 × M50 × 1.5
	2BB2, 2BB6, 2BC2, 2BC6, 2BD6	848				
	1LE15.3-, 1LE16.3-					
	2BA2	818	338	225	436	2 × M50 × 1.5
	2BB2, 2BC2	848				
250 M	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	2CA2, 2CA6, 2CB2, 2CC2, 2CC6, 2CD2, 2CD6	887	410	250	490	2 × M63 × 1.5
	2CB6	957				
	1LE15.3-, 1LE16.3-					
	2CA2, 2CB2, 2CC2	887	410	250	490	2 × M63 × 1.5
280 S	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	2DA0, 2DB0, 2DC0, 2DD0	960	433	280	540	2 × M63 × 1.5
	1LE15.3, 1LE16.3					
	2DA0, 2DB0, 2DC0	960	433	280	540	2 × M63 × 1.5
280 M	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	2DA2, 2DB2, 2DC2, 2DC6, 2DD2, 2DD6	960	433	280	540	2 × M63 × 1.5
	2DA6, 2DB6	1070				
	1LE15.3-, 1LE16.3-					
	2DC2	960	433	280	540	2 × M63 × 1.5
	2DA2, 2DB2	1070				
315 S	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	3AA0	1052	515	315	610	2 × M63 × 1.5
	3AB0, 3AC0, 3AD0	1082				
	1LE15.3-, 1LE16.3-					
	3AA0	1052	515	315	610	2 × M63 × 1.5
	3AB0, 3AC0	1082				
315 M	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	3AC2, 3AD2	1082	515	315	610	2 × M63 × 1.5
	3AA2	1217				
	3AB2	1247				
	1LE15.3-, 1LE16.3-					
	3AA2	1217	515	315	610	2 × M63 × 1.5
	3AB2, 3AC2	1247				
315 L	Cast-iron series, self-ventilated					
	1LE15.1-, 1LE16.1-					
	3AA4	1217	515	315	610	2 × M63 × 1.5
	3AB4, 3AC4, 3AC5, 3AD4, 3AD5, 3AD6	1247				
	3AA5, 3AA6	1372				
	3AB5, 3AB6, 3AC6	1402				
	1LE15.3-, 1LE16.3-					
	3AA4	1217	515	315	610	2 × M63 × 1.5
	3AB4, 3AC4	1247	515	315	610	2 × M63 × 1.5
	3AA5	1372				
	3AB5, 3AC5, 3AC6	1402				

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Notes on the dimensions

Overview (continued)

- Dimensional drawings according to DIN EN 50347 and IEC 60072.
- Fits
  The shaft extensions specified in the dimension tables (DIN 748) and centering spigot diameters (DIN EN 50347) are machined with the following fits:

Dimension designation	ISO fit DIN ISO 286-2	
D, DA	to 30	j6
	over 30 to 50	k6
	over 50	m6
N	to 250	j6
	over 250	h6
F, FA		h9
K		H17
S	Flange (FF)	H17

  The drilled holes of couplings and belt pulleys should have an ISO fit of at least H7.

- Dimension tolerances
  For the following dimensions, the admissible deviations are given below:

Dimension designation	Dimension	Admissible deviation
H	to 250	– 0.5
	over 250	– 1.0
E, EA		– 0.5

  Keyways and feather keyways (dimensions GA, GC, F and FA) are made in compliance with DIN 6885 Part 1.
- All dimensions are specified in mm.

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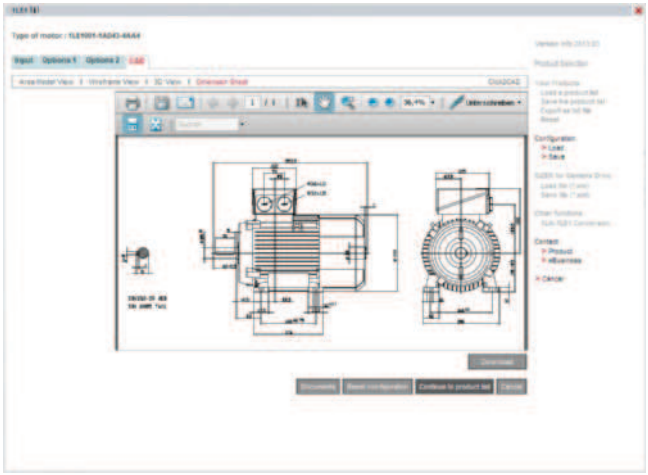
Overview (continued)

A dimensional drawing can be created in the DT Configurator for every configurable motor. A dimensional drawing can be requested for every other motor.



When a complete Article No. is entered with or without order codes, a dimensional drawing can be called up under the "Documentation" tab.
These dimensional drawings can be presented in different views and sections and printed.
The corresponding dimension sheets can be exported, saved and processed further in DXF format (interchange/ import format for CAD systems) or as bitmap graphics.

Online access in the Siemens Industry Mall

The DT Configurator is integrated into the Siemens Industry Mall and can be used on the Internet without installation.
German: www.siemens.de/dt-konfigurator
English: www.siemens.com/dt-configurator

Offline access in the Interactive Catalog CA 01

The DT Configurator is also part of the Interactive Catalog CA 01 on DVD – the offline version of Siemens Industry Mall. CA 01 can be ordered from the relevant Siemens sales office or via the Internet: www.siemens.com/automation/CA01

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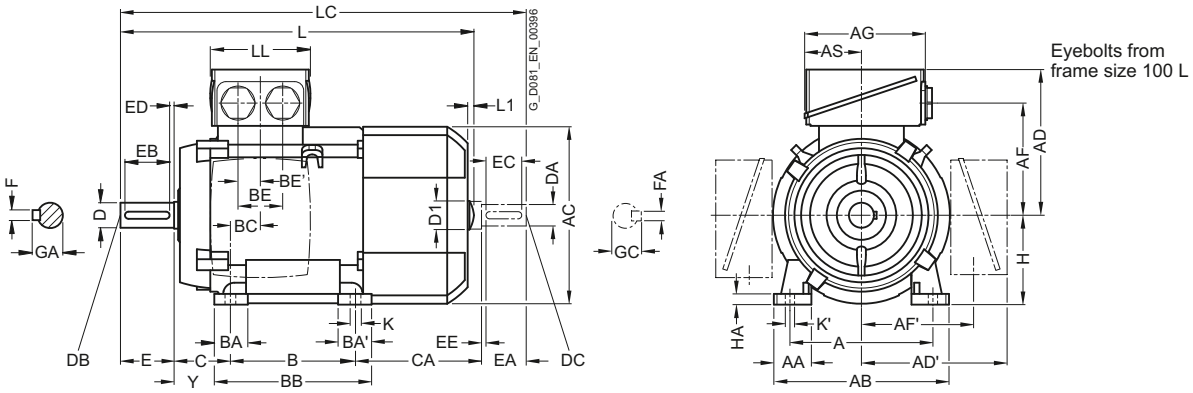
Dimensions

Aluminum series 1LE1001, 1LE1002, 1LE1011, 1LE1012, 1LE1021
Self-ventilated, frame sizes 100 L (80 M) to 160 L

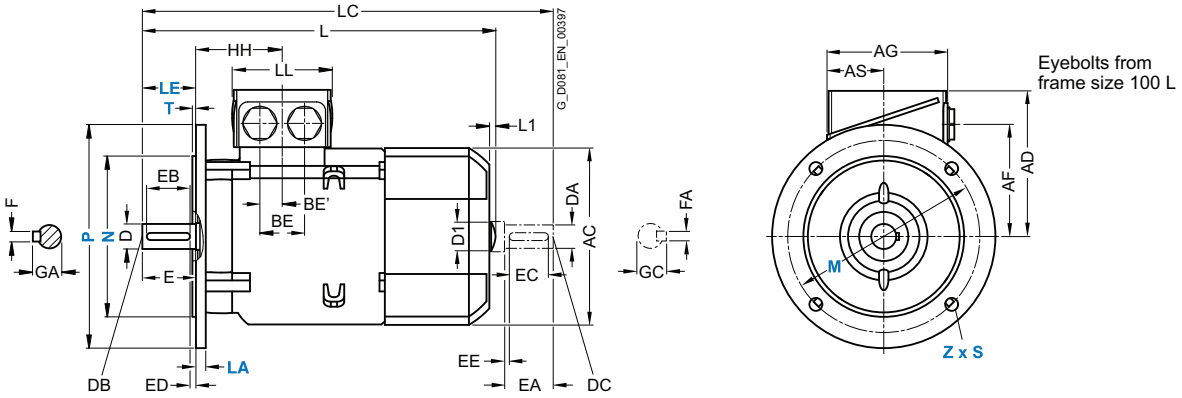
Dimensional drawings

Type of construction IM B3



Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																				
Frame size	Motor type 1LE100. 1LE101. 1LE1021	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*	H	HA	Y
80 M	1LE1001	2, 4, 6	125	30.5	150	159	121	–	96.5	–	93	43	100	32	–	118	23	–	18[1]	50	–	80	8	41
90 S		2, 4, 6	140	30.5	165	178	126	–	101.5	–	93	43	100	33	–	143	22.5	–	18[1]	56	–	90	10	47
90 L		2, 4, 6	140	30.5	165	178	126	–	101.5	–	93	43	125	33	–	143	22.5	–	18[1]	56	–	90	10	47
100 L	All	2, 4, 6, 8	160	42	196	198	166	166	125.5	125.5	135	63.5	140	37.5	–	176	33.5	50	25	63	141	100	12	45
112 M	All	2, 4, 6, 8	190	46	226	222	177	177	136.5	136.5	135	63.5	140	35.4	–	176	26	50	25	70	129.7	112	12	52
132 S	All	2, 4, 6, 8	216	53	256	262	202	202	159.5	159.5	155	70.5	140	38	76	218	26.5	48	24	89	128.5	132	15	69
132 M	All	2, 4, 6, 8	216	53	256	262	202	202	159.5	159.5	155	70.5	178	38	76	218	26.5	48	24	89	128.5	132	15	69
160 M	All	2, 4, 6, 8	254	60	300	314	236.5	236.5	190	190	175	77.5	210	44	89	300	47	57	28.5	108	148	160	18	85
160 L	All	2, 4, 6, 8	254	60	300	314	236.5	236.5	190	190	175	77.5	254	44	89	300	47	57	28.5	108	148[2]	160	18	85

* This dimension is assigned in DIN EN 50347 to the frame size listed.

[1] Connecting hole for terminal box is on the side at the rear of the terminal box.

[2] Only for pole-changing types 1LE1011-1DP6 and 1LE1012-1DQ6 the dimension CA* is 208 mm.

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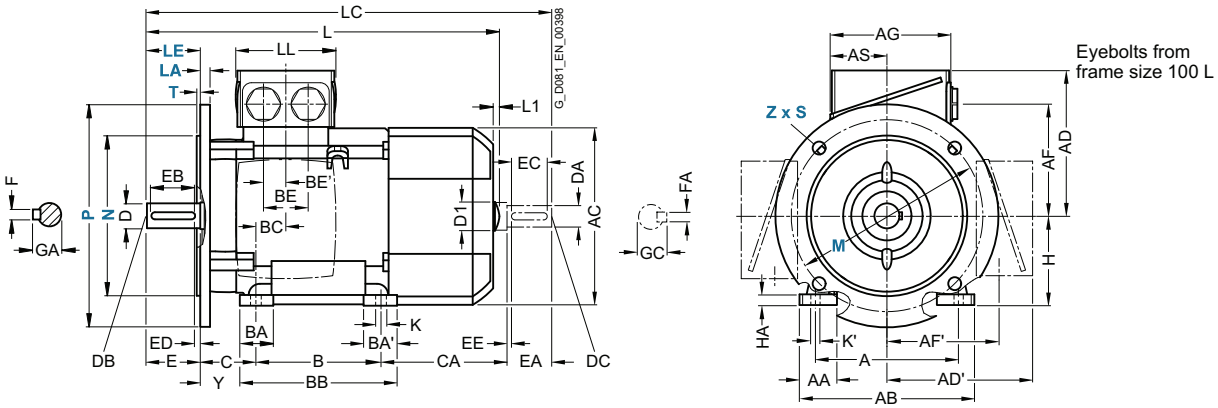
Dimensional drawings (continued)

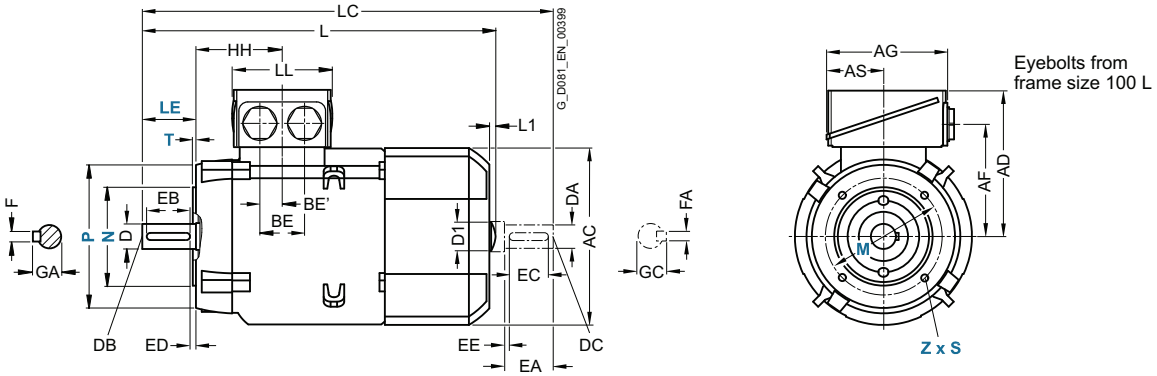
Type of construction IM B35

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



Type of construction IM B14

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**							DE shaft extension								NDE shaft extension						
Frame size	Motor type 1LE100. 1LE101. 1LE1021	No. of poles	HH	K	K'	L	L1	D1	LC	LL	D	DB	E	EB	ED	F	GA	DA	DC	EA	EC	EE	FA	GC
80 M	1LE1001	2, 4, 6	73	9.5	13.5	292	–	–	–	79	19	M6	40	32	4	6	21.5	19	M6	40	32	4	6	21.5
90 S		2, 4, 6	78.5	10	14	347	–	–	–	79	24	M8	50	40	5	8	27	19	M6	40	32	4	6	21.5
90 L		2, 4, 6	78.5	10	14	347	–	–	–	79	24	M8	50	40	5	8	27	19	M6	40	32	4	6	21.5
100 L	All	2, 4, 6, 8	96.5	12	16	395.5[1]	7	32	454	112	28	M10	60	50	5	8	31	24	M8	50	40	5	8	27
112 M	All	2, 4, 6, 8	96	12	16	389[1]	7	32	450	112	28	M10	60	50	5	8	31	24	M8	50	40	5	8	27
132 S	All	2, 4, 6, 8	115.5	12	16	465[1]	8.5	39	535.5	130	38	M12	80	70	5	10	41	28	M10	60	50	5	8	31
132 M	All	2, 4, 6, 8	115.5	12	16	465[1]	8.5	39	535.5	130	38	M12	80	70	5	10	41	28	M10	60	50	5	8	31
160 M	All	2, 4, 6, 8	155	15	19	604[1]	10	45	730	145	42	M16	110	90	10	12	45	42	M16	110	90	10	12	45
160 L	All	2, 4, 6, 8	155	15	19	604[1][2]	10	45	730[3]	145	42	M16	110	90	10	12	45	42	M16	110	90	10	12	45

[1] The length is specified as far as the tip of the fan cover.

[2] Only for pole-changing types 1LE1011-1DP6 and 1LE1012-1DQ6 the dimension L is 664 mm.

[3] Only for pole-changing types 1LE1011-1DP6 and 1LE1012-1DQ6 the dimension LC is 790 mm.

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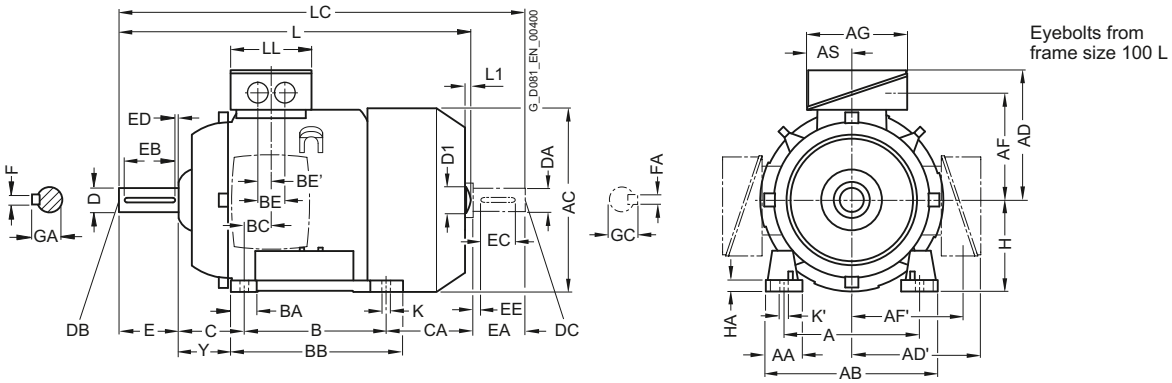
Dimensions

Aluminum series 1LE1001, 1LE1002
Self-ventilated, with increased output, frame sizes 100 L to 160 L

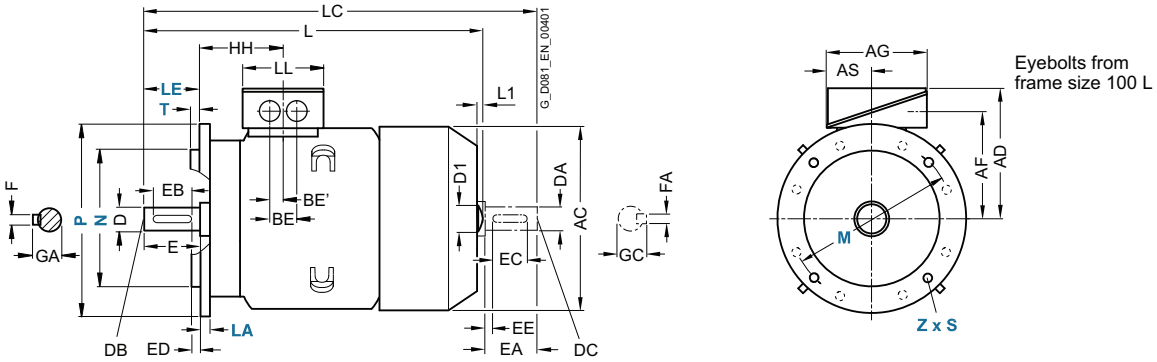
Dimensional drawings

Type of construction IM B3



Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																				
Frame size	Motor type 1LE1001 1LE1002	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*	H	HA	Y
100 L	All	2, 4, 6, 8	160	42	196	198	166	166	125.5	125.5	135	63.5	140	37.5	–	176	33.5	50	25	63	176	100	12	45
112 M	All	2, 4, 6, 8	190	46	226	222	177	177	136.5	136.5	135	63.5	140	35.4	–	176	26	50	25	70	155	112	12	52
132 M	All	2, 4, 6, 8	216	53	256	262	202	202	159.5	159.5	155	70.5	178	38	–	218	26.5	48	24	89	178.5	132	15	69
160 L	All	2, 4, 6, 8	254	60	300	314	236.5	236.5	190	190	175	77.5	254	44	–	300	47	57	28.5	108	208	160	18	85

* This dimension is assigned in DIN EN 50347 to the frame size listed.

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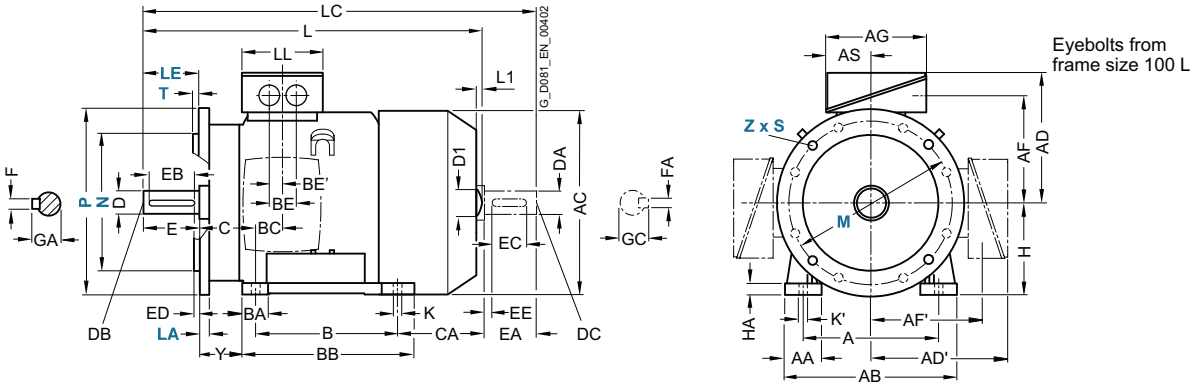
Dimensional drawings (continued)

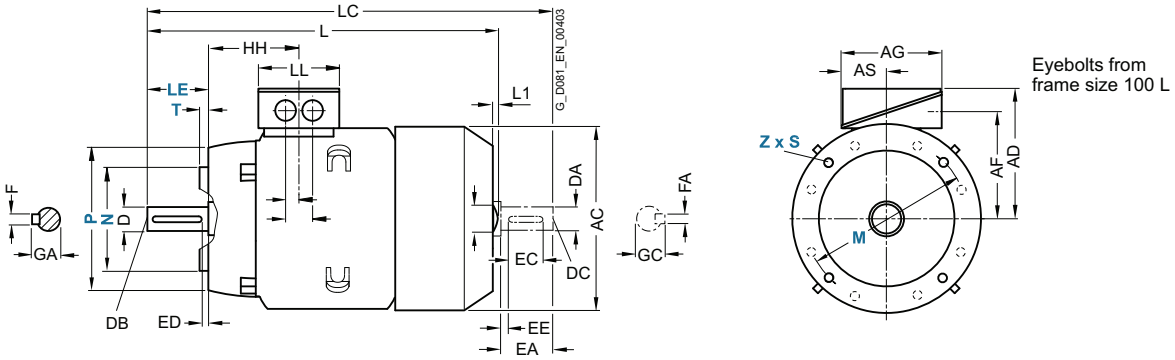
Type of construction IM B35

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



Type of construction IM B14

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**								DE shaft extension							NDE shaft extension						
Frame size	Motor type 1LE1001 1LE1002	No. of poles	HH	K	K'	L [1)]	L1	D1	LC	LL	D	DB	E	EB	ED	F	GA	DA	DC	EA	EC	EE	FA	GC
100 L	All	2, 4, 6, 8	96.5	12	16	430.5	7	32	489	112	28	M10	60	50	5	8	31	24	M8	50	40	5	8	27
112 M	All	2, 4, 6, 8	96	12	16	414	7	32	475	112	28	M10	60	50	5	8	31	24	M8	50	40	5	8	27
132 M	All	2, 4, 6, 8	115.5	12	16	515	8.5	39	585.5	130	38	M12	80	70	5	10	41	28	M10	60	50	5	8	31
160 L	All	2, 4, 6, 8	155	15	19	664	10	45	790	145	42	M16	110	90	10	12	45	42	M16	110	90	10	12	45

[1)] The length is specified as far as the tip of the fan cover.

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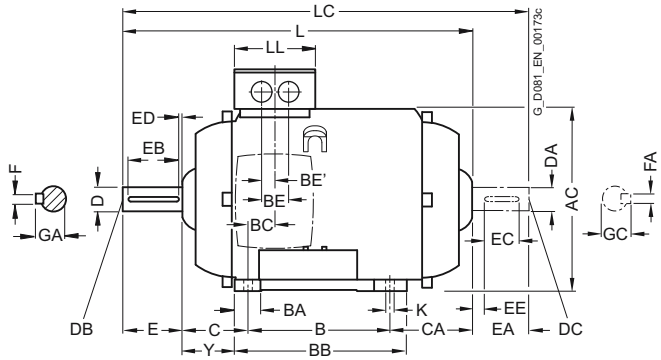
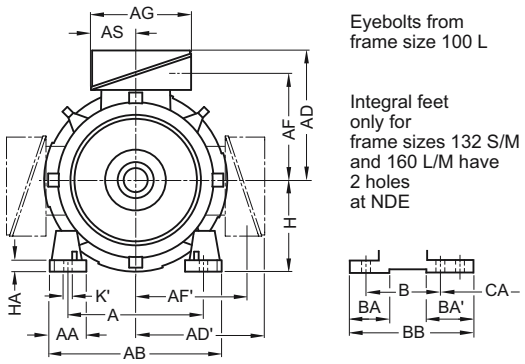
Dimensions

Aluminum series 1LE1001, 1PC1001, 1LE1002, 1PC1002, 1LE1021
Forced-air cooled or naturally cooled, frame sizes 80 M to 160 L

Dimensional drawings

Type of construction IM B3



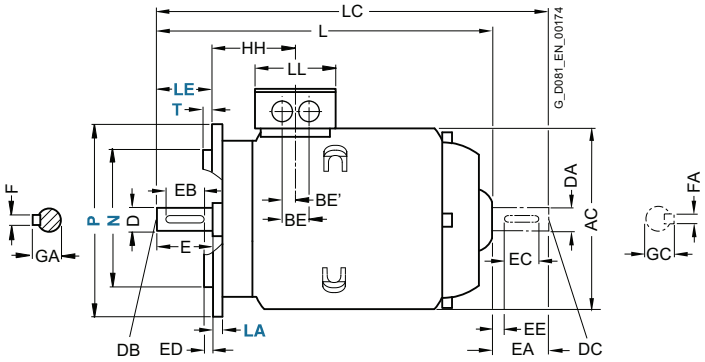


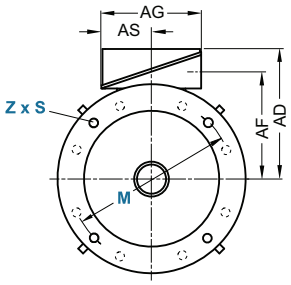
Eyebolts from frame size 100 L

Integral feet only for frame sizes 132 S/M and 160 L/M have 2 holes at NDE

Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)





Eyebolts from frame size 100 L

For motor			Dimension designation acc. to **IEC**																				
Frame size	Motor type 1LE100. 1PC100. 1LE1021	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*	H	HA	Y
80 M	1LE1001	2, 4, 6	125	30.5	150	159	121	–	96.5	–	93	43	100	32	–	118	23	–	18	50	–	80	8	41
	1LE1021	2, 4, 6					149		112		119.5	61.5												
90 S	1LE1001	2, 4, 6	140	30.5	165	178	126	–	101.5	–	93	43	100	33	–	143	22.5	–	18	56	–	90	10	47
	1LE1021	2, 4, 6					154		117		119.5	61.5												
90 L	1LE1001	2, 4, 6	140	30.5	165	178	126	–	101.5	–	93	43	125	33	–	143	22.5	–	18	56	–	90	10	47
	1LE1021	2, 4, 6					154		117		119.5	61.5												
100 L	All	2, 4, 6, 8	160	42	196	197	166	166	125.5	125.5	135	63.5	140	37.5	–	176	33.5	50	25	63	–	100	12	45
112 M	All	2, 4, 6, 8	190	46	226	221	177	177	136.5	136.5	135	63.5	140	35.4	–	176	26	50	25	70	–	112	12	52
132 S	All	2, 4, 6, 8	216	53	256	261	202	202	159.5	159.5	155	70.5	140	38	76[1]	218[2]	26.5	48	24	89	–	132	15	69
132 M	All	2, 4, 6, 8	216	53	256	261	202	202	159.5	159.5	155	70.5	178	38	76	218	26.5	48	24	89	–	132	15	69
160 M	All	2, 4, 6, 8	254	60	300	314	236.5	236.5	190	190	175	77.5	210	44	89[3]	300[4]	47	57	28.5	108	–	160	18	85
160 L	All	2, 4, 6, 8	254	60	300	314	236.5	236.5	190	190	175	77.5	254	44	89	300	47	57	28.5	108	–	160	18	85

* This dimension is assigned in DIN EN 50347 to the frame size listed.

[1] With screwed-on feet, dimension BA' is 38 mm.

[2] With screwed-on feet, dimension BB is 180 mm.

[3] With screwed-on feet, dimension BA' is 44 mm.

[4] With screwed-on feet, dimension BB is 256 mm.

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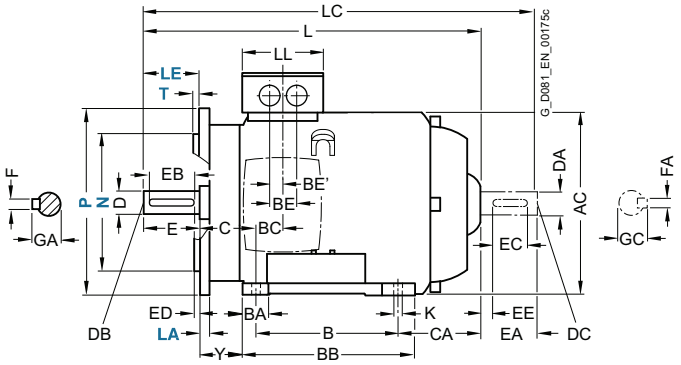
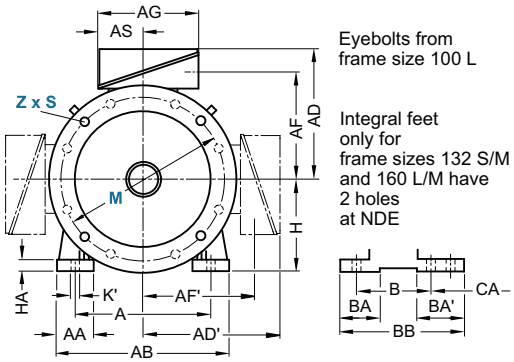
Dimensional drawings (continued)

Type of construction IM B35

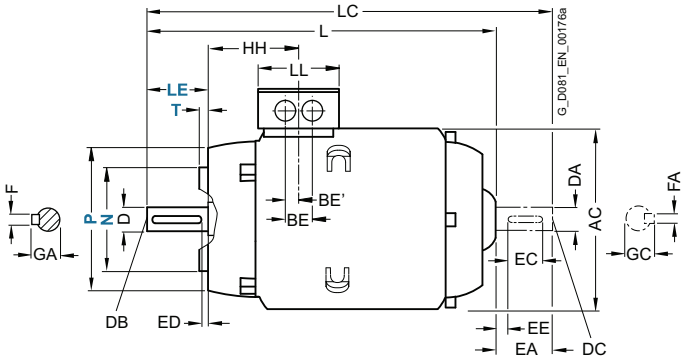
For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)

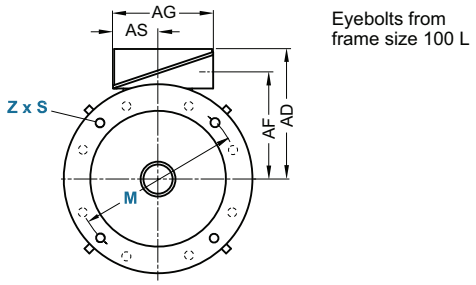




Type of construction IM B14

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)





For motor			Dimension designation acc. to **IEC**						DE shaft extension							NDE shaft extension						
Frame size	Motor type 1LE100. 1PC100. 1LE1021	No. of poles	HH	K	K'	L	LC	LL	D	DB	E	EB	ED	F	GA	DA	DC	EA	EC	EE	FA	GC
80 M	1LE1001	2, 4, 6	73	9.5	13.5	253	–	79	19	M6	40	32	4	6	21.5	–	–	–	–	–	–	–
	1LE1021	2, 4, 6						123														
90 S	1LE1021	2, 4, 6	78.5	10	14	295	–	79	24	M8	50	40	5	8	27	–	–	–	–	–	–	–
	1LE1021	2, 4, 6						123														
90 L	1LE1021	2, 4, 6	78.5	10	14	295	–	123	24	M8	50	40	5	8	27	–	–	–	–	–	–	–
	1LE1021	2, 4, 6						123														
100 L	All	2, 4, 6, 8	96.5	12	16	321.5	–	112	28	M10	60	50	5	8	31	–	–	–	–	–	–	–
112 M	All	2, 4, 6, 8	96	12	16	311	–	112	28	M10	60	50	5	8	31	–	–	–	–	–	–	–
132 S	All	2, 4, 6, 8	115.5	12	16	380.5	–	130	38	M12	80	70	5	10	41	–	–	–	–	–	–	–
132 M	All	2, 4, 6, 8	115.5	12	16	380.5	–	130	38	M12	80	70	5	10	41	–	–	–	–	–	–	–
160 M	All	2, 4, 6, 8	155	15	19	510	–	145	42	M16	110	90	10	12	45	–	–	–	–	–	–	–
160 L	All	2, 4, 6, 8	155	15	19	510	–	145	42	M16	110	90	10	12	45	–	–	–	–	–	–	–

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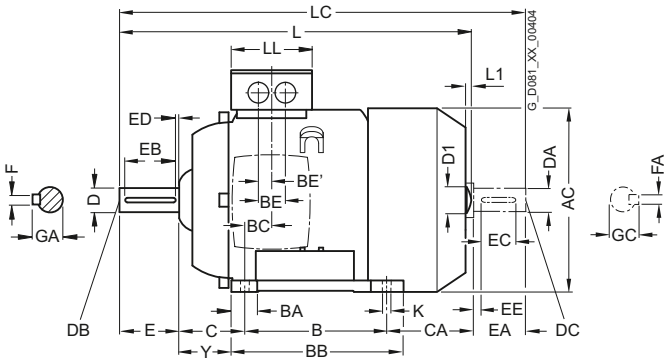
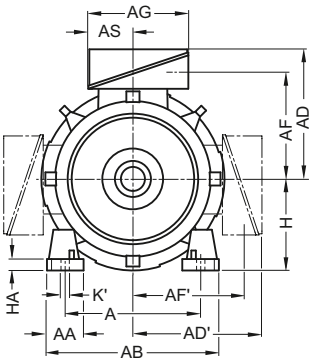
Dimensions

Aluminum series 1LE1003, 1LE1023
Self-ventilated, frame sizes 80 M to 90 L

Dimensional drawings

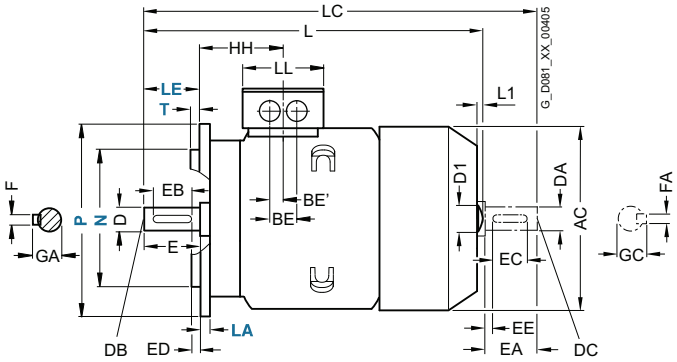
Type of construction IM B3

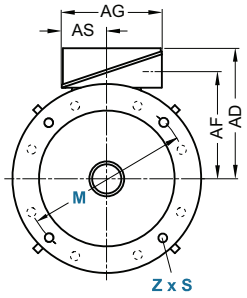




Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)





For motor			Dimension designation acc. to **IEC**																				
Frame size	Motor type	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AS	B*	BA	BB	BC	BE	BE'	C	CA*	H	HA	Y
80 M	1LE1003-0DA2, -0DB2, -0DC2 -0DA3, -0DB3, -0DC3	2, 4, 6	125	30.5	150	159	121	–	96.5	–	93	43	100	32	118	23	–	18	50	–	80	8	41
	1LE1023-0DA2, -0DB2, -0DC2 -0DA3, -0DB3, -0DC3	2, 4, 6					149		112		119.5	61.5											
90 S	1LE1003-0EA0, -0EB0, -0EC0	2, 4, 6	140	30.5	165	178	126	–	101.5	–	93	43	100	33	143	22.5	–	18	56	–	90	10	47
	1LE1023-0EA0, -0EB0, -0EC0	2, 4, 6					154		117		119.5	61.5											
90 L	1LE1003-0EA4, -0EB4, -0EC4	2, 4, 6	140	30.5	165	178	126	–	101.5	–	93	43	100	33	143	22.5	–	18	56	–	90	10	47
	1LE1023-0EA4, -0EB4, -0EC4	2, 4, 6					154		117		119.5	61.5											

* This dimension is assigned in DIN EN 50347 to the frame size listed.

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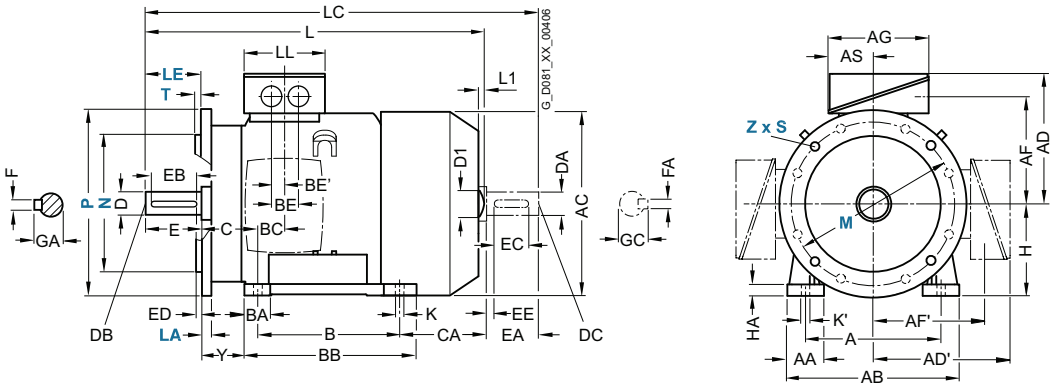
Dimensional drawings (continued)

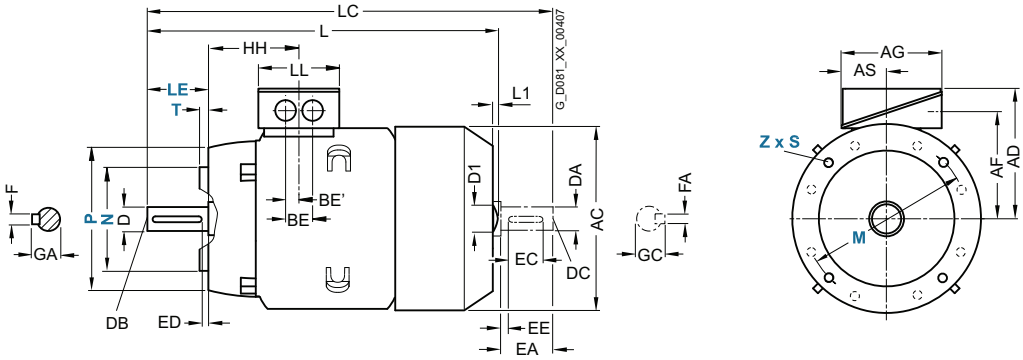
Type of construction IM B35

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



Type of construction IM B14

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																				
Frame size	Motor type	No. of poles	HH	K	K'	L[1]	L1	D1	LC	LL	D	DB	E	EB	ED	F	GA	DA	DC	EA	EC	EE	FA	GC
80 M	1LE1003-0DA2, -0DB2, -0DC2, -0DA3, -0DB3, -0DC3	2, 4, 6	73	9.5	13.5	292	–	–	–	79	19	M6	40	32	4	6	21.5	19	M6	40	32	4	6	21.5
						327																		
	1LE1023-0DA2, -0DB2, -0DC2, -0DA3, -0DB3, -0DC3	2, 4, 6								123														
90 S	1LE1003-0EA0, -0EB0, -0EC0	2, 4, 6	78.5	10	14	347	–	–	–	79	24	M8	50	40	5	8	27	19	M6	40	32	4	6	21.5
	1LE1023-0EA0, -0EB0, -0EC0	2, 4, 6								123														
90 L	1LE1003-0EA4, -0EB4, -0EC4	2, 4, 6	78.5	10	14	387	–	–	–	79	24	M8	50	40	5	8	27	19	M6	40	32	4	6	21.5
	1LE1023-0EA4, -0EB4, -0EC4	2, 4, 6								123														

[1] The length is specified as far as the tip of the fan cover.

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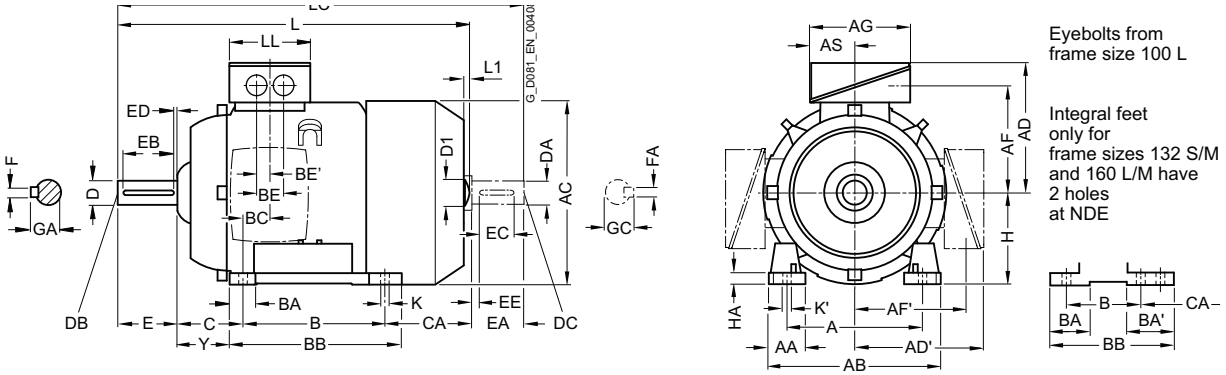
Dimensions

Aluminum series 1LE1003, 1LE1023
Self-ventilated, frame sizes 100 L to 160 L

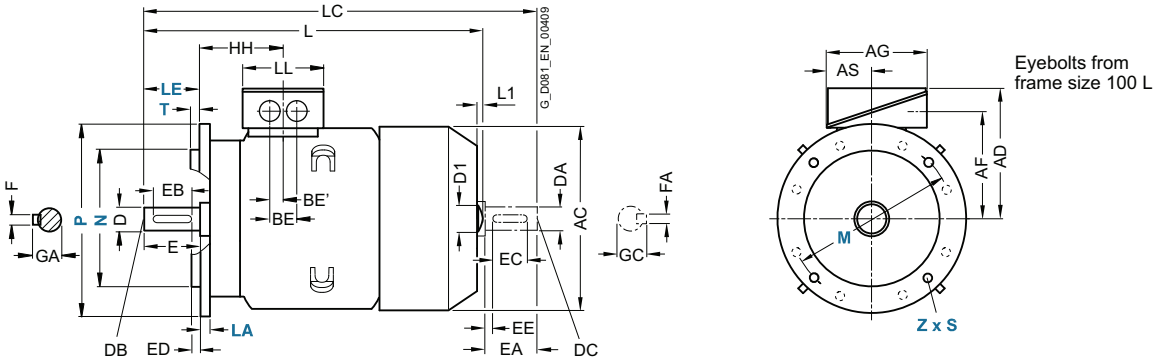
Dimensional drawings

Type of construction IM B3



Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																				
Frame size	Motor type 1LE1003- 1LE1023-	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*	H	HA	Y
100 L	1AA4, 1AB4, 1AB5	2, 4, 6	160	42	196	198	166	166	125.5	125.5	135	63.5	140	37.5	–	176	33.5	50	25	63	176	100	12	45
112 M	1BA2, 1BB2	2, 4, 6	190	46	226	222	177	177	136.5	136.5	135	63.5	140	35.4	–	176	26	50	25	70	155	112	12	52
132 S	1CA0, 1CC0	2, 6	216	53	256	262	202	202	159.5	159.5	155	70.5	140	38	76[1]	218[2]	26.5	48	24	89	128.5[3]	132	15	69
	1CA1, 1CB0	2, 4													–						178.5			
132 M	1CC2	6	216	53	256	262	202	202	159.5	159.5	155	70.5	178	38	76	218	26.5	48	24	89	128.5[3]	132	15	69
	1CB2, 1CC3	4, 6, 8													–						178.5			
160 M	1DA2, 1DA3, 1DB2, 1DC2	2, 4, 6	254	60	300	314	236.5	236.5	190	190	175	77.5	210	44	89[4]	300[5]	47	57	28.5	108	148[6]	160	18	85
160 L	1DA4, 1DB4, 1DC4	2, 4, 6	254	60	300	314	236.5	236.5	190	190	175	77.5	254	44	–	300	47	57	28.5	108	208	160	18	85

* This dimension is assigned in DIN EN 50347 to the frame size listed.

1) With screwed-on feet, dimension BA' is 38 mm.

2) With screwed-on feet, dimension BB is 180 mm.

3) With screwed-on feet, dimension CA is 166.5 mm.

4) With screwed-on feet, dimension BA' is 44 mm.

5) With screwed-on feet, dimension BB is 256 mm.

6) With screwed-on feet, dimension CA is 192 mm.

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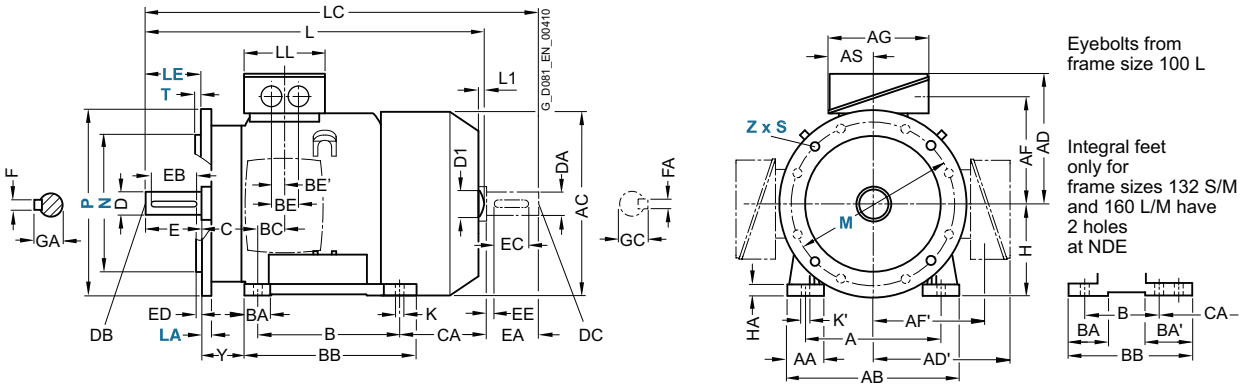
Dimensional drawings (continued)

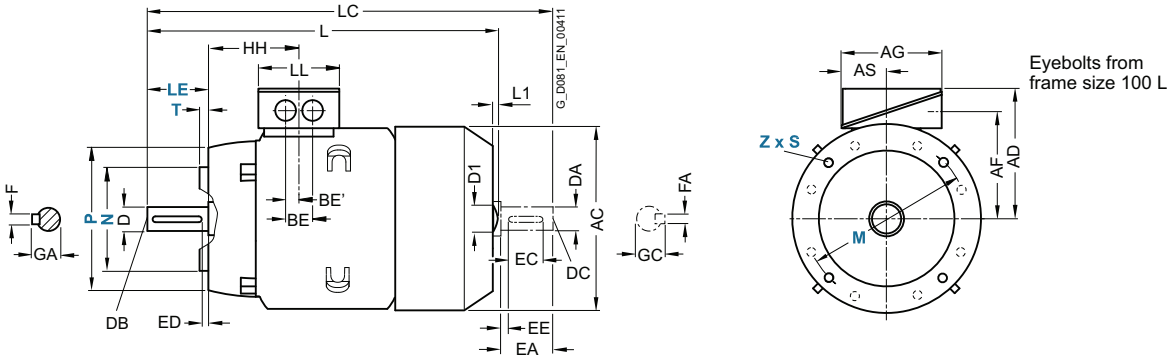
Type of construction IM B35

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



Type of construction IM B14

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**								DE shaft extension							NDE shaft extension						
Frame size	Motor type 1LE1003- 1LE1023-	No. of poles	HH	K	K'	L[1)]	L1	D1	LC	LL	D	DB	E	EB	ED	F	GA	DA	DC	EA	EC	EE	FA	GC
100 L	1AA4, 1AB4, 1AB5	2, 4, 6	96.5	12	16	430.5	7	32	489	112	28	M10	60	50	5	8	31	24	M8	50	40	5	8	27
112 M	1BA2, 1BB2	2, 4, 6	96	12	16	414	7	32	475	112	28	M10	60	50	5	8	31	24	M8	50	40	5	8	27
132 S	1CA0, 1CC0	2, 6	115.5	12	16	465	8.5	39	535.5	130	38	M12	80	70	5	10	41	28	M10	60	50	5	8	31
	1CA1, 1CB0	2, 4				515			585.5															
132 M	1CC2	6	115.5	12	16	465	8.5	39	535.5	130	38	M12	80	70	5	10	41	28	M10	60	50	5	8	31
	1CB2, 1CC3	4, 6				515			585.5															
160 M	1DA2, 1DA3, 1DB2, 1DC2	2, 4, 6	155	15	19	604	10	45	730	145	42	M16	110	90	10	12	45	42	M16	110	90	10	12	45
160 L	1DA4, 1DB4, 1DC4	2, 4, 6	155	15	19	664	10	45	790	145	42	M16	110	90	10	12	45	42	M16	110	90	10	12	45

[1)] The length is specified as far as the tip of the fan cover.

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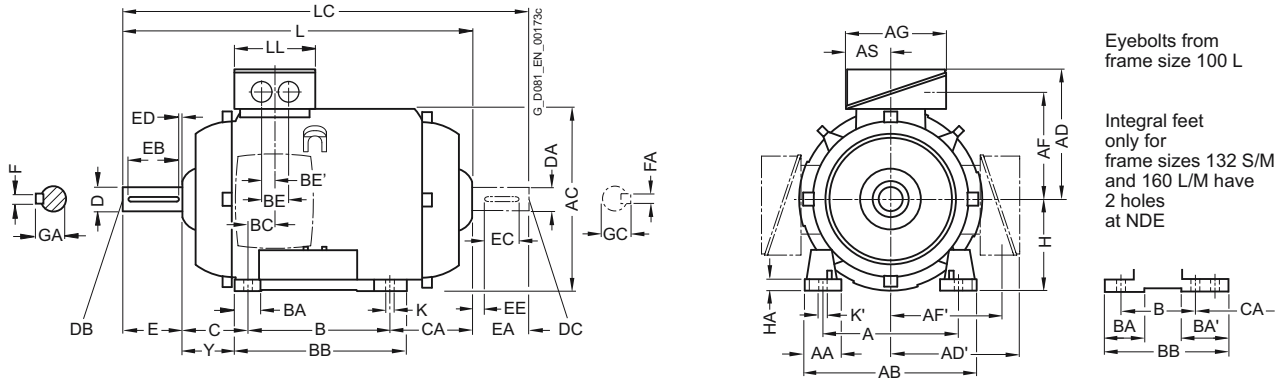
Dimensions

Aluminum series 1LE1023
Forced-air cooled, frame sizes 80 M to 90 L

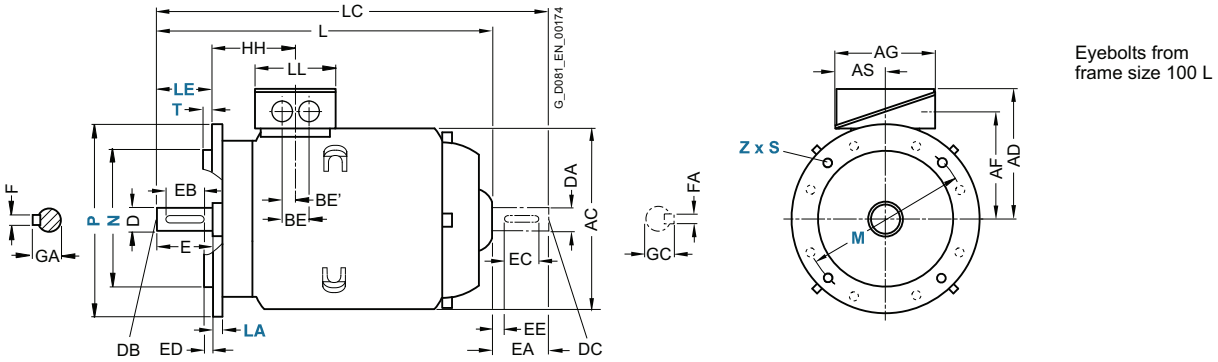
Dimensional drawings

Type of construction IM B3



Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																				
Frame size	Motor type 1LE1023-	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*	H	HA	Y
80 M	0DA2, 0DB2, 0DC2	2, 4, 6	125	30.5	150	159	121	–	96.5	–	93	43	100	32	–	118	23	–	18	50	–	80	8	41
	0DA3, 0DB3, 0DC3	2, 4, 6																						
90 S	0EA0, 0EB0, 0EC0	2, 4, 6	140	30.5	165	178	126	–	101.5	–	93	43	100	33	–	143	22.5	–	18	56	–	90	10	47
90 L	0EA4, 0EB4, 0EC4	2, 4, 6	140	30.5	165	178	126	–	101.5	–	93	43	100	33	–	143	22.5	–	18	56	–	90	10	47

* This dimension is assigned in DIN EN 50347 to the frame size listed.

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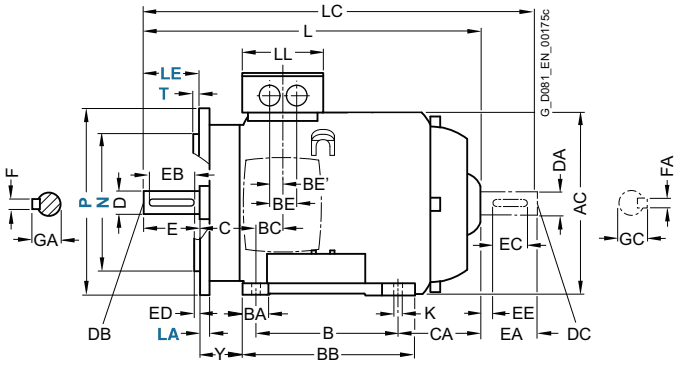
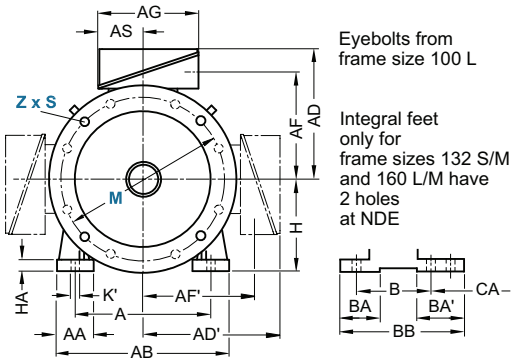
Dimensional drawings (continued)

Type of construction IM B35

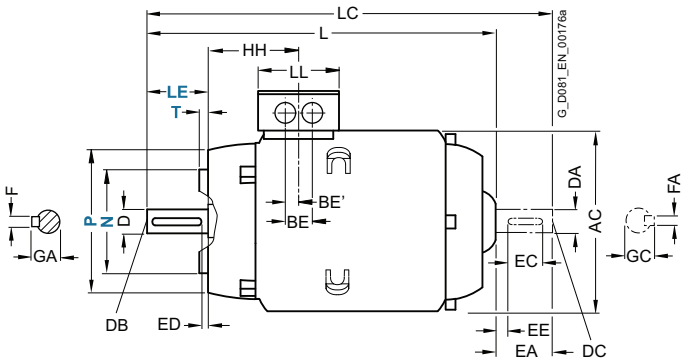
For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)

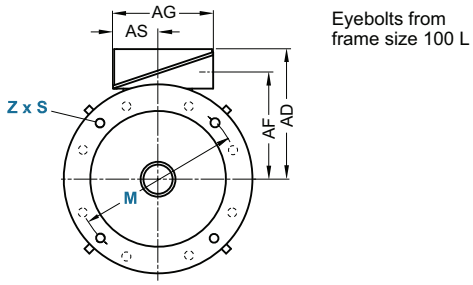




Type of construction IM B14

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)





For motor			Dimension designation acc. to **IEC**						DE shaft extension							NDE shaft extension						
Frame size	Motor type 1LE1023-	No. of poles	HH	K	K'	L[1)]	LC	LL	D	DB	E	EB	ED	F	GA	DA	DC	EA	EC	EE	FA	GC
80 M	0DA2, 0DB2, 0DC2	2, 4, 6	73	9.5	13.5	292	–	79	19	M6	40	32	–	–	–	–	–	–	–	–	–	–
	0DA3, 0DB3, 0DC3	2, 4, 6				327																
90 S	0EA0, 0EB0, 0EC0	2, 4, 6	78.5	10	14	347	–	79	19	M6	40	32	–	–	–	–	–	–	–	–	–	–
90 L	0EA4, 0EB4, 0EC4	2, 4, 6	78.5	10	14	387	–	79	19	M6	40	32	–	–	–	–	–	–	–	–	–	–

[1)] The length is specified as far as the tip of the fan cover.

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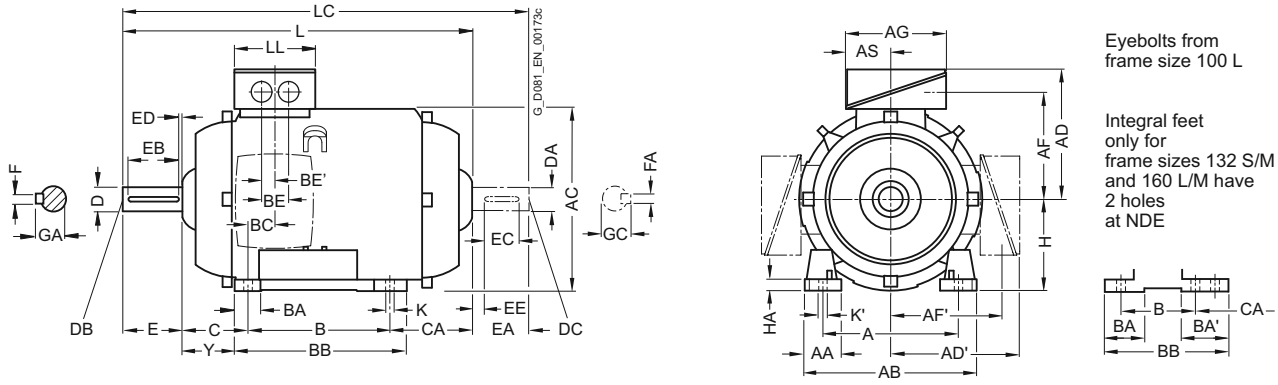
Dimensions

Aluminum series 1LE1023
Forced-air cooled, frame sizes 100 L to 160 L

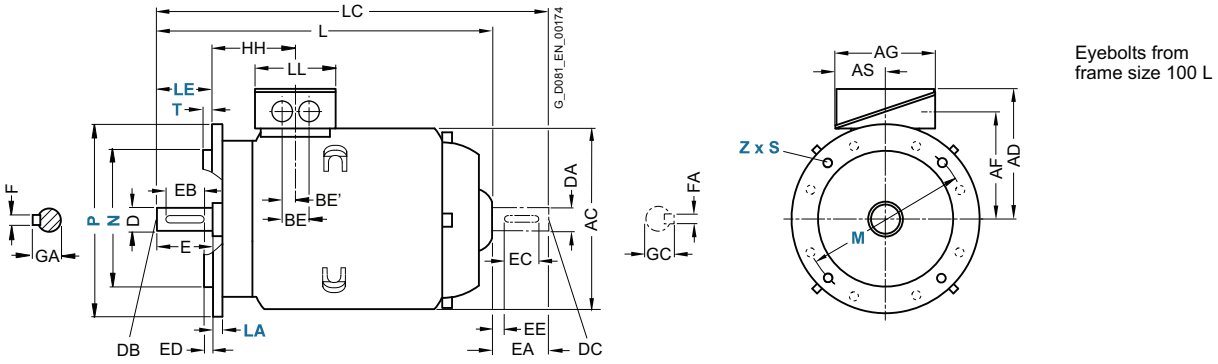
Dimensional drawings

Type of construction IM B3



Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																					
Frame size	Motor type 1LE1023-	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AQ	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*	H	HA	Y
100 L	1AA4, 1AB4, 1AB5	2, 4	160	42	196	198	166	166	125.5	125.5	135	195	63.5	140	37.5	–	176	33.5	50	25	63	–	100	12	45
112 M	1BA2, 1BB2	2, 4	190	46	226	222	177	177	136.5	136.5	135	195	63.5	140	35.4	–	176	26	50	25	70	–	112	12	52
132 S	1CA0, 1CC0	2, 6	216	53	256	262	202	202	159.5	159.5	155	260	70.5	140	38	76[1]	218[2]	26.5	48	24	89	–	132	15	69
	1CA1, 1CB0	2, 4														–									
132 M	1CC2	6	216	53	256	262	202	202	159.5	159.5	155	260	70.5	178	38	76	218	26.5	48	24	89	–	132	15	69
	1CB2, 1CC3	4, 6, 8														–									
160 M	1DA2, 1DA3, 1DB2, 1DC2	2, 4, 6	254	60	300	314	236.5	236.5	190	190	175	260	77.5	210	44	89[3]	300[4]	47	57	28.5	108	–	160	18	85
160 L	1DA4, 1DB4, 1DC4	2, 4, 6	254	60	300	314	236.5	236.5	190	190	175	260	77.5	254	44	–	300	47	57	28.5	108	–	160	18	85

* This dimension is assigned in DIN EN 50347 to the frame size listed.

1) With screwed-on feet, dimension BA' is 38 mm.

2) With screwed-on feet, dimension BB is 180 mm.

3) With screwed-on feet, dimension BA' is 44 mm.

4) With screwed-on feet, dimension BB is 256 mm.

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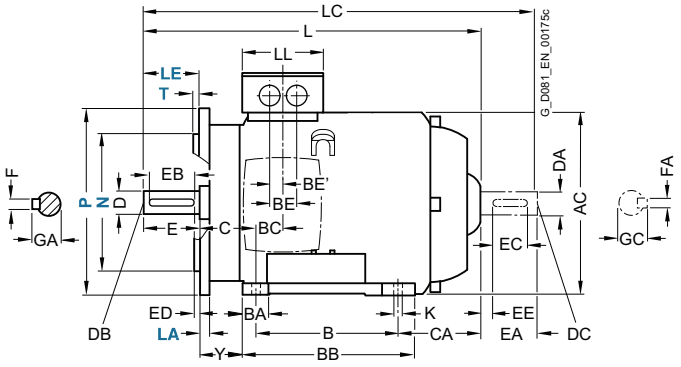
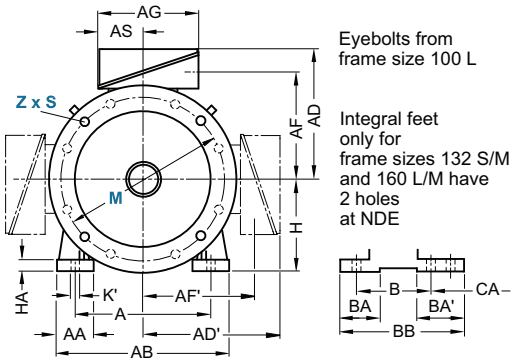
Dimensional drawings (continued)

Type of construction IM B35

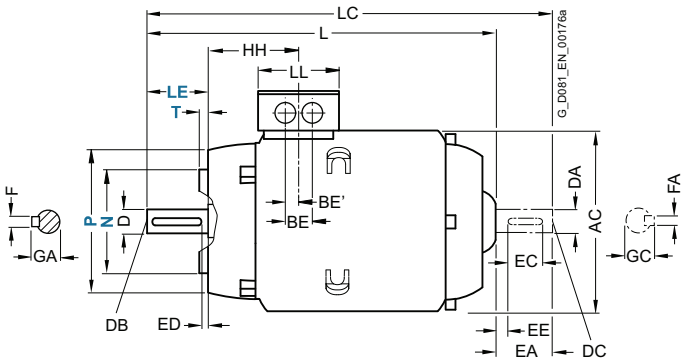
For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)

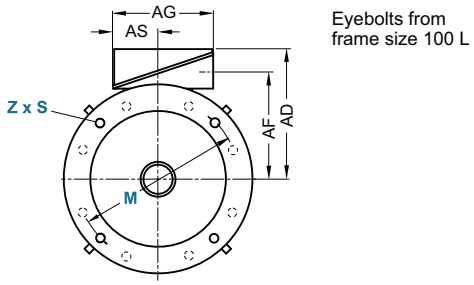




Type of construction IM B14

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)





For motor			Dimension designation acc. to **IEC**						DE shaft extension							NDE shaft extension						
Frame size	Motor type 1LE1023-	No. of poles	HH	K	K'	L[1)]	LC	LL	D	DB	E	EB	ED	F	GA	DA	DC	EA	EC	EE	FA	GC
100 L	1AA4, 1AB4, 1AB5	2, 4	96.5	12	16	356.5	–	112	28	M10	60	50	5	8	31	–	–	–	–	–	–	–
112 M	1BA2, 1BB2	2, 4	96	12	16	336	–	112	28	M10	60	50	5	8	31	–	–	–	–	–	–	–
132 S	1CA0, 1CC0	2, 6	115.5	12	16	380.5	–	130	38	M12	80	70	5	10	41	–	–	–	–	–	–	–
	1CA1, 1CB0	2, 4				430.5	–									–	–	–	–	–	–	–
132 M	1CC2	6	115.5	12	16	380.5	–	130	38	M12	80	70	5	10	41	–	–	–	–	–	–	–
	1CB2, 1CC3	4, 6				430.5	–									–	–	–	–	–	–	–
160 M	1DA2, 1DA3, 1DB2, 1DC2	2, 4, 6	155	15	19	510	–	145	42	M16	110	90	10	12	45	–	–	–	–	–	–	–
160 L	1DA4, 1DB4, 1DC4	2, 4, 6	155	15	19	570	–	145	42	M16	110	90	10	12	45	–	–	–	–	–	–	–

[1)] The length is specified as far as the tip of the fan cover.

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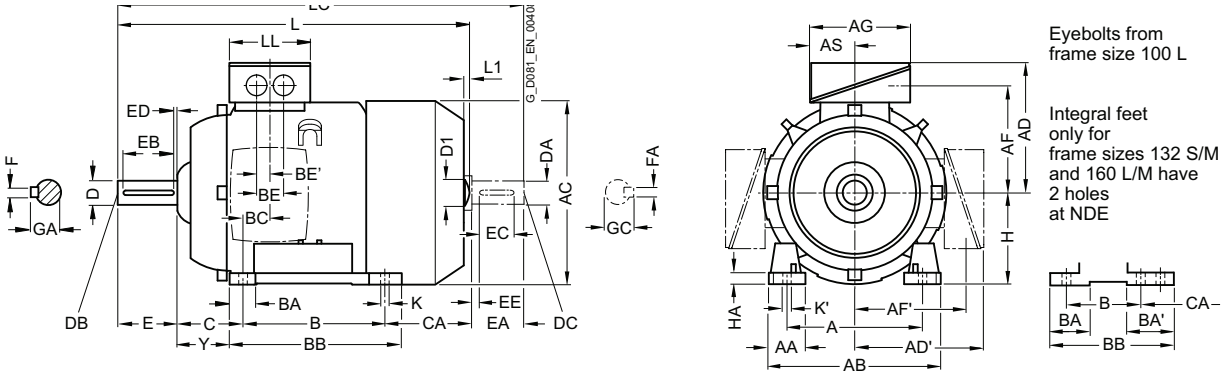
Dimensions

Cast-iron series 1LE1501, 1LE1521, 1LE1601, 1LE1621
Self-ventilated, frame sizes 71 M to 160 L

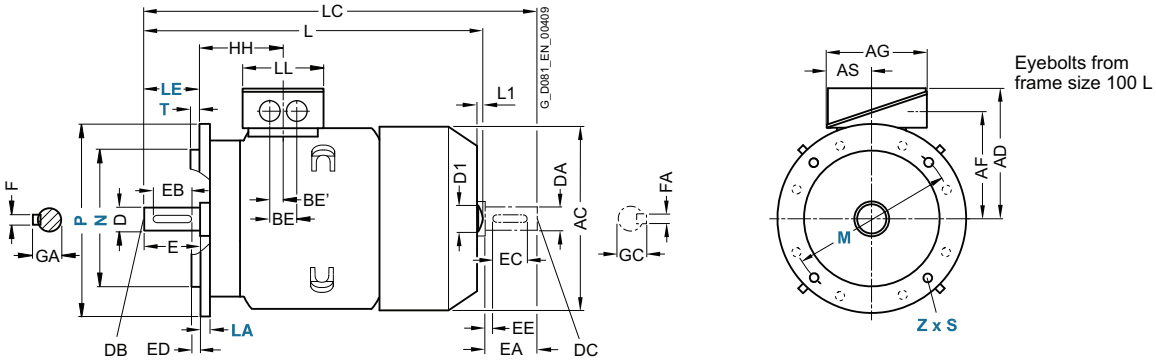
Dimensional drawings

Type of construction IM B3



Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																				
Frame size	Motor type 1LE15.1, 1LE16.1	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*	H	HA	Y
71 M	1LE15.1	2, 4, 6	112	27	132	138.5	148	148	112	112	126	62	90	–	–	106	20.5	36	18	45	–	71	7	37
80 M	1LE15.1	2, 4, 6	125	27	150	156	158	158	122	122	129	62	100	–	–	118	21.5	36	18	50	–	80	8.5	41
90 S	1LE15.1	2, 4, 6	140	30.5	165	173.5	163	163	127	127	129	62	100	–	–	143	23.5	36	18	56	–	90	11	47
90 L	1LE15.1	2, 4, 6	140	30.5	165	173.5	163	163	127	127	129	62	125	–	–	143	23.5	36	18	56	–	90	11	47
100 L	All	2, 4, 6, 8	160	42	196	198	193	193	147	147	163	80.5	140	40	–	176	37.5	48	24	63	141	100	12	45
112 M	All	2, 4, 6, 8	190	46	226	222	195	195	150	150	163	80.5	140	40	–	176	30	48	24	70	129.7	112	12	52
132 S	All	2, 4, 6, 8	216	53	256	262	214.5	214.5	169	169	163	80.5	140	44	81[1)]	218[3)]	26.5	48	24	89	–	132	15	69
132 M	All	2, 4, 6, 8	216	53	256	262	214.5	214.5	169	169	163	80.5	178	44	81[1)]	218	26.5	48	24	89	–	132	15	69
160 M	All	2, 4, 6, 8	254	60	300	314	265	265	213	213	190	92	210	51	95[2)]	300[4)]	37	60	30	108	–	160	18	85
160 L	All	2, 4, 6, 8	254	60	300	314	265	265	213	213	190	92	254	51	95[2)]	300	37	60	30	108	–	160	18	85

* This dimension is assigned in DIN EN 50347 to the frame size listed.

[1)] With screwed-on feet, dimension BA' is 43 mm.

[2)] With screwed-on feet, dimension BA' is 51 mm.

[3)] With screwed-on feet, dimension BB is 180 mm.

[4)] With screwed-on feet, dimension BB is 256 mm.

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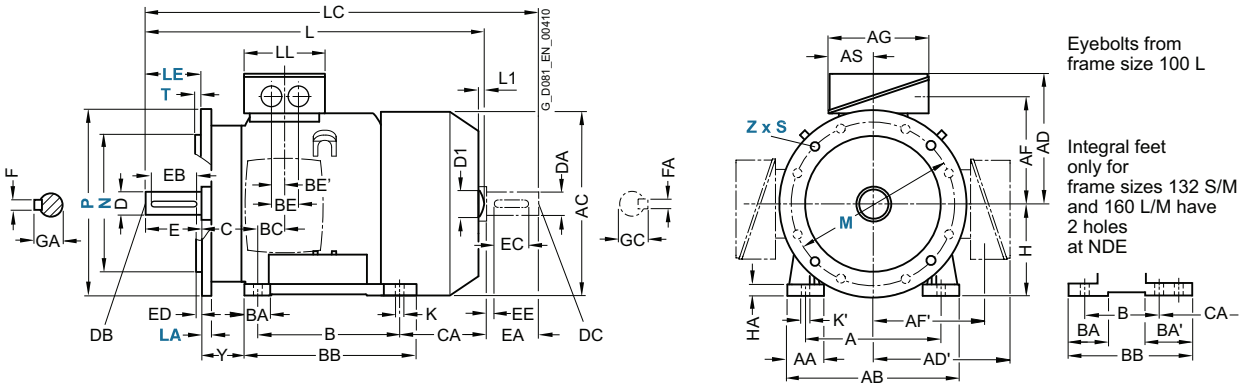
Dimensional drawings (continued)

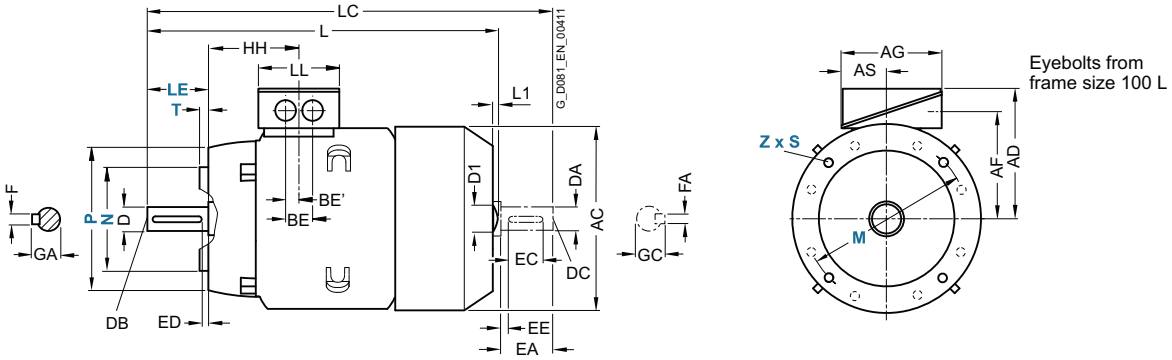
Type of construction IM B35

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



Type of construction IM B14

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**								DE shaft extension						NDE shaft extension							
Frame size	Motor type 1LE15.1, 1LE16.1	No. of poles	HH	K	K'	L[1]	L1[2]	D1	LC	LL	D	DB	E	EB	ED	F	GA	DA	DC	EA	EC	EE	FA	GC
71 M	1LE15.1	2, 4, 6	64.5	7	7	240	–	–	278	102	14	M5	30	22	4	5	16	14	M5	30	16	4	5	16
80 M	1LE15.1	2, 4, 6	71.5	10	10	292	–	–	343	102	19	M6	40	32	4	6	21.5	19	M6	40	22	4	6	21.5
90 S	1LE15.1	2, 4, 6	79.5	10	10	347	–	–	405	102	24	M8	50	40	5	8	27	19	M6	40	32	4	6	21.5
90 L	1LE15.1	2, 4, 6	79.5	10	10	347	–	–	405	102	24	M8	50	40	5	8	27	19	M6	40	32	4	6	21.5
100 L	All	2, 4, 6, 8	100.5	12	16	388.5	7	32	454	134	28	M10	60	50	5	8	31	24	M8	50	40	5	8	27
112 M	All	2, 4, 6, 8	100.5	12	16	382	7	32	450	134	28	M10	60	50	5	8	31	24	M8	50	40	5	8	27
132 S	All	2, 4, 6, 8	115.5	12	16	456.5	8.5	39	535.5	134	38	M12	80	70	5	10	41	28	M10	60	50	5	8	31
132 M	All	2, 4, 6, 8	115.5	12	16	456.5	8.5	39	535.5	134	38	M12	80	70	5	10	41	28	M10	60	50	5	8	31
160 M	All	2, 4, 6, 8	145	15	19	594	10	45	730	165	42	M16	110	90	10	12	45	42	M16	110	90	10	12	45
160 L	All	2, 4, 6, 8	145	15	19	594	10	45	730	165	42	M16	110	90	10	12	45	42	M16	110	90	10	12	45

[1] For 1LE15 motors, plus dimension L1.

[2] Only for 1LE15 motors.

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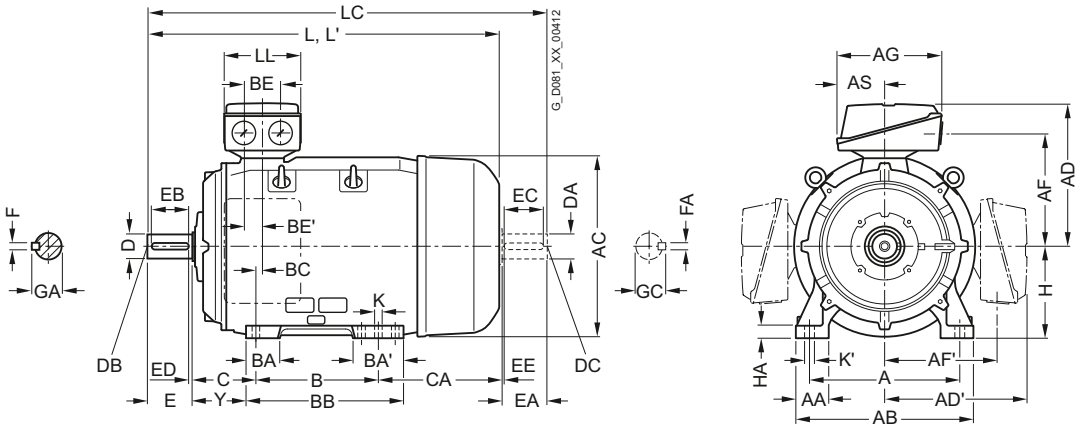
Dimensions

Cast-iron series 1LE1501, 1LE1521, 1LE1601, 1LE1621
Self-ventilated, frame sizes 180 M to 250 M

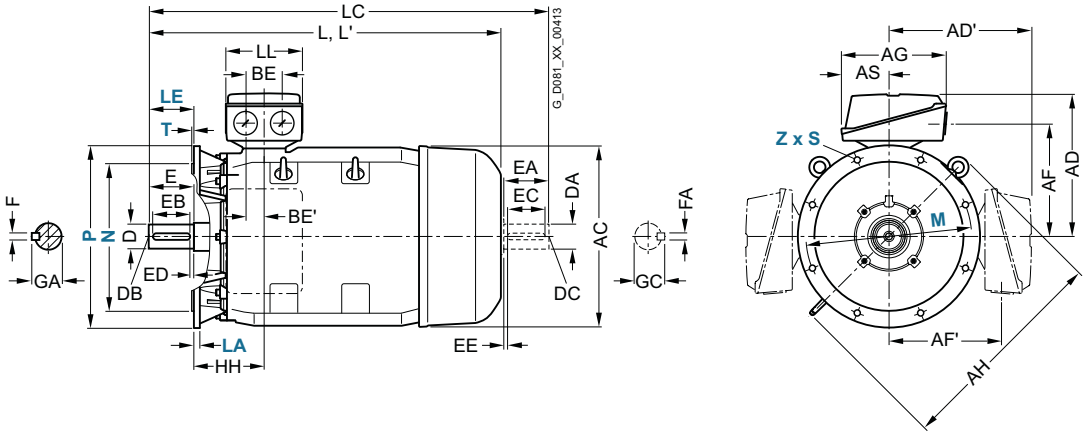
Dimensional drawings

Type of construction IM B3



Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																		
Frame size	Motor type 1LE1501-, 1LE1521- 1LE1601-, 1LE1621-	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AH	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*
180 M/ 180 L	1EA2, 1EB2, 1EC6	2, 4, 6	279	65	339	356	286	286	234	234	190	468	92	241	85	120	328	34	60	30	121	202
	1EB4, 1EC4, 1EA6, 1EB6	2, 4, 6												279								
200 L	2AA4,2AA5, 2AB5, 2AC4,2AC5	2, 4, 6	318	70	378	396	315	315	259	259	266	533	112	305	104	104	355	31	85	42.5	133	177
	2AA6, 2AB6, 2AC6	2, 4, 6																				
225 S/ 225 M	2BB0, 2BD0,	4, 8	356	80	436	449	338	338	282	282	266	556	112	311	92	117	361	15	85	42.5	149	253
	2BB2, 2BC2, 2BD2, 2BB6, 2BC6, 2BD6	4, 6, 8																				
	2BA2, 2BA6	2																				
250 M	2CA2, 2CA6	2	406	100	490	497	410	410	322	322	319	620	145	349	102	102	409	24	110	55	168	230
	2CB2, 2CC2, 2CD2, 2CC6, 2CD6,	4, 6, 8																				
	2CB6	4																				300

* This dimension is assigned in DIN EN 50347 to the frame size listed.

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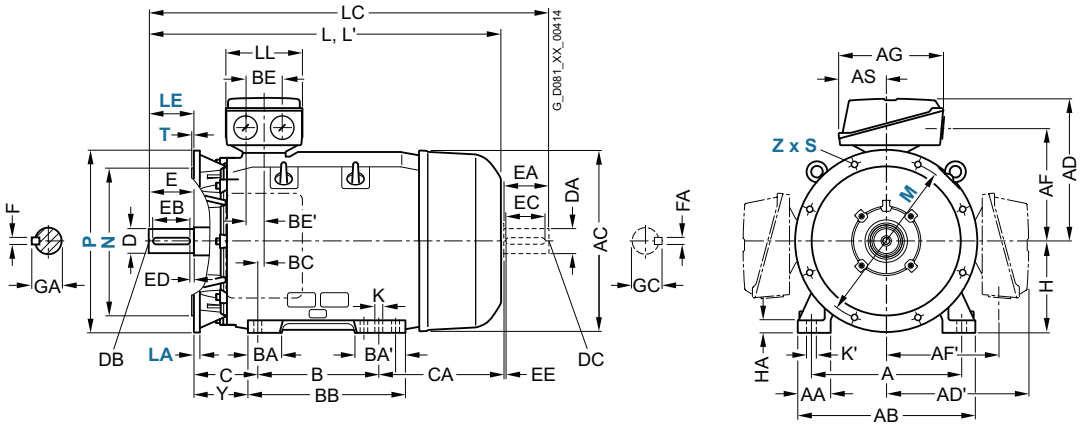
Dimensional drawings (continued)

Type of construction IM B35

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



Motor type 1LE1501-, 1LE1521- 1LE1601-, 1LE1621-	H	HA	Y	HH	K	K'	L	L' [1]	LC [2]	LL	DE shaft extension D	DB	E	EB	ED	F	GA	NDE shaft extension DA	DC	EA	EC	EE	FA	GC
1EA2, 1EB2, 1EC6	180	20	95	155	15	19	668	668	784	165	48	M16	110	100	5	14	52	48	M16	110	100	5	14	51.5
1EB4, 1EC4, 1EA6, 1EB6							698	698	814															
2AA4,2AA5, 2AB5, 2AC4,2AC5	200	25	108	164	19	25	721	755	835	197	55	M20	110	100	5	16	59	55	M20	110	100	5	16	59
2AA6, 2AB6, 2AC6							746	780	860															
2BB0, 2BD0	225	34	124	164	19	25	788	–	903	197	60	M20	140	125	10	18	64	55	M20	110	100	5	16	59
2BB2, 2BC2, 2BD2, 2BB6, 2BC6, 2BD6							848		963															
2BA2, 2BA6							818	852	933		55		110	100	5	16	59	48	M16				14	51.5
2CA2, 2CA6	250	40	138	192	24	30	887	924	1002	233	60	M20	140	125	10	18	64	55	M20	110	100	5	16	59
2CB2, 2CC2, 2CD2, 2CC6, 2CD6								–	1032		65						69	60		140	125	10	18	64
2CB6							957		1072															

[1] For version with low-noise fan for 2-pole motors.

[2] In the low-noise version, a second shaft extension and/or mounted encoder is not possible.

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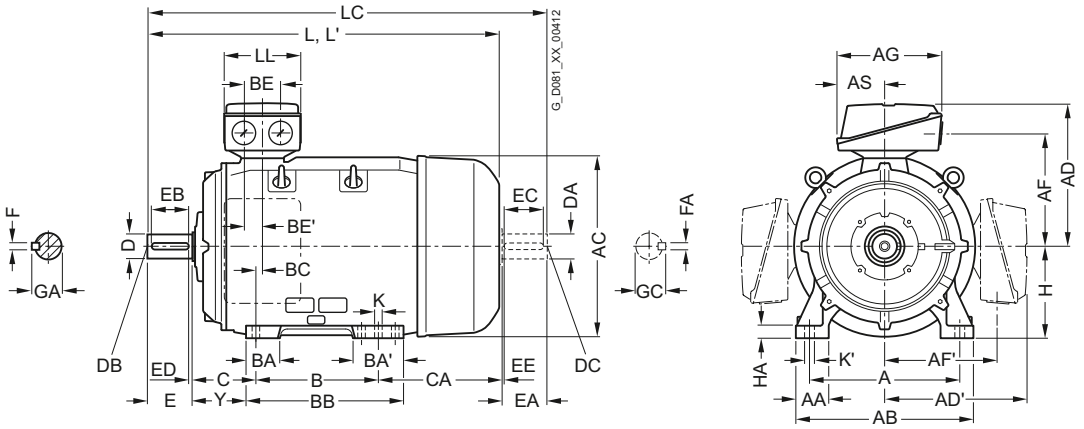
Dimensions

Cast-iron series 1LE1501, 1LE1521, 1LE1601, 1LE1621
Self-ventilated, frame sizes 280 S to 315 L

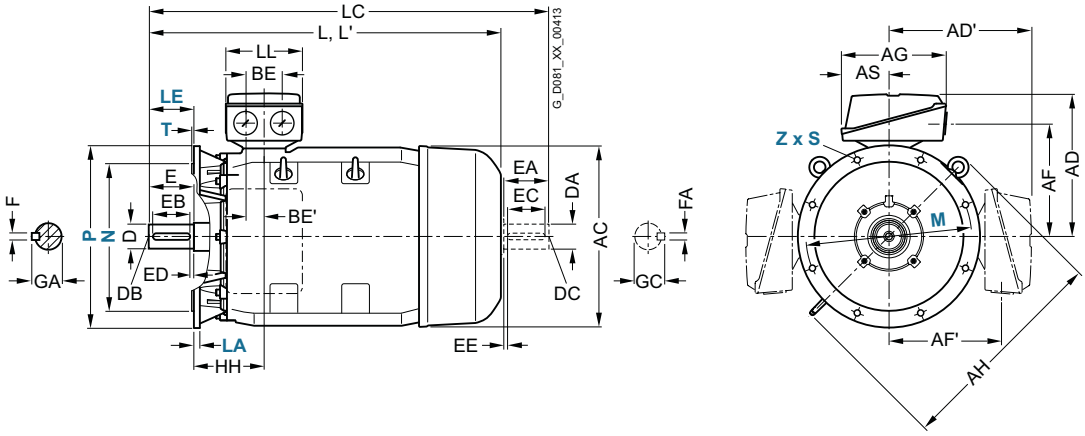
Dimensional drawings

Type of construction IM B3



Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																			
Frame size	Motor type 1LE1501-, 1LE1521- 1LE1601-, 1LE1621-	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AH	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*
280 S	2DA0	2	457	100	540	551	433	433	345	345	319	672	145	368	101	152	479	20	110	55	190	267
	2DB0, 2DC0, 2DD0	4, 6, 8												368								267
280 M	2DA6	2												419								326
	2DA2																					216
	2DB2, 2DC2, 2DD2, 2DC6, 2DD6	4, 6, 8																				
	2DB6	4																				326
315 S	3AA0	2	508	120	610	616	515	515	404	404	374	780	164	406	113	170	527	22	110	55	216	295
	3AB0, 3AC0, 3AD0	4, 6, 8																				
315 M	3AA2[1)]	2												457			578					409
	3AB2[1)]	4																				
	3AC2, 3AD2	6, 8												406			527					244
315 L[1)]	3AA4	2												508			578					358
	3AB4, 3AC4, 3AD4, 3AC5, 3AD5, 3AD6	4, 6, 8																				
	3AA5, 3AA6	2												508	176	227	648					513
	3AB5, 3AB6, 3AC6	4, 6																				

* This dimension is assigned in DIN EN 50347 to the frame size listed.

[1)] With terminal box position right, terminal box left, and with order code **H01** only screwed-on feet with 3 drilled holes with dimension "B" (406, 457 and 508 mm). The dimension "BB" will then be 666 mm.

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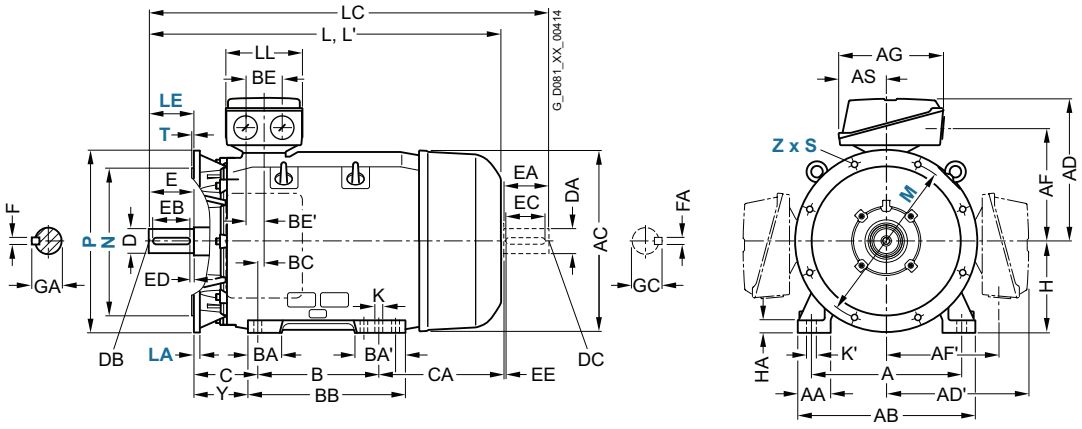
Dimensional drawings (continued)

Type of construction IM B35

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



Motor type 1LE1501-, 1LE1521- 1LE1601-, 1LE1621-	H	HA	Y	HH	K	K'	L	L' [1]	LC [2]	LL	DE shaft extension D	DB	E	EB	ED	F	GA	NDE shaft extension DA	DC	EA	EC	EE	FA	GC
2DA0	280	40	160	210	24	30	960	998	1105	233	65	M20	140	125	10	18	69	60	M20	140	125	10	18	64
2DB0, 2DC0, 2DD0								–			75					20	79.5	65						69
2DA6							1070	1108	1215		65					18	69	60						64
2DA2							960	998	1105															
2DB2, 2DC2, 2DD2, 2DC6, 2DD6								–			75					20	79.5	65						69
2DB6							1070		1215															
3AA0	315	50	181	238	28	35	1052	1122	1197	299	65	M20	140	125	10	18	69	60	M20	140	125	10	18	64
3AB0, 3AC0, 3AD0							1082	–	1227		80		170	140	25	22	85	70					20	74.5
3AA2							1217	1287	1362		65		140	125	10	18	69	60					18	64
3AB2							1247	–	1392		80		170	140	25	22	85	70					20	74.5
3AC2, 3AD2							1082		1227															
3AA4							1217	1287	1362		65		140	125	10	18	69	60					18	64
3AB4, 3AC4, 3AD4, 3AC5, 3AD5, 3AD6							1247	–	1392		80		170	140	25	22	85	70					20	74.5
3AA5, 3AA6			146				1372	1442	1517		65		140	125	10	18	69	60					18	64
3AB5, 3AB6, 3AC6							1402	–	1547		80		170	140	25	22	85	70					20	74.5

[1] For version with low-noise fan for 2-pole motors.

[2] In the low-noise version, a second shaft extension and/or mounted encoder is not possible.

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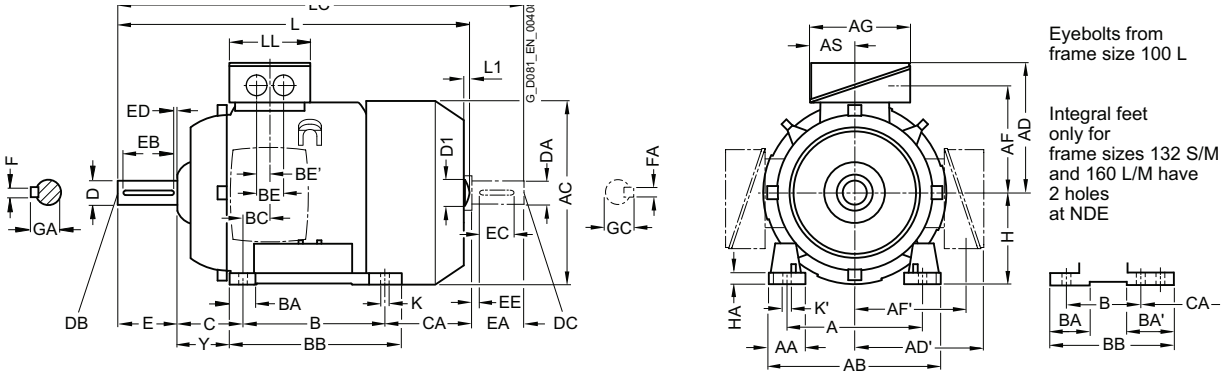
Dimensions

Cast-iron series 1LE1503, 1LE1523, 1LE1603, 1LE1623
Self-ventilated, frame sizes 71 M to 160 L

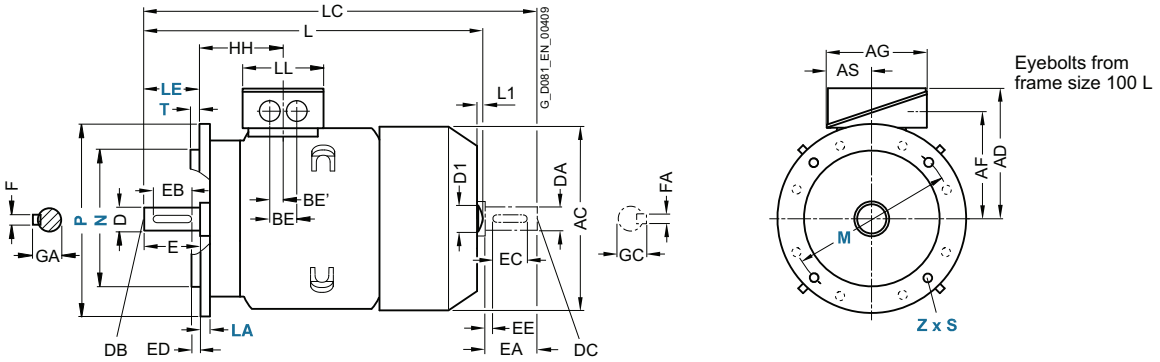
Dimensional drawings

Type of construction IM B3



Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																				
Frame size	Motor type 1LE15.3- 1LE16.3-	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*	H	HA	Y
71 M	1LE15.3- 0..0, 0..2 0..3, 0..4	2, 4, 6	112	27	132	138.5	148	148	112	112	126	62	90	–	–	106	20.5	36	18	45	–	71	7	37
80 M	1LE15.3- 0..0, 0..2 0..3, 0..4	2, 4, 6	125	27	150	156	158	158	122	122	129	62	100	–	–	118	21.5	36	18	50	–	80	8.5	41
90 S	1LE15.3- 0..0, 0..2 0..3, 0..4	2, 4, 6	140	30.5	165	173.5	163	163	127	127	129	62	100	–	–	143	23.5	36	18	56	–	90	11	47
90 L	1LE15.3	2, 4, 6	140	30.5	165	173.5	163	163	127	127	129	62	125	–	–	143	23.5	36	18	56	–	90	11	47
100 L	All	2, 4, 6	160	42	196	198	193	193	147	147	163	80.5	140	40	–	176	37.5	48	24	63	176	100	12	45
112 M	All	2, 4, 6	190	46	226	222	195	195	150	150	163	80.5	140	40	–	176	30	48	24	70	155	112	12	52
132 S	1CA0, 1CC0	2, 6	216	53	256	262	214.5	214.5	169	169	163	80.5	140	44	81[1]	218[2]	26.5	48	24	89	128.5	132	15	69
	1CA1, 1CB0	2, 4													–						178.5			
132 M	1CC2	6	216	53	256	262	214.5	214.5	169	169	163	80.5	178	44	81[1]	218	26.5	48	24	89	128.5	132	15	69
	1CB2, 1CC3	4, 6, 8													–						178.5			
160 M	All	2, 4, 6	254	60	300	314	261	261	213	213	190	92	210	51	95[3]	300[4]	37	60	30	108	148	160	18	85
160 L	All	2, 4, 6	254	60	300	314	261	261	213	213	190	92	254	51	95[3]	300	37	60	30	108	208	160	18	85

* This dimension is assigned in DIN EN 50347 to the frame size listed.

[1] With screwed-on feet, dimension BA' is 43 mm.

[2] With screwed-on feet, dimension BB is 180 mm.

[3] With screwed-on feet, dimension BA' is 51 mm.

[4] With screwed-on feet, dimension BB is 256 mm.

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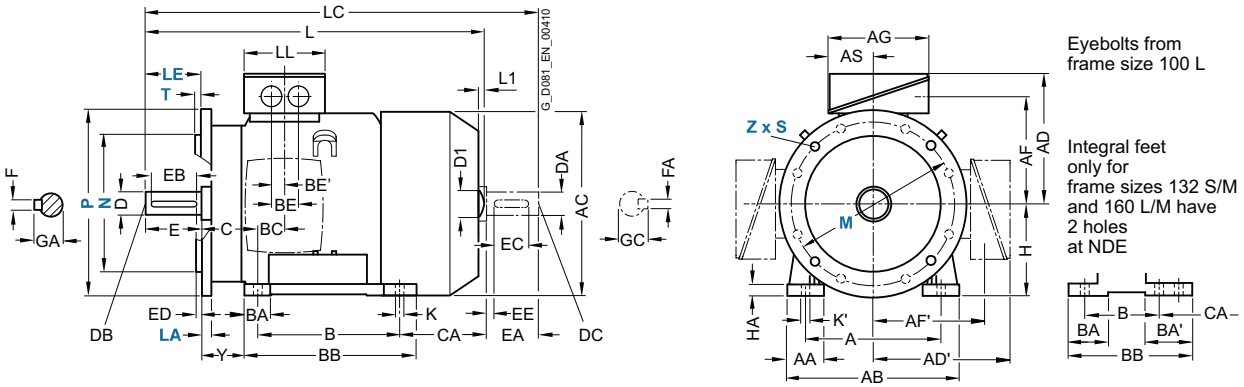
Dimensional drawings (continued)

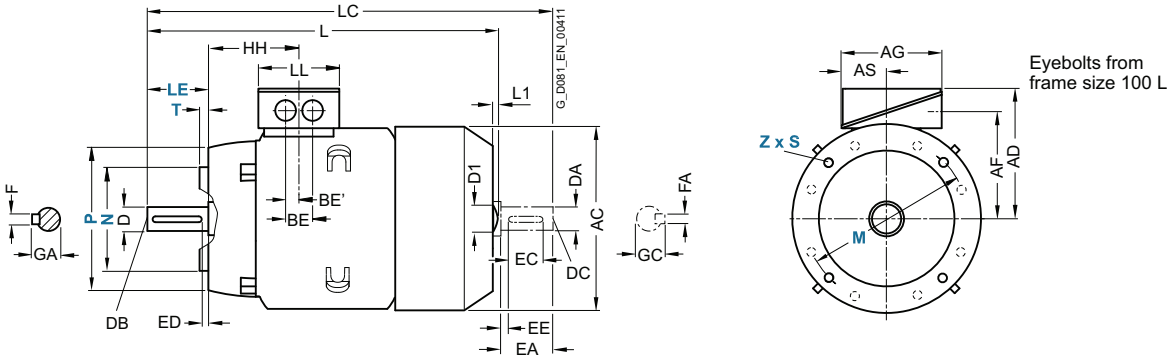
Type of construction IM B35

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



Type of construction IM B14

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**								DE shaft extension							NDE shaft extension						
Frame size	Motor type 1LE15.3- 1LE16.3-	No. of poles	HH	K	K'	L[1]	L1[2]	D1	LC	LL	D	DB	E	EB	ED	F	GA	DA	DC	EA	EC	EE	FA	GC
71 M	1LE15.3- 0..0, 0..2 0..3, 0..4	2, 4, 6	64.5	7	7	240 280	–	–	278 318	102	14	M5	30	22	4	5	16	14	M5	30	16	4	5	16
80 M	1LE15.3- 0..0, 0..2 0..3, 0..4	2, 4, 6	71.5	10	10	292 327	–	–	343 378	102	19	M6	40	32	4	6	21.5	19	M6	40	16	4	6	21.5
90 S	1LE15.3- 0..0, 0..2 0..3, 0..4	2, 4, 6	79.5	10	10	347 387	–	–	405 445	102	24	M8	50	40	5	8	27	19	M6	40	27	4	6	21.5
90 L	1LE15.3	2, 4, 6	79.5	10	10	387	–	–	445	102	24	M8	50	40	5	8	27	19	M6	40	27	4	6	21.5
100 L	All	2, 4, 6	100.5	12	16	425	7	32	489	134	28	M10	60	50	5	8	31	24	M8	50	40	5	8	27
112 M	All	2, 4, 6	100.5	12	16	408.5	7	32	475	134	28	M10	60	50	5	8	31	24	M8	50	40	5	8	27
132 S	1CA0, 1CC0	2, 6	115.5	12	16	458	8.5	39	535.5	134	38	M12	80	70	5	10	41	28	M10	60	50	5	8	31
	1CA1, 1CB0	2, 4				508			585.5															
132 M	1CC2	6	115.5	12	16	458	8.5	39	535.5	134	38	M12	80	70	5	10	41	28	M10	60	50	5	8	31
	1CB2, 1CC3	4, 6				508			585.5															
160 M	All	2, 4, 6	145	15	19	596	10	45	730	165	42	M16	110	90	10	12	45	42	M16	110	90	10	12	45
160 L	All	2, 4, 6	145	15	19	656	10	45	790	165	42	M16	110	90	10	12	45	42	M16	110	90	10	12	45

[1] For 1LE15 motors, plus dimension L1.

[2] Only for 1LE15 motors.

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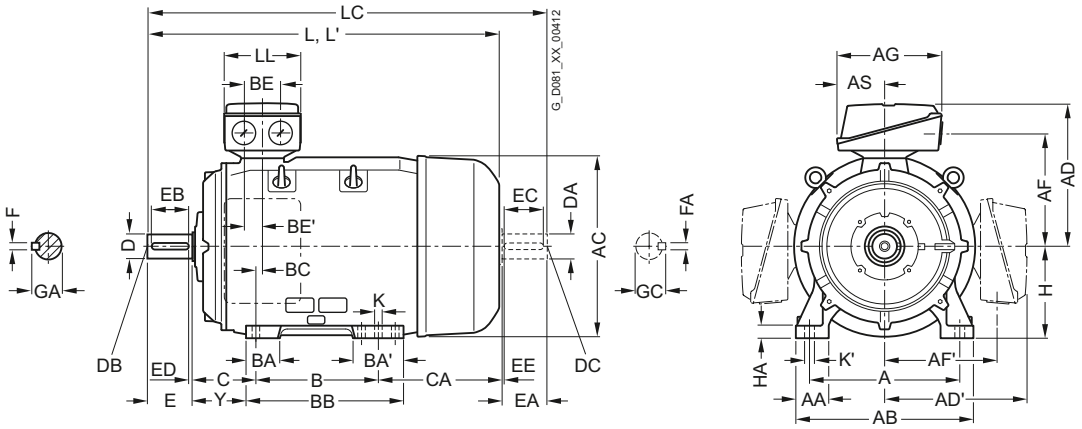
Dimensions

Cast-iron series 1LE1503, 1LE1523, 1LE1603, 1LE1623
Self-ventilated, frame sizes 180 M to 315 L

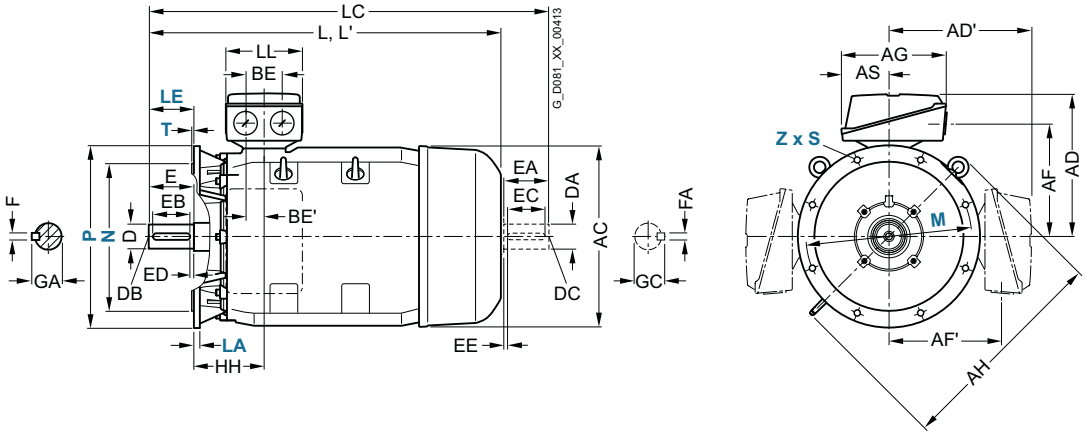
Dimensional drawings

Type of construction IM B3



Types of construction IM B5 and IM V1

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



For motor			Dimension designation acc. to **IEC**																			
Frame size	Motor type 1LE1503-, 1LE1523- 1LE1603-, 1LE1623-	No. of poles	A	AA	AB	AC	AD	AD'	AF	AF'	AG	AH	AS	B*	BA	BA'	BB	BC	BE	BE'	C	CA*
180 M/ 180 L	1EB2, 1EC4 1EA2, 1EB4	4, 6 2, 4	279	65	339	356	286	286	234	234	190	468	92	241	85	120	328	34	60	30	121	202
200 L	2AA4, 2AC4 2AA5. 2AB5.2AC5	2, 6 2, 4, 6	318	70	378	396	315	315	259	259	266	533	112	305	104	104	355	31	85	42.5	133	177
225 S 225 M	2BB0 2BA2 2BB2, 2BC2	4 2 4, 6	356 356	80 80	436 436	449 449	338 338	338 338	282 282	282 282	266 266	556 556	112 112	286 311	92 92	117 117	361 361	15 15	85 85	42.5 42.5	149 149	218 253
250 M	2CA2 2CB2, 2CC2	2 4, 6	406	100	490	497	410	410	322	322	319	620	145	349	102	102	409	24	110	55	168	230
280 S	2DA0 2DB0, 2DC0	2 4, 6	457	100	540	551	433	433	345	345	319	672	145	368	101	152	479	20	110	55	190	267
280 M	2DC2 2DA2 2DB2	6 2 4	457	100	540	551	433	433	345	345	319	672	145	419	101	152	479	20	110	55	190	216 326
315 S	3AA0 3AB0, 3AC0	2 4, 6	508	120	610	616	515	515	404	404	374	780	164	406	113	170	527	22	110	55	216	295
315 M[1)]	3AA2 3AB2, 3AC2	2 4, 6	508	120	610	616	515	515	404	404	374	780	164	457	113	170	578	22	110	55	216	409
315 L[1)]	3AA4 3AB4, 3AC4 3AA5 3AB5, 3AC5, 3AC6	2 4, 6 2 4, 6	508	120	610	616	515	515	404	404	374	780	164	508	113 176	170 227	578 648	22	110	55	216	358 513

* This dimension is assigned in DIN EN 50347 to the frame size listed.

[1)] With order codes for terminal box positions (**K05**, **K06**, **H01**) only screwed-on feet with 3 drilled holes with dimension "B" (406, 457 and 506 mm). The dimension "BB" will then be 666 mm.

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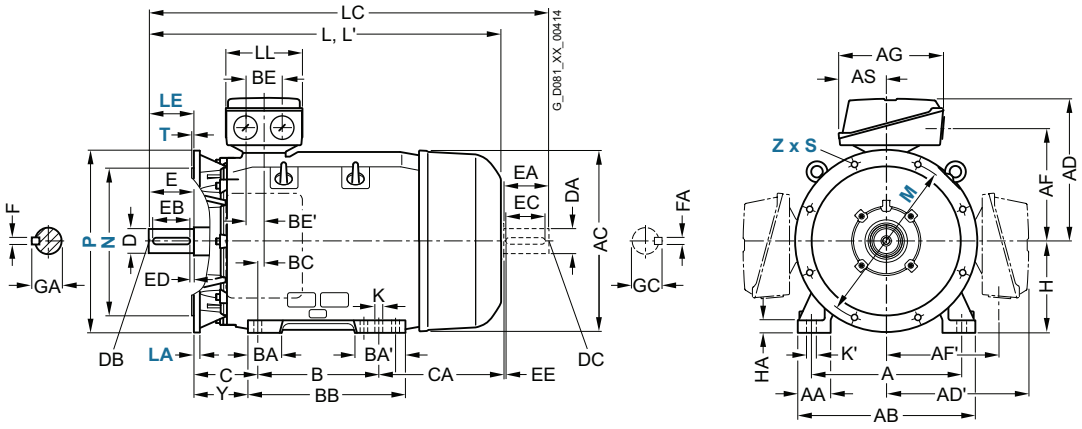
Dimensional drawings (continued)

Type of construction IM B35

For flange dimensions, see Page 2/94 (**Z** = the number of retaining holes)



Motor type 1LE1503-, 1LE1523- 1LE1603-, 1LE1623-	H	HA	Y	HH	K	K'	L	L' [1]	LC[2]	LL	DE shaft extension D	DB	E	EB	ED	F	GA	NDE shaft extension DA	DC	EA	EC	EE	FA	GC
1EB2, 1EC4	180	20	95	155	15	19	668	668	784	165	48	M16	110	100	5	14	52	48	M16	110	100	5	14	51.5
1EA2, 1EB4							698	698	814															
2AA4, 2AC4	200	25	108	164	19	25	721	755	835	197	55	M20	110	100	5	16	59	55	M20	110	100	5	16	59
2AA5. 2AB5.2AC5							746	780	860															
2BB0	225	34	124	164	19	25	788	–	903	197	60	M20	140	125	10	18	64	55	M20	110	100	5	16	59
2BA2	225	34	124	164	19	25	818	852	933	197	55	M20	110	100	5	16	59	48	M16	110	100	5	14	51.5
2BB2, 2BC2							848	–	963		60		140	125	10	18	64	55	M20				16	59
2CA2	250	40	138	192	24	30	887	924	1002	233	60	M20	140	125	10	18	64	55	M20	110	100	5	16	59
2CB2, 2CC2								–	1032		65						69	60		140	125	10	18	64
2DA0	280	40	160	210	24	30	960	998	1105	233	65	M20	140	125	10	18	69	60	M20	140	125	10	18	64
2DB0, 2DC0								–			75					20	79.5	65						69
2DC2	280	40	160	210	24	30	960	–	1105	233	75	M20	140	125	10	20	79.5	65	M20	140	125	10	18	69
2DA2							1070	1108	1215		65					18	69	60						64
2DB2								–			75					20	79.5	65						69
3AA0	315	50	181	238	28	35	1052	1122	1197	299	65	M20	140	125	10	18	69	60	M20	140	125	10	18	64
3AB0, 3AC0							1082	–	1227		80		170	140	25	22	85	70					20	74.5
3AA2	315	50	181	238	28	35	1217	1287	1362	299	65	M20	140	125	10	18	69	60	M20	140	125	10	18	64
3AB2, 3AC2							1247	–	1392		80		170	140	25	22	85	70					20	74.5
3AA4	315	50	181	238	28	35	1217	1287	1362	299	65	M20	140	125	10	18	69	60	M20	140	125	10	18	64
3AB4, 3AC4							1247	–	1392		80		170	140	25	22	85	70					20	74.5
3AA5			146				1372	1442	1517		65		140	125	10	18	69	60					18	64
3AB5, 3AC5, 3AC6							1402	–	1547		80		170	140	25	22	85	70					20	74.5

[1] For version with low-noise fan for 2-pole motors.

[2] In the low-noise version, a second shaft extension and/or mounted encoder is not possible.

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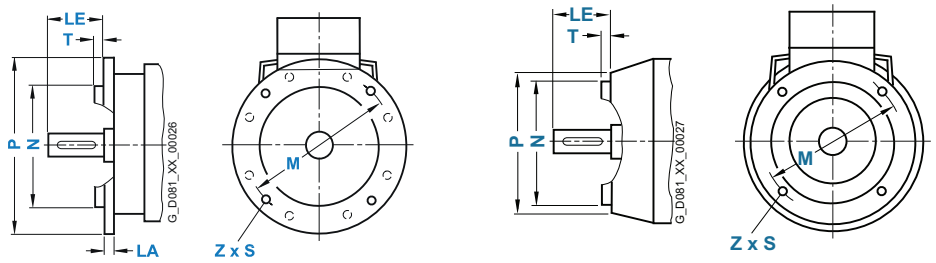
Dimensions

Flange dimensions

Dimensional drawings



In DIN EN 50347, the frame sizes are allocated flange FF with through holes and flange FT with tapped holes.
The designation of flange A and C according to DIN 42948 (invalid since September 2003) are also listed for information purposes. See the table below.
(**Z** = the number of retaining holes)

Frame size	Type of construction	Flange type	Flange with through holes (**FF**/A) tapped holes (**FT**/C) acc. to DIN EN 50347	acc. to DIN 42948	Dimension designation acc. to **IEC** LA	LE	M	N	P	S	T	Z
71 M	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF 130**	A 160	9	30	130	110	160	10	3.5	4
	IM B14, IM B34, IM V18, IM V19	Standard flange	**FT 85**	C 105	–	30	85	70	105	M6	2.5	4
80 M	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF 165**	A 200	10	40	165	130	200	12	3.5	4
	IM B14, IM B34, IM V18, IM V19	Standard flange	**FT 100**	C 120	–	40	100	80	120	M6	3	4
90 S, 90 L	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF 165**	A 200	10	50	165	130	200	12	3.5	4
	IM B14, IM B34, IM V18, IM V19	Standard flange	**FT 115**	C 140	–	50	115	95	140	M8	3	4
100 L	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF 215**	A 250	11	60	215	180	250	14.5	4	4
	IM B5, IM B35, IM V1, IM V3	Next larger standard flange – Order code **P01**	**FF 265**	A 300	12	60	265	230	300	14.5	4	4
	IM B5, IM B35, IM V1, IM V3	Next smaller standard flange – Order code **P02**	**FF 165**	A 200	11	60	165	130	200	12	3.5	4
	IM B14, IM B34, IM V18, IM V19	Standard flange	**FT 130**	C 160	–	60	130	110	160	M8	3.5	4
	IM B14, IM B34, IM V18, IM V19	Next larger standard flange – Order code **P01**	**FT 165**	C 200	–	60	165	130	200	M10	3.5	4
112 M	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF 215**	A 250	11	60	215	180	250	14.5	4	4
	IM B5, IM B35, IM V1, IM V3	Next larger standard flange – Order code **P01**	**FF 265**	A 300	12	60	265	230	300	14.5	4	4
	IM B5, IM B35, IM V1, IM V3	Next smaller standard flange – Order code **P02**	**FF 165**	A 200	11	60	165	130	200	12	3.5	4
	IM B14, IM B34, IM V18, IM V19	Standard flange	**FT 130**	C 160	–	60	130	110	160	M8	3.5	4
	IM B14, IM B34, IM V18, IM V19	Next larger standard flange – Order code **P01**	**FT 165**	C 200	–	60	165	130	200	M10	3.5	4
132 S, 132 M	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF 265**	A 300	12	80	265	230	300	14.5	4	4
	IM B5, IM B35, IM V1, IM V3	Next larger standard flange – Order code **P01**	**FF 300**	A 350	13	80	300	250	350	18.5	5	4
	IM B5, IM B35, IM V1, IM V3	Next smaller standard flange – Order code **P02**	**FF 215**	A 250	11	80	215	180	250	14.5	4	4
	IM B14, IM B34, IM V18, IM V19	Standard flange	**FT 165**	C 200	–	80	165	130	200	M10	3.5	4
	IM B14, IM B34, IM V18, IM V19	Next larger standard flange – Order code **P01**	**FT 215**	C 250	–	80	215	180	250	M12	4	4
160 M, 160 L	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF 300**	A 350	13	110	300	250	350	18.5	5	4
	IM B5, IM B35, IM V1, IM V3	Next smaller standard flange – Order code **P02**	**FF 265**	A 300	12	110	265	230	300	14.5	4	4
	IM B14, IM B34, IM V18, IM V19	Standard flange	**FT 215**	C 250	–	110	215	180	250	M12	4	4
180 M, 180 L	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF300**	A350	13	110	300	250	350	18.5	5	4
	IM B5, IM B35, IM V1, IM V3	Next smaller standard flange – Order code **P02**	**FF 265**	A 300	12	110	265	230	300	14.5	4	4
200 L	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF350**	A400	15	110	350	300	400	18.5	5	4
	IM B5, IM B35, IM V1, IM V3	Next smaller standard flange – Order code **P02**	**FF300**	A350	13	110	300	250	350	18.5	5	4
225 S, 225 M 2-pole 4-pole to 8-pole	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF400**	A450	16	110 140	400	350	450	18.5	5	8
250 M	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF500**	A550	18	140	500	450	550	18.5	5	8
280 S, 280 M	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF500**	A550	18	140	500	450	550	18.5	5	8
315 S, 315 M, 315 L 2-pole 4-pole to 8-pole	IM B5, IM B35, IM V1, IM V3	Standard flange	**FF600**	A660	22	140 170	600	550	660	24	6	8